MASSLESS PHOTONS AND MAXWELL INFRARED LIMITS IN THREE-DIMENSIONAL QED

Abstract. We prove further properties of the continuum curvature laws constructed from noncompact lattice QED with chamber fermions. At any fixed charge, assume that the positive lattice masses are eventually at most one and bounded below by a fixed positive multiple of the spacing. A meson correlator estimate then gives a positive massless photon weight in almost every translation-ergodic component of every volume-first limit point. The full infrared law of each such component converges to a free Maxwell law. The previously established Osterwalder–Schrader axioms already yield a local relativistic field; we give a direct realization with quantitative energy and analytic-continuation bounds. We verify the mass hypothesis for the tuned construction at sufficiently small charge. Combined with its previously proved positive fourth cumulant, this gives non-Gaussian component theories with nondegenerate Maxwell infrared limits.

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Introduction

Let $P_{N,T}$ be the all-flux curvature laws of the chamber construction of [4], with standard weights, Wilson parameter $0 < \rho < 1$, mass parameter $\bar{m} \in (0, 1]$, spacing $\eta = L^{-N}$ and torus side $8hL^T$, where L is its fixed odd scale factor and $h = L^{\lfloor \log_L \bar{m}^{-1} \rfloor}$. A mass assignment is a real sequence $m_{0,N}$ independent of T; its lattice mass is $\boldsymbol{\mu}_N = \bar{m}\eta + m_{0,N}$. Write Π_c for the orthogonal projection of $L^2(\mathbb{R}^3; \Lambda^2)$ onto the closed forms. Existence of the limit points and the properties in clause (i) below are recalled from [4, Section 6, Section 7.5, and Theorem 8.6].

Theorem A. Fix $e > 0$ and a mass assignment for which, with a constant $c_{\boldsymbol{\mu}} > 0$,

$$c_{\boldsymbol{\mu}} \bar{m} L^{-N} \le \boldsymbol{\mu}_N \le 1 \qquad \text{for all sufficiently large } N.$$

Volume-first limit points exist. Let μ be any such limit point: $\mu = \lim_i \mu^{(N_i)}$ weakly on the Hermite–Sobolev space H_{-4}, where $N_i \to \infty$ and each $\mu^{(N_i)}$ is a weak limit of $P_{N_i,T}$ as $T \to \infty$. Let $\mu = \int \mu_\xi \, d\nu(\xi)$ be its decomposition under translations of $\mathbb{R}^3$. There is a measurable set of full ν-measure on which every law $\lambda = \mu_\xi$ has the following properties.

(i) *Euclidean field and OS axioms.* The law λ is a probability measure on the real tempered distributional two-forms. Its field extends to a continuous linear map $X : L^2(\mathbb{R}^3; \Lambda^2) \to L^2(\lambda)$ and obeys

$$\lambda(\exp X(g)) \le \exp(\tfrac{1}{2}\|\Pi_{\mathrm{c}} g\|^2).$$

It is concentrated on $dX = 0$. Put $S_0 = 1$. For complex tensor Schwartz tests $\Psi \in \mathcal{S}(\mathbb{R}^{3n}; (\Lambda^2\mathbb{C}^3)^{\otimes n})$,

$$|S_n^\xi(\Psi)| \le 2\,66^n\sqrt{n!}\, q_{6n}(\Psi), \qquad q_j(\Psi) = \max_{|\alpha|\le j} \sup_y (1+|y|^2)^{j/2} |\partial^\alpha \Psi(y)|,$$

where the maximum includes tensor components. Its Schwinger functions are tempered, symmetric and Euclidean covariant under the two-form action $((a,R)\cdot g)(y) = (\wedge^2 R) g(R^{-1}(y-a))$; the law's characteristic functional extends to an entire function of every finite collection of test coefficients. For every plane with reflection r and either open half space O bounded by that plane,

$$\lambda(\overline{\widehat{r}A}\, A) \ge 0 \qquad (A \in L^2(\Sigma_O, \lambda)),$$

where Σ_O is generated by $X(g)$ with $g \in C_c^\infty(O; \Lambda^2)$ and $\widehat{r}X(g) = X(r\cdot g)$. For $A, B \in L^2(\lambda)$,

$$\lambda(\bar{A}\, U(a) B) \longrightarrow \overline{\lambda(A)}\lambda(B) \qquad (|a| \to \infty).$$

These are the regularity, covariance, positivity, symmetry and cluster axioms for the field strength; the tensor reflection is part of positivity.

(ii) *Photon weight and Fisher information.* The two-point function has a unique positive Källén–Lehmann measure kl_ξ on $[0,\infty)$, with the integration variable denoting mass, determined for real $g, g' \in L^2(\mathbb{R}^3; \Lambda^2)$ by

$$S_2^\xi(g,g') = \int_0^\infty \left\langle \Pi_{\mathrm{c}} g, \frac{-\Delta}{-\Delta + t^2} \Pi_{\mathrm{c}} g' \right\rangle d\mathrm{kl}_\xi(t).$$

Its total mass is at most one, and $z_\xi := \mathrm{kl}_\xi(\{0\}) > 0$. There are measurable constants C_ξ such that

$$z_\xi^{-1} \le C_\xi < \infty, \qquad \int C_\xi \, d\nu \le 1 + \tfrac{1}{2} c_F, \qquad \nu\{z_\xi \le s\} \le \frac{c_F s}{2(1-s)} \quad (0 < s < 1),$$

where

$$c_F = 15\pi^4 [3(1+\rho)]^4 C_\sharp(\rho) \frac{e^2}{c_{\boldsymbol{\mu}} \bar{m}} \max\left\{1, \frac{e^2}{c_{\boldsymbol{\mu}} \bar{m}}\right\}^3.$$

Here $C_\sharp(\rho) < \infty$ is the constant of Theorem 5.14. For every real closed $f \in L^2$,

$$z_\xi \|f\|^2 \le S_2^\xi(f,f) \le \|f\|^2, \qquad I_\xi(f) \le C_\xi \|f\|^2.$$

In this formula $I_\xi(f) = \sup_\Phi [2\lambda(\partial_f \Phi) - \lambda(\Phi^2)]$, where $\Phi = \phi(X(g_1), \dots, X(g_k))$, $g_j \in L^2$ real, $\phi \in C_b^\infty(\mathbb{R}^k)$ (all derivatives bounded), and $\partial_f \Phi = \sum_j \langle f, g_j \rangle \partial_j \phi$.

(iii) *Relativistic reconstruction.* Time is the first Euclidean coordinate. The reflection-positive reconstruction has a Hilbert space $\mathcal{H}$, a unique vacuum Ω, commuting translation generators (H, P_2, P_3) with $\mathrm{spec}(H, \vec{P}) \subset \{\omega \ge |\vec{p}|\}$, and a strongly continuous unitary representation $\mathcal{U}$ of $\mathbb{R}^{1+2} \rtimes SO_0(1,2)$. On the invariant dense domain $D = \bigcap_{j\ge 0} D(H^j)$ there is a tempered two-form field $F(h) : D \to D$ which is symmetric for real Schwartz tests, satisfies $dF = 0$, is covariant under $\mathcal{U}$, and commutes at spacelike separated supports for the metric $(x^0)^2 - (x^2)^2 - (x^3)^2$. Products of the field applied to Ω span a dense subspace. Its vacuum

functions analytically continue the Euclidean Schwinger functions, with a factor i for every time index.

Let E be the joint spectral measure of $(H, \vec{P})$ and put $\partial V = \{\omega = |\vec{p}|\}$ and $\mathcal{H}^{(1)} = \overline{\{F(h)\Omega : h \in \mathcal{S}(\mathbb{R}^{1+2}; \Lambda^2\mathbb{C}^3)\}}$. The subspace $\mathcal{H}_\gamma = E(\partial V)\mathcal{H}^{(1)}$ is invariant and irreducible under $\mathcal{U}$. It is unitarily identified with

$$L^2\left(\mathbb{R}^2, \frac{d^2k}{(2\pi)^2\, 2|k|}\right),$$

with $(H, \vec{P})$ acting as $(|k|, -k)$. If $B_0(h)$ is the magnetic field at time zero, as defined in Proposition 4.2, this identification sends $E(\partial V)[B_0(h)]$ to $z_\xi^{1/2}|k|\hat{h}(k)$. Thus $\operatorname{spec} H = [0, \infty)$ and the field produces a massless particle of multiplicity one.

(iv) *Infrared limit.* For every real $g \in L^2(\mathbb{R}^3; \Lambda^2)$, with $(\delta_l g)(y) = l^{-3/2} g(y/l)$,

$$\lambda(\exp iX(\delta_l g)) \longrightarrow \exp(-\tfrac{1}{2} z_\xi \|\Pi_{\mathrm{c}} g\|^2) \qquad (l \to \infty).$$

The limit is the free Maxwell law of weight z_ξ.

■□

The additional conclusions concern the positive photon weight and component Fisher bounds in (ii), the identified massless subspace in (iii), and the infrared limit of the full law in (iv). The local Wightman field associated with (i) is supplied by the Osterwalder–Schrader reconstruction theorem [2, Section IV.1], with the two-form covariance and reflection conventions used here; see also [1, Section 6] for nonscalar fields. Section 3 gives a direct realization on $D = \bigcap_m D(H^m)$, including the energy bound of Theorem 3.3(a) and the tube bounds of Theorem 3.3(f). These estimates also enter the proof of the infrared limit.

The cluster assertion concerns the components. The theorem gives an integral bound on the reciprocals of their photon weights; it does not give a positive lower bound independent of ξ. The massless subspace in (iii) is the one generated by a single field applied to the vacuum.

The mass hypothesis holds for every assignment $c_-\eta \le \boldsymbol{\mu}_N \le c_+\eta$ eventually, with $0 < c_- \le c_+ < \infty$. We call such an assignment physical. Here this term refers only to the two-sided bounds on the lattice mass. We prove these bounds for the tuned shifts of [4] in the following range and combine them with that construction's fourth-cumulant estimate.

Corollary B. Let (β, x_0, r_β) be data of Section F.2, let $f = dd^*\beta$ and $b(f) > 0$ be its one-loop coefficient, and let $e_2 > 0$ be the charge ceiling specified there. Use the tuned mass assignment. Put $e_7 = (\bar{m}/(4C_\natural))^{1/2}$, where $C_\natural$ is defined in Section 6. If $0 < e \le \min\{e_2, e_7\}$, Theorem A applies and, for every volume-first limit point and almost every component,

$$\kappa_4^\mu(f) \ge \tfrac{1}{2} e^4 b(f) > 0, \qquad \kappa_4^\xi(f) \ge \tfrac{1}{2} e^4 b(f) > 0, \qquad 0 < z_\xi < 1.$$

In particular these component theories are not Gaussian, while their infrared limits are nondegenerate free Maxwell laws. The tuned lattice masses obey $\frac{1}{2}\bar{m}\eta \le \boldsymbol{\mu}_N \le 2\bar{m}\eta$ for every $N \ge N_1(e)$, with $N_1(e) < \infty$. ■□

The proof. The preceding construction supplies the OS properties, but those properties alone do not ensure a positive massless photon weight. The estimate that yields this weight is a bound on the second spatial moment of the meson correlator in one box of size comparable with the inverse lattice mass. The determinant supplies a Ward identity, and the identity turns this moment into a bound on the Fisher information of curvature shifts. An eigenvalue count for the chamber operator removes the logarithmic loss in an earlier estimate. With a lattice mass bounded below by a positive multiple of the spacing, the Fisher bound survives the limit.

A bound for the limit law must still be transferred to its components. Invariant functions behave as constants under compactly supported field shifts. This gives a common score after disintegration.

Large dilates of one test then isolate the atom at zero in the spectral measure. To obtain a constant valid at all tests in almost every component, we approximate the indicators of invariant partition cells by weights at the cutoffs. The lattice clustering estimate controls their ergodic averages uniformly through the volume and spacing limits. The weighted inequality survives those limits; taking the supremum over finite partitions controls the mean of the component constants. This gives the integral estimate in Theorem A(ii).

The direct reconstruction in Sections 2 and 3 provides the quantitative realization used below. Products of the curvature field generate the probabilistic OS Hilbert space. Bounded insertions in a time slab give holomorphic point-field vectors with a bound near the boundary of the forward tube. Their boundary values define the Minkowski field on D with the energy estimate in Theorem 3.3(a). The construction also identifies the massless subspace produced by one field.

Finally, dilating an ergodic law sends its spectral measure to its atom at zero. To identify the law, rather than just its two-point function, we prove that an ergodic law in this class with a purely massless spectral measure is Gaussian. Reflection positivity makes the higher cumulant kernels curl free away from the diagonals. Their harmonic singularities are removed by the L^2 bounds, closedness and clustering. The contact-term argument then identifies the full law. Applied to subsequential limits of dilates, this proves (iv).

The general photon theorem uses the mass hypothesis in Theorem A and holds at every charge. For Corollary B, Section 6 verifies that hypothesis for the tuned masses. The positive fourth cumulant is imported from Section F.2; the corollary establishes that the same non-Gaussian components have the photon and infrared properties of Theorem A.

Notation and the preceding construction

All Euclidean test forms are two-forms on $\mathbb{R}^3$. The real L^2 norm is denoted by $\|\cdot\|$, and complexification is understood when taking moments as multilinear forms. The Hodge star identifies two-forms with vector fields; under this identification closed forms are divergence free and Π_c has Fourier multiplier $\Pi(p) = 1 - pp^T/|p|^2$. We use $\hat{f}(p) = \int e^{-ip\cdot y} f(y)\, dy$. The law of weight $0 \le a \le 1$ with characteristic functional $\exp(-a\|\Pi_c g\|^2/2)$ is denoted by γ_a; γ_0 is the point mass at zero, and $\gamma = \gamma_1$.

Write $(\mathcal{S}', \Sigma)$ for the space of tempered distributional two-forms with its evaluation σ-algebra, and $X(g)(\omega) = \omega(g)$. For a law λ, write $\Phi_\lambda(g) = \lambda(\exp iX(g))$ for its characteristic functional. Gaussian domination, denoted by (G), is the inequality

$$\lambda(e^{X(g)}) \le e^{\|g\|^2/2} \qquad (g \in \mathcal{S} \text{ real}).$$

It implies that X extends continuously from L^2 to $L^p(\lambda)$ for every finite $p \ge 1$, with a consistent complex-linear extension, and that $\lambda(X(g)) = 0$ [4, Proposition 5.1]. Condition (dF) means that $X(d^*u) = 0$ almost surely for every smooth compactly supported three-form u, equivalently that the law is concentrated on closed distributional two-forms. For complex tests put $S_0 = 1$ and

$$\begin{aligned} S_n(f_1, \dots, f_n) &= \lambda\left(\prod_{j=1}^{n} X(f_j)\right), \\ \kappa_n(f_1, \dots, f_n) &= \sum_{\pi} (-1)^{|\pi|-1}(|\pi|-1)! \prod_{B\in\pi} S_{|B|}((f_j)_{j\in B}), \end{aligned} \tag{1}$$

where π ranges over the partitions of $\{1, \dots, n\}$ into nonempty blocks. These cumulants are the derivatives at zero of $\log \lambda(\exp \sum_j z_j X(f_j))$; this logarithm is analytic when $\sum_j |z_j|\|f_j\| < 2/5$ [4, Section 5.2]. In particular $\kappa_2 = S_2$ and $\kappa_4(f) = S_4(f,f,f,f) - 3S_2(f,f)^2$. A law is Gaussian if each real $X(g)$ is normally distributed, allowing variance zero. A mixture $\int \lambda_\xi \, d\nu(\xi)$ means that $\xi \mapsto \lambda_\xi(E)$ is measurable for every $E \in \Sigma$ and that its integral is the probability of E. The translation of a test is $(\tau_a g)(y) = g(y-a)$, and $U(a)X(g) = X(\tau_a g)$. The action of a Euclidean map

on tests is the two-form action in Theorem A. Its induced operator on random variables is denoted by a hat. The local σ-algebra Σ_O is generated by the smooth tests supported in the open set O.

Write $O_{\mathrm{cub}} = O(3) \cap \mathrm{GL}(3,\mathbb{Z})$. Condition (H) consists of (G), translation and O_{cub} invariance, and reflection positivity at the three coordinate planes. For a reflection r bounding a half-space O, reflection positivity means

$$\sum_{p,q} \overline{c_p} c_q \, \lambda\Big(e^{i(X(f_q) - X(r\cdot f_p))}\Big) \geq 0 \tag{2}$$

for every finite family of complex c_p and real $f_p \in C_c^\infty(O;\Lambda^2)$. The class $\mathcal{C}_9$ consists of translation-invariant laws that are reflection invariant and reflection positive at every plane

$$\{y_i = c\} \quad (1 \leq i \leq 3), \qquad \{y_i \pm y_j = c\} \quad (1 \leq i < j \leq 3), \qquad c \in \mathbb{R}.$$

We set

$$\mathcal{K} = \{\lambda : \lambda \text{ satisfies (H), (dF), and belongs to } \mathcal{C}_9\}.$$

Every law in $\mathcal{K}$ is Euclidean invariant, reflection positive at every plane, and has the round spectral cone and the Källén–Lehmann representation of Theorem A(ii) [4, Propositions 7.1–7.3, Section 7.3, and Theorem 8.6], Proposition D.1, and Section B.2. Every volume-first limit point and almost every component belong to $\mathcal{K}$, at every charge and for every mass assignment [4, Section 6, Section 7.5, and Theorem 8.6]. An ergodic law in this class is mixing, with the local clustering estimate of Proposition C.1.

For these laws the reflection-positive reconstruction at $y_1 = 0$ is $(\mathcal{H}, [\,\cdot\,], H, P_2, P_3, \Omega)$, where $\langle [A], [B] \rangle = \lambda(\widehat{\overline{\theta_1 A}}\, B)$, $e^{-tH}[A] = [U(te_1)A]$ for $t \geq 0$, and spatial translations are $e^{i\vec{a}\cdot\vec{P}}$. In a picture with time axis e_j we use subscripts j. The symbols (T_{123}), (R_j) and (RP_j) mean, respectively, translation invariance, reflection invariance at $y_j = 0$, and positivity there. The one-plane hypotheses for (j,i) include (G), (dF), these invariances and positivity, the quarter turn in the (j,i) plane, and the spectral bound $H_j \geq |P_i|$, as specified explicitly in Section 2 below.

A probe is a real $\beta \in C_c^\infty(\mathbb{R}^3;\Lambda^2)$, with $a = d^*\beta$ and $f = dd^*\beta$; a quotient by $\|f\|$ always requires $f \neq 0$. In the tuned construction,

$$b(f) = -6\,\mathrm{Tr}\big[(S_{\bar m}\, i\gamma\cdot a)^4\big], \qquad S_{\bar m} = (\bar m + \gamma\cdot\partial)^{-1},$$

where $\gamma_j = \mathrm{diag}(\sigma_j, -\sigma_j)$ and σ_j are the Pauli matrices. The data of Corollary B are precisely the probe, separation point and support radius of Section F.2. The set Ξ_μ of tuned components and the exponent $\sigma > 0$ are specified there. Set $g_0 = f$ and $g_\pm = f \pm \tau_{x_0} f$. We also write $\mathrm{Ext}_{x_0}(f) = \kappa_4(f) - 3\kappa_4(f, f, \tau_{x_0} f, \tau_{x_0} f)$. The tuned shifts and all their constants are imported from that construction; no choice of a new mass prescription is made here.

The sets $\mathcal{V}_N$ consist of the fixed-spacing volume limits. All weak limits in the definition of Theorem A use the Hermite–Sobolev space H_{-4}. Explicitly, let $h_\kappa = h_k e_{ij}$, $k \in \mathbb{N}_0^3$, $i < j$, be the real Hermite basis for two-forms, with $(1 + |y|^2 - \Delta)h_\kappa = \lambda_\kappa h_\kappa$ and $\lambda_\kappa = 2|k|_1 + 4$. Set

$$\|\omega\|_s^2 = \sum_\kappa \lambda_\kappa^{2s} |\omega(h_\kappa)|^2, \qquad H_s = \{\omega \in \mathcal{S}' : \|\omega\|_s < \infty\}. \tag{3}$$

The trace of Σ on H_s is its Hilbert-space Borel σ-algebra. Indeed, evaluations on Schwartz tests are continuous on H_s, and the Hermite coordinates generate its Borel sets. Moreover $\sum_\kappa \lambda_\kappa^{-2s} < \infty$ for $s > 3/2$: there are $3\binom{m+2}{2}$ indices with $|k|_1 = m$. The separate notation for the finite lattice below is that of [4, Sections 2–4].

1. Contact terms and separated supports

Conventions. For a law with (G), the moments and cumulants are the bounded multilinear forms defined in (1). Since $S_1 = 0$, one has $\kappa_2 = S_2$ and

$$\kappa_4(f_1,\dots,f_4) = S_4(f_1,\dots,f_4) - S_2(f_1,f_2)\,S_2(f_3,f_4)$$

$$- S_2(f_1, f_3)\, S_2(f_2, f_4) - S_2(f_1, f_4)\, S_2(f_2, f_3).$$

As in the notation fixed above, a two-form g is identified with the vector field $*g$. For $b_1, \dots, b_N \in L^2(\mathbb{R}^3; \mathbb{C}^3)$ one accordingly writes $S_N(b_1, \dots, b_N)$ and $\kappa_N(b_1, \dots, b_N)$ in place of $S_N(*^{-1}b_1, \dots, *^{-1}b_N)$ and $\kappa_N(*^{-1}b_1, \dots, *^{-1}b_N)$. Thus $S_N(b_1, \dots, b_N) = \lambda(\mathbb{B}(b_1) \cdots \mathbb{B}(b_N))$, where $\mathbb{B}(b) = X(*^{-1}b)$.

Condition (H) and O_{cub} are defined above. For $j \neq i$, let R_t^{ji} fix the remaining coordinate and satisfy $R_t^{ji} e_j = \cos t\, e_j + \sin t\, e_i$ and $R_t^{ji} e_i = -\sin t\, e_j + \cos t\, e_i$. The reconstruction with time direction e_j is called the picture e_j. The cone condition (C_{ji}) means that the joint spectral measure of $(\mathsf{H}_j, \mathsf{P}_i)$ is carried by $\{(\omega, p_i) : \omega \geq |p_i|\}$; equivalently $\mathsf{H}_j \geq |\mathsf{P}_i|$ with constant one.

The two-dimensional linear subspaces E of $\mathbb{R}^3$ are called *planes*. Let p_E be the orthogonal projection onto E, and let $E_0 := \mathbb{R}e_1 + \mathbb{R}e_2$. Compactly supported forms or vector fields $f_1, \dots, f_N$ are said to be *E-separated* when no two of the sets $p_E(\operatorname{supp} f_i)$ intersect; in that case no two of their supports intersect either. One says that an N-linear form *vanishes at non-coinciding arguments* when its value is zero at all real $f_1, \dots, f_N \in C_c^\infty$ with pairwise disjoint supports, and that it *vanishes at E-separated arguments* when its value is zero at all real E-separated $f_1, \dots, f_N \in C_c^\infty$. Of these two properties the second is the weaker one.

Lemma 1.1. Let $N \geq 1$, and let D be a bounded N-linear form on $L^2(\mathbb{R}^3; \mathbb{C}^3)$ with the property that $D(b_1, \dots, b_N) = 0$ as soon as some entry is equal to ∇u for a function $u \in C_c^\infty(\mathbb{R}^3; \mathbb{C})$. Let E be a plane, and let $r > 0$. If $D \neq 0$, one can find real E-separated $b_1, \dots, b_N \in C_c^\infty(\mathbb{R}^3; \mathbb{R}^3)$, each of them with support in a ball of radius r, such that $D(b_1, \dots, b_N) \neq 0$. ■

Proof. 1. *Reduction to E_0.* Choose $g \in SO(3)$ with $gE_0 = E$, and put $(g \cdot b)(x) = g\, b(g^{-1}x)$. The map $b \mapsto g \cdot b$ is an isometry of $L^2(\mathbb{R}^3; \mathbb{C}^3)$; it preserves real fields and the space C_c^∞, it maps every ball onto a ball of the same radius, it maps ∇u to $\nabla(u \circ g^{-1})$, and it satisfies

$$p_E(\operatorname{supp} g \cdot b) = g\, p_{E_0}(\operatorname{supp} b).$$

Consequently the hypotheses of the lemma are fulfilled by the form $(b_i) \mapsto D(g \cdot b_1, \dots, g \cdot b_N)$, and the conclusion for this form and the plane E_0 coincides with the conclusion for D and the plane E. One may therefore assume $E = E_0$.

2. One applies [4, Proposition 8.4] with $U = \mathbb{R}^2$, so that $L^2_U = L^2(\mathbb{R}^3; \mathbb{C}^3)$. The hypothesis (b) of that proposition is satisfied and its conclusion $D = 0$ is false, so that its hypothesis (a) fails: there exist pairwise disjoint compact sets $K_1, \dots, K_N \subset \mathbb{R}^2$ and fields $b_i \in L^2(\mathbb{R}^3; \mathbb{C}^3)$ such that $b_i = 0$ almost everywhere outside $K_i \times \mathbb{R}$ and $D(b_1, \dots, b_N) \neq 0$.
3. *Real fields.* Write $b_i = \operatorname{Re} b_i + i \operatorname{Im} b_i$ and expand $D(b_1, \dots, b_N)$ by N-linearity. At least one of the resulting 2^N terms differs from zero, and each entry of that term vanishes, as the corresponding b_i does, outside $K_i \times \mathbb{R}$. Real fields b_i may therefore be used.
4. *Smooth fields.* Choose $\delta > 0$ so that the closed δ-neighbourhoods V_i of the K_i in $\mathbb{R}^2$ are pairwise disjoint, and take a mollifier ψ_ε supported in the ball of radius $\varepsilon < \delta$. The fields

$$\psi_\varepsilon * \left(1_{\{|x_3| \leq 1/\varepsilon\}}\, b_i\right)$$

are real and smooth and have compact supports contained in $V_i \times \mathbb{R}$; hence they are E_0-separated. They tend to b_i in L^2 as $\varepsilon \to 0$. The continuity of D implies that its value at these fields is not zero for all sufficiently small ε.
5. *Small supports.* Choose a smooth partition of unity (φ_k) of $\mathbb{R}^3$ which is locally finite and in which every $\operatorname{supp} \varphi_k$ is contained in a ball of radius r. The sum $b_i = \sum_k \varphi_k b_i$ has only finitely many non-zero terms, and expanding by N-linearity shows that at least one of the numbers $D(\varphi_{k_1} b_1, \dots, \varphi_{k_N} b_N)$ differs from zero.

□

Proposition 1.2. Consider a law lw with (G) and (dF) and a plane E.

(a) Let $N \geq 2$, and let $\kappa_N(g_1, \dots, g_N) \neq 0$ for some $g_1, \dots, g_N \in L^2(\mathbb{R}^3; \Lambda^2) \otimes \mathbb{C}$. Then for each $r > 0$ one can find real E-separated $f_1, \dots, f_N \in C_c^\infty(\mathbb{R}^3; \Lambda^2)$, each of them supported in some ball of radius r, such that $\kappa_N(f_1, \dots, f_N) \neq 0$. The same holds with S_N in place of κ_N.

(b) The three conditions below are mutually equivalent:
  (i) the law lw is Gaussian;
  (ii) κ_N vanishes at non-coinciding arguments whenever $N \geq 3$;
  (iii) κ_N vanishes at E-separated arguments whenever $N \geq 3$.

(c) Suppose that lw′, a second law with (G) and (dF), is such that, whatever $N \geq 2$, the form $S_N^{\mathrm{lw}'} - S_N^{\mathrm{lw}}$ vanishes at E-separated arguments. Then $\mathrm{lw}' = \mathrm{lw}$. Thus the Schwinger functions at non-coinciding arguments of a law having (G) and (dF) determine that law.

■

Proof. 1. *The form.* For $b_1, \dots, b_N \in L^2(\mathbb{R}^3; \mathbb{C}^3)$ put $D(b_1, \dots, b_N) := \kappa_N(*^{-1}b_1, \dots, *^{-1}b_N)$. The partition formula (1) shows that D is symmetric and N-linear, also for complex entries. It is bounded: complex entries included, one has

$$|D(b_1, \dots, b_N)| \leq (\log 2) \left(\frac{5 \exp(1)}{2}\right)^N N! \prod_{i=1}^N \|b_i\|$$

by [4, Section 5.2, equation (5.7)], the inequality $N^N \leq e^N N!$, and the pointwise isometry of $*$. Only (G) is used here.

2. *Gradients.* Let $u \in C_c^\infty(\mathbb{R}^3; \mathbb{C})$. Condition (dF), applied to the real and imaginary parts of u, gives $X(*^{-1}\nabla u) = 0$ almost surely. If $b_\ell = \nabla u$, every summand in (1) contains a moment whose integrand has this zero factor. Hence $D = 0$ at such entries.

3. Part (a) is Lemma 1.1 applied to D, with $f_i := *^{-1}b_i$. In the case of S_N, the form is bounded by [4, Proposition 5.1], and its vanishing on gradients is seen as in step 2.

4. *Part (b), (i) implies (ii).* Let lw be Gaussian. Every real linear combination of a finite family $X(f_1), \dots, X(f_N)$ is then normal. Thus the family is jointly Gaussian; it is centered by (G). Therefore

$$\log \mathrm{lw}\Big(\exp \sum_a z_a X(f_a)\Big) = \frac{1}{2} \sum_{a,c} z_a z_c \, S_2(f_a, f_c),$$

Its derivative $\partial_{z_1} \cdots \partial_{z_N}$ at zero is $\kappa_N(f_1, \dots, f_N)$ by (1), and vanishes for $N \geq 3$. That (ii) implies (iii) follows from the fact that E-separated arguments have pairwise disjoint supports.

5. *Part (b), (iii) implies (i).* Lemma 1.1, together with steps 1 and 2, shows that $\kappa_N = 0$ identically for each $N \geq 3$. For real $f \in \mathcal{S}$, put $w(z) = \log \mathrm{lw}(e^{zX(f)})$. By [4, Section 5.2], this is analytic for $|z|\|f\| < 2/5$ and $w^{(n)}(0) = \kappa_n(f, \dots, f)$. It follows that $w(z) = \frac{1}{2} S_2(f, f) z^2$ for these z. Both sides of the identity

$$\mathrm{lw}(\exp zX(f)) = \exp(\tfrac{1}{2} S_2(f, f) z^2)$$

are entire functions of z [4, Proposition 5.1 and Section 5.2], so that the identity holds at $z = it$, and $X(f)$ is normal.

6. *Part (c).* Lemma 1.1 applies to the form $S_N^{\mathrm{lw}'} - S_N^{\mathrm{lw}}$, which is bounded [4, Proposition 5.1] and zero on gradients, and shows that this form vanishes identically. Thus all moments of lw′ and of lw at real tests in C_c^∞ coincide, the products being lw′-integrable by (G), and [4, Proposition 5.1 and Section 5.2] gives $\mathrm{lw}' = \mathrm{lw}$.

□

2. Cyclicity of the curvature field

In this section a *law* is a probability measure on $(\mathcal{S}', \Sigma)$, where $\mathcal{S}' = \mathcal{S}'(\mathbb{R}^3; \Lambda^2)$. Let $j \neq i$ be two elements of $\{1, 2, 3\}$. The *one-plane hypotheses for* (j, i) are:

- (G) and (dF);
- (T_{123}): the law is invariant under every translation of $\mathbb{R}^3$;
- (R_j) and (RP_j): the law is invariant under the reflection θ_j in the plane $\{x_j = 0\}$, and it is reflection positive with respect to this plane;
- (Q_{ji}): the law is invariant under the quarter turn $e_j \mapsto e_i \mapsto -e_j$;
- (C_{ji}): $\mathfrak{E}_j(\{\omega < |p_i|\}) = 0$; equivalently, the inequality $\mathsf{H}_j \geq |\mathsf{P}_i|$ holds with the constant equal to one.

Write $(\mathcal{H}_j, [\,\cdot\,]_j, \mathsf{H}_j, V_j, \Omega_j)$ for the reconstruction in direction e_j. The space $\mathcal{H}_j$ is the completion of $L^2(\Sigma_j^+, \mu)$ modulo null vectors, where $\Sigma_j^+ = \Sigma_{\{x_j > 0\}}$, for the form

$$\langle [\Phi]_j, [\Phi']_j \rangle = \mu\big((\Theta_j \Phi) \cdot \Phi'\big), \qquad \Theta_j \Phi = \overline{\hat{\theta}_j \Phi};$$

the operators H_j and $V_j(y)$ and the vector Ω_j are given by

$$e^{-t\mathsf{H}_j}[\Phi]_j = [U(te_j)\Phi]_j \quad (t \geq 0), \qquad V_j(y)[\Phi]_j = [U(y)\Phi]_j \quad (y \perp e_j), \qquad \Omega_j = [1]_j.$$

For $i \neq j$, let P_i denote the generator of $u \mapsto V_j(ue_i)$; one writes $\mathfrak{E}_j$ for the joint spectral measure of $(\mathsf{H}_j, (\mathsf{P}_i)_{i \neq j})$, which is a measure on the space $\mathbb{R} \times e_j^{\perp}$ whose points are written $(\omega, p_\perp)$ [4, Proposition 7.2]. The definitions of (H) and $\mathcal{C}_9$ remain those fixed above.

We use the tube and plate notation of [4, Sections 8.2–8.3], recalled here. By a *label* one means a pair $\lambda = (\sigma, \chi)$ with $\sigma \in \{+, -, 3\}$ and $\chi \in L^2(\mathbb{R}; \mathbb{R})$. Set $w(+) = 1$, $w(-) = -1$, $w(3) = 0$, and $w(\lambda) = \sum_a w(\sigma_a)$ for a tuple $\lambda_a = (\sigma_a, \chi_a)$. The forward tube is

$$\mathcal{W} = \{\xi \in \mathbb{C}^2 : \operatorname{Re} \xi^0 > |\operatorname{Im} \xi^1|\},$$

with closure $\overline{\mathcal{W}}$, and one puts

$$\mathcal{W}^{(n)} = \{x \in (\mathbb{C}^2)^n : x_1 \in \mathcal{W},\ x_{a+1} - x_a \in \mathcal{W}\}.$$

For real β, let Λ_β be the linear map of $\mathbb{C}^2$ with

$$(\Lambda_\beta \xi)^0 = \xi^0 \cosh\beta - i\xi^1 \sinh\beta, \qquad (\Lambda_\beta \xi)^1 = \xi^1 \cosh\beta + i\xi^0 \sinh\beta;$$

on configurations it acts entry by entry [4, Proposition 8.2 and Section 8.3]. Under (C_{12}), the operators $\mathsf{T}(\kappa)$ of [4, Proposition 8.2 and Section 8.3] are given for $\kappa \in \overline{\mathcal{W}}$ by

$$\mathsf{T}(\kappa) = \int d\mathfrak{E}_1(\omega, p_\perp)\, e^{-\kappa^0 \omega + i\kappa^1 p_2}.$$

Throughout this section, β is the boost parameter and σ is a label. They do not denote a probe or the exponent in the tuned estimates.

Theorem 2.1. Let the law μ fulfil the one-plane hypotheses for the pair $(1, 2)$.

(a) The linear span of Ω_1 and of the vectors

$$[X(g_1) \cdots X(g_N)]_1,$$

where $N \geq 1$ and $g_1, \ldots, g_N \in C_c^\infty(\{x_1 > 0\}; \Lambda^2)$ are real and satisfy

$$\sup\{x_1 : x \in \operatorname{supp} g_a\} < \inf\{x_1 : x \in \operatorname{supp} g_{a+1}\} \qquad (1 \leq a < N),$$

is dense in $\mathcal{H}_1$.

(b) One has $\mathcal{H}_{12}^{\Psi} = \mathcal{H}_1$, where $\mathcal{H}_{12}^{\Psi}$ denotes the closed span of the plate vectors $\Psi(x; \lambda)$ with $n \geq 0$, $x \in \mathcal{W}^{(n)}$ and λ a tuple of n labels as in Section B.1.

(c) The vector Ω_1 is cyclic for the field strength in two forms, one for each of the two families described next. The first family consists of the point plates $\mathcal{B}_\lambda(\xi,\xi')$, $\xi,\xi' \in \mathcal{W}$, each of which stands for $\mathsf{T}(\xi)\,\mathbb{B}_\lambda(\delta_0)\,\mathsf{T}(\xi')$ [4, Proposition 8.2 and Section 8.3]. The second family consists of the operators $\mathcal{B}_l(f)$, where $l > 0$ and where $f \in L^2(\mathbb{R}^3;\mathbb{C}^3)$ vanishes outside $\{0 \le x_1 \le l\}$ [4, Proposition 8.1]. For each of the two families, Ω_1 and the vectors obtained by applying finite products of members of the family to Ω_1 form a total set in $\mathcal{H}_1$.

■□

Corollary 2.2. (a) Let the law μ satisfy the hypotheses of Theorem 2.1. There is a unique strongly continuous unitary group $(\mathfrak{L}(\beta))_{\beta\in\mathbb{R}}$, defined on all of $\mathcal{H}_1$, such that $\mathfrak{L}(\beta)\Psi(x;\lambda) = e^{w(\lambda)\beta}\,\Psi(\Lambda_\beta x;\lambda)$ for every plate vector. For $\beta, u \in \mathbb{R}$, $t \ge 0$, $\kappa \in \overline{\mathcal{W}}$, $\xi,\xi' \in \mathcal{W}$ and every label $\lambda_0 = (\sigma,\chi)$, the following relations hold on $\mathcal{H}_1$:

$$\mathfrak{L}(\beta)\Omega_1 = \Omega_1, \qquad \mathfrak{L}(\beta)\,\mathsf{T}(\kappa)\,\mathfrak{L}(\beta)^{-1} = \mathsf{T}(\Lambda_\beta\kappa),$$
$$\mathfrak{L}(\beta)\,\mathcal{B}_{\lambda_0}(\xi,\xi')\,\mathfrak{L}(\beta)^{-1} = e^{w(\sigma)\beta}\,\mathcal{B}_{\lambda_0}(\Lambda_\beta\xi,\Lambda_\beta\xi'),$$

and

$$\mathfrak{L}(\beta)\,e^{-t\mathsf{H}_1}\,\mathfrak{L}(\beta)^{-1} = e^{-t(\mathsf{H}_1\cosh\beta+\mathsf{P}_2\sinh\beta)}, \tag{2.1}$$
$$\mathfrak{L}(\beta)\,e^{iu\mathsf{P}_2}\,\mathfrak{L}(\beta)^{-1} = e^{iu(\mathsf{P}_2\cosh\beta+\mathsf{H}_1\sinh\beta)}, \tag{2.2}$$

the right-hand sides of (2.1) and (2.2) being integrals against $\mathfrak{E}_1$.

(b) Under the assumptions of (a), the operators $\mathfrak{L}(\beta)$ and $V_1(ue_3)$ commute for every $u \in \mathbb{R}$, and one has

$$\mathfrak{L}(\beta)\,\mathfrak{E}_1(A)\,\mathfrak{L}(\beta)^{-1} = \mathfrak{E}_1(\ell_{-\beta}A)$$

for every Borel set $A \subset \mathbb{R}\times e_1^\perp$, where

$$\ell_\beta(\omega,p_2,p_3) := (\omega\cosh\beta + p_2\sinh\beta,\ p_2\cosh\beta+\omega\sinh\beta,\ p_3).$$

(c) Assume that the law μ belongs to $\mathcal{C}_9$ and has (H) and (dF). Then, for every ordered pair $j \ne i$, the law μ satisfies the one-plane hypotheses for (j,i), one has $\mathcal{H}_{ji}^\Psi = \mathcal{H}_j$, and the group $\mathfrak{L}_{ji}$ of Lemma B.1 acts on all of $\mathcal{H}_j$ and satisfies the relations of (a) and (b) for $(\mathsf{H}_j,\mathsf{P}_i)$. Furthermore, let $\mathbb{R}\times e_j^\perp$ carry the quadratic form $\omega^2 - |p_\perp|^2$; then to each Λ in its restricted Lorentz group there corresponds a unitary operator W on $\mathcal{H}_j$ satisfying $W\Omega_j = \Omega_j$, as well as $W\,\mathfrak{E}_j(A)\,W^{-1} = \mathfrak{E}_j(\Lambda A)$ for each Borel set A. In particular, $\mathfrak{E}_j$ has a Lorentz invariant support, which is a closed subset of the cone $\{\omega \ge |p_\perp|\}$.

(d) The hypotheses of (c) are satisfied in the following cases:
- for every charge $e > 0$ and every mass assignment: by every volume-first limit point μ of [4, Section 6, Section 7.5, and Theorem 8.6], and by ν-almost every component μ_ξ;
- for $0 < e \le e_2$ and the tuned mass assignment of Section F.2: by every volume-first limit point μ, and by every μ_ξ with $\xi \in \Xi_\mu$.

■□

Corollary 2.3. Let μ be as in Theorem 2.1, and suppose that $\ker\mathsf{H}_1 = \mathbb{C}\,\Omega_1$. Then the only bounded operators on $\mathcal{H}_1$ that commute with all point plates $\mathcal{B}_\lambda(\xi,\xi')$ are the multiples of the identity. This applies to every tuned component μ_ξ, $\xi \in \Xi_\mu$, and to almost every component in the arbitrary-charge construction: the invariant-projection identity of [4, Section 7.3] identifies $\ker\mathsf{H}_1$ with the translation-invariant vectors, which are constants in an ergodic law. ■□

Proof of Theorem 2.1. The notation is that of the definitions in Section 2. For $b = *g$ one writes $\mathbb{B}(b) = X(g)$, and $\mathbb{B}$ is extended complex linearly to $b \in L^2(\mathbb{R}^3;\mathbb{C}^3)$. Points of $\mathbb{R}^3$ are written $x = (y,x_3)$ with $y = (y^0,y^1) = (x_1,x_2)$, and $\mathbb{H} = \{y \in \mathbb{R}^2 : y^0 > 0\}$. One writes $L^2_{\mathbb{H}}$ for the subspace formed by the b that are equal to zero almost everywhere outside $\mathbb{H}\times\mathbb{R}$. For a label $\lambda = (\sigma,\chi)$ and

$\alpha \in L^2(\mathbb{R}^2;\mathbb{C})$ one puts $(\alpha\otimes\lambda)(x)=\alpha(y)\chi(x_3)e^\sigma$ and $\mathbb{B}_\lambda(\alpha)=\mathbb{B}(\alpha\otimes\lambda)$. The closed linear span of the vectors listed in (a) is denoted by $\mathcal{H}^<$.

Step 1 (a bound). Let $b_1,\dots,b_N\in L^2_{\mathbb{H}}$. The product $\prod_a \mathbb{B}(b_a)$ belongs to $L^2(\Sigma_1^+,\mu)$, and

$$\Big\|\Big[\prod_a \mathbb{B}(b_a)\Big]_1\Big\| \le \Big\|\prod_a \mathbb{B}(b_a)\Big\|_{L^2(\mu)} \le c_{2N}^N \prod_a \|b_a\|.$$

This follows from [4, Proposition 5.1] and Hölder's inequality, together with the fact that the class map $\Phi\mapsto[\Phi]_1$ is a contraction [4, Sections 5.2 and 7.4]. Consequently, the class of a product depends N-linearly and continuously on its entries.

Step 2 (plates in time order). 2a. Consider $\alpha_1,\dots,\alpha_N\in L^2(\mathbb{R}^2;\mathbb{C})$ whose supports are time-ordered in $\mathbb{H}$ (Section B.1). Cutting the plates $\alpha_a\otimes\lambda_a$ off in x_3, mollifying at a scale smaller than the gaps between the time supports and than their distance from $\{y^0=0\}$, and splitting into real and imaginary parts, one exhibits each plate as an L^2-limit of fields $*g+i*g'$ in which g,g' are real, smooth and compactly supported, the time order being preserved. Step 1 then yields $[\prod_a \mathbb{B}_{\lambda_a}(\alpha_a)]_1\in\mathcal{H}^<$.

2b. Conversely, $\mathcal{H}^<\subset\mathcal{H}^\Psi_{12}$. This inclusion is used for (b), in Step 6, and the proof of (a) does not rely on it. Take g_a as in (a) and pick compact rectangles $A_a\subset\mathbb{H}$ in time order so that $*g_a$ vanishes outside $A_a\times\mathbb{R}$; by [4, Section 8.2], each $*g_a$ is then an L^2-limit of finite sums of plates $\alpha\otimes\lambda$ in which $\alpha=0$ outside A_a. By Lemma B.1, the closed space $\mathcal{H}^\Psi_{12}$ contains the class of every product of such plates, and Step 1 gives the inclusion.

Step 3 (entries over disjoint compact subsets of the plane). Claim. Assume that the sets $K_1,\dots,K_N\subset\mathbb{H}$ are compact and pairwise disjoint, that each $\alpha_a\in L^2(\mathbb{R}^2;\mathbb{C})$ is equal to zero outside K_a, and that $\lambda=(\lambda_1,\dots,\lambda_N)$ is a tuple of labels. Then $[\prod_a \mathbb{B}_{\lambda_a}(\alpha_a)]_1\in\mathcal{H}^<$.

3a (decomposition). Given $d>0$ and $k\in\mathbb{Z}$, set $\alpha_a^k:=\alpha_a\cdot 1_{\{kd\le y^0<(k+1)d\}}$; for $k=(k_1,\dots,k_N)\in\mathbb{Z}^N$, set $\Phi_k:=\prod_a \mathbb{B}_{\lambda_a}(\alpha_a^{k_a})$. Then $\prod_a \mathbb{B}_{\lambda_a}(\alpha_a)=\sum_k\Phi_k$, and the sum is finite.

3b (distinct intervals). For pairwise distinct integers k_a, one has $[\Phi_k]_1\in\mathcal{H}^<$. Indeed, let the strips be replaced by $\{kd+\varepsilon\le y^0\le(k+1)d-\varepsilon\}$; since the factors commute, they may be taken in the order of increasing k_a, and the resulting functions then have time-ordered supports in $\mathbb{H}$, so that Step 2 applies. Letting $\varepsilon\to0$ and using Step 1, one obtains the assertion.

3c (the rest). Denote by Φ_d^{r} the sum of the other Φ_k, and by E_d the set of configurations $Y=(y_1,\dots,y_N)\in(\mathbb{R}^2)^N$ such that two of the times y_a^0 fall into a common interval $[kd,(k+1)d)$. Besides being antilinear, the map Θ_1 is multiplicative; furthermore, according to [4, Section 8.2], one has almost surely

$$\Theta_1\mathbb{B}_{\lambda_a}(\alpha)=s(\sigma_a)\,\mathbb{B}_{\lambda_a}(\overline{\alpha}\circ\theta_1).$$

It follows that $\|[\Phi_d^{\mathrm{r}}]_1\|^2=\mu\big((\Theta_1\Phi_d^{\mathrm{r}})\cdot\Phi_d^{\mathrm{r}}\big)$ is $s(\lambda)=\pm1$ times a finite sum of moments S_{2N} whose entries are $(\overline{\alpha_a^{k_a}}\circ\theta_1)\otimes\lambda_a$ and $\alpha_a^{k'_a}\otimes\lambda_a$, $1\le a\le N$.

3d (no singular part). The supports of these $2N$ functions are compact subsets of the $2N$ pairwise disjoint sets $\theta_1K_1,\dots,\theta_1K_N,K_1,\dots,K_N$. Hence [4, Section 8.10, equation (8.42)], applied with $2N$ in place of N, writes each moment as the integral, against the plate function $\tilde G_{2N}(\,\cdot\,;\lambda,\lambda)$, of the product of these functions. The plate function is real analytic on the set of configurations of $2N$ distinct points [4, Section 8.10, equation (8.42)]; hence there is a constant C, independent of d, which bounds it on the compact set $\prod_a\theta_1K_a\times\prod_a K_a$. Since

$$\sum_k\prod_a\alpha_a^{k_a}(y_a)=1_{E_d}(Y)\prod_a\alpha_a(y_a),$$

in which k runs through the tuples having a repetition, one obtains after summation

$$\|[\Phi_d^{\mathrm{r}}]_1\|^2\le C\cdot\Big(\int_{E_d}dY\prod_a|\alpha_a(y_a)|\Big)^2.$$

3e (limit). Here $\prod_a |\alpha_a(y_a)|$ is an integrable function of Y, and

$$E_d \subset \{Y : |y_a^0 - y_b^0| < d \text{ for some } a \neq b\},$$

where the sets on the right-hand side decrease, as $d \to 0$, to a set of measure zero. The dominated convergence theorem therefore gives $[\Phi_d^{\mathrm{r}}]_1 \to 0$. Together with 3a and 3b this proves the claim, because $\mathcal{H}^<$ is closed.

Step 4 (no contact terms). 4a. Let $\psi \perp \mathcal{H}^<$ and $N \geq 1$, and put

$$D(b_1, \dots, b_N) := \Big\langle \psi, \Big[\prod_a \mathbb{B}(b_a) \Big]_1 \Big\rangle, \qquad b_a \in L^2_{\mathbb{H}}.$$

Step 1 shows that D is N-linear and bounded. If $b_\ell = \nabla u$ for $u \in C_c^\infty(\mathbb{H} \times \mathbb{R}; \mathbb{C})$, condition (dF) gives $\mathbb{B}(b_\ell) = 0$ almost surely, and hence $D = 0$. The form D also vanishes when each b_a is zero outside $K_a \times \mathbb{R}$, the sets $K_a \subset \mathbb{H}$ being compact and pairwise disjoint: for plates this is Step 3, and the general case follows from [4, Section 8.2] with $A = K_a$, multilinearity and Step 1.

4b. These two properties are the hypotheses of [4, Proposition 8.4] for the open set $U = \mathbb{H}$. Consequently $D = 0$, for every $N \geq 1$.

Step 5 (polynomials suffice). Let $g \in C_c^\infty(\{x_1 > 0\}; \Lambda^2)$ be real. Each partial sum of $\sum_n (iX(g))^n/n!$ is bounded in modulus by $\exp |X(g)|$, and (G), applied to $\pm 2g$, shows that $\exp |X(g)|$ belongs to $L^2(\mu)$. The partial sums therefore converge to $\exp(iX(g))$ in $L^2(\Sigma_1^+, \mu)$; combined with $\psi \perp \Omega_1$, Step 4 then yields $\langle \psi, [\exp(iX(g))]_1 \rangle = 0$. According to [4, Sections 5.2 and 7.4], the linear hull of such exponentials is dense in $L^2(\Sigma_1^+, \mu)$, and according to [4, Sections 5.2 and 7.4], the class map has a dense image. Thus $\psi = 0$, and (a) is established.

Step 6 (assertion (b), and assertion (c) for the point plates). Combining (a) with 2b gives $\mathcal{H}_1 = \mathcal{H}^< \subset \mathcal{H}_{12}^\Psi \subset \mathcal{H}_1$. By Section B.1, the plate vectors consist of Ω_1 together with the vectors obtained from Ω_1 by applying products of point plates.

Step 7 (assertion (c) for the operators $\mathcal{B}_l(f)$). With g_a as in (a), write $b_a = *g_a$, set $l_0 = 0$, and pick $l_1 < \cdots < l_N$ such that each b_a vanishes outside $\{l_{a-1} \leq x_1 \leq l_a\}$. Iterating the identity $\mathcal{B}_l(f)[\Phi]_1 = [\mathbb{B}(f) \cdot U(le_1)\Phi]_1$ of [4, Proposition 8.1], one finds

$$[X(g_1) \cdots X(g_N)]_1 = \mathcal{B}_{l_1}(b_1)\, \mathcal{B}_{l_2 - l_1}(\tau_{-l_1 e_1} \cdot b_2) \cdots \mathcal{B}_{l_N - l_{N-1}}(\tau_{-l_{N-1} e_1} \cdot b_N)\, \Omega_1.$$

The assertion then follows from (a). □

Proof of Corollary 2.2. (a) The assertion follows from Lemma B.1, since $\mathcal{H}_{12}^\Psi = \mathcal{H}_1$ by Theorem 2.1 (b).

(b1) Smeared translations along e_3. For a tuple λ of labels let λ^u be the tuple with the profiles $\chi_a(\,\cdot - u)$; then $\tau_{ue_3} \cdot (\alpha \otimes \lambda_a) = \alpha \otimes \lambda_a^u$ and $w(\lambda^u) = w(\lambda)$. For time-ordered supports in $\mathbb{H}$, the identities $V_1(y)[\Phi]_1 = [U(y)\Phi]_1$ and $U(a)\mathbb{B}(b) = \mathbb{B}(\tau_a b)$ give

$$V_1(ue_3) \Big[\prod_a \mathbb{B}_{\lambda_a}(\alpha_a) \Big]_1 = \Big[\prod_a \mathbb{B}_{\lambda_a^u}(\alpha_a) \Big]_1.$$

Expressed by means of the second identity of [4, Proposition 8.3], this equality becomes

$$\int dY \prod_a \alpha_a(y_a)\, (V_1(ue_3)\Psi(Y;\lambda) - \Psi(Y;\lambda^u)) = 0.$$

(b2) The pointwise identity, and the commutation. On the convex open set $\mathcal{W}^{(n)}$, the real points of which are the time-ordered configurations in $\mathbb{H}$, the vectors $V_1(ue_3)\Psi(x;\lambda)$ and $\Psi(x;\lambda^u)$ depend holomorphically on x [4, Proposition 8.3]. Applying [4, Section 8.3 and Proposition 8.3] to the scalar products of these vectors with an arbitrary fixed vector, one therefore obtains $V_1(ue_3)\Psi(x;\lambda) = \Psi(x;\lambda^u)$ on $\mathcal{W}^{(n)}$. One deduces

$$\mathfrak{L}(\beta) V_1(ue_3)\Psi(x;\lambda) = e^{w(\lambda)\beta} \Psi(\Lambda_\beta x; \lambda^u) = V_1(ue_3)\mathfrak{L}(\beta)\Psi(x;\lambda),$$

and the set of plate vectors is total.

(b3) The spectral measure. Let $t \geq 0$ and $y = u_2 e_2 + u_3 e_3$, and put $F_{t,y}(\omega, p_\perp) = e^{-t\omega + i p_\perp \cdot y}$. From the relations (2.1) and (2.2), from part (b2), and from the multiplicativity of integrals against $\mathfrak{E}_1$, one obtains

$$\mathfrak{L}(\beta)\, e^{-t\mathsf{H}_1} V_1(y)\, \mathfrak{L}(\beta)^{-1} = \int d\mathfrak{E}_1\, F_{t,y} \circ \ell_\beta .$$

The measure $\mathfrak{E}_1$ is carried by the set $\{\omega \geq |p_2|\}$, and this set is invariant under the map ℓ_β. For each ψ, the finite positive measures

$$\langle \psi, \mathfrak{L}(\beta)\, \mathfrak{E}_1(\cdot)\, \mathfrak{L}(\beta)^{-1} \psi \rangle \qquad \text{and} \qquad \langle \psi, \mathfrak{E}_1(\ell_\beta^{-1}\, \cdot\,) \psi \rangle$$

on $[0,\infty) \times \mathbb{R}^2$ therefore give rise to one and the same Laplace–Fourier transform. By Lemma D.3 the two measures are equal; moreover, $\ell_\beta^{-1} = \ell_{-\beta}$.

(c1) All planes. Membership in $\mathcal{C}_9$ supplies translation invariance, coordinate reflection positivity and the corresponding diagonal reflection positivity. Therefore [4, Proposition 7.2] gives (C_{ji}). Condition (H) supplies (G) and invariance under O_{cub}, which contains the quarter turn in (Q_{ji}). Condition (dF) is assumed. The coordinate transport in Section B.2 carries the one-plane hypotheses to $\mu' = (g_{ji}^{-1})_* \mu$. It supplies a unitary $J : \mathcal{H}_1(\mu') \to \mathcal{H}_j(\mu)$ carrying $\mathsf{H}_1', V_1'(y), \mathfrak{E}_1'$ to $\mathsf{H}_j, V_j(g_{ji} y), \mathfrak{E}_j$; explicitly,

$$\mathfrak{E}_j(A) = J\, \mathfrak{E}_1'(g_{ji}^{-1} A)\, J^{-1},$$

where g_{ji} also denotes the map $(\omega, p_\perp) \mapsto (\omega, g_{ji} p_\perp)$ of $\mathbb{R} \times e_1^\perp$ onto $\mathbb{R} \times e_j^\perp$. In Section B.2 this relation is stated for Borel subsets A of $[0,\infty) \times e_j^\perp$. It extends to all Borel subsets of $\mathbb{R} \times e_j^\perp$: both $\mathfrak{E}_j$ and $\mathfrak{E}_1'$ vanish on $\{\omega < 0\}$ [4, Proposition 7.2], while g_{ji} leaves ω unchanged, so that replacing A by $A \cap \{\omega \geq 0\}$ changes neither side. Since $\mathcal{H}_{ji}^{\Psi} = J \mathcal{H}_{12}^{\Psi}(\mu')$ and $\mathfrak{L}_{ji} = J \mathfrak{L} J^{-1}$ by definition, Theorem 2.1 (b) and parts (a) and (b) for μ' give the first three assertions of (c).

(c2) Rotations. Here $\mu' = \mu$, because $g_{ji} \in O_{\mathrm{cub}}$, and by (c1) one has $\mathfrak{E}_j(A) = J\, \mathfrak{E}_1(g_{ji}^{-1} A)\, J^{-1}$ for all Borel sets $A \subset \mathbb{R} \times e_j^\perp$; it therefore suffices to take $j = 1$. Indeed, for Λ acting on $\mathbb{R} \times e_j^\perp$, the conjugate $\Lambda' := g_{ji}^{-1} \Lambda g_{ji}$ belongs to the restricted Lorentz group of $\mathbb{R} \times e_1^\perp$. Suppose that W' is a unitary operator on $\mathcal{H}_1$ for which $W' \Omega_1 = \Omega_1$ and $W' \mathfrak{E}_1(A') W'^{-1} = \mathfrak{E}_1(\Lambda' A')$ for each Borel set A'; then $W := J W' J^{-1}$ fixes the vector $\Omega_j = J \Omega_1$, and one has

$$W\, \mathfrak{E}_j(A) W^{-1} = J\, \mathfrak{E}_1(\Lambda' g_{ji}^{-1} A)\, J^{-1} = \mathfrak{E}_j(\Lambda A).$$

The group O(3) leaves μ invariant [4, Theorem 8.6]. Accordingly, Section B.2 provides, for $g \in \mathrm{O}(3)$ with $g e_1 = e_1$, unitary operators $\mathcal{U}(g)$ such that $\mathcal{U}(g) \Omega_1 = \Omega_1$ and

$$\mathcal{U}(g)\, \mathfrak{E}_1(A)\, \mathcal{U}(g)^{-1} = \mathfrak{E}_1(gA), \qquad gA := \{(\omega, g p_\perp) : (\omega, p_\perp) \in A\}.$$

This relation is stated there for Borel subsets of $[0,\infty) \times e_1^\perp$; the argument of (c1) extends it to all Borel subsets of $\mathbb{R} \times e_1^\perp$, because $(\omega, p_\perp) \mapsto (\omega, g p_\perp)$ leaves ω unchanged.

(c3) Lorentz transformations. Let Λ be given. Since the point $\Lambda(1,0,0)$ lies on $\{\omega = (1 + |p_\perp|^2)^{1/2}\}$, it can be written as $g \ell_\beta (1,0,0)$ with a rotation g of $e_1^\perp$ and a real β. As a restricted Lorentz transformation leaving $(1,0,0)$ fixed, the map $g' := (g \ell_\beta)^{-1} \Lambda$ is a rotation. Parts (b) and (c2) show that the operator $W := \mathcal{U}(g)\, \mathfrak{L}(-\beta)\, \mathcal{U}(g')$ fixes Ω_1 and that

$$W\, \mathfrak{E}_1(A)\, W^{-1} = \mathfrak{E}_1(g \ell_\beta g' A) = \mathfrak{E}_1(\Lambda A).$$

Finally, Section B.2 shows that $\mathfrak{E}_1$ is carried by the round cone.

(d) By [4, Section 6, Section 7.5, and Theorem 8.6], the law μ has (G), (dF), the invariance under $\mathbb{R}^3 \rtimes O_{\mathrm{cub}}$ and reflection positivity at the coordinate planes, and lies in $\mathcal{C}_9$; by [4, Section 6, Section 7.5, and Theorem 8.6] the same holds for almost every component. The tuned laws in Section F.2 have (G), (dF), cubic invariance and the nine reflection positivities; the same properties hold on the specified set Ξ_μ of components. Thus the preceding argument applies to them as well. □

Proof of Corollary 2.3. Step 1 (two stabilities). Let C be a bounded operator that commutes with all point plates, and denote by $\mathcal{P}$ the set of all finite products of point plates. If $A \in \mathcal{P}$, then A^* or $-A^*$ belongs to $\mathcal{P}$ [4, Proposition 8.2 and Section 8.3]; hence C^* also commutes with every element of $\mathcal{P}$. Applied with $\kappa = -it\,n_0$ (where $n_0 = (1,0)$ is the time direction fixed in Section 2) and $\kappa' = it\,n_0$, both of which lie in $\overline{\mathcal{W}}$, the same clause shows that $A \in \mathcal{P}$ implies $A_t := e^{it\mathsf{H}_1} A e^{-it\mathsf{H}_1} \in \mathcal{P}$; here one uses $\mathsf{T}(\mp it\,n_0) = e^{\pm it\mathsf{H}_1}$ and $\mathsf{T}(\kappa)\Omega_1 = \Omega_1$ [4, Proposition 8.2 and Section 8.3].

Step 2 (positivity of the energy). Let $A \in \mathcal{P}$, and let t be real; then

$$\langle C^*\Omega_1, e^{it\mathsf{H}_1} A \Omega_1\rangle = \langle \Omega_1, C A_t \Omega_1\rangle = \langle \Omega_1, A_t C \Omega_1\rangle = \langle A^*\Omega_1, e^{-it\mathsf{H}_1} C\Omega_1\rangle.$$

As functions of t, the first and the last member of this chain are Fourier transforms of finite complex measures, carried by $[0,\infty)$ and by $(-\infty,0]$ respectively. Because a finite complex measure is determined by its Fourier transform, both measures are carried by $\{0\}$. Let P_0 denote the projection onto $\ker \mathsf{H}_1 = \mathbb{C}\,\Omega_1$; evaluation at $t=0$ then yields $\langle C^*\Omega_1, A\Omega_1\rangle = \langle C^*\Omega_1, P_0 A\Omega_1\rangle$.

Step 3 (cyclicity). Put $c := \langle \Omega_1, C^*\Omega_1\rangle$. By Step 2, the vector $C^*\Omega_1 - c\,\Omega_1$ is orthogonal to Ω_1 and to every $A\Omega_1$ with $A \in \mathcal{P}$, and it vanishes by Theorem 2.1 (c). On a total set of vectors one thus has $C^*A\Omega_1 = AC^*\Omega_1 = c\,A\Omega_1$, whence $C^* = c$. That μ_ξ fulfils the hypotheses for every $\xi \in \Xi_\mu$ follows from Corollary 2.2 (c), (d) and Section F.2. □

3. The local relativistic field

The OS axioms and their growth condition give a local Wightman theory by [2, Section IV.1]. We retain a direct reconstruction to identify its field on the probabilistic OS Hilbert space and prove the energy and tube estimates on its common domain in Theorem 3.3. These estimates are used in Section 4 to identify the full infrared law.

Conventions. We retain the notation above. Time is the first coordinate, so that points are written $y = (y_1, \vec{y})$, and one sets

$$\theta := \theta_1, \qquad \Sigma_+ := \Sigma_{\{y_1>0\}}, \qquad L^2 := L^2(\mathbb{R}^3;\Lambda^2)\otimes\mathbb{C}.$$

On Minkowski space $\mathbb{R}^3$, with points $x = (x^0, \vec{x})$ and basis e_0, e_2, e_3, the quadratic form is $x\cdot x = (x^0)^2 - |\vec{x}|^2$ and the forward cone is $V_+ = \{x^0 > |\vec{x}|\}$. Put

$$d_{\mathcal{T}}(\zeta) := \operatorname{Im}\zeta^0 - |\operatorname{Im}\vec{\zeta}|;$$

this function may be compared with the function $d_{\mathcal{W}}(\xi) = \operatorname{Re}\xi^0 - |\operatorname{Im}\xi^1|$ on $\mathbb{C}^2$ defined in the definitions in Section 2, whereas the letters d and d' without index stand for half widths of slabs, as in [4, Proposition 8.1]. The sets

$$\mathcal{T} := \mathbb{R}^3 + iV_+ = \{d_{\mathcal{T}} > 0\},$$
$$\mathcal{T}_n := \{z \in \mathbb{C}^{3n} : z_{a+1} - z_a \in \mathcal{T}\}, \qquad \mathcal{T}_n^+ := \{z \in \mathcal{T}_n : z_1 \in \mathcal{T}\}$$

are open and convex. The matrix $J := \operatorname{diag}(i,1,1)$ sends y to the point $Jy = (iy_1, \vec{y})$, called a *Euclidean point*, and sends e_1 to ie_0. Every linear map M of $\mathbb{C}^3$ acts on $b \in \Lambda^2\mathbb{C}^3$ through $\wedge^2 M$, abbreviated Mb; it acts componentwise on tuples, and on fields through $(M\cdot g)(y) = M\,g(M^{-1}y)$. In particular $J(e_1\wedge e_k) = i\,e_0\wedge e_k$ and $|Jb| = |b|$; moreover

$$\overline{Jy} = J\theta y, \qquad \overline{Jb} = J\theta\bar{b}. \tag{K}$$

Write $\mathrm{Sw}_n = \mathcal{S}(\mathbb{R}^{3n};(\Lambda^2)^{\otimes n})$, with complex values. By a Minkowski test one means an $h \in \mathrm{Sw}_1$; its values lie in $\Lambda^2\mathbb{C}^3$, and its components are the h^α with $\alpha \in \{02,03,23\}$. For such an index α let $\mathrm{ti}(\alpha)$ be the number of its entries equal to 0, and put $\mathrm{ti}(\boldsymbol{\alpha}) := \sum_a \mathrm{ti}(\alpha_a)$. The group $\mathcal{P}_+^\uparrow = \mathbb{R}^3 \rtimes SO_0(1,2)$, and in Proposition 3.5 the group $\mathbb{R}^3 \rtimes O(1,2)$, act on tests by $((a,\Lambda)\cdot h)(x) = \Lambda\,h(\Lambda^{-1}(x-a))$; the matrix groups $SO_0(1,2)$ and $O(1,2)$ are introduced in Fact 6. For $j \in \{2,3\}$ put $R_t^{(j)} = R_t^{1j}$; thus

$$R_t^{(j)} e_1 = \cos t\, e_1 + \sin t\, e_j, \qquad R_t^{(j)} e_j = -\sin t\, e_1 + \cos t\, e_j.$$

This matrix is an entire function of t, and $\Lambda_s^{(j)} := JR_{is}^{(j)}J^{-1}$ is the boost

$$e_0 \mapsto \cosh s\, e_0 + \sinh s\, e_j, \qquad e_j \mapsto \sinh s\, e_0 + \cosh s\, e_j.$$

Compact sets $K_1, \dots, K_n$ satisfying $\max_{K_a} y_1 < \min_{K_{a+1}} y_1$ are said to be *time-ordered*. Euclidean tests are denoted g and g_a; for $y = (y_1, \dots, y_n)$ one puts $g(y) := (g_1(y_1), \dots, g_n(y_n))$, and $h(x)$ has the analogous meaning.

Proposition 3.1. *The hypotheses.* Fix the tuned data of Section F.2 and a charge $0 < e \le e_2$. Let μ be a volume-first limit point of the laws $P_{N,T}$ at the cutoffs (N, T), and let $\xi \in \Xi_\mu$. Then the law $\mathrm{lw} := \mu_\xi$ enjoys the properties below.

(G), (dF) Both hold by Section F.2.

(Inv) Every Euclidean map, including the reflections, leaves lw invariant, by Section F.2.

(RP) For every plane $\{\hat{n} \cdot y = \mathrm{po}\}$ and all $A \in L^2(\Sigma_{\{\hat{n}\cdot y > \mathrm{po}\}}, \mathrm{lw})$ one has $\langle \hat{r}_{\hat{n},\mathrm{po}} A, A\rangle_{\mathrm{lw}} \ge 0$, by Section F.2.

(Cl) For $A, B \in L^2(\mathrm{lw})$ one has $\mathrm{lw}(\bar{A}\, U(a) B) \to \overline{\mathrm{lw}(A)}\, \mathrm{lw}(B)$ as $|a| \to \infty$ in any manner, by Section F.2.

(Rec) The reconstruction at the plane $\{y_1 = 0\}$ is denoted

$$(\mathcal{H}, [\,\cdot\,], H, P_2, P_3, \Omega) := (\mathcal{H}_1, [\,\cdot\,]_1, \mathsf{H}_1, \mathsf{P}_2, \mathsf{P}_3, \Omega_1).$$

For $A, B \in L^2(\Sigma_+, \mathrm{lw})$, for $t \ge 0$ and for real a_2, a_3 it satisfies

$$\langle [A], [B] \rangle = \langle \hat{\theta} A, B \rangle_{\mathrm{lw}},$$
$$e^{-tH}[A] = [U(te_1)A], \qquad H = H^* \ge 0,$$
$$e^{i(a_2 P_2 + a_3 P_3)}[A] = [U(a_2 e_2 + a_3 e_3)A],$$

by [4, Proposition 7.1]. Its hypotheses hold because lw is translation invariant and reflection invariant and positive at $\{y_1 = 0\}$. Gaussian domination supplies the moment and continuity bounds used for the field insertions. The joint spectral measure of (H, P_2, P_3) is denoted by E; see [4, Section 7.1].

(Sp) The measure E is carried by $\{(\omega, \vec{p}) : \omega \ge |\vec{p}|\}$, by Section F.2.

(Vac) The kernel of H is $\mathbb{C}\,\Omega$, by Section F.2. This also follows from (Cl), as

$$\langle [A], e^{-tH}[B] \rangle = \mathrm{lw}(\overline{\hat{\theta} A}\, U(te_1) B) \to \langle [A], \Omega \rangle \langle \Omega, [B] \rangle \qquad (t \to \infty).$$

■

Free use is made of the following facts, taken from [4, Proposition 5.1] and [4, Sections 5.2 and 7.4]. The field X extends to L^2, with $\|X(g)\|_{L^N(\mathrm{lw})} \le c_N \|g\|$; and $X(g)$ belongs to $L^2(\Sigma_+, \mathrm{lw})$ whenever g vanishes on $\{y_1 < 0\}$. If Gi is a Euclidean map, its pullback is a unitary multiplicative operator on $L^2(\mathrm{lw})$ by (Inv). The test action gives $\widehat{\mathrm{Gi}} X(g) = X(\mathrm{Gi} \cdot g)$ and sends the generators of Σ_O to those of $\Sigma_{\mathrm{Gi}O}$; hence $\widehat{\mathrm{Gi}} L^2(\Sigma_O, \mathrm{lw}) = L^2(\Sigma_{\mathrm{Gi}O}, \mathrm{lw})$. Each S_n is a bounded symmetric form on L^2; moreover $S_1 = 0$ and $\overline{S_n(g_1, \dots)} = S_n(\bar{g}_1, \dots)$. Condition (dF) also gives $X(g) = 0$ almost surely if $*g = \nabla u$ for a scalar Schwartz function u, by approximation from compactly supported tests. In particular, Proposition 3.1 includes the one-plane hypotheses of Section 2 for $(1, 2)$.

The hypotheses listed above may instead be imposed on one law. They hold for almost every component of every volume-first limit point, at every charge and for every mass assignment [4, Section 6, Section 7.5, and Theorem 8.6]. In the proofs of Theorems 3.2 and 3.3, clustering is used through the unique-vacuum property. *Localisation.* If $g \in L^2$ vanishes off an open set O, then $X(g) \in L^p(\Sigma_O)$ for $p < \infty$: approximate g by smooth tests supported in O, use the Gaussian moment bound, and use that $L^2(\Sigma_O)$ is closed. Products of finitely many such fields belong to $L^2(\Sigma_O)$ by Hölder's inequality.

Theorems 3.2 and 3.3 apply to every law with the properties listed in Proposition 3.1. Every ergodic law in $\mathcal{K}$ has those properties. For the analytic-kernel statement (f2), Steps 1–10 use only (G), (Inv), (RP), (Rec) and (Sp); the unique-vacuum property is not needed.

Theorem 3.2. *The Poincaré group.* On $\mathcal{H}$ there is a unique strongly continuous unitary representation $\mathcal{U}$ of $\mathcal{P}_+^\uparrow$ satisfying (a), (b) and (c); this representation satisfies (d) and (e) as well.

(a) One has $\mathcal{U}(a,1) = e^{i(a^0H + a^2P_2 + a^3P_3)}$, and the joint spectrum of (H, P_2, P_3) is contained in $\{\omega \geq |\vec{p}|\}$.

(b) For $A \in L^2(\Sigma_+, \mathrm{lw})$ and for every rotation ϱ about e_1, that is, on Minkowski space, about e_0, one has $\mathcal{U}(0,\varrho)[A] = [\hat{\varrho}A]$.

(c) For $j = 2, 3$ one has $\mathcal{U}(0, \Lambda_s^{(j)}) = e^{isL_j}$, where L_j denotes the unique self-adjoint operator with the following property: *whenever O is open, $0 < t_0 < \frac{\pi}{2}$, $R_t^{(j)}O \subset \{y_1 > 0\}$ for $|t| \leq t_0$, and $A \in L^2(\Sigma_O, \mathrm{lw})$, the vector $[A]$ lies in $D(e^{tL_j})$ and $e^{tL_j}[A] = [\hat{R}_t^{(j)}A]$ for $|t| < t_0$.* Consequently the map $t \mapsto [\hat{R}_t^{(j)}A]$ has the holomorphic extension $\zeta \mapsto e^{\zeta L_j}[A]$ to the strip $\{|\operatorname{Re}\zeta| < t_0\}$, whose value at $\zeta = is$ is $e^{isL_j}[A]$; in other words, a rotation through the imaginary angle is is the boost of rapidity s. For every fixed t_0 the vectors $[A]$ just described form a total set.

(d) Each $\mathcal{U}(a,\Lambda)$ fixes Ω, that is, $\mathcal{U}(a,\Lambda)\Omega = \Omega$; and every vector that is invariant under all e^{itH} is a scalar multiple of Ω.

(e) The unitary group e^{isL_2} coincides with the group $\mathfrak{L}(s)$ of Lemma B.1; by Theorem 2.1 (b), $\mathfrak{L}(s)$ acts on the whole space $\mathcal{H}$.

■□

Because $\mathcal{U}$ is a representation, every relation valid in $\mathcal{P}_+^\uparrow$ becomes an identity between bounded operators; an example is

$$e^{isL_j}\,\mathcal{U}(a,1)\,e^{-isL_j} = \mathcal{U}(\Lambda_s^{(j)}a, 1).$$

Theorem 3.3. *The field.* Put $D := \bigcap_m D(H^m)$ and $\|\psi\|_m := \|(1+H)^m\psi\|$. One can find operators $F(h)\colon D \to D$, depending linearly on $h \in \mathrm{Sw}_1$, such that the following hold.

(a) *Field.* For $\psi \in D$ and $m \geq 0$ one has

$$\|F(h)\psi\|_m \leq |h|_m \|\psi\|_{m+4}, \qquad |h|_m := C \sum_{k \leq m+2} \binom{m+2}{k} \|\partial_0^k h\|_{L^1},$$

where C is an absolute constant; in particular $h \mapsto \langle\psi, F(h)\psi'\rangle$ is tempered. The relation $\langle\psi, F(h)\psi'\rangle = \langle F(\bar{h})\psi, \psi'\rangle$ holds, and therefore $F(h)$ is symmetric when h is real. Further, Ω belongs to D, and $\mathcal{U}(a,\Lambda)D = D$.

(b) *Covariance.* For $(a,\Lambda) \in \mathcal{P}_+^\uparrow$ one has $\mathcal{U}(a,\Lambda)F(h)\mathcal{U}(a,\Lambda)^{-1} = F((a,\Lambda)\cdot h)$; thus F is an antisymmetric tensor field of rank two.

(c) *Locality.* The commutator $[F(h), F(h')]$ vanishes on D if $(x-x')\cdot(x-x') < 0$ for all $x \in \operatorname{supp} h$ and $x' \in \operatorname{supp} h'$.

(d) *$dF = 0$.* For $u \in \mathcal{S}(\mathbb{R}^3; \mathbb{C})$ one has $F(d^*u) = 0$ on D, where

$$(d^*u)^{23} = -\partial_0 u, \qquad (d^*u)^{03} = \partial_2 u, \qquad (d^*u)^{02} = -\partial_3 u.$$

(e) *Cyclicity.* The vector Ω and the vectors $F(h_1)\cdots F(h_n)\Omega$ span a dense subspace of $\mathcal{H}$.

(f) *Relation to* lw. Put $\check{R} := JRJ^{-1}$ and

$$\mathcal{T}_n^{R,\pi} := \{z : (\check{R}^{-1}z_{\pi(a)})_a \in \mathcal{T}_n\}, \qquad \mathcal{D}_n := \bigcup_{R,\pi} \mathcal{T}_n^{R,\pi},$$

the union being taken over $R \in SO(3)$ and over all permutations π. There exist functions $\mathcal{W}_n(z;b)$, holomorphic in $z \in \mathcal{D}_n$ and n-linear in $b \in (\Lambda^2\mathbb{C}^3)^n$, with the properties (f1), (f2),

(f3) below. Here an n-linear form in b is identified with a linear form on $(\Lambda^2\mathbb{C}^3)^{\otimes n}$, so that $\mathcal{W}_n(z;c)$ makes sense for every c in the tensor product, for instance when c is the value at x of an element of Sw_n.

(f1) On $\mathcal{T}_n$ the function $\mathcal{W}_n$ is a function of the differences $z_{a+1}-z_a$ alone. For $n\geq 2$ it obeys

$$|\mathcal{W}_n(z;b)| \leq \prod_a (c_0|b_a|)\left(\tfrac{1}{2}\min_a d_{\mathcal{T}}(z_{a+1}-z_a)\right)^{-3n/2},$$

c_0 being an absolute constant, while $\mathcal{W}_1=0$. For $\Lambda\in SO_0(1,2)$ one has $\mathcal{W}_n(\Lambda z;\Lambda b)=\mathcal{W}_n(z;b)$. The form $W_n(h_1,\dots,h_n):=\langle\Omega,F(h_1)\cdots F(h_n)\Omega\rangle$ has a continuous extension to Sw_n, and for $\mathbf{h}\in\mathrm{Sw}_n$

$$\int dx\,\mathcal{W}_n(x+i\eta;\mathbf{h}(x)) \to W_n(\mathbf{h}) \qquad \text{as } \eta\to 0 \text{ with } \eta_{a+1}-\eta_a\in V_+.$$

(f2) Whenever the y_a are pairwise distinct, the point $Jy=(Jy_1,\dots,Jy_n)$ lies in $\mathcal{D}_n$. Permuting the pairs (z_a,b_a) leaves $\mathcal{W}_n$ unchanged, and so does the map $(z,b)\mapsto(\check{R}z,\check{R}b)$. If $g_1,\dots,g_n\in L^2$ have compact supports which are pairwise disjoint, then

$$S_n(g_1,\dots,g_n)=\int dy\,\mathcal{W}_n(Jy;Jg(y)).$$

Written out in components, with the index 1 identified with 0, this says that $S_n^{\boldsymbol{\alpha}}(y)=i^{\mathrm{ti}(\boldsymbol{\alpha})}\,\mathcal{W}_n^{\boldsymbol{\alpha}}(Jy)$, so that every time index contributes a factor i. It follows that the S_n are real analytic wherever no two arguments coincide, configurations with equal times included.

(f3) From the W_n one recovers the functions $\mathcal{W}_n(z;b)$ on $\mathcal{D}_n$, and from these the S_n at pairwise disjoint compact supports; by Proposition 1.2 (c), no law with (G) and (dF) other than lw has these S_n at such supports. In particular, two laws having the properties of Proposition 3.1 and the same W_n for all $n\geq 1$ are equal.

■□

Proof of Theorems 3.2 and 3.3. We first construct bounded slab insertions and holomorphic point-field vectors (Steps 1–7). Continuation, totality and the rotation identities then give the Poincaré representation (Steps 8–15). Boundary forms and Hamiltonian estimates turn the kernels into operators on D (Steps 16–19). Covariance, locality, closedness, cyclicity and the boundary values are established in Steps 20–25.

Auxiliary classical facts.

Fact 1 (mean value). For u holomorphic in a neighbourhood of the closed polydisc $\{a\in\mathbb{C}^M : |a_i|\leq\delta\}$,

$$u(0)=(\pi\delta^2)^{-M}\int_{\{|a_i|<\delta\}} u\,d\lambda^{2M}.$$

Fact 2 (identity theorem on a real slice). Suppose that $I\subset\mathbb{C}^M$ is a connected open set, that L is an invertible complex linear map, and that u is a holomorphic function on I with values in a Hilbert space. If $u=0$ on some non-empty subset of $I\cap L\mathbb{R}^M$ which is open relative to $I\cap L\mathbb{R}^M$, then u vanishes identically.

Fact 3 (Schwarz reflection). Suppose that u is continuous on the closed half-plane $\{\mathrm{Im}\,\lambda\geq 0\}$ and holomorphic on the open half-plane $\{\mathrm{Im}\,\lambda>0\}$, and that u vanishes on some non-empty open interval $I\subset\mathbb{R}$; then $u=0$. For the proof, choose an open disc Δ with centre in I such that $\Delta\cap\mathbb{R}\subset I$, and let $\tilde{u}$ be the function on Δ that coincides with u on $\Delta\cap\{\mathrm{Im}\,\lambda\geq 0\}$ and is zero on $\Delta\cap\{\mathrm{Im}\,\lambda<0\}$. As $u=0$ on $\Delta\cap\mathbb{R}$, the function $\tilde{u}$ is continuous on Δ; it is holomorphic away from the real axis, and by Morera's theorem it is holomorphic on Δ. Being zero on the open lower half of Δ, it is zero on all of Δ. Hence u is zero on the non-empty open set $\Delta\cap\{\mathrm{Im}\,\lambda>0\}$; the identity theorem, applied on the connected set $\{\mathrm{Im}\,\lambda>0\}$, and then continuity yield $u=0$.

Restricting to the complex lines $x+\lambda\eta$, $\eta\in V_+$, and treating one variable at a time, one deduces a statement in several variables. Let $B_1,\dots,B_k$ be non-empty open subsets of $\mathbb{R}^3$, and let G be a function on $\overline{\mathcal{T}}^k$ such that, for each a and for arbitrary values of the other variables in $\overline{\mathcal{T}}$, G is continuous on $\overline{\mathcal{T}}$ and holomorphic on $\mathcal{T}$ as a function of its a-th variable. If $G=0$ on $B_1\times\cdots\times B_k$, then $G=0$ on $\overline{\mathcal{T}}^k$. The proof is an induction on $a=1,\dots,k$; at the stage a it is known that G vanishes on $\overline{\mathcal{T}}^{a-1}\times B_a\times\cdots\times B_k$.

(1) Give the variables of index smaller than a arbitrary values in $\overline{\mathcal{T}}$ and those of index larger than a arbitrary values in $B_{a+1},\dots,B_k$, and let $x\in B_a$ and $\eta\in V_+$. Inserting $x+\lambda\eta$ as the a-th variable, one obtains a function of λ that is continuous for $\operatorname{Im}\lambda\geq 0$ and holomorphic for $\operatorname{Im}\lambda>0$. It is zero at each real λ with $x+\lambda\eta\in B_a$; these λ form an open subset of $\mathbb{R}$ which contains 0, and so contains a non-empty open interval. By the case of one variable, this function of λ vanishes identically.

(2) Taking $\lambda=i$, one sees that G, as a function of its a-th variable, is zero on B_a+iV_+, which is a non-empty open subset of the convex set $\mathcal{T}$. By the identity theorem it is zero on $\mathcal{T}$, and by continuity on $\overline{\mathcal{T}}$. Consequently G vanishes on $\overline{\mathcal{T}}^{\,a}\times B_{a+1}\times\cdots\times B_k$, which completes the stage a.

Steps 24 and 25 and Part 3 of the proof of Corollary 3.4 use the case $B_1=\cdots=B_k=\mathbb{R}^3$; the proof of Proposition 3.6 uses balls.

Fact 4 (spectral theorem). Let K be a self-adjoint operator.

(i) Suppose $v\in D(e^{tK})$ for all $|t|<t_0$; then $\gamma\mapsto e^{i\gamma K}v$ is holomorphic in the strip $\{|\operatorname{Im}\gamma|<t_0\}$.

(ii) Conversely, if $s\mapsto e^{isK}v$ has a holomorphic extension V to that strip, then $v\in D(e^{tK})$ and $e^{tK}v=V(-it)$; to see this, one pairs with the vectors of compact spectral support, which constitute a core.

(iii) Every dense subspace $D'\subset D(e^{tK})$ satisfying $e^{isK}D'\subset D'$ for all real s is a core for e^{tK}. Indeed, if ϕ is orthogonal to D' with respect to the graph inner product, then each of the finite measures $(1+e^{2t\lambda})\,d\langle\phi,E_K(\lambda)\psi\rangle$, $\psi\in D'$, has vanishing Fourier transform.

Fact 5 (nuclear theorem with a bound). Let M be an n-linear form on Sw_1 such that $|M(h_1,\dots,h_n)|\leq\prod_a p(h_a)$ for some continuous seminorm p. Then $M(h_1,\dots,h_n)=W(h_1\otimes\cdots\otimes h_n)$ for a unique continuous linear form W on Sw_n, and $|W|\leq P$, where the continuous seminorm P is determined by p and n alone. The proof proceeds by expansion in Hermite functions, as in [4, Proposition 5.1 and Section 5.3].

Fact 6 (Lorentz group). In the basis e_0,e_2,e_3 the form $x\cdot x$ has the matrix $-J^2=\operatorname{diag}(1,-1,-1)$. One defines

$$O(1,2):=\{\Lambda\text{ real}:\Lambda^TJ^2\Lambda=J^2\},\qquad SO_0(1,2):=\{\Lambda\in O(1,2):\det\Lambda=1,\ (\Lambda e_0)^0>0\},$$
$$so(1,2):=\{Y\text{ real}:Y^TJ^2+J^2Y=0\}.$$

An element $\Lambda\in O(1,2)$ permutes the two convex components $\pm V_+$ of $\{x\cdot x>0\}$; hence $(\Lambda e_0)^0>0$ holds if and only if $\Lambda V_+=V_+$, and $SO_0(1,2)$ is a group.

(i) For $\Lambda\in O(1,2)$ one has $(\Lambda e_0)\cdot(\Lambda e_0)=1$, whence $|(\Lambda e_0)^0|\geq 1$. Let $Y\in so(1,2)$. Then $e^{sY}\in O(1,2)$, and both $\det e^{sY}\in\{\pm1\}$ and the sign of $(e^{sY}e_0)^0$ depend continuously on s; therefore $e^{sY}\in SO_0(1,2)$.

(ii) Each $\Lambda\in SO_0(1,2)$ can be written $\Lambda=\varrho\Lambda_s^{(2)}\varrho'$ with $s\geq 0$ and rotations ϱ,ϱ' about e_0. To see this, write $\Lambda e_0=\cosh s\,e_0+\sinh s\,u$, with u a unit vector in $\operatorname{span}(e_2,e_3)$, which is possible by (i). For a rotation ϱ with $\varrho e_2=u$ one has $\varrho\Lambda_s^{(2)}e_0=\Lambda e_0$. The matrix $\varrho':=(\varrho\Lambda_s^{(2)})^{-1}\Lambda$ preserves the form and fixes e_0; it therefore maps $\operatorname{span}(e_2,e_3)$ isometrically onto itself, and since its determinant is 1, it is a rotation.

(iii) Each of the three factors in (ii) has the form e^{sY} with $Y\in so(1,2)$. Hence these one-parameter groups generate $SO_0(1,2)$, and $SO_0(1,2)$ is connected.

(iv) Let $\theta_2 = \mathrm{diag}(1,-1,1)$ and $\theta = \mathrm{diag}(-1,1,1)$. The group $O(1,2)$ is the disjoint union of the four cosets $SO_0(1,2)$, $SO_0(1,2)\,\theta_2$, $SO_0(1,2)\,\theta$ and $SO_0(1,2)\,\theta_2\theta$. Indeed, if $(\Lambda e_0)^0 < 0$ one multiplies Λ on the right by θ, and afterwards by θ_2 if the determinant equals -1; multiplication by θ_2 leaves $(\Lambda e_0)^0$ unchanged.

Step 1 (semigroup). For $\zeta \in \overline{\mathcal{T}}$ define

$$T(\zeta) := \int e^{i(\zeta^0\omega + \vec{\zeta}\cdot\vec{p})}\, dE.$$

According to (Sp), the integrand is bounded in modulus by $e^{-d_{\mathcal{T}}(\zeta)\omega}$. Consequently T is a contraction semigroup with $T(\zeta)^* = T(-\bar{\zeta})$ and $T(\zeta)\Omega = \Omega$, and $T(\mathcal{T})\mathcal{H} \subset D$. On each $D(H^m)$ equipped with $\|\cdot\|_m$, it is a contraction, strongly continuous on $\overline{\mathcal{T}}$ and holomorphic on $\mathcal{T}$. At real points a one has $T(a) = e^{i(a^0 H + \vec{a}\cdot\vec{P})}$, and for $y_1 \geq 0$ one has $T(Jy)[A] = [U(y)A]$.

Step 2 (insertions; the single bound that is uniform in the number of fields). The hypotheses (G), (T_{123}), (R_1), (RP_1) of [4, Proposition 8.1] are among those of Proposition 3.1. That proposition therefore yields, for each $g \in L^2$ vanishing outside $\{|y_1| \leq d\}$, a unique bounded operator $\mathcal{B}_d(g)$, depending linearly on g, such that

$$\mathcal{B}_d(g)[A] = [X(\tau_{de_1} g)\, U(2de_1) A] \qquad \text{for } A \in \textstyle\bigcap_p L^p(\Sigma_+, \mathrm{lw});$$

in [4] this operator is denoted $\mathcal{B}^{\mathrm{c}}_d(*g)$. It satisfies the following.

(i) $\|\mathcal{B}_d(g)\| \leq k_1 \|g\|$ with $k_1 = (2\exp 1)^{1/2}$.
(ii) $\mathcal{B}_d(g)^* = \mathcal{B}_d(\overline{\theta \cdot g})$.
(iii) For real a and $s \geq |a_1|$,

$$T(ise_0 + Ja)\, \mathcal{B}_d(g)\, T(ise_0 - Ja) = \mathcal{B}_{d+s}(\tau_a g).$$

If $a_3 = 0$, this is part (c) of [4, Proposition 8.1]. If $a = a_3 e_3$ and $s = 0$, it results from the identity that defines $\mathcal{B}_d$, from (Rec) applied to $e^{ia_3 P_3} = T(a_3 e_3)$, and from the facts that $U(a_3 e_3)$ is multiplicative and that $U(a_3 e_3) X(g) = X(\tau_{a_3 e_3} g)$. For an arbitrary real a, the semigroup law of Step 1 gives $T(ise_0 \pm Ja) = T(\pm a_3 e_3)\, T(ise_0 \pm J(a_1, a_2, 0))$, and one has $\tau_a = \tau_{a_3 e_3} \tau_{(a_1,a_2,0)}$; the identity follows on applying the case $a_3 = 0$ first, and then the case $a = a_3 e_3$, $s = 0$ with $d + s$ and $\tau_{(a_1,a_2,0)} g$ in place of d and g, the latter function vanishing outside $\{|y_1| \leq d + s\}$ because $s \geq |a_1|$.

The constant k_1 is independent of the vectors and so of the number of fields they contain; it is obtained from (G) through repeated reflections and Cauchy's estimate [4, Section 8.1]. Accordingly, no estimate in what follows involves a sum over n.

Step 3 (point fields). Fix $\delta > 0$. For $v \in Q := (-\delta, \delta)^3$ set $\chi_v(u) := \prod_i 1[u_i^2 + v_i^2 < \delta^2]$, and for $d_{\mathcal{T}}(\zeta), d_{\mathcal{T}}(\zeta') \geq 3\delta$ set

$$\Phi_b(\zeta, \zeta') := (\pi\delta^2)^{-3} \int_Q dv\, T(\zeta - i\delta e_0 + iJv)\, \mathcal{B}_\delta(\chi_v J^{-1} b)\, T(\zeta' - i\delta e_0 - iJv).$$

The arguments of T belong to $\mathcal{T}$, because $iJv = (-v_1, i\vec{v})$.

(a) Since $\|\chi_v\|_2 \leq (2\delta)^{3/2}$, one has $\|\Phi_b\| \leq 2^{9/2}\pi^{-3} k_1 |b|\, \delta^{-3/2}$.
(b) For $\kappa, \kappa' \in \overline{\mathcal{T}}$ one has $T(\kappa)\Phi_b(\zeta, \zeta')T(\kappa') = \Phi_b(\kappa + \zeta, \zeta' + \kappa')$.
(c) $\Phi_b(\zeta, \zeta')^* = \Phi_{\bar{b}}(-\bar{\zeta}', -\bar{\zeta})$; this follows from Step 2 (ii), from (K), and from the change of variables $v \to -\theta v$.

Step 4 (the point fields are the kernel of the insertions).

(d) *If $g \in L^2$ has compact support in $\{|y_1| \leq d'\}$ and $d - d' \geq 3\delta$, then*

$$\mathcal{B}_d(g) = \int dy\, \Phi_{Jg(y)}(ide_0 + Jy,\, ide_0 - Jy).$$

To prove (d), consider

$$Z(a) := T(i(d-d')e_0 + Ja)\,\mathcal{B}_{d'}(g)\,T(i(d-d')e_0 - Ja).$$

This operator depends weakly holomorphically on a in the region $|\operatorname{Re} a_1| + |\operatorname{Im}\vec{a}| < d - d'$, which contains a neighbourhood of $\{|a_i| \le \delta\}$; moreover $Z(0) = \mathcal{B}_d(g)$. One applies Fact 1 with $a = u + iv$. The imaginary part iv of the shift yields the factors $T(\,\cdot \pm iJv)$ appearing in Step 3, while Step 2 (iii) converts the real part u into the translation $\tau_u g$. In L^2 one has $\int du\,\chi_v(u)\,\tau_u g = \int dy\,(\tau_y\chi_v)\,g(y)$, and a second use of Step 2 (iii) converts τ_y into factors T.

Step 5 (independence of δ). Take two values of δ. The corresponding weakly continuous functions $y \mapsto \Phi_b(ide_0 + Jy, ide_0 - Jy)$ yield, by (d), the same integral against every g, and are therefore equal. It follows from (b) that the two functions Φ_b coincide where $\min(d_{\mathcal{T}}(\zeta), d_{\mathcal{T}}(\zeta')) > 3\delta$, with δ the larger of the two values, and by continuity also where equality holds. Hence Φ_b is a norm holomorphic function on $\mathcal{T}\times\mathcal{T}$ which has the properties (b) and (c) of Step 3 and, for compact support in $\{|y_1| < d\}$, the property (d) of Step 4. Moreover, the choice $3\delta = \min(d_{\mathcal{T}}(\zeta), d_{\mathcal{T}}(\zeta'))$ in Step 3 (a) gives

$$\|\Phi_b(\zeta,\zeta')\| \le c_0|b|\min(d_{\mathcal{T}}(\zeta), d_{\mathcal{T}}(\zeta'))^{-3/2}, \qquad c_0 = 2^{9/2}3^{3/2}\pi^{-3}k_1.$$

From (b) one also sees that $\Phi_b(\cdot,\zeta')\Phi_{b'}(\zeta'',\cdot)$ is a function of $\zeta'+\zeta''$ alone, and that neither $\Phi_b(\zeta,\kappa)\Omega$ nor $\langle\Omega, \Phi_b(\kappa,\zeta)\,\cdot\,\rangle$ depends on κ.

Step 6 (vectors and kernels). On $\mathcal{T}_n^+$ put

$$\Psi_n(z;b) := \Phi_{b_1}\Big(z_1, \frac{z_2-z_1}{2}\Big)\Phi_{b_2}\Big(\frac{z_2-z_1}{2}, \frac{z_3-z_2}{2}\Big)\cdots\Phi_{b_n}\Big(\frac{z_n-z_{n-1}}{2}, \kappa\Big)\Omega,$$

and put $\Psi_0 := \Omega$ and $D_1 := \operatorname{span}\{\Psi_n(z;b)\}$; on $\mathcal{T}_n$ put $\mathcal{W}_n(z;b) := \langle\Omega, \Psi_n(z+\kappa';b)\rangle$. Here $\kappa \in \mathcal{T}$ is arbitrary, and κ' is any vector with $z_1 + \kappa' \in \mathcal{T}$. Step 5 implies the following.

(a) The bound of (f1) holds for $\mathcal{W}_n$. For $n = 1$, letting $d_{\mathcal{T}}(\kappa)$ and $d_{\mathcal{T}}(\kappa')$ tend to ∞ in the bound of Step 5 gives $\mathcal{W}_1 = 0$; this is also a consequence of Step 7.

(b) $T(\kappa)\Psi_n(z;b) = \Psi_n(z+\kappa;b)$. Hence $D_1 \subset D$ and $e^{-sH}D_1 \subset D_1$, and $\mathcal{W}_n$ is a function of the differences only.

(c) For the inner products one finds

$$\langle\Psi_m(z';b'), \Psi_n(z;b)\rangle = \mathcal{W}_{m+n}(\bar{z}'_m, \ldots, \bar{z}'_1, z_1, \ldots, z_n; \bar{b}'_m, \ldots, \bar{b}'_1, b_1, \ldots, b_n).$$

Step 7 (ordered times). *For $g_a \in L^2$ with time-ordered compact supports in $\{y_1 > 0\}$,*

$$\Big[\prod_a X(g_a)\Big] = \int dy\,\Psi_n(Jy; Jg(y)).$$

Choose slabs $\{|y_1 - c_a| < d_a\}$ containing $\operatorname{supp} g_a$ such that $0 \le c_1 - d_1$ and $c_a + d_a \le c_{a+1} - d_{a+1}$. By induction over the definition of $\mathcal{B}$ one obtains

$$\Big[\prod_a X(g_a)\Big] = e^{-(c_1-d_1)H}\mathcal{B}_{d_1}(\tau_{-c_1e_1}g_1)\,e^{-(c_2-d_2-c_1-d_1)H}\mathcal{B}_{d_2}(\tau_{-c_2e_1}g_2)\cdots\Omega.$$

Substituting Step 4 (d) for each factor $\mathcal{B}$ and combining the arguments by means of Step 3 (b), one sees that the first argument turns into Jy_1, and the argument between two consecutive factors into $J(y_{a+1} - y_a)$. Consequently $S_n(g_1,\ldots,g_n) = \int dy\,\mathcal{W}_n(Jy; Jg(y))$ for time-ordered compact supports wherever they lie, and $\mathcal{W}_1 = 0$.

Step 8 (rotated and permuted tubes). Define on $\mathcal{T}_n^{R,\pi}$

$$\mathcal{W}_n^{R,\pi}(z;b) := \mathcal{W}_n\big((\check{R}^{-1}z_{\pi(a)})_a; (\check{R}^{-1}b_{\pi(a)})_a\big).$$

Since $\mathcal{T} = \{Jw : \operatorname{Re} w_1 > |\operatorname{Im}\vec{w}|\}$ and since R and π are real, the set $\mathcal{T}_n^{R,\pi}$ is open and convex, and the map $Jw \mapsto J\operatorname{Re} w$ sends it into itself. The Euclidean points in it are exactly the Jy for which

$u \cdot y_{\pi(1)} < \cdots < u \cdot y_{\pi(n)}$, with $u = Re_1$. For $g_a \in L^2$ whose compact supports lie over such points one has

$$S_n(g_1, \ldots, g_n) = S_n(R^{-1} \cdot g_{\pi(1)}, \ldots, R^{-1} \cdot g_{\pi(n)}) = \int dy\, \mathcal{W}_n^{R,\pi}(Jy; Jg(y)).$$

The first equality holds because the $X(g)$ commute and because of (Inv), tensor indices included; the second follows from Step 7 and the substitution $y \to Ry$.

Step 9 (one function; (f2)). Let two of the tubes have non-empty intersection. This intersection is convex, and the Euclidean points in it include a relatively open set; there the two functions are both continuous densities of S_n, hence equal. Fact 2, applied with $L = J$, extends the equality to the whole intersection. The $\mathcal{W}_n^{R,\pi}$ are therefore restrictions of a single holomorphic function $\mathcal{W}_n$ on $\mathcal{D}_n$, whose construction gives it the invariance properties stated in (f2). If the y_a are pairwise distinct, there is a unit vector u for which the numbers $u \cdot y_a$ are distinct as well; thus $Jy \in \mathcal{D}_n$. For disjoint compact supports, one cuts the g_a into pieces by multiplying with indicator functions of small cubes, which is legitimate in L^2; by the Lebesgue-number lemma, each product of pieces then lies over a single tube.

Step 10 (Lorentz invariance). Let $z \in \mathcal{T}_n$. Owing to the holomorphy in z and the multilinearity in b, the derivative $\frac{d}{ds}\mathcal{W}_n(e^{sY}z; e^{sY}b)|_{s=0}$ depends complex linearly on the matrix Y; by Step 9 it vanishes for $Y \in J\,so(3)J^{-1}$. The complex span of $J\,so(3)J^{-1}$ contains $so(1,2)$, since $J^2 = -\operatorname{diag}(1,-1,-1)$ implies that $Y \in so(1,2)$ if and only if $J^{-1}YJ$ is antisymmetric. For such Y the matrices e^{sY} preserve $\mathcal{T}_n$ by Fact 6 (i), and $\mathcal{W}_n(e^{sY}z; e^{sY}b)$ is thus independent of s; by Fact 6 (iii), these one-parameter groups generate $SO_0(1,2)$.

Step 11 (totality; here (dF) is indispensable). *The subspace D_1 is dense.* Proposition 3.1 makes Theorem 2.1 (a) applicable; according to it, Ω together with the vectors $[\prod_a X(g_a)]$ with time-ordered supports in $\{y_1 > 0\}$ is a total set, and Step 7 places these vectors in $\overline{D_1}$. In the proof of Theorem 2.1 (a), the hypothesis (dF) enters through [4, Proposition 8.4], applied to the forms $\langle \psi, [\prod_a X(g_a)]\rangle$, which vanish on gradients by (dF). Step 7 with shrinking supports, combined with Fact 2 on $\mathcal{T}_n^+$, shows that a smaller set is total as well: Ω and the $[\prod_a X(g_a)]$ whose g_a are supported near one time-ordered configuration in $\{y_1 > 0\}$, one configuration being fixed for each n.

Step 12 (the representation; Theorem 3.2 (a), (b), (d)). Step 10, Step 6 (c) and the identity $\overline{\Lambda z} = \Lambda \bar{z}$ show that the Gram matrices of the vectors $\Psi_n(z;b)$ do not change under $(z,b) \mapsto (\Lambda z, \Lambda b)$. The prescription $\mathcal{U}(0,\Lambda)\Psi_n(z;b) := \Psi_n(\Lambda z; \Lambda b)$ therefore defines an isometry of D_1 onto D_1. On D_1 one reads off the group law, the strong continuity and the relation $\mathcal{U}(0,\Lambda)T(a)\mathcal{U}(0,\Lambda)^{-1} = T(\Lambda a)$. Consequently $\mathcal{U}(a,\Lambda) := T(a)\mathcal{U}(0,\Lambda)$ is a representation of the group, and it leaves $\Omega = \Psi_0$ fixed. What (a) and (d) assert beyond this is contained in (Sp) and (Vac). Since J commutes with every rotation ϱ about e_1, the relation $\mathcal{U}(0,\varrho)[B] = [\hat{\varrho}B]$ holds on the total set formed by the time-ordered products of Step 7; moreover $[A] \mapsto [\hat{\varrho}A]$ is unitary by Section B.2.

Step 13 (continuation in the angle: products). Let j and $0 < t_0 < \frac{\pi}{2}$ be fixed; write $R_t := R_t^{(j)}$ and $\Lambda_s := \Lambda_s^{(j)}$, and let L be the self-adjoint operator, provided by Stone's theorem, for which $\mathcal{U}(0,\Lambda_s) = e^{isL}$. A product $B = \prod_a X(g_a)$ is called *admissible* when the $g_a \in L^2$ have compact supports and, for every $|t| \le t_0$, the sets $R_t \operatorname{supp} g_a$ are time-ordered in $\{y_1 > 0\}$. Denote by K the union of the supports. On the compact set $[-t_0, t_0] \times K$ the function $(t,y) \mapsto (R_t y)_1$ is continuous and positive; hence it is bounded below there by some $2\epsilon > 0$. Let O be the open ϵ-neighbourhood of K. Each R_t being an isometry, $(R_t y)_1 > \epsilon$ for $y \in O$, so that $R_t O \subset \{y_1 > 0\}$ for $|t| \le t_0$; and $B \in L^2(\Sigma_O, \mathrm{lw})$ by the statement *Localisation* following Proposition 3.1. The vectors $[B]$, as well as $\Omega = [1]$, are thus among the vectors $[A]$ of Theorem 3.2 (c). Step 11, used at the configuration $(e_1, 2e_1, \ldots, ne_1)$, shows that these $[B]$ together with Ω form a total set; this is the totality claimed in Theorem 3.2 (c). By Steps 7 and 12, $e^{isL}[B] = V(s)$ with

$$V(\gamma) := \int dy\, \Psi_n(JR_{i\gamma}y; JR_{i\gamma}g(y)).$$

If $\gamma = s - it$, then $JR_{i\gamma}y = \Lambda_s JR_t y$ lies in $\mathcal{T}_n^+$. Therefore V is holomorphic in $\{|\operatorname{Im}\gamma| < t_0\}$ with $V(-it) = [\hat{R}_t B]$, and Fact 4 (ii) yields $e^{tL}[B] = [\hat{R}_t B]$.

Step 14 (all A; Theorem 3.2 (c)). Take O, t_0 and A as in Theorem 3.2 (c), an admissible B, and s, t with $|s| + |t| < t_0$. From $\theta R_t = R_{-t}\theta$ and (Inv) one gets

$$\langle [\hat{R}_t \hat{R}_s B], [A]\rangle = \langle \hat{\theta}\hat{R}_s B, \hat{R}_t A\rangle_{\mathrm{lw}}, \quad \text{i.e.} \quad \langle e^{(s+t)L}[B], [A]\rangle = \langle e^{sL}[B], [\hat{R}_t A]\rangle.$$

By Fact 4 (i) both members are antiholomorphic in s; evaluating at $s = is'$ with real s' gives $\langle e^{tL}v, [A]\rangle = \langle v, [\hat{R}_t A]\rangle$ for $v = e^{is'L}[B]$. The linear span of these v and of Ω is a dense subspace contained in $D(e^{tL})$ and stable under every e^{isL}; by Fact 4 (iii) it is a core for the self-adjoint operator e^{tL}. It follows that $[A] \in D(e^{tL})$ and $e^{tL}[A] = [\hat{R}_t A]$.

Step 15 (uniqueness; Theorem 3.2 (e)). Suppose that a self-adjoint operator L' has the property stated in Theorem 3.2 (c). Applying this property to an admissible B and to the constant function 1, with the set O of Step 13, one obtains $e^{tL'}[B] = [\hat{R}_t B]$ and $e^{tL'}\Omega = \Omega$ for $|t| < t_0$. By Steps 13 and 12, L satisfies the same two identities. Let v be $[B]$ or Ω; the latter belongs to the total set as well. By Fact 4 (i), $e^{i\gamma L'}v$ and $e^{i\gamma L}v$ are holomorphic in the strip, and they coincide at $\gamma = -it$; hence they coincide for real γ. Thus $e^{isL'} = e^{isL}$, and Stone's theorem gives $L' = L$. The uniqueness of $\mathcal{U}$ follows, because $SO_0(1,2) = \{\varrho\Lambda_s^{(2)}\varrho'\}$ by Fact 6 (ii). For (e), consider admissible products B of plates; by Step 11 and [4, Section 8.2], these $[B]$ and Ω form a total set. In the definitions in Section 2 the symbol Λ_s denotes the matrix that is written R_{is} in this section. For such B, the map $s \mapsto \mathfrak{L}(s)[B]$ continues to the strip and takes the value $[\hat{R}_t B]$ at $s = -it$, by Section B.1. Here, for $\gamma = s - it$ with $|t| < t_0$, one has $R_{i\gamma} = R_{is}R_t$; the matrix R_t carries each point of the product of the supports to a real configuration that is time-ordered in $\{y_1 > 0\}$ and hence lies in $\mathcal{W}^{(n)}$, and R_{is} maps $\mathcal{W}^{(n)}$ into itself. The conclusion then follows from Fact 4 (ii), as in the case of L'.

Step 16 (the field as a form). For $t, t' > 0$ consider the operator

$$F_{t,t'}(h) := \int dx\, \Phi_{h(x)}(x + ite_0, -x + it'e_0).$$

Its norm does not exceed $c_0\|h\|_{L^1}\min(t,t')^{-3/2}$, and $e^{-sH}F_{t,t'}(h)e^{-s'H} = F_{t+s,t'+s'}(h)$. Consequently the form

$$q_h(\psi,\psi') := \langle e^{tH}\psi, F_{t,t'}(h)e^{t'H}\psi'\rangle \qquad \text{on } D_0 := \bigcup_M E(\{\omega \le M\})\mathcal{H}$$

is independent of t and t'. Insert $(1+H)^{-2} = \int_0^\infty te^{-t}e^{-tH}dt$ on either side and use $\iint tt'e^{-t-t'}\min(t,t')^{-3/2}\,dt\,dt' \le 2\sqrt{\pi}$; this gives

$$|q_h(\psi,\psi')| \le C\|h\|_{L^1}\|\psi\|_2\|\psi'\|_2, \qquad C = 2\sqrt{\pi}c_0.$$

The exponent $-3/2$ coincides with that of the free field. If instead s powers are taken on each side, by means of

$$(1+H)^{-s} = \int_0^\infty t^{s-1}e^{-t}e^{-tH}dt \Big/ \int_0^\infty t^{s-1}e^{-t}dt,$$

the relevant double integral, that of $(tt')^{s-1}e^{-t-t'}\min(t,t')^{-3/2}$, is finite exactly when $s > \frac{3}{2}$; the estimate above uses $s = 2$. Since the estimate involves h only through $\|h\|_{L^1}$, and since $x \mapsto \Phi_b(x + ite_0, -x + it'e_0)$ is bounded and norm continuous, q_h extends to finite measures h. Put $e_{02} := e_0 \wedge e_2$, $e_{03} := e_0 \wedge e_3$ and $e_{23} := e_2 \wedge e_3$, so that $h = \sum_\alpha h^\alpha e_\alpha$. For $x \in \mathbb{R}^3$ and each index α, the point field $F_\alpha(x) := q_{\delta_x e_\alpha}$ is therefore a form on $D(H^2) \times D(H^2)$, and

$$q_h(\psi,\psi') = \sum_\alpha \int dx\, h^\alpha(x)\, F_\alpha(x)(\psi,\psi') \qquad (\psi,\psi' \in D(H^2));$$

as q_h is the form of $F(h)$ (Step 17), this says that $F(h) = \sum_\alpha \int dx\, h^\alpha(x)\, F_\alpha(x)$ in the sense of forms. For $f \in L^1(\mathbb{R}^2; \Lambda^2\mathbb{C}^3)$ the field $F(\delta_c \otimes f)$ at the sharp time $x^0 = c$ is likewise a form on $D(H^2) \times D(H^2)$. Derivatives of h in x^0 are needed only in Step 17, where the form is turned into

an operator. No assertion is made on whether $F(\delta_c \otimes f)$ is an operator on D. **Step 17** (the field as an operator; Theorem 3.3 (a)). The relation $e^{isH} F_{t,t'}(h) e^{-isH} = F_{t,t'}(\tau_{se_0} h)$ implies

$$q_h(H\psi, \psi') - q_h(\psi, H\psi') = q_{i\partial_0 h}(\psi, \psi'),$$

and consequently

$$q_h((1+H)^m \psi, \psi') = \sum_k \binom{m}{k} q_{(i\partial_0)^k h}(\psi, (1+H)^{m-k}\psi').$$

Applying this identity with $m+2$ in place of m and with $(1+H)^{-2}\psi$ in place of ψ, and estimating each term by Step 16, one arrives at

$$|q_h((1+H)^m \psi, \psi')| \le |h|_m \|\psi\| \, \|\psi'\|_{m+4}.$$

Riesz representation provides $F(h)\psi'$ with $q_h(\psi, \psi') = \langle \psi, F(h)\psi' \rangle$. Since D_0 is a core for $(1+H)^m$, the vector $F(h)\psi'$ lies in $D(H^m)$ and satisfies the bound in (a); by continuity, $F(h)$ extends to D. The equality $\mathcal{U}(a,\Lambda)D = D$ holds because (Sp) implies

$$\mathcal{U}(0,\Lambda) H \, \mathcal{U}(0,\Lambda)^{-1} = (\Lambda e_0)^0 H + \overrightarrow{\Lambda e_0} \cdot \vec{P} \le 2(\Lambda e_0)^0 H.$$

More precisely, $\|\mathcal{U}(0,\Lambda)\psi\|_m \le (2(\Lambda e_0)^0)^m \|\psi\|_m$. To prove this, write $\mathcal{U}(0,\Lambda)^{-1} H \, \mathcal{U}(0,\Lambda) = \int f \, dE$, where $f = (\Lambda^{-1} e_0)^0 \omega + \overrightarrow{\Lambda^{-1} e_0} \cdot \vec{p}$. On the cone one has $0 \le f \le c\,\omega$ with $c = 2(\Lambda^{-1}e_0)^0 = 2(\Lambda e_0)^0 \ge 2$; here $(\Lambda^{-1}e_0)^0$ and $(\Lambda e_0)^0$ coincide because Fact 6, together with $J^4 = 1$, gives $\Lambda^{-1} = J^2 \Lambda^T J^2$. Hence $1 + f \le c\,(1+\omega)$, and

$$\|(1+H)^m \, \mathcal{U}(0,\Lambda)\psi\| = \Big\| \int (1+f)^m dE \, \psi \Big\| \le c^m \|(1+H)^m \psi\|.$$

Step 18 (rules).

(e1) $T(\zeta)F(h)T(\zeta') = \int dx \, \Phi_{h(x)}(\zeta + x, \zeta' - x)$ on $\mathcal{T} \times \mathcal{T}$.

(e2) $F_{t,t}(h)^* = F_{t,t}(\bar{h})$ by Step 3 (c), that is, by (K); this is the symmetry in Theorem 3.3 (a).

(e3) $T(a)F(h)T(-a) = F(\tau_a h)$ for real a.

(e4) For $\kappa, \kappa_a \in \mathcal{T}$, the operator $T(\kappa)F(h) \subset (F(\bar{h})T(\kappa)^*)^*$ is bounded,

$$T(\kappa)F(h)\Psi_n(z;b) = \int dx \, \Psi_{n+1}(x+\kappa, z+\kappa; h(x), b),$$

and

$$T(\kappa_1)F(h_1)\cdots T(\kappa_n)F(h_n)\Omega = \int dx \, \Psi_n(x_1 + c_1, \ldots, x_n + c_n; h(x)), \qquad c_a = \kappa_1 + \cdots + \kappa_a.$$

Step 19 (D_1 is a core). Let $s > 0$. The operator $(1+H)^m e^{-sH}$ is bounded, self-adjoint and injective, and hence has dense range; it therefore carries the dense set D_1 onto a dense set, which is contained in $(1+H)^m D_1$. It follows that D_1 is dense in $D(H^m) \supset D$ for the norm $\|\cdot\|_m$.

Step 20 (covariance; Theorem 3.3 (b)). Apply $\mathcal{U}(0,\Lambda)$ to (e4), change variables by $x \to \Lambda^{-1}x$, and let $\kappa \to 0$. This gives $\mathcal{U}(0,\Lambda)F(h) = F(\Lambda \cdot h)\mathcal{U}(0,\Lambda)$ on D_1. Both sides are bounded from $\|\cdot\|_4$ to $\mathcal{H}$: the left one by Theorem 3.3 (a), the right one by Theorem 3.3 (a) together with the bound $\|\mathcal{U}(0,\Lambda)\psi\|_4 \le (2(\Lambda e_0)^0)^4 \|\psi\|_4$ of Step 17. By Step 19 the identity therefore extends to D. For translations the statement is (e3).

Step 21 (locality: a formula). Suppose that h and h' have compact supports which are space-like separated. Put $\Psi' = \Psi_m(z';b')$, $\Psi = \Psi_n(z;b)$, $N = m+n+2$, and

$$p_\epsilon := (\bar{z}'_m, \ldots, \bar{z}'_1, x, x' + i\epsilon e_0, z_1 + i\epsilon e_0, \ldots, z_n + i\epsilon e_0),$$

which lies in $\mathcal{T}_N$ when $\epsilon > 0$. Two applications of (e4), together with Step 6 (b), (c), give

$$\langle \Psi', F(h)T(i\epsilon e_0)F(h')\Psi \rangle = \iint dx \, dx' \, \mathcal{W}_N(p_\epsilon; \bar{b}'_m, \ldots, \bar{b}'_1, h(x), h'(x'), b_1, \ldots, b_n).$$

As $\epsilon \to 0$, the left-hand side converges to $\langle \Psi', F(h)F(h')\Psi\rangle$.

Step 22 ($p_0 \in \mathcal{D}_N$; Theorem 3.3 (c)). With one exception, the consecutive differences of the entries of p_0 belong to $\mathcal{T}$; the exception is the real vector $w = x' - x$, which satisfies $|\vec{w}| > |w^0|$. Choose a rotation ϱ taking $\vec{w}$ to $|\vec{w}|e_2$. Then

$$\operatorname{Im}(JR_t^{(2)}J^{-1}\varrho\, w) = -\sin t\,(|\vec{w}|, w^0, 0) \in V_+$$

for small $t < 0$, whereas, $\mathcal{T}$ being open, the images of the other differences belong to $\mathcal{T}$ for small $|t|$; therefore $p_0 \in \mathcal{D}_N$. When x ranges over $\operatorname{supp} h$, x' over $\operatorname{supp} h'$ and ϵ over $[0,1]$, the points p_ϵ form a compact subset of $\mathcal{D}_N$, and $\mathcal{W}_N$ is continuous there. It follows that $\langle \Psi', F(h)F(h')\Psi\rangle = \iint \mathcal{W}_N(p_0; \dots)$, an expression which, by (f2), does not change when $(x, h(x))$ and $(x', h'(x'))$ are interchanged. The relation $[F(h), F(h')] = 0$ extends to D by Step 19 and Theorem 3.3 (a), and to supports that are not compact by means of cut-offs.

Step 23 ($dF = 0$; Theorem 3.3 (d)). Put $G_b(a) := \Phi_b(ide_0 + a, ide_0 - a)$. This function is holomorphic in the convex region $\{|\operatorname{Im} a^0| + |\operatorname{Im}\vec{a}| < d\}$, which includes $\mathbb{R}^3$ as well as every point Jy with $|y_1| < d$. Let $u \in C_c^\infty(\{|y_1| < d\})$ and

$$g = \partial_1 u\, e_2 \wedge e_3 + \partial_2 u\, e_3 \wedge e_1 + \partial_3 u\, e_1 \wedge e_2,$$

so that $*g = \nabla u$; then (dF) yields $\mathcal{B}_d(g) = 0$. Combining Step 4 (d), an integration by parts and the relation $\partial_{y_1}[G(Jy)] = i(\partial_0 G)(Jy)$, one deduces

$$i\,(\partial_0 G_{e_2\wedge e_3} + \partial_2 G_{e_3\wedge e_0} + \partial_3 G_{e_0\wedge e_2})(Jy) = 0,$$

where a single factor i is common to the three terms. Fact 2 shows that the bracket is zero on $\mathbb{R}^3$. Hence $F_{d,d}(d^*u) = 0$, and so $q_{d^*u} = 0$, for every $u \in \mathcal{S}$.

Step 24 (cyclicity; Theorem 3.3 (e)). Suppose that ψ is orthogonal to $F(h_1)\cdots F(h_n)\Omega$ for all choices of the h_a, and consider

$$G(c) := \langle \psi, T(c_1)F(h_1)\cdots T(c_n)F(h_n)\Omega\rangle.$$

For real c this vanishes, by (e3). Step 1 and Theorem 3.3 (a) show that G fulfils the assumptions of Fact 3 on $\overline{\mathcal{T}}^n$, whence $G \equiv 0$. Letting $h_a \to \delta_0 b_a$ in (e4), one finds $\psi \perp D_1$, and therefore $\psi = 0$.

Step 25 (Wightman distributions; (f1), (f3)). Theorem 3.3 (a) gives, for $\zeta_a \in \overline{\mathcal{T}}$,

$$|\langle \Omega, F(h_1)T(\zeta_1)F(h_2)\cdots T(\zeta_{n-1})F(h_n)\Omega\rangle| \le \prod_a |h_a|_{4n}.$$

Fact 5 then shows that W_n is continuous on Sw_n. Fix η with $\eta_{a+1} - \eta_a \in V_+$, and fix a norm $|c|$ on $(\Lambda^2\mathbb{C}^3)^{\otimes n}$. By (f1), $|\mathcal{W}_n(x + i\eta; c)| \le C_\eta |c|$ for real x and $c \in (\Lambda^2\mathbb{C}^3)^{\otimes n}$, with a constant C_η depending on η and n; hence the functional $\mathbf{h} \mapsto \int dx\, \mathcal{W}_n(x + i\eta; \mathbf{h}(x))$ on Sw_n is bounded by a multiple of $\|\mathbf{h}\|_{L^1}$, and it is continuous. By (e4), its value at $h_1 \otimes \cdots \otimes h_n$ is the above form at $\zeta_a = i(\eta_{a+1} - \eta_a)$, an n-linear form to which Fact 5 applies with $p = |\cdot|_{4n}$. By the uniqueness clause of Fact 5, the functional is the continuous linear form W furnished there, and it is therefore bounded by the seminorm P, which does not depend on η. Since $T(i\eta) \to 1$ strongly for every norm $\|\cdot\|_m$, these functionals tend to W_n on products of tests, and consequently on all of Sw_n. For (f3), observe that by (e3) the forms at real ζ_a equal $W_n(h_1, \tau_{\zeta_1}h_2, \tau_{\zeta_1+\zeta_2}h_3, \dots)$. Arguing with Fact 3 as in Step 24, one concludes that the W_n determine these forms on $\overline{\mathcal{T}}^{n-1}$; when two laws are compared, Fact 3 is applied to the difference of their forms. Therefore the W_n determine $\mathcal{W}_n$ on $\mathcal{T}_n$, by (e4) with $h_a \to \delta_0 b_a$, and on $\mathcal{D}_n$, by the definition of the $\mathcal{W}_n^{R,\pi}$; through (f2) they determine the S_n at pairwise disjoint compact supports. Proposition 1.2 (c) completes the argument. □

Corollary 3.4. *F is not a generalised free field.* Take the tuned data of Section F.2, a charge $0 < e \le e_2$, and $\mathrm{lw} = \mu_\xi$ with $\xi \in \Xi_\mu$. Define W_n^{tr} from the W_k by the partition formula (1), retaining the order of the arguments within each block. Then $W_4^{\mathrm{tr}} \neq 0$; in particular, it is not the case that $W_n^{\mathrm{tr}} = 0$ for every $n \neq 2$. Furthermore, given non-empty open sets $O_1, \dots, O_4 \subset \mathbb{R}^3$ which are pairwise disjoint, one can choose real $f_a \in C_c^\infty(O_a; \Lambda^2)$ such that $\kappa_4^\xi(f_1, \dots, f_4) \neq 0$. ■□

Proof. **Part 1.** Since $\mathcal{W}_1 = 0$, the truncated function is $\mathcal{W}_4^{\mathrm{tr}} := \mathcal{W}_4 - \sum \mathcal{W}_2\mathcal{W}_2$, the sum being over the three pairings. As $\mathcal{T}$ is a convex cone, $\mathcal{W}_4^{\mathrm{tr}}$ is holomorphic on $\mathcal{D}_4$; and (f2) identifies it, at pairwise disjoint compact supports, with the density of κ_4^{ξ}.

Part 2. The positive fourth cumulant of Section F.2, together with Proposition 1.2(a), gives disjoint compact supports at which $\kappa_4^{\xi} \neq 0$. The function $y \mapsto \mathcal{W}_4^{\mathrm{tr}}(Jy;\cdot)$ is real analytic on the set of configurations of four distinct points, which is connected; consequently it does not vanish on any non-empty open subset. This proves the last assertion, and shows that $\mathcal{W}_4^{\mathrm{tr}}$ is not identically zero on $\mathcal{T}_4$.

Part 3. Let $c \in \mathcal{T}^3$. By (e4), the integral

$$\int dx\, \mathcal{W}_4^{\mathrm{tr}}(x_1, x_2 + c_1, x_3 + c_1 + c_2, x_4 + c_1 + c_2 + c_3; h(x))$$

can be written as a polynomial in the matrix elements $\langle \Omega, F(h_a)T(\cdot)F(h_{a'})\cdots\Omega\rangle$. As a function of c it extends to $\overline{\mathcal{T}}^3$ in the manner required by Fact 3, and at real c its value is $W_4^{\mathrm{tr}}(h_1, \tau_{c_1}h_2, \tau_{c_1+c_2}h_3, \tau_{c_1+c_2+c_3}h_4)$. If W_4^{tr} vanished, it would follow, on letting $h_a \to \delta_0 b_a$, that $\mathcal{W}_4^{\mathrm{tr}} = 0$ on $\mathcal{T}_4$. □

The Euclidean fourth-cumulant bound used here is part of the tuned construction. Corollary 3.4 gives its Minkowski continuation and its consequence for arbitrary disjoint open sets.

Proposition 3.5. *Discrete symmetries and the parity-even field-strength two-point function.* Let $\mathrm{lw} = \mu_\xi$ be the law of Proposition 3.1.

(a) *Component reflection invariance, recalled from (Inv).* Every reflection r of $\mathbb{R}^3$ satisfies $r_*\mu_\xi = \mu_\xi$. In other words, although the reflections, under which μ is invariant, could permute the components of μ, each component with $\xi \in \Xi_\mu$ is fixed by them. Denoting by Rl the linear part of r, the action of r on $b = *f$ is $(r \cdot b)(y) = -\mathrm{Rl}\, b(r^{-1}y)$; thus the dual field is a pseudovector.

(b) *Implementation.* The representation $\mathcal{U}$ extends to a representation of $\mathbb{R}^3 \rtimes O(1,2)$ on $\mathcal{H}$ fixing Ω. If $(\Lambda e_0)^0 > 0$, that is, if Λ preserves the direction of time, then $\mathcal{U}(a,\Lambda)$ is unitary and

$$\mathcal{U}(a,\Lambda)F(h)\,\mathcal{U}(a,\Lambda)^{-1} = F((a,\Lambda)\cdot h);$$

otherwise $\mathcal{U}(a,\Lambda)$ is antiunitary and the right-hand side is replaced by $F(\overline{(a,\Lambda)\cdot h})$. When $R \in O(3)$ satisfies $Re_1 = e_1$, which on Minkowski space reads $Re_0 = e_0$, the operator $\mathcal{U}(0,R)$ is $[A] \mapsto [\hat{R}A]$. For a reflection R of space one has $\mathcal{U}(0,R)F_{23}(x)\,\mathcal{U}(0,R)^{-1} = -F_{23}(Rx)$; that is, under parity the magnetic field changes sign.

(c) CT, *not* T. One has $\mathcal{U}(0,\theta) = \Gamma$, where $\Gamma[A] := [\bar{A}]$; moreover $\Gamma H \Gamma = H$, $\Gamma P_i \Gamma = -P_i$ and $\Gamma F_\alpha(x)\Gamma = (-1)^{\mathrm{ti}(\alpha)}F_\alpha(\theta x)$. Thus F_{23} is even and the electric field (F_{02}, F_{03}) is odd. This transformation is CT and not the time reversal T, since T reverses the magnetic field.

(d) *Two-point function.* Let f', f'' be real, and put $b' = *f'$ and $b'' = *f''$. Then

$$S_2^{\xi}(f', f'') = (2\pi)^{-3}\int dp\, \overline{\hat{b}'(p)}\cdot M(p)\,\hat{b}''(p), \qquad M(p) = \Pi(p)\int d\mathrm{kl}_\xi(t)\,\frac{|p|^2}{|p|^2 + t^2}.$$

Boundedness on L^2 and translation invariance give a bounded measurable multiplier M by [9, Chapter I, Theorem 3.18]. Symmetry, (dF) and $SO(3)$ invariance then permit, for almost every p,

$$M(p) = M_+(|p|)\,\Pi(p) + M_-(|p|)\,\epsilon(p), \qquad \epsilon(p)_{ab} := \sum_c \epsilon_{abc}p_c/|p|,$$

where M_- is purely imaginary in the case of a law. The term containing $\epsilon(p)$ changes sign under reflections. For μ_ξ this term does not occur.

(e) *Any order of the chamber limits.* Consider the chamber cutoff laws of Section E.6, for any charge $e > 0$ and any real mass assignment independent of volume. Every joint subsequential limit, with both cutoff indices tending to infinity, is O_{cub}-invariant. Almost every translation-ergodic component is invariant under $\{\pm 1\}^3$, including each coordinate reflection and inversion. Iterated subsequential limits in either order are among these joint limits. The two-point multipliers of the parent law and almost every component are therefore real and symmetric almost everywhere. These assertions, which require only coordinate reflection positivity, are proved in Proposition E.14.

(f) *Charge conjugation.* Let $\mathrm{Cc}\colon \omega \mapsto -\omega$. The image law $\mathrm{Cc}_*\mu_\xi$ is a component $\mu_{\xi'}$ as well, with $\xi' \in \Xi_\mu$, and its Schwinger functions are $(-1)^n S_n^\xi$; either it equals μ_ξ, or the two laws are mutually singular. The following three statements are equivalent.

(i) $\mathrm{Cc}_*\mu_\xi = \mu_\xi$.

(ii) $W_n = 0$ for every odd n.

(iii) Some unitary operator C on $\mathcal{H}$ satisfies $\mathsf{C}\Omega = \Omega$, $\mathsf{C}D = D$ and $\mathsf{C}F(h)\mathsf{C}^{-1} = -F(h)$.

One says that T is *implemented* if some antiunitary operator T_0 on $\mathcal{H}$ satisfies $\mathsf{T}_0\Omega = \Omega$, $\mathsf{T}_0 D = D$ and $\mathsf{T}_0 F(h)\mathsf{T}_0^{-1} = -F(\overline{\theta \cdot h})$, that is, maps $F_\alpha(x)$ to $-(-1)^{\mathrm{ti}(\alpha)}F_\alpha(\theta x)$; these signs are those of T. The operator Γ is antiunitary, with $\Gamma^2 = 1$, $\Gamma\Omega = \Omega$, $\Gamma D = D$ and $\Gamma F(h)\Gamma = F(\overline{\theta \cdot h})$; hence multiplication by Γ turns each such T_0 into an operator C as in (iii), and conversely. Consequently T is implemented if and only if (i), (ii), (iii) hold.

■

Proof. **Part 1.** Assertion (a) is a restatement of (Inv); the action on $*f$ is described in the notation above.

Part 2. Let $R \in O(3)$ satisfy $Re_1 = e_1$. Then R commutes with J, maps $\mathcal{T}_n$ onto itself and respects the order of the times. Hence (Inv), Step 7 and Fact 2 yield $\mathcal{W}_n(Rz; Rb) = \mathcal{W}_n(z; b)$ on $\mathcal{T}_n$. At the Euclidean points, where $\theta\bar{z} = z$, the relation $\overline{S_n(g_1, \dots)} = S_n(\bar{g}_1, \dots)$, Step 7 and (K) give $\overline{\mathcal{W}_n(z; b)} = \mathcal{W}_n(\theta\bar{z}; \theta\bar{b})$. Both members are antiholomorphic on $\mathcal{T}_n$, because $-\theta V_+ = V_+$; so the identity holds on all of $\mathcal{T}_n$.

Part 3. By Fact 6 (iv), the group $O(1,2)$ is generated by $SO_0(1,2)$, by one reflection of space and by θ. Let $\Lambda \in O(1,2)$. If Λ preserves the direction of time, Step 10, Part 2 and Step 6 (c) show that $\Psi_n(z; b) \mapsto \Psi_n(\Lambda z; \Lambda b)$ extends to a unitary operator; if it reverses that direction, they show that $\Psi_n(z; b) \mapsto \Psi_n(\Lambda\bar{z}; \Lambda\bar{b})$ extends to an antiunitary operator. On D_1 one sees the group law directly, together with the relation $\mathcal{U}(0,\Lambda)T(a)\mathcal{U}(0,\Lambda)^{-1} = T(\Lambda a)$ for real a. Applied with $a \in \mathbb{R}e_0$, this relation shows that $\mathcal{U}(0,\varrho)$ commutes with H whenever $\varrho e_0 = \pm e_0$, the operator being antiunitary in the case of the sign $-$. By Fact 6 (iv) one may write $\Lambda = \Lambda_0\varrho$ with $\Lambda_0 \in SO_0(1,2)$ and $\varrho \in \{1, \theta_2, \theta, \theta_2\theta\}$; then $\Lambda e_0 = \pm\Lambda_0 e_0$, so that $(\Lambda_0 e_0)^0 = |(\Lambda e_0)^0|$. As $\mathcal{U}(0,\Lambda) = \mathcal{U}(0,\Lambda_0)\mathcal{U}(0,\varrho)$ and $\mathcal{U}(0,\varrho)$ preserves each norm $\|\cdot\|_m$, the bound of Step 17, applied to Λ_0, gives

$$\|\mathcal{U}(0,\Lambda)\psi\|_m \le \big(2|(\Lambda e_0)^0|\big)^m \|\psi\|_m.$$

Hence $\mathcal{U}(a,\Lambda)D = D$, and (e4) yields the action on $F(h)$ exactly as in Step 20.

Part 4. For the R of Part 2, Step 7 yields $\mathcal{U}(0,R)[B] = [\hat{R}B]$, and also $\mathcal{U}(0,\theta)[B] = [\bar{B}]$, for all B in the total set of time-ordered products. The map $[A] \mapsto [\hat{R}A]$ is unitary by Section B.2, and $[A] \mapsto [\bar{A}]$ is antiunitary because $\hat{\theta}$ commutes with complex conjugation. From $\Gamma T(a)\Gamma = T(\theta a)$ one infers $\Gamma H\Gamma = H$ and $\Gamma P_i\Gamma = -P_i$, and the signs are those of $\wedge^2 R$ and of $\wedge^2\theta$. The identities for the components $F_\alpha(x)$ in (b), (c) and (f) are identities between forms; they follow from the corresponding identities for $F(h)$ as h tends to $\delta_x e_\alpha$, by the bound of Step 16. In (b) and (c) they hold on $D(H^2) \times D(H^2)$, which $\mathcal{U}(0,R)$, for $Re_0 = e_0$, and Γ map onto itself, since these operators commute with H by Part 3; in (f) they hold on $D \times D$, since $\mathsf{T}_0 D = D$.

Part 5. For (d), the covariance formula is Proposition D.1 applied to μ_ξ. Consider any law with (G), (dF) and invariance under translations. The covariance of such a law is a bounded positive

operator on L^2 commuting with translations. The multiplier theorem recalled in the next sentence therefore gives a measurable Hermitian matrix M with $0 \le M \le 1$; closedness gives $M(p)p = 0$ almost everywhere. This part uses no reflection or rotation hypothesis. The multiplier theorem used here is [9, Chapter I, Theorem 3.18]: a bounded operator on $L^2(\mathbb{R}^n)$ commuting with all translations is a Fourier multiplier with bounded measurable symbol. For almost every p one has

$$M(-p) = M(p)^T, \qquad M(p)p = 0, \qquad M(-p) = \overline{M(p)},$$

by symmetry, by (dF) and by reality respectively. Likewise, for each $R \in O(3)$ leaving the law invariant, $M(Rp) = RM(p)R^T$ for almost every p, by invariance, with $\det R$ entering squared; here the exceptional null set depends on R. Recall Tonelli's theorem: if $u \ge 0$ is measurable on the product of two σ-finite measure spaces, then both iterated integrals of u exist in $[0,\infty]$ and coincide with the integral of u over the product. Assume that the law is invariant under $SO(3)$. Choose a bounded Borel representative M, equip $SO(3)\times\mathbb{R}^3$ with the product of Haar measure and Lebesgue measure, and let u be the indicator function of $\{(R,p) \in SO(3)\times\mathbb{R}^3 : M(Rp) \ne RM(p)R^T\}$. For each fixed R the corresponding section of this set is a null set. By Tonelli's theorem, for almost every p the identity $M(Rp) = RM(p)R^T$ holds for almost every R; consequently $\tilde{M}(p) := \int dR\, R^T M(Rp) R$ coincides with $M(p)$ for almost every p. The right invariance of Haar measure gives $\tilde{M}(R'p) = R'\tilde{M}(p)R'^T$ for all $R' \in SO(3)$ and all p. The relations $\tilde{M}(p)p = 0$ and $\tilde{M}(-p) = \tilde{M}(p)^T = \overline{\tilde{M}(p)}$ are valid almost everywhere, on a set that is invariant under rotations; hence they are valid at each p whose length $|p|$ avoids a certain null set of $(0,\infty)$. At such p the matrix $\tilde{M}(p)$ annihilates p from both sides and commutes with the rotations about the axis p. On $p^\perp$, the matrices commuting with these rotations are precisely the linear combinations of $\Pi(p)$ and $\epsilon(p)$. Therefore $\tilde{M}(p) = M_+(p)\,\Pi(p) + M_-(p)\,\epsilon(p)$. The exact covariance of $\tilde{M}$, together with $\epsilon(Rp) = \det R \cdot R\,\epsilon(p)R^T$, shows that M_+ and M_- depend on $|p|$ only; and M_- is purely imaginary, since $\tilde{M}(p)$ is Hermitian whereas $\epsilon(p)$ is real and antisymmetric. With $\tilde{M}$ in place of M, a single improper R forces $M_- = 0$.

Part 6. For (e), take $R = -1$ in Part 5: it gives $M(-p) = M(p)$ for almost every p, so that M is symmetric as well as Hermitian, and therefore real.

Part 7. For (f): the law μ is Cc-invariant by Section F.2, and Cc commutes with all Euclidean maps. Thus Cc_* maps ergodic laws to ergodic laws, and uniqueness of the ergodic decomposition implies that it preserves ν; see [4, Section 7.5]. The set Ξ_μ in Section F.2 is specified by translation invariance and ergodicity, (G), (dF), the nine reflection conditions, and the stated centered-probe bounds. Each condition is preserved by Cc_*: the symmetry requirements commute with $F \mapsto -F$, (G) and reflection positivity range over all real tests, closedness is linear, and Pw_2 and Pw_4 are even in their first variable. Two distinct ergodic laws lw' and lw'' are mutually singular. Indeed, let λ be their mean; one-point sets of laws are measurable by [4, Section 7.5], so that $\{\lambda_\omega = \mathrm{lw}'\}$ is an event, and by [4, Section 7.5] this event separates lw' from lw''.

Part 8. Assume (i). On $L^2(\mathrm{lw})$ the operator $A \mapsto A \circ \mathrm{Cc}$ is then unitary; it commutes with $\hat{\theta}$ and maps $L^2(\Sigma_+, \mathrm{lw})$ onto itself. The operator C it induces on $\mathcal{H}$ satisfies $\mathsf{C}\Psi_n = (-1)^n\Psi_n$, by Step 7 and Fact 2, and (e4) yields the other properties required in (iii). If (iii) holds, then $W_n = (-1)^n W_n$, which is (ii). Finally assume (ii). Step 25 gives $\mathcal{W}_n = 0$ for odd n, so that $\mathrm{Cc}_*\mathrm{lw}$, a law with (G) and (dF), has the same moments as lw at disjoint supports; Proposition 1.2 (c) then yields (i). □

The next proposition is the Reeh–Schlieder consequence of positive energy and locality [3]. We include its proof for the common domain D used in Theorem 3.3.

Proposition 3.6. *The property of Reeh and Schlieder.* Let lw be a law having the properties listed in Proposition 3.1, and let $\mathcal{H}$, Ω, D and F be attached to it as in Theorems 3.2 and 3.3.

(a) *Cyclicity.* Let O be a non-empty open subset of Minkowski space $\mathbb{R}^3$. The vector Ω, together with the vectors $F(h_1)\cdots F(h_n)\Omega$ for which $n \ge 1$ and each h_a lies in $C_c^\infty(O; \Lambda^2\mathbb{C}^3)$, spans a dense subspace of $\mathcal{H}$.

(b) *Separation.* Let $O_1 \subset \mathbb{R}^3$ be an open set for which there exists a non-empty open set O_2 such that $(x - x') \cdot (x - x') < 0$ whenever $x \in O_1$ and $x' \in O_2$; every bounded open set O_1 is of this kind. Consider a finite sum $P = c_0 + \sum_k F(h_{k,1}) \cdots F(h_{k,n_k})$, where $c_0 \in \mathbb{C}$, each $h_{k,a}$ belongs to Sw_1, and $\operatorname{supp} h_{k,a} \subset O_1$. If $P\Omega = 0$, then $P\chi = 0$ for all $\chi \in D$.

■

Proof. Write $|x|_{\mathrm{E}} := ((x^0)^2 + |\vec{x}|^2)^{1/2}$ for $x \in \mathbb{R}^3$.

Part 1 (the function G). Suppose that $\psi \in \mathcal{H}$ is orthogonal to Ω and to all the vectors described in (a). Fix $x_0 \in \mathbb{R}^3$ and $r > 0$ with $\{|x - x_0|_{\mathrm{E}} < 2r\} \subset O$, let $n \geq 1$, take $h_1, \ldots, h_n$ in $C_c^\infty(\{|x - x_0|_{\mathrm{E}} < r\}; \Lambda^2\mathbb{C}^3)$, and define, as in Step 24,

$$G(c) := \langle \psi, T(c_1)F(h_1) \cdots T(c_n)F(h_n)\Omega \rangle \qquad (c \in \overline{\mathcal{T}}^n).$$

According to Step 1, for each norm $\| \cdot \|_m$ the operators $T(\zeta)$ are contractions that depend strongly continuously on $\zeta \in \overline{\mathcal{T}}$ and holomorphically on $\zeta \in \mathcal{T}$; according to Theorem 3.3 (a), $F(h)$ is a bounded map from the norm $\| \cdot \|_{m+4}$ to the norm $\| \cdot \|_m$. It follows that, in each variable and for arbitrary values of the other variables in $\overline{\mathcal{T}}$, the function G is continuous on $\overline{\mathcal{T}}$ and holomorphic on $\mathcal{T}$.

Part 2 (real arguments). For real c, the rule (e3) of Step 18 and the equality $T(a)\Omega = \Omega$ of Step 1 give

$$G(c) = \langle \psi, F(\tau_{c_1} h_1)F(\tau_{c_1+c_2} h_2) \cdots F(\tau_{c_1+\cdots+c_n} h_n)\Omega \rangle.$$

If every c_a satisfies $|c_a|_{\mathrm{E}} < r/n$, the supports of the tests occurring in this expression lie in $\{|x - x_0|_{\mathrm{E}} < 2r\} \subset O$, and therefore $G(c) = 0$.

Part 3 (continuation). Fact 3, applied with the balls $B_a = \{|c_a|_{\mathrm{E}} < r/n\}$, shows that $G = 0$ on $\overline{\mathcal{T}}^n$.

Part 4 (point fields). Let $\kappa \in \mathcal{T}^n$ and $z_a := \kappa_1 + \cdots + \kappa_a$. The rule (e4) of Step 18 expresses $G(\kappa)$ as

$$G(\kappa) = \int dx \, \langle \psi, \Psi_n(x_1 + z_1, \ldots, x_n + z_n; h(x)) \rangle.$$

Choose $h_a := \phi_\epsilon(\cdot - x_0)\, b_a$ with $b_a \in \Lambda^2\mathbb{C}^3$ and $0 < \epsilon < r$, where $\phi_\epsilon \geq 0$ belongs to $C_c^\infty(\{|x|_{\mathrm{E}} < \epsilon\})$ and $\int \phi_\epsilon = 1$. Since Ψ_n is norm continuous on $\mathcal{T}_n^+$ and n-linear in b (Steps 5 and 6), the limit $\epsilon \to 0$ yields

$$\langle \psi, \Psi_n(x_0 + z_1, \ldots, x_0 + z_n; b) \rangle = 0,$$

and $z = (z_1, \ldots, z_n)$ ranges over $\mathcal{T}_n^+$ as κ ranges over $\mathcal{T}^n$.

Part 5 (density). Step 6 (b) gives $\Psi_n(x_0 + z_1, \ldots, x_0 + z_n; b) = T(x_0)\Psi_n(z; b)$, and $T(x_0)\Omega = \Omega$. Thus $T(x_0)^*\psi$ is orthogonal to D_1. The operator $T(x_0)$ being unitary and the subspace D_1 being dense by Step 11, one concludes that $\psi = 0$, and (a) is proved.

Part 6 (a space-like separated set). If $O_1 \subset \{|x|_{\mathrm{E}} < r'\}$ for some $r' > 0$, then, with the basis vector e_2 of Minkowski space, the ball $O_2 := \{|x - 4r'e_2|_{\mathrm{E}} < r'\}$ has the property required in (b), since $|x^0 - x'^0| < 2r' < |\vec{x} - \vec{x}'|$ for $x \in O_1$ and $x' \in O_2$. In the general case, let O_2 be as in (b), and let Q be either 1 or a product $F(h'_1) \cdots F(h'_k)$ with $h'_a \in C_c^\infty(O_2; \Lambda^2\mathbb{C}^3)$. By Theorem 3.3 (c) one has $PQ = QP$ on D, a subspace which each of these operators maps into itself; applied to Ω, this yields $PQ\Omega = QP\Omega = 0$.

Part 7 (the adjoint). Let $P^+ := \bar{c}_0 + \sum_k F(\bar{h}_{k,n_k}) \cdots F(\bar{h}_{k,1})$. Theorem 3.3 (a) shows that $\langle \phi, P\chi \rangle = \langle P^+\phi, \chi \rangle$ for $\phi, \chi \in D$. Hence $\langle P^+\phi, Q\Omega \rangle = \langle \phi, PQ\Omega \rangle = 0$ for every Q of Part 6, and (a), applied to the set O_2, gives $P^+\phi = 0$. Consequently $\langle \phi, P\chi \rangle = 0$ for all $\phi, \chi \in D$, and the density of D yields $P\chi = 0$. □

4. The spectral measure and the Maxwell infrared limit

4.1. **The spectral multiplier.** For $\lambda \in \mathcal{K}$, let kl be the measure in Theorem A(ii) and put

$$\mathsf{D}(s) = \int \frac{s}{s+t^2}\, d\mathrm{kl}(t), \qquad Y_t(y) = \frac{e^{-t|y|}}{4\pi|y|}.$$

The measure variable t is mass. The two-point kernel off the origin is $(-\delta_{ab}\Delta + \partial_a\partial_b)\int Y_t\, d\mathrm{kl}(t)$ after Hodge duality, by Proposition D.1.

Proposition 4.1. Let $0 \neq g \in L^2$ be real and closed, and put $g_l(y) = l^{-3/2} g(y/l)$. Then

$$\frac{S_2(g_l, g_l)}{\|g\|^2} \downarrow \mathrm{kl}(\{0\}) \quad (l \to \infty), \qquad \frac{S_2(g_l, g_l)}{\|g\|^2} \uparrow \mathrm{kl}([0,\infty)) \quad (l \downarrow 0).$$

In particular these two masses are the infimum and supremum, respectively, of $S_2(g,g)/\|g\|^2$ over nonzero real closed tests. ∎

Proof. The Fourier transform of g_l is $l^{3/2}\hat{g}(lp)$. The representation in Theorem A(ii) therefore gives

$$\frac{S_2(g_l, g_l)}{\|g\|^2} = \int \mathsf{D}(|p|^2/l^2)\, \frac{|\hat{g}(p)|^2\, dp}{(2\pi)^3 \|g\|^2}.$$

The last factor is a probability measure. The function D is nondecreasing, bounded between the atom at zero and the total mass, and tends to those two numbers at zero and infinity. Monotonicity and dominated convergence give the assertion.

□

4.2. **The magnetic field at a fixed time.**

Proposition 4.2. *Notation.* Throughout this proposition the letter t stands for a time, while s serves as the variable of kl. Momenta are written $p = (p_1, k)$ with $k \in \mathbb{R}^2$, and $\omega_s(k) := (|k|^2 + s^2)^{1/2}$. Section 2 supplies the conditions (T_{123}), (R_1), (RP_1), the reflection θ_1 in the plane $\{x_1 = 0\}$, its operator $\hat{\theta}_1$ on functions on $\mathcal{S}'$, the antilinear map Θ_1 given by $\Theta_1\Phi = \overline{\hat{\theta}_1\Phi}$, the reconstruction $(\mathcal{H}_1, [\,\cdot\,]_1, \mathsf{H}_1, V_1, \Omega_1)$ along e_1, and the joint spectral measure $\mathfrak{E}_1$ of this reconstruction, written in the variables $(\omega, p_\perp)$; see [4, Proposition 7.2]. For a real $h \neq 0$ in $C_c^\infty(\mathbb{R}^2)$ put

$$\|h\|_{1/2}^2 := \int \frac{d^2k}{(2\pi)^2}\, |k|\, |\hat{h}(k)|^2.$$

Given a real number t, let the real closed form $g_{t,h}$ in $L^2(\mathbb{R}^3; \Lambda^2)$ be determined by

$$\hat{g}_{t,h}(p) = \exp(-ip_1 t)\, \hat{h}(k)\, \Pi(p)\, e_1,$$

so that $\|g_{t,h}\|^2 = \frac{1}{2}\|h\|_{1/2}^2$, and put $B_t(h) := X(g_{t,h})$. Fix an even function $\varrho \geq 0$ in $C_c^\infty(\mathbb{R})$ with $\int \varrho = 1$, set $\varrho_\varepsilon(x_1) := \varepsilon^{-1}\varrho(x_1/\varepsilon)$, and define the two-form $u_{\varepsilon,t}$ by ${*u_{\varepsilon,t}}(x) = \varrho_\varepsilon(x_1 - t)\, h(x_2, x_3)\, e_1$; among its components only the component 23 differs from zero. The letter λ stands, in (a), (b) and (c), for a probability measure on $(\mathcal{S}', \Sigma)$.

(a) Suppose that λ satisfies (G) and (dF). As $\varepsilon \to 0$, one has $X(u_{\varepsilon,t}) \to B_t(h)$ in $L^p(\lambda)$ for every $p < \infty$. In this sense, $B_t(h)$ is the smearing with h, at the time t, of the component F_{23}, that is, of the magnetic field of the picture e_1. Moreover $\lambda(\exp B_t(h)) \leq \exp(\frac{1}{4}\|h\|_{1/2}^2)$.

(b) If, besides (G) and (dF), the measure λ satisfies (T_{123}), (R_1) and (RP_1), then $B_0(h)$ belongs to both $L^2(\Sigma_{\{x_1>0\}}, \lambda)$ and $L^2(\Sigma_{\{x_1<0\}}, \lambda)$, and $\Theta_1 B_0(h) = B_0(h)$. The symbol $\mathcal{E}_0$ denotes the $L^2(\lambda)$-closure of the linear space generated by all bounded Borel functions of a finite number of the variables $B_0(h)$. Restricted to $\mathcal{E}_0$, the class map is an isometry: $\|[\Phi]_1\| = \|\Phi\|_{L^2(\lambda)}$.

(c) Suppose moreover that S_2 has the representation in Proposition D.1 for a positive measure kl of mass at most one, and put $\psi_h = [B_0(h)]_1$. For $t \geq 0$ and $y \perp e_1$,

$$\begin{aligned}&\langle \psi_h, \exp(-t\,\mathsf{H}_1)\, V_1(y)\, \psi_h\rangle \\ &\qquad = \int d\mathrm{kl}(s) \int \frac{d^2k}{(2\pi)^2}\, |\hat h(k)|^2\, |k|^2\, \frac{\exp(-t\,\omega_s(k) + ik\cdot y)}{2\,\omega_s(k)}.\end{aligned} \tag{4.1}$$

Consequently $\langle \psi_h, \mathfrak{E}_1(\cdot)\,\psi_h\rangle$ coincides with the image of the measure

$$\mathrm{kl}(ds)\, \frac{|\hat h(k)|^2\, |k|^2}{2\,\omega_s(k)}\, \frac{d^2k}{(2\pi)^2}$$

under the map $(s,k) \mapsto (\omega_s(k), k)$. Furthermore, ψ_h is in the domain of the quadratic form of H_1, and

$$2\,\langle \psi_h, \mathsf{H}_1\, \psi_h\rangle = \mathrm{kl}([0,\infty))\, \|\nabla h\|^2_{L^2(\mathbb{R}^2)}. \tag{4.2}$$

(d) For a law satisfying (a)–(c), suppose that its joint spectral measure $\mathfrak{E}_1$ is carried by $\{\omega \geq |p_\perp|\}$. The mass operator is defined as $\mathsf{M} := \int (\omega^2 - |p_\perp|^2)^{1/2}\, d\mathfrak{E}_1$. In this situation $\psi_h \perp \Omega_1$, the operator M is reduced by the subspace $\Omega_1^\perp$, and M has in ψ_h the spectral measure $c_h(s)\,\mathrm{kl}(ds)$, with

$$c_h(s) := \int \frac{d^2k}{(2\pi)^2}\, \frac{|\hat h(k)|^2\, |k|^2}{2\,\omega_s(k)} \in (0,\infty).$$

Consequently the spectrum of M on $\Omega_1^\perp$ contains the support of kl; one has $\|\mathfrak{E}_1(\{\omega = |p_\perp|\})\,\psi_h\|^2 = \frac{1}{2}\|h\|^2_{1/2}\,\mathrm{kl}(\{0\})$;

■

Proof. 1. *Assertion (a).* One has

$$\hat u_{\varepsilon,t}(p) = \exp(-ip_1 t)\, \hat\varrho(\varepsilon p_1)\, \hat h(k)\, e_1, \qquad e_1 \cdot \Pi(p)\, e_1 = \frac{|k|^2}{|p|^2},$$

$$\int \frac{dp_1}{2\pi}\, \frac{1}{p_1^2 + |k|^2} = \frac{1}{2|k|}.$$

These formulas yield the norm of $g_{t,h}$. The reality of h implies that of $g_{t,h}$, and $\Pi^2 = \Pi$ shows that $g_{t,h}$ is closed. As $|\hat\varrho| \leq 1$ and $\hat\varrho(\varepsilon p_1) \to 1$, the convergence $\Pi_{\mathrm c} u_{\varepsilon,t} \to g_{t,h}$ in L^2 follows by dominated convergence. If (G) and (dF) hold, then every real $f' \in L^2$ satisfies $X(f') = X(\Pi_{\mathrm c} f')$ almost surely and $\lambda(\exp X(f')) \leq \exp(\frac{1}{2}\langle f', \Pi_{\mathrm c} f'\rangle)$ [4, Proposition 5.1 and Section 5.2], [4, Section 8.1]. Moreover $\|X(f')\|_{L^N(\lambda)} \leq c_N\, \|f'\|$ for $N = 1, 2, \ldots$ [4, Proposition 5.1].

2. *Assertion (b).* When $t > 0$ and ε is small, $u_{\varepsilon,t}$ is zero on $\{x_1 < 0\}$; hence $X(u_{\varepsilon,t}) \in L^2(\Sigma_{\{x_1>0\}}, \lambda)$ [4, Proposition 5.1]. This subspace is closed. By (a) it therefore contains $B_t(h)$; it contains $B_0(h)$ as well, since dominated convergence gives $\|g_{t,h} - g_{0,h}\| \to 0$ as $t \to 0$. The same holds for the half space $\{x_1 < 0\}$.

Under θ_1, the normal component of the pseudovector $*f'$ is even; the function ϱ has the same parity. Hence $\theta_1 \cdot u_{\varepsilon,0} = u_{\varepsilon,0}$, and, since this test is real, $\Theta_1 X(u_{\varepsilon,0}) = X(u_{\varepsilon,0})$. By (R_1) the map Θ_1 is isometric on $L^2(\lambda)$, and the limit $\varepsilon \to 0$ gives $\Theta_1 B_0(h) = B_0(h)$.

Further, $\hat\theta_1$ preserves products, and it commutes with the operation of composing with a function [4, Sections 5.2 and 7.4]; under (G) and (R_1) it maps functions that agree λ-almost surely to functions that agree λ-almost surely [4, Proposition 5.1]. It follows that $\hat\theta_1 \Phi = \Phi$ for every bounded Borel function Φ of finitely many $B_0(h)$, and thus on $\mathcal{E}_0$; moreover $\mathcal{E}_0 \subset L^2(\Sigma_{\{x_1>0\}}, \lambda)$. Therefore $\|[\Phi]_1\|^2 = \lambda((\Theta_1\Phi)\cdot\Phi) = \lambda(|\Phi|^2)$.

3. *Assertion (c).* Combining (b) with the definition of the reconstruction and with the relation $U(a)X(f') = X(\tau_a \cdot f')$ [4, Proposition 5.1], one finds that the left side of (4.1) equals

$$\lambda(B_0(h) \cdot X(\tau_{te_1+y} \cdot g_{0,h})) = S_2(g_{0,h}, \tau_{te_1+y} \cdot g_{0,h}).$$

Using Proposition D.1, one can write this as

$$(2\pi)^{-3} \int dp\, |\hat{h}(k)|^2\, \frac{|k|^2}{|p|^2}\, \exp(-ip_1 t - ik \cdot y) \int d\mathrm{kl}(s)\, \frac{|p|^2}{|p|^2 + s^2}.$$

A majorant of the integrand is $|\hat{h}(k)|^2\, |k|^2/|p|^2$, and step 1 shows that it is integrable. Fubini's theorem, the formula

$$\int \frac{dp_1}{2\pi}\, \frac{\exp(-ip_1 t)}{p_1^2 + \omega^2} = \frac{\exp(-t\omega)}{2\omega}$$

and the equality $|\hat{h}(-k)| = |\hat{h}(k)|$ give (4.1).

The two sides of (4.1) are the Laplace–Fourier transforms of $\langle \psi_h, \mathfrak{E}_1(\cdot)\, \psi_h \rangle$ [4, Proposition 7.2] and of the image measure, respectively. Both being finite measures on $[0,\infty) \times \mathbb{R}^2$, they are equal by Lemma D.3. The integral of ω against the image measure equals $\frac{1}{2}\,\mathrm{kl}([0, \infty)) \int \frac{d^2k}{(2\pi)^2}\, |k|^2\, |\hat{h}(k)|^2 < \infty$; this is (4.2).

4. *Assertion (d).* By the definition of the reconstruction, $\exp(-t\,\mathsf{H}_1)\, V_1(y)\, \Omega_1 = \Omega_1$. Hence, by Lemma D.3, the measure $\langle \Omega_1, \mathfrak{E}_1(\cdot)\, \Omega_1 \rangle$ is the unit mass at the origin, and $\mathfrak{E}_1(A)\, \Omega_1 = 1_A(0,0)\, \Omega_1$; consequently $\Omega_1^{\perp}$ reduces M. Further, $\langle \Omega_1, \psi_h \rangle = \lambda(B_0(h)) = 0$ [4, Proposition 5.1]. Under the map $(\omega, p_\perp) \mapsto (\omega^2 - |p_\perp|^2)^{1/2}$ the measure of (c) is carried onto $c_h(s)\, \mathrm{kl}(ds)$. Since $\hat{h}$ is not identically zero, c_h is a continuous positive function; consequently this measure and kl have the same support. The point 0 has the preimage $\{\omega = |p_\perp|\}$ under this map, and $c_h(0) = \frac{1}{2}\|h\|_{1/2}^2$.

□

The magnetic variable is defined on the Euclidean probability space. The proposition identifies its vector in the reconstructed space; it does not define an operator at a sharp real time.

4.3. **A purely massless spectral measure and the infrared limit.** The following harmonic and probability facts will be used.

(K20) (Harmonic Laurent expansion; [10, Theorems 10.1 and 1.21; Chapter 5].) Let u satisfy $\operatorname{div} u = \operatorname{curl} u = 0$ on $\{0 < |z| < R\} \subset \mathbb{R}^3$. Then, with locally uniform convergence,

$$u = \sum_{l \geq 1} \nabla P_l + \sum_{l \geq 0} \nabla H_l,$$

where each P_l is a homogeneous harmonic polynomial of degree l, and $H_l = |z|^{-2l-1} \tilde{P}_l$ for a polynomial $\tilde{P}_l$ of the same kind; every derivative of order m of an H_l has the form of an H_{l+m}. Let P_l, P_k be homogeneous harmonic polynomials of the degrees l, k, and let H_l, H_k be functions of the form just described, taken from the same expansion or from different ones. On every sphere $\{|z| = r\}$ the fields ∇P_l and ∇H_k are orthogonal in L^2 for all l and k, and so are ∇P_l and ∇P_k, as well as ∇H_l and ∇H_k, whenever $l \neq k$. A harmonic function on $\mathbb{R}^3$ whose square is integrable outside some ball is zero. (Poisson.) Let φ be harmonic in a neighbourhood of the closed ball $\{|z| \leq s\} \subset \mathbb{R}^3$; then

$$\varphi(z) = \int_{|z'|=s} \frac{s^2 - |z|^2}{4\pi s\, |z - z'|^3}\, \varphi(z')\, dS(z') \qquad (|z| < s).$$

(K21) (Minlos; [11, Theorem 1].) On $\mathcal{S}$, every continuous positive definite function Φ with $\Phi(0) = 1$ is the characteristic functional of a unique law on $(\mathcal{S}', \Sigma)$. (Lévy.) Let Q and Q_k, $k \geq 1$, be probability measures on $\mathbb{R}$; if the characteristic function of Q_k converges at every point to

that of Q, then $Q_k \to Q$ weakly. (Fatou.) If $Q_k \to Q$ weakly and $\varphi \geq 0$ is continuous, then $Q(\varphi) \leq \liminf_k Q_k(\varphi)$. Let finitely many random variables have moments of every order, and let them fall into two non-empty groups such that the two groups are independent of each other; then the joint cumulant of all of them vanishes.

Theorem 4.3. *The massless part of the field.*

(a) An ergodic law lw of $\mathcal{K}$ with $\mathrm{kl} = c_0\delta_0$ equals γ_{c_0}.

(b) Let lw be an ergodic law of $\mathcal{K}$; at every charge $e > 0$ and for every mass assignment, ν-almost all components of each volume-first limit point μ are laws of this kind. Put $c_0 := \mathrm{kl}(\{0\})$ and $(\delta_l g)(y) := l^{-3/2} g(y/l)$. Then, for real $g \in L^2$,

$$\mathrm{lw}(\exp iX(\delta_l g)) \longrightarrow \exp(-\tfrac{c_0}{2}\|\Pi_{\mathrm{c}} g\|^2) \qquad (l \to \infty).$$

■□

Proof. We first prove (a), assuming $\mathrm{kl} = c_0\delta_0$; Steps 7 and 8 prove (b). Throughout, two-forms are identified with vector fields by means of $*$; a closed form then corresponds to a divergence free field, and $\Pi_{\mathrm{c}} \operatorname{curl} = \operatorname{curl}$.

Step 1 (kernels). By Corollary 2.2 (c), every ergodic law of $\mathcal{K}$ satisfies the one-plane hypotheses for $(1, 2)$; it is, moreover, invariant under $SO(3)$ [4, Theorem 8.6]. For pairwise disjoint compact supports, Theorem 3.3(f2) represents each S_n by a kernel real analytic where the points are distinct. The partition formula (1) gives the same conclusion for κ_n. Write k_n for the kernel of κ_n. Step 1 of the proof of Proposition 1.2 gives $|\kappa_n(b_1, \dots, b_n)| \leq C_n \prod_i \|b_i\|$ on L^2. Condition (dF) yields $\operatorname{div}_{y_a} \mathsf{k}_n = 0$.

Step 2 ($\operatorname{curl}_{y_a} \mathsf{k}_n = 0$). Let $n \geq 2$, and let $a, b_2, \dots, b_n \in C_c^\infty$ be supported in balls of radius δ any two of whose centres are farther apart than 4δ. Set $a' := a(\cdot - y)$ and $f(y) := S_n(\operatorname{curl} a', b_2, \dots, b_n)$. The function f is real analytic on the set of those y for which the closed balls do not meet; this set is the complement of finitely many disjoint closed balls and is therefore connected. Suppose that some plane, with reflection r, has the first ball on one side and the other balls on the other side; the points y with this property fill a neighbourhood of infinity. From reflection positivity at this plane, invariance under r [4, Theorem 8.6] and two applications of the Schwarz inequality one obtains

$$|f(y)|^2 \leq S_2(\mathsf{r} \cdot \operatorname{curl} a', \operatorname{curl} a') \left\| \prod_j X(b_j) \right\|_2^2,$$

where $\mathsf{r}\cdot$ is the two-form action of r. Because $\mathrm{kl} = c_0\delta_0$ and the two supports are disjoint,

$$S_2(\mathsf{r} \cdot \operatorname{curl} a', \operatorname{curl} a') = c_0 \langle \mathsf{r} \cdot \operatorname{curl} a', \operatorname{curl} a' \rangle = 0$$

by Proposition D.1. Thus f is zero on a non-empty open subset of the connected open set on which it is real analytic, and the identity theorem for real analytic functions [4, Section 8.3 and Proposition 8.3] shows that f vanishes identically. In the formula for the cumulant, the block containing $\operatorname{curl} a$ is either a singleton, whose factor is $S_1 = 0$, or has two elements or more, in which case its factor is the value at $y = 0$ of the function f of that order, which vanishes as well; hence $\kappa_n(\operatorname{curl} a, b_2, \dots, b_n) = 0$. Since a, the b_j and δ were arbitrary, $\operatorname{curl}_{y_1} \mathsf{k}_n = 0$, and the symmetry of k_n gives the same for the other arguments.

Step 3 (the singularity at $y_1 = y_2$). Fix $n \geq 3$ and $b_3, \dots, b_n$ supported in disjoint closed balls, and let O be the complement of these balls, which is connected. Define

$$k_{ij}(y_1, y_2) := \int \mathsf{k}_n \, b_3 \cdots b_n, \qquad B(b_1, b_2) := \kappa_n(b_1, \dots, b_n).$$

To $z \mapsto k_{\cdot j}(y_2 + z, y_2)$ apply (K20), and call $H_{l,j}(z; y_2)$ the terms of the resulting expansion. The orthogonality in (K20) expresses each coefficient of these terms as the integral, over a small sphere $|z| = s$, of $k_{\cdot j}(y_2 + z, y_2)$ paired with a fixed field, so that the coefficients depend real analytically

on y_2. Fix a point $y_* \in O$, let $\varepsilon_0(y_*) > 0$ be such that the closed ball of radius $3\varepsilon_0(y_*)$ about y_* lies in O, let $0 < \varepsilon \le \varepsilon_0(y_*)$, and set

$$b_i^\varepsilon := \varepsilon^{-3/2}\beta_i((\cdot - y_*)/\varepsilon),$$

where $\beta_2 = \chi e_j$ for some $0 \le \chi \in C_c^\infty$ with $\int \chi = 1$ and support in $|\eta| < \frac{1}{2}$, and where $\beta_1 = \psi(|\eta|)\nabla H_{l,j}(\cdot; y_*)$ for some $0 \le \psi \in C_c^\infty((1,2))$ with $\psi \neq 0$. Then b_2^ε is supported in $\{|y - y_*| < \varepsilon/2\}$ and b_1^ε in $\{\varepsilon < |y - y_*| < 2\varepsilon\}$; on the product of these supports $|y_1 - y_2| < 5\varepsilon/2$, which is less than the distance from y_2 to the complement of O, so that the expansion about y_2 applies there. Uniformly for $|\eta_2| \le \frac{1}{2}$ and $1 \le |\eta_1| \le 2$, expand $H_{l'}(\eta_1 - \eta_2)$ in homogeneous terms of degree m in η_2. In η_1, the term of degree m is an $H_{l'+m}$, so that about the centre 0 there occur only orders $\ge l'$; and the regular part, once shifted by η_2, becomes a series of fields $\nabla P_k(\eta_1)$. From the orthogonality on spheres one thus gets

$$B(b_1^\varepsilon, b_2^\varepsilon) = \sum_{l' \le l} \varepsilon^{1-l'} \int \chi\, F_{l'},$$

where each $F_{l'} = F_{l'}(\eta_2; \varepsilon)$ extends continuously to $\varepsilon = 0$, and $F_l \to \int \psi\, |\nabla H_{l,j}(\cdot; y_*)|^2$ as $\varepsilon \to 0$. As $\|b_i^\varepsilon\| = \|\beta_i\|$, the bound on B of Step 1 then forces $H_{l,j}(\cdot; y_*) = 0$ for every $l \ge 2$; since the point $y_* \in O$ and the index j were arbitrary, $H_{l,j}(\cdot; y_2)$ vanishes for $l \ge 2$ at each $y_2 \in O$ and for each j.

Step 4 ($H_{0,j} = 0$). Write $H_{0,j} = q_j(y_2)/(4\pi|z|)$. Take $b_2 \in C_c^\infty(O)$, and $u \in C_c^\infty(O)$ with $u = 1$ near $\operatorname{supp} b_2$. Condition (dF) and the divergence theorem yield

$$0 = B(\nabla u, b_2) = \sum_j \int b_{2,j}(y_2)\, q_j(y_2)\, dy_2,$$

since the flux of $k_{\cdot j}(\cdot, y_2)$ across small spheres centred at y_2 equals $-q_j(y_2)$, and $\nabla H_{0,j}$ is the only term of the expansion with a non-zero flux. As this is true for every b_2, one concludes that $q_j \equiv 0$. Consequently, with $G(z) := \frac{1}{4\pi|z|}$,

$$k_{ij} = \sum_m T_{mj}(y_2)\, \partial_i \partial_m G(y_1 - y_2) + R_{ij},$$

where $z \mapsto R_{\cdot j}(y_2 + z, y_2)$ is the regular part. On each sphere $|z| = s$ the three fields $\nabla\partial_m G$ are mutually orthogonal, and $\int_{|z|=s} |\nabla \partial_m G|^2\, dS = 1/(2\pi s^4)$; therefore

$$T_{mj}(y_2) = 2\pi s^4 \int_{|z|=s} \sum_i k_{ij}(y_2 + z, y_2)\, \partial_i \partial_m G(z)\, dS,$$

and T_{mj} is real analytic. For $|z| < s$, Poisson's formula for the ball $\{|z| < s\}$ of $\mathbb{R}^3$, applied to each component of the harmonic field $z \mapsto R_{\cdot j}(y_2 + z, y_2)$, gives

$$R_{ij}(y_2 + z, y_2) = \int_{|z'|=s} \frac{s^2 - |z|^2}{4\pi s\, |z - z'|^3} \Big[k_{ij}(y_2 + z', y_2) - \sum_m T_{mj}(y_2)\, \partial_i \partial_m G(z') \Big]\, dS(z'),$$

an expression which is smooth in (z, y_2); thus R is smooth on $O \times O$ as a function of both variables jointly. By symmetry, k is free of divergence and of curl also in the pair (y_2, j). Write $y = y_2$ and $z = y_1 - y$. For $n \in S^2$ put

$$Q(n) = 3nn^T - 1, \qquad \partial_i \partial_m G(rn) = \frac{Q(n)_{im}}{4\pi r^3}.$$

The degree -4 part of $\partial_{y_a} k_{ib} - \partial_{y_b} k_{ia} = 0$ is

$$\sum_m (T_{mb} \partial_i \partial_m \partial_a G - T_{ma} \partial_i \partial_m \partial_b G) = 0.$$

Contracting in i with n_i gives $Q(n)T = T^T Q(n)$ for every $n \in S^2$. For $a \neq b$, taking $n = e_a$ and $n = e_b$ yields $2T_{ab} = -T_{ba}$ and $-T_{ab} = 2T_{ba}$, so both off-diagonal entries vanish. Taking

$n = (e_a + e_b)/\sqrt{2}$ then gives $T_{aa} = T_{bb}$. Thus $T = \tau 1$. Since $\Delta G = 0$ away from zero and the derivatives of R are bounded near $z = 0$, the degree -3 part of $\sum_j \partial_{y_j} k_{ij} = 0$ now gives

$$Q(n)\nabla\tau = 0 \qquad (n \in S^2).$$

The eigenvalues of $Q(n)$ are $2, -1, -1$, so $\nabla\tau = 0$. As O is connected, τ is constant there.

Step 5 ($\tau = 0$). The functions $\partial\partial G$ are homogeneous of degree -3, whereas R is a bounded function near (y_*, y_*). It follows that, whenever the supports of β_1 and β_2 are disjoint,

$$B(b_1^\varepsilon, b_2^\varepsilon) = \tau\, c_\beta + O(\varepsilon^3), \qquad c_\beta := \sum_{i,m} \iint \beta_{1,i}(\eta_1)\, \partial_i\partial_m G(\eta_1 - \eta_2)\, \beta_{2,m}(\eta_2)\, d\eta_1\, d\eta_2.$$

Choose the β_i as in Step 3, but with $\partial_j G$ instead of $H_{l,j}(\cdot; y_*)$; the computation made in Step 3 yields

$$c_\beta = \int \psi\, |\nabla\partial_j G|^2 = \frac{1}{2\pi} \int \psi(x)\, x^{-4}\, dx > 0.$$

Assume that the balls carrying the b_j, $j \geq 3$, are contained in the cube $(-r, r)^3$, that $|y_*|_\infty \geq 4r$, and that $\varepsilon < r/2$; the functions $b_i^\varepsilon(\cdot + y_*)$ are then supported in that cube as well. Every mixed moment occurring in the cumulant formula is to be replaced by the product of its far part and its near part. Since lw is ergodic, Proposition C.1 bounds the error so made by $Cr/|y_*|_\infty$ with a constant C independent of ε and of y_*: that bound is expressed through the $L^2(\mathrm{lw})$ norms of the two parts, Hölder's inequality and [4, Proposition 5.1] give, for real $g_1, \dots, g_k \in L^2$,

$$\Big\| \prod_{a=1}^{k} X(g_a) \Big\|_{L^2(\mathrm{lw})} \leq c_{2k}^k \prod_{a=1}^{k} \|g_a\|,$$

and $\|b_i^\varepsilon\| = \|\beta_i\|$. After the replacement one is left with the joint cumulant of $(X(b_1^\varepsilon), X(b_2^\varepsilon))$ and of an independent copy of $(X(b_3), \dots, X(b_n))$, neither family being empty since $n \geq 3$, and the moments of every order of these variables being finite [4, Proposition 5.1]. This mixed cumulant is zero: independence makes the logarithm of the joint generating function the sum of the logarithms for the two families. Hence $|\tau|\, c_\beta \leq Cr/|y_*|_\infty + O(\varepsilon^3)$. Letting first $\varepsilon \to 0$ and afterwards $|y_*|_\infty \to \infty$, which is legitimate because τ is constant on O, one arrives at $\tau = 0$.

Step 6. With $\tau = 0$ one has $k = R$, which is smooth on $O \times O$ and harmonic in y_1; so k has a harmonic extension across $y_1 = y_2$. Let $b_2, \dots, b_n \in C_c^\infty$ have supports in pairwise disjoint closed balls $\bar B_j$, denote by V the complement of the union of these balls, put $O_j := V \cup \bar B_j$, and define on V

$$w(y_1) := \int \mathsf{k}_n(y_1, \cdot)\, b_2 \cdots b_n.$$

Steps 3–5, applied with $O = O_2$, show that $\int R(\cdot, y_2)\, b_2(y_2)\, dy_2$ is harmonic on O_2 and agrees with w on V. Since κ_n is symmetric, Steps 3–5 apply to every pair $(1, j)$, the other entries being held fixed, and they furnish a harmonic extension of w to O_j. The sets O_j cover $\mathbb{R}^3$, and any two of them intersect in V; so w extends to a harmonic field on all of $\mathbb{R}^3$. For $b_1 \in C_c^\infty(V)$, Step 1 gives $|\int w \cdot b_1| \leq C_n \|b_1\| \prod_{j\geq 2} \|b_j\|$; hence w is square integrable outside the balls, and (K20) gives $w = 0$. A partition of unity shows that κ_n vanishes at non-coinciding arguments for each $n \geq 3$, and Proposition 1.2 (b) shows that lw is Gaussian. Its mean is zero and its covariance is $c_0\langle \cdot, \Pi_{\mathrm{c}} \cdot \rangle$, as in Step 2. The definition of γ_{c_0} then gives $\Phi_{\mathrm{lw}} = \Phi_{\gamma_{c_0}}$ on $\mathcal{S}$, and (K21) gives $\mathrm{lw} = \gamma_{c_0}$. This establishes (a).

Step 7 (part (b)). For $\omega \in \mathcal{S}'$ put $\delta_l^*\omega := \omega \circ \delta_l$, and let $\mathrm{lw}_l := (\delta_l^*)_* \mathrm{lw}$, that is, the law of $g \mapsto X(\delta_l g)$; then $\Phi_{\mathrm{lw}_l} = \Phi_{\mathrm{lw}} \circ \delta_l$. The map δ_l is an isometry, it sends gradients to gradients, it commutes with Π_{c}, and it normalises the Euclidean group:

$$\delta_l(\mathrm{Gi} \cdot f) = \mathrm{Gi}^{(l)} \cdot \delta_l f, \qquad \text{where } \mathrm{Gi}^{(l)} \colon y \mapsto \mathrm{Rl}\, y + la \ \text{ for } \ \mathrm{Gi} \colon y \mapsto \mathrm{Rl}\, y + a.$$

In particular, δ_l transforms the reflection in $\{\hat n\cdot y=\mathrm{po}\}$, and the forms supported in $\{\hat n\cdot y>\mathrm{po}\}$, into the corresponding objects for the plane $\{\hat n\cdot y=l\,\mathrm{po}\}$; within each of the nine families, po takes every real value. By their definitions above, (H), (dF), and membership in $\mathcal{C}_9$ are expressed by inequalities and identities for the characteristic functional:

(i) (G) and (dF), taken together, are equivalent to the inequality $\mathrm{lw}(\exp X(f))\le\exp(\frac12\|\Pi_{\mathrm c}f\|^2)$ for $f\in\mathcal{S}$;
(ii) $\Phi_{\mathrm{Gi}_*\mathrm{lw}}(f)=\Phi_{\mathrm{lw}}(\mathrm{Gi}^{-1}\cdot f)$, and therefore $\mathrm{Gi}_*\mathrm{lw}=\mathrm{lw}$ holds exactly when $\Phi_{\mathrm{lw}}(\mathrm{Gi}\cdot f)=\Phi_{\mathrm{lw}}(f)$ for all $f\in\mathcal{S}$ (K21); this applies to the translations, to O_{cub} and to the reflections in the planes of the nine families;
(iii) reflection positivity at a plane, including (RP_j) of Section 2, is by its definition a family of inequalities among values of Φ_{lw}.

Hence $\mathrm{lw}_l\in\mathcal{K}$. This law is ergodic, because $\Psi\mapsto\Psi\circ\delta_l^*$ is a unitary map of $L^2(\mathrm{lw}_l)$ onto $L^2(\mathrm{lw})$ under which $U(a)$ corresponds to $U(la)$.

According to [4, Proposition 5.1], $|\Phi_{\mathrm{lw}_l}(g)-\Phi_{\mathrm{lw}_l}(g')|\le\|g-g'\|$, so that every sequence $l\to\infty$ admits a subsequence l_k such that $\Phi_{\mathrm{lw}_{l_k}}$ has a limit at every point of real L^2. This limit, Φ_∞, is a positive definite function with $\Phi_\infty(0)=1$, Lipschitz with respect to $\|\cdot\|$, and hence continuous on $\mathcal{S}$. By (K21) (Minlos), there is one law lw_∞ with $\Phi_\infty=\Phi_{\mathrm{lw}_\infty}$ on $\mathcal{S}$. For $f\in\mathcal{S}$, the law of $X(f)$ under lw_{l_k} converges weakly to its law under lw_∞, by Lévy's continuity theorem, since $\Phi_{\mathrm{lw}_{l_k}}(tf)\to\Phi_{\mathrm{lw}_\infty}(tf)$ for every real t. Applied to lw_{l_k} at $2f$, (i) gives $\sup_k\mathrm{lw}_{l_k}(\exp 2X(f))\le\exp(2\|\Pi_{\mathrm c}f\|^2)$, so that uniform integrability of the exponentials yields $\mathrm{lw}_\infty(\exp X(f))=\lim_k\mathrm{lw}_{l_k}(\exp X(f))\le\exp(\frac12\|\Pi_{\mathrm c}f\|^2)$; in this way (i) carries over to lw_∞, and (ii) and (iii) are preserved under pointwise limits. Thus $\mathrm{lw}_\infty\in\mathcal{K}$. As a law of $\mathcal{K}$, lw_∞ has (G), so that $\Phi_{\mathrm{lw}_\infty}$ is defined on L^2; there it coincides with Φ_∞, both functions being Lipschitz and $\mathcal{S}$ being dense.

Step 8. Let $S_2^{(l)}$ denote the second moment of lw_l, put $b:=*g$, let $P^\perp(p):=1-pp^T/|p|^2$, and let

$$\mathsf{D}(s):=\int d\mathrm{kl}(t)\,\frac{s}{s+t^2},$$

a function which decreases to $\mathrm{kl}(\{0\})$ as $s\downarrow0$. By Proposition D.1,

$$S_2^{(l)}(g,g)=(2\pi)^{-3}\int|P^\perp\hat b|^2\,\mathsf{D}(|p|^2/l^2)\,dp\longrightarrow c_0\,\|\Pi_{\mathrm c}g\|^2.$$

Consider a law with (G), a real $g\in L^2$ and $t>0$. The elementary inequality $|u^2-2(1-\cos u)|\le u^4/12$ for real u, together with [4, Proposition 5.1] and $c_4^4=64$, yields

$$\bigl|S_2(g,g)-2t^{-2}(1-\mathrm{Re}\,\Phi(tg))\bigr|\le\frac{t^2}{12}\,\mathrm{lw}(X(g)^4)\le\frac{c_4^4}{12}\,t^2\|g\|^4.$$

Every lw_l, and lw_∞ too, satisfies (G) with the constant one, and $\Phi_{\mathrm{lw}_{l_k}}(tg)\to\Phi_{\mathrm{lw}_\infty}(tg)$. It follows that $\limsup_k|S_2^{(l_k)}(g,g)-S_2^{\mathrm{lw}_\infty}(g,g)|\le\frac{c_4^4}{6}t^2\|g\|^4$ for each $t>0$, which means $S_2^{\mathrm{lw}_\infty}(g,g)=c_0\|\Pi_{\mathrm c}g\|^2$. Polarisation and the uniqueness assertion of Proposition D.1 lead to $\mathrm{kl}_\infty=c_0\delta_0$. The constants in the bound of Proposition C.1 are absolute:

$$\bigl|\Phi_{\mathrm{lw}_l}(g'(\cdot-a)-g)-\overline{\Phi_{\mathrm{lw}_l}(g)}\,\Phi_{\mathrm{lw}_l}(g')\bigr|\le C_0\,r/|a|_\infty$$

for every l, every $r>0$, all $g,g'\in C_c^\infty((-r,r)^3)$ and all a with $|a|_\infty\ge4r$. This inequality carries over to lw_∞. Accordingly lw_∞ clusters on the linear span of the functions $\exp iX(g)$, $g\in C_c^\infty$, a span which is dense in $L^2(\mathrm{lw}_\infty)$; so lw_∞ is mixing and ergodic [4, Sections 5.2 and 7.4]. Part (a) then yields $\mathrm{lw}_\infty=\gamma_{c_0}$, whichever subsequence was chosen, and this proves (b). □

For laws $\mathrm{lw}_1,\mathrm{lw}_2$ and $0<\epsilon<1$, let $\mathrm{lw}_1*_\epsilon\mathrm{lw}_2$ be the law of $(1-\epsilon)^{1/2}F+\epsilon^{1/2}G$, where F and G are independent with these respective laws. The constants C_U, $C_{\bar\kappa}$, $C'_{\bar\kappa}$, c_{cl} and $\mathfrak{e}_2$ in the next proposition are those of Section F.2.

4.4. **The properties (D1)–(D6) do not determine a law.**

Proposition 4.4. With the data of Section F.2 held fixed, let $e \in (0, e_2]$ be a charge, μ a volume-first limit point, and $\xi \in \Xi_\mu$. Define

$$\epsilon_0 := \frac{(C'_{\bar\kappa} - \frac{1}{2}C_{\bar\kappa}) \cdot e^{\sigma}}{2 \max_{\mathrm{ix}} |b(g_{\mathrm{ix}})|};$$

then $0 < \epsilon_0 \le \frac{1}{20}$. Set $\mu^0_\xi := \mu_\xi$, and $\mu^\epsilon_\xi := \mu_\xi *_\epsilon \gamma$ when $0 < \epsilon \le \epsilon_0$. For each $\epsilon \in (0, \epsilon_0]$ the following assertions are true.

(a) The law μ^ϵ_ξ has properties (D1)–(D6) of Section F.2, with the same tests g_{ix}, point x_0, and constants $c_{\mathrm{cl}}, C'_{\bar\kappa}, \sigma, \mathfrak{e}_2$.

(b) For $n \ge 3$ the cumulants of order n of μ^ϵ_ξ are $(1-\epsilon)^{n/2}\kappa^\xi_n$. In particular, $\kappa_4(f) = (1-\epsilon)^2\kappa^\xi_4(f)$ for the fourth cumulant of μ^ϵ_ξ; hence the laws μ^ϵ_ξ, $0 \le \epsilon \le \epsilon_0$, differ from one another.

(c) The Källén–Lehmann measure of μ^ϵ_ξ is $(1-\epsilon)\,\mathrm{kl}_\xi + \epsilon\,\delta_0$.

(d) Consider a positive measure ϱ, defined on $[0,\infty)$, of total mass at most 1; the symbol ϱ is unrelated to the Wilson parameter ρ. Let γ^ϱ denote the centred Gaussian law with covariance

$$C_\varrho(g,g') := (2\pi)^{-3} \int \mathrm{d}p\, \widehat{*\bar g}(p) \cdot \Pi(p)\, \widehat{*g'}(p) \int \mathrm{d}\varrho(t)\, \frac{|p|^2}{|p|^2 + t^2};$$

thus $\gamma^{a\delta_0} = \gamma_a$. For γ^ϱ the properties (H) and (dF) hold; this law belongs to $\mathcal{C}_9$, is ergodic, and its Källén–Lehmann measure is ϱ. Assume in addition that

$$\epsilon \cdot \max_{\mathrm{ix}} (\|g_{\mathrm{ix}}\|^2 - C_\varrho(g_{\mathrm{ix}}, g_{\mathrm{ix}})) \le (C'_{\bar\kappa} - 2C_U) \cdot e^{2-\mathfrak{e}_2}.$$

The law $\mu_\xi *_\epsilon \gamma^\varrho$ then fulfils (a) together with the first assertion of (b), and its Källén–Lehmann measure equals $(1-\epsilon)\,\mathrm{kl}_\xi + \epsilon\,\varrho$. In the case $\varrho = a\delta_0$, $0 < a \le 1$, the assumption takes the form

$$\epsilon\,(1-a) \max_{\mathrm{ix}} \|g_{\mathrm{ix}}\|^2 \le (C'_{\bar\kappa} - 2C_U)\, e^{2-\mathfrak{e}_2},$$

and it is fulfilled for all ϱ as soon as

$$\epsilon \max_{\mathrm{ix}} \|g_{\mathrm{ix}}\|^2 \le (C'_{\bar\kappa} - 2C_U)\, e^{2-\mathfrak{e}_2}.$$

■

Proof. Step 1: the product formula. For real $g \in \mathcal{S}$, independence in the definition of $*_\epsilon$ gives

$$\Phi_{\mu^\epsilon_\xi}(g) = \Phi_{\mu_\xi}((1-\epsilon)^{1/2}g) \cdot \Phi_\gamma(\epsilon^{1/2}g);$$

the same factorisation is valid for $\mathrm{lw}(\exp X(g))$.

Step 2: (H), (dF) and $\mathcal{C}_9$. The two factors share the following properties: (G) and (dF); invariance under $\mathbb{R}^3$ and under O_{cub}; and, for each plane in one of the nine families, invariance under the reflection in that plane together with reflection positivity at it. For μ_ξ these are contained in (D1), and for γ in Lemma D.2. These properties pass to μ^ϵ_ξ for the following three reasons. First, the map $(F,G) \mapsto (1-\epsilon)^{1/2}F + \epsilon^{1/2}G$ commutes with every Euclidean map and sends closed forms to closed forms. Second, $\frac{1-\epsilon}{2}\|g\|^2 + \frac{\epsilon}{2}\|g\|^2 = \frac{1}{2}\|g\|^2$. Third, the character matrix in (2) is the entrywise product of the corresponding positive semidefinite matrices for the two independent fields; the Schur product theorem preserves positivity. This argument applies to every one of the specified reflection planes.

Step 3: ergodicity. Take real $g, g' \in C^\infty_c$. By Step 1, $\mu^\epsilon_\xi(\exp(iX(g)) \cdot U(a) \exp(iX(g')))$ splits into two factors: the corresponding expression for μ_ξ, evaluated at $(1-\epsilon)^{1/2}g$ and $(1-\epsilon)^{1/2}g'$, and the corresponding expression for γ_ϵ, evaluated at g and g'. Each factor factorises when $|a| \to \infty$ in any manner: for the first this is the second sentence of (D4), for the second it is Lemma D.2.

Characters span L^2 and translations are isometries, so approximation extends this factorization to all L^2 functions. Thus μ_ξ^ϵ is mixing and therefore ergodic; compare [4, Section 7.4].

Step 4: (D1)–(D5). Steps 2 and 3 together establish (H) and (dF) for μ_ξ^ϵ, its membership in $\mathcal{C}_9$, and its ergodicity. These are the structural properties (D1). Their consequences (D2)–(D5) require no small-charge or tuned-mass assumption on the individual law. Euclidean invariance and reflection positivity at every plane follow from [4, Theorem 8.6]; Section B.2 and Proposition D.1 give the spectral conclusions. The invariant-projection identity in [4, Section 7.3], together with ergodicity, gives the unique vacuum. Proposition C.1 supplies the rate in (D4), and the moment bounds of [4, Section 5] and the tensor argument of [4, Section 27] give the Schwinger-function properties in (D5).

Step 5: moments and cumulants. For a sum of independent random variables, the logarithm of the generating function equals the sum of the logarithms of the generating functions of the summands; furthermore, all cumulants of γ of order 3 or more vanish. Hence (b) holds, and the second moment S_2^ϵ of μ_ξ^ϵ is given by

$$S_2^\epsilon(g,g') = (1-\epsilon)\, S_2^\xi(g,g') + \epsilon\, \langle g, \Pi_{\mathrm{c}} g' \rangle$$

by independence and centering of the two summands. For closed g, the projection satisfies $\Pi_{\mathrm{c}} g = g$.

Step 6: (D6). Because $\xi \in \Xi_\mu$, the component has (G) and the centered-probe bounds specified in Section F.2. Fix ix, put $z' = \|g_{\mathrm{ix}}\|^2$, and let $\mathrm{dv}_m = \mu_\xi(\mathrm{Wk}_{m,\mathrm{ix}})$ for $m = 2, 4$. The centered-probe bounds give $|\mathrm{dv}_m| \le \mathfrak{u}^{\inf}_{m,\mathrm{ix}}$. Here $\mathrm{Wk}_{m,\mathrm{ix}} = \mathrm{Pw}_m(X(g_{\mathrm{ix}}); z', c^{\inf}_{m,\mathrm{ix}})$, with

$$\mathrm{Pw}_2(z; z', \mathrm{cf}) = e^{-2}(z^2 - z') - \mathrm{cf}, \qquad \mathrm{Pw}_4(z; z', \mathrm{cf}) = e^{-4}(z^4 - 6z'z^2 + 3z'^2) - \mathrm{cf}.$$

Gaussian domination gives $S_1^\xi = 0$, so (1) reduces to $\kappa_4^\xi(g) = S_4^\xi(g,g,g,g) - 3S_2^\xi(g,g)^2$. Integrating $\mathrm{Wk}_{2,\mathrm{ix}}$ and $\mathrm{Wk}_{4,\mathrm{ix}}$ against μ_ξ, one finds

$$e^{-2}(S_2^\xi(g_{\mathrm{ix}}, g_{\mathrm{ix}}) - z') = c^{\inf}_{2,\mathrm{ix}} + \mathrm{dv}_2, \qquad e^{-4}\kappa_4^\xi(g_{\mathrm{ix}}) = c^{\inf}_{4,\mathrm{ix}} + \mathrm{dv}_4 - 3(c^{\inf}_{2,\mathrm{ix}} + \mathrm{dv}_2)^2.$$

The parameter conditions in Section F.2 give $0 < e \le e_2 \le 1$, $\sigma > 0$, and $\mathfrak{e}_2 \in [0,1)$. Thus $e^{2\sigma} \le e^\sigma \le 1 \le e^{-\mathfrak{e}_2}$. The probe bounds in Section F.2 give $\mathfrak{u}^{\inf}_{2,\mathrm{ix}} \le C_U e^\sigma$ and $|c^{\inf}_{2,\mathrm{ix}}|\mathfrak{u}^{\inf}_{2,\mathrm{ix}} \le C_U e^\sigma$, whence

$$|S_2^\xi(g_{\mathrm{ix}}, g_{\mathrm{ix}}) - \|g_{\mathrm{ix}}\|^2| \le e^2(|c^{\inf}_{2,\mathrm{ix}}| + \mathfrak{u}^{\inf}_{2,\mathrm{ix}}) \le e^2(C_U e^{-\mathfrak{e}_2} + C_U e^\sigma) \le 2C_U\, e^{2-\mathfrak{e}_2},$$

$$\begin{aligned} |e^{-4}\kappa_4^\xi(g_{\mathrm{ix}}) - b(g_{\mathrm{ix}})| &\le |c^{\inf}_{4,\mathrm{ix}} - 3(c^{\inf}_{2,\mathrm{ix}})^2 - b(g_{\mathrm{ix}})| + |\mathrm{dv}_4| + 6\,|c^{\inf}_{2,\mathrm{ix}}|\,|\mathrm{dv}_2| + 3\,|\mathrm{dv}_2|^2 \\ &\le (1+1+6)\, C_U e^\sigma + 3C_U^2 e^{2\sigma} \le (8C_U + 3C_U^2)e^\sigma = \tfrac{1}{2}C_{\bar\kappa}\, e^\sigma, \end{aligned}$$

where $C_{\bar\kappa} = 2(8C_U + 3C_U^2)$ by definition in Section F.2. Thus $|\kappa_4^\xi(g_{\mathrm{ix}}) - e^4 b(g_{\mathrm{ix}})| \le \frac{1}{2}C_{\bar\kappa} e^{4+\sigma}$, and the fourth cumulant $(1-\epsilon)^2\kappa_4^\xi(g_{\mathrm{ix}})$ of μ_ξ^ϵ at g_{ix} (Step 5) satisfies

$$|(1-\epsilon)^2\kappa_4^\xi(g_{\mathrm{ix}}) - e^4 b(g_{\mathrm{ix}})| \le \tfrac{1}{2}C_{\bar\kappa} e^{4+\sigma} + 2\epsilon\, e^4 |b(g_{\mathrm{ix}})| \le C'_{\bar\kappa} e^{4+\sigma}.$$

As $g_{\mathrm{ix}} = dd^*\beta_{\mathrm{ix}}$ is closed, Step 5 leads to

$$S_2^\epsilon(g_{\mathrm{ix}}, g_{\mathrm{ix}}) - \|g_{\mathrm{ix}}\|^2 = (1-\epsilon)(S_2^\xi(g_{\mathrm{ix}}, g_{\mathrm{ix}}) - \|g_{\mathrm{ix}}\|^2).$$

Step 7: (D6), continued. Every translation invariant law with (G) whose fourth cumulant obeys the bound established for μ_ξ^ϵ in Step 6, namely

$$|\kappa_4(g_{\mathrm{ix}}) - e^4 b(g_{\mathrm{ix}})| \le C'_{\bar\kappa} e^{4+\sigma} \qquad (\mathrm{ix} = 0, +, -),$$

falls under Lemma F.4, which yields $\mathrm{Ext}_{x_0}(f) \ge \frac{1}{2}e^4 b(f)$ and $\kappa_4(f) \ge \frac{1}{2}e^4 b(f) > 0$. The equality $\kappa_4(\mathrm{Gi}\cdot f) = \kappa_4(f)$ follows from the Euclidean invariance established in Step 4.

Step 8: the size of ϵ_0. The choices in Section F.2 satisfy $C'_{\bar\kappa} = 8C_{\bar\kappa} + 3C_{\bar\kappa}^2 > \frac{1}{2}C_{\bar\kappa} > 0$, $\max_{\mathrm{ix}} |b(g_{\mathrm{ix}})| \ge b(f) > 0$, and $C'_{\bar\kappa} e^\sigma \le b(f)/10$.

Step 9: (c). By Proposition D.1, the Maxwell covariance has spectral measure δ_0. Insert this representation and that of μ_ξ into the covariance formula of Step 5. Uniqueness in the same proposition gives the asserted spectral measure of μ_ξ^ϵ.

Step 10: (d), the law γ^ϱ. Restricted to real tests, C_ϱ is bilinear, symmetric and positive semidefinite, and it satisfies $C_\varrho(g,g) \le \|\Pi_c g\|^2 \le \|g\|^2$. Euclidean invariance follows from the pseudovector identity $*(\mathrm{Gi}\cdot g) = \det(\mathrm{Rl})\,\mathrm{Rl}(*g)(\mathrm{Gi}^{-1}\cdot)$ and $\Pi(\mathrm{Rl}p) = \mathrm{Rl}\Pi(p)\mathrm{Rl}^T$. By Lemma D.2, there is one and only one law γ^ϱ with $\Phi_{\gamma^\varrho} = \exp(-\frac12 C_\varrho)$. It is a centred Gaussian law whose covariance is C_ϱ, and $\gamma^\varrho(\exp X(g)) = \exp(\frac12 C_\varrho(g,g))$. Moreover, it is invariant under all Euclidean maps; and since it is mixing, it is ergodic. The bound $C_\varrho(g,g) \le \|g\|^2$ and the vanishing of C_ϱ on the closedness-null tests give (G) and (dF), respectively, by Lemma D.2.

Step 11: (d), reflection positivity. Let $g \in C_c^\infty(\{y_1 > 0\}; \Lambda^2)$ be real, and set $j := \mathrm{curl}\, *g$. For $t \ge 0$ and $k = (k_2, k_3) \in \mathbb{R}^2$ define $E_t := (|k|^2 + t^2)^{1/2}$ and

$$v(k) := \int \mathrm{d}y\, \exp(-i(k_2 y_2 + k_3 y_3) - E_t y_1)\, j(y) \in \mathbb{C}^3.$$

The identity $\overline{(p \times u)} \cdot (p \times u') = \bar u \cdot (|p|^2 - pp^T)\, u'$ shows that the form whose multiplier is $\Pi(p)\,\frac{|p|^2}{|p|^2+t^2}$ can be written as

$$(g, g') \mapsto \langle \mathrm{curl}\, *g, (-\Delta + t^2)^{-1}\, \mathrm{curl}\, *g' \rangle.$$

Further, $\mathrm{curl}\, *(\theta_1 \cdot g) = \theta_1\, j(\theta_1 \cdot)$, and one has

$$\int \frac{\mathrm{d}p_1}{2\pi}\, \frac{\exp(ip_1 s)}{p_1^2 + E_t^2} = \frac{\exp(-E_t|s|)}{2E_t}.$$

Combining these facts, one arrives at

$$C_\varrho(\theta_1 \cdot g, g) = \int \mathrm{d}\varrho(t) \int \frac{\mathrm{d}^2 k}{(2\pi)^2}\, \frac{|v_2|^2 + |v_3|^2 - |v_1|^2}{2E_t}.$$

Since $\mathrm{div}\, j = 0$, integrating by parts yields $E_t v_1 + i(k_2 v_2 + k_3 v_3) = 0$; hence $|v_1|^2 \le \frac{|k|^2}{E_t^2}(|v_2|^2 + |v_3|^2)$, and the integrand cannot be negative. Reflection positivity of γ^ϱ at $\{y_1 = 0\}$ is then a consequence of Lemma D.2. Euclidean invariance transports this reflected form to every plane, so γ^ϱ is reflection positive at every plane. Accordingly γ^ϱ has (H) and belongs to $\mathcal{C}_9$. Uniqueness in Proposition D.1 gives $\mathrm{kl}_{\gamma^\varrho} = \varrho$.

Step 12: (d), the law $\mu_\xi *_\epsilon \gamma^\varrho$. Concerning γ, Steps 1–5, 7 and 9 require nothing beyond the properties proved for γ^ϱ in Steps 10 and 11 and the vanishing of the cumulants of order at least 3. In Step 3 the second factor is the corresponding expression for $\gamma^{\epsilon\varrho}$, which factorises by Lemma D.2; in Steps 5 and 9 one puts $C_\varrho(g,g')$ and ϱ in the places of $\langle g, \Pi_c g'\rangle$ and δ_0. The bound of Step 6 for $(1-\epsilon)^2\kappa_4^\xi(g_{\mathrm{ix}})$ holds for this law as well. For the second moment, Step 6 gives $|S_2^\xi(g_{\mathrm{ix}}, g_{\mathrm{ix}}) - \|g_{\mathrm{ix}}\|^2| \le 2C_U\, e^{2-\mathfrak{e}_2}$, and therefore the second moment S_2 of $\mu_\xi *_\epsilon \gamma^\varrho$ satisfies

$$|S_2(g_{\mathrm{ix}}, g_{\mathrm{ix}}) - \|g_{\mathrm{ix}}\|^2| \le (1-\epsilon)\, 2C_U\, e^{2-\mathfrak{e}_2} + \epsilon(\|g_{\mathrm{ix}}\|^2 - C_\varrho(g_{\mathrm{ix}}, g_{\mathrm{ix}})) \le C'_{\bar\kappa}\, e^{2-\mathfrak{e}_2}.$$

The definitions in Section F.2 give $C'_{\bar\kappa} \ge 8C_{\bar\kappa} \ge 128 C_U > 2C_U$. Also $0 \le \|g\|^2 - C_\varrho(g,g) \le \|g\|^2$. □

5. The lattice Fisher estimate

We use the unit-mesh chamber model of Definition E.1. Thus $\beta \in C_{\mathrm{c}}^2$ is a lattice two-cochain, ν is a law on $\Omega = \{F : dF = 0\}$, and $W = \mathbb{Z}^3 \rtimes O_{\mathrm{cub}}$. The lattice charge is $q > 0$. Put $\mathrm{sw} = \sum_i w_i$ and $\mathbf{m} = M - \rho\,\mathrm{sw}/2$; the matrix D_A includes this mass. For $\rho \ge 0$, the norm of a type-three hopping matrix is $\mathsf{h}_3 = w_3(1+\rho)/2$, equal to $3(1+\rho)$ for the standard weights. At a continuum cutoff, $q = e\eta^{1/2}$ and $\mathbf{m} = \boldsymbol{\mu}_N$. The graph distance is denoted $d_{\mathbb{T}}$. Only type-three hops change π, by a unit coordinate vector. Hence

$$d_{\mathbb{T}}(x,y) \ge |\pi(x) - \pi(y)|_1,$$

the right side being the ℓ^1 distance of $\mathbb{Z}^3/\Gamma$.

5.1. **The positive-mass inverse.**

Lemma 5.1. Let $\rho \geq 0$, $w_0, \dots, w_3 > 0$, $q > 0$, and let $M \in \mathbb{R}$ be such that $\mathbf{m} > 0$. Consider a real one-cochain A and a space $\mathcal{H}$ from among those of Lemma E.2; thus $\mathcal{H}$ is $\ell^2(\mathbb{X}; \mathbb{C}^4)$ or the space of spinors of a pair (Γ, χ), and in the latter case A is assumed to be Γ-periodic. Put

$$\vartheta := \log\Big(1 + \frac{\mathbf{m}}{(1+\rho)\,\mathrm{sw}}\Big).$$

Then D_A is invertible on $\mathcal{H}$, $\|D_A^{-1}\| \leq \mathbf{m}^{-1}$, and for all sites x, y the 4×4 blocks of the inverse satisfy

$$\|D_A^{-1}(x,y)\| \leq \frac{2}{\mathbf{m}}\, e^{-\vartheta\, d_{\mathbb{T}}(x,y)}.$$

The same statements hold for $\check{D}_A$ and its 2×2 blocks. The constants are independent of A, q, Γ and χ. ∎

Proof. Step 1. By Lemma E.2 one has $\mathrm{Re}\, D_A \geq \mathbf{m}$. Therefore

$$\mathbf{m}\|\psi\|^2 \leq \mathrm{Re}\langle \psi, D_A \psi\rangle \leq \|\psi\|\, \|D_A\psi\|,$$

and the same inequalities hold with D_A^* in place of D_A. It follows that D_A is invertible and that $\|D_A^{-1}\| \leq \mathbf{m}^{-1}$. Only the lower bound on the real part enters this argument.

Step 2. Fix a site y and $N < \infty$, and let $\phi := \min(d_{\mathbb{T}}(\cdot, y), N)$. This function is bounded and Γ-periodic, and $|\phi(x) - \phi(x^{(i)})| \leq 1$. One has $e^{\vartheta\phi} D_A e^{-\vartheta\phi} = D_A - R$, where

$$(R\psi)(x) = \sum_i \big(e^{\vartheta(\phi(x) - \phi(x^{(i)}))} - 1\big)\, w(x, x^{(i)})\, U_A(x, x^{(i)})\, \psi(x^{(i)}).$$

Repeating the computation in the proof of Lemma E.2 with the additional factors, each of modulus at most $e^{\vartheta} - 1$, one obtains

$$\|R\| \leq (e^{\vartheta} - 1) \cdot \frac{1+\rho}{2}\, \mathrm{sw} = \frac{\mathbf{m}}{2}.$$

Step 3. Consequently $\mathrm{Re}(D_A - R) \geq \frac{\mathbf{m}}{2}$, and Step 1 gives $\|(D_A - R)^{-1}\| \leq \frac{2}{\mathbf{m}}$. Since $\phi(y) = 0$ and

$$D_A^{-1}(x,y) = e^{-\vartheta(\phi(x) - \phi(y))}\, (D_A - R)^{-1}(x,y),$$

the bound on the blocks follows on letting $N \to \infty$. The statements for $\check{D}_A$ follow, since $\check{D}_A$ is a diagonal block of D_A (Lemma E.2). □

The proof uses exponential conjugation, the norm bound for the four hops, and accretivity; self-adjointness is not required.

5.2. **The finite-mass polarisation bound.**

Lemma 5.2. Let $\rho \geq 0$, $w_0, \dots, w_3 > 0$ and $\mathbf{m} > 0$. For $q > 0$, an integer $n \geq 4$ and $\Gamma = n\mathbb{Z}^3$, let $\ell_n(A) := \log\det D_A^{C(n)}$, and define the polarisation form

$$K_n(u, u') := -\,\mathbb{E}_{C(n)}\big[\partial_u \partial_{u'} \ell_n(A)\big] \qquad (u, u' \in C^1(\Gamma)),$$

where ∂_u is the derivative along $A \mapsto A + tu$. There exists $c_* = c_*(\mathbf{m}, \rho, w) < \infty$, independent of q and of n, such that

$$K_n(u,u) \leq c_*\, q^2 \Big(\|du\|^2 + n^{-5} \Big|\sum_x u(x)\Big|^2\Big)$$

for all $u \in C^1(\Gamma)$, where $\sum_x u(x) \in \mathbb{R}^3$ denotes the vector with the components $\sum_{x \in \mathbb{Z}^3/\Gamma} u_\mu(x)$. ∎

Proof. Lemma 5.8 gives

$$|K_n(u,u)| \le \mathsf{c}_n \left(\frac{\pi^2}{4} \|du\|^2 + 2\pi^2 n^{-5} \left| \sum_x u(x) \right|^2 \right).$$

Apply Theorem 5.7(i). One may take $c_* = 1152\pi^4 \mathsf{h}_3^4 \mathbf{m}^{-4}$. □

5.3. **Shift identities at a fixed spacing.**

Proposition 5.3. Let $0 < \rho < 1$, $w_0, \dots, w_3 > 0$, $q > 0$, and let $M \in \mathbb{R}$ be such that $\mathbf{m} > 0$. Fix integers $n_i \ge 4$ with $n_i \to \infty$ and, as in Proposition E.7, a weak limit point ν of the laws $\mu_{C(n_i)}$. On the space of square summable two-cochains, let P_{ex} be the orthogonal projection whose range is the closure of dC^1_{c}. Let c_* be the constant of Lemma 5.2.

(a) For every $\beta \in C^2_{\mathrm{c}}$,

$$(1 + c_* q^2)^{-1} \, \|P_{\mathrm{ex}}\beta\|^2 \le \int \langle \beta, F \rangle^2 \, \mathrm{d}\nu \le \|P_{\mathrm{ex}}\beta\|^2.$$

The right side is the variance of $\langle \beta, F \rangle$ under the Gaussian weight $\exp(-\frac{1}{2}\|dA\|^2)$.

(b) For every plaquette σ,

$$\sum_{v \in \mathbb{Z}^3} \left| \int F_\sigma F_{\sigma+v} \, \mathrm{d}\nu \right| = \infty.$$

Exponential decay of the two-point function is thereby excluded.

(c) Let T_μ be the transfer operator of Lemma E.8, where $\mu = 1, 2, 3$. For every $\varepsilon \in (0,1)$ the spectral projection of T_μ belonging to the interval $(1-\varepsilon, 1)$ is different from zero. In particular 1 is a limit point of the spectrum of T_μ.

■

In the proof below, Steps 6 and 7 give the upper bound in (a), Steps 1 to 5 the lower bound, and Steps 8 to 15 parts (b) and (c).

Proof of Proposition 5.3. Step 1 (flow on $C(n)$). In Steps 1 to 3 the assumption $\rho \ge 0$ suffices. The density of $Z_{C(n)}$ is e^{-V}, where $V := \frac{1}{2}\|F\|^2 - \ell_n(A)$. Let $u \in C^1(\Gamma)$ and $t \in \mathbb{R}$. The shift $(A, z) \mapsto (A + tu, z)$ commutes with each of the maps $(A, z) \mapsto (A + \frac{2\pi}{q}k, \, z - dk)$, $k \in C^1(\Gamma; \mathbb{Z})$, by which the equivalence relation defining $\mathrm{Conf}(\Gamma)$ is given, and its inverse is the shift by $-tu$; consequently it maps each equivalence class onto an equivalence class, and it descends to $\mathrm{Conf}(\Gamma)$. Since Lebesgue measure is invariant under translations and z is left unchanged, the shift preserves the product of $\mathrm{d}A$ with the counting measure on the closed z. For each closed z, the section at z of the image of the fundamental domain given by the box is the box translated by tu; this image is thus a Borel set of pairs, and each equivalence class meets it in a single pair, because the fundamental domain given by the box has this property and the shift is a bijection of the pairs which maps equivalence classes onto equivalence classes. The clause of Definition E.1 on such sets allows it to take the place of the fundamental domain given by the box in the definition of $\mathrm{d}(A, z)$, and therefore the shift preserves $\mathrm{d}(A, z)$.

Step 2 (integration by parts). Let Φ be smooth in A, and let Φ and $\partial_u \Phi$ grow at most polynomially in F. The integral $\int \mathrm{d}(A, z) \, (\Phi e^{-V})(A + tu, z)$ is independent of t. For $|t| \le 1$ the t-derivative of the integrand is bounded by a constant multiple of $\prod_\sigma \exp(-\frac{1}{4}F_\sigma^2)$, which is integrable (Proposition E.5). Differentiating at $t = 0$ one obtains $\mathbb{E}[\partial_u \Phi] = \mathbb{E}[\Phi \, \partial_u V]$.

Step 3 (the Schwarz inequality). For $\Phi = \langle \beta, F \rangle$ and for $\Phi = \partial_u V = \langle F, du \rangle - \partial_u \ell_n$, Step 2 reads

$$\mathbb{E}[\langle \beta, F \rangle \, \partial_u V] = \langle \beta, du \rangle, \qquad \mathbb{E}[(\partial_u V)^2] = \|du\|^2 + K_n(u,u).$$

Consequently, for all $\beta \in C^2(\Gamma)$ and $u \in C^1(\Gamma)$,

$$\langle \beta, du \rangle^2 \le \mathbb{E}_{C(n)} \langle \beta, F \rangle^2 \cdot \left(\|du\|^2 + K_n(u,u) \right).$$

Step 4 (limit of the moments). Let $\beta \in C^2_{\mathrm{c}}$ and $u \in C^1_{\mathrm{c}}$. Once i is large, the periodisations of β and u have the same norms and pairings as β and u themselves, and $\langle \beta, \widetilde{F} \rangle$ coincides with the pairing on the torus (Definition E.1). On every $C(n)$ one has $\mathbb{E}\langle \beta, F\rangle^4 \le \frac{3}{2}\exp(2)\,\|\beta\|^4$ (see (13)), and therefore the second moments converge along the sequence.

Step 5 (lower bound in (a)). Since $\sum_x u(x)$ is fixed, Lemma 5.2 gives $\limsup_i K_{n_i}(u,u) \le c_* q^2 \|du\|^2$. Hence

$$\langle \beta, du\rangle^2 \le (1 + c_* q^2)\,\|du\|^2 \int \langle \beta, F\rangle^2 \,\mathrm{d}\nu,$$

and taking the supremum over $u \in C^1_{\mathrm{c}}$ yields the lower bound.

Step 6 (upper bound in (a)). By Proposition E.7, the law ν satisfies (G) and is symmetric under $F \mapsto -F$. Expanding (G) for $t\beta$ to the order t^2 one finds $\int \langle \beta, F\rangle^2 \,\mathrm{d}\nu \le \|\beta\|^2$. Since $\langle d^*\gamma, F\rangle = 0$ on Ω for every $\gamma \in C^3_{\mathrm{c}}$ (Definition E.1), the cochain β can be replaced by $\beta - d^*\gamma$.

Step 7 (Hodge decomposition). The infimum of $\|\beta - d^*\gamma\|^2$ over γ equals $\|P_{\mathrm{ex}}\beta\|^2$. Put $\mathsf{s}_\mu(p) = 2\sin(p_\mu/2)$. Indeed, the square summable two-cochains form the orthogonal sum of the closures of dC^1_{c} and of $d^*C^3_{\mathrm{c}}$, because a square summable β with $d\beta = 0$ and $d^*\beta = 0$ satisfies $|\mathsf{s}(p)|^2\hat\beta(p) = 0$.

Step 8 (Fourier form). Let β be identified with the triple $(\beta_{23}, -\beta_{13}, \beta_{12})$, and let Fourier transforms be taken with respect to the centres of the plaquettes. Then P_{ex} acts as multiplication by $1 - \mathsf{s}\,\mathsf{s}^T/|\mathsf{s}|^2$. Write $\varpi := 1 - \mathsf{s}_1^2/|\mathsf{s}|^2$.

Step 9 (reduction of (b)). By the invariance under W it suffices to take $\sigma = (0;23)$; let $g(v) := \int F_\sigma F_{\sigma+v}\,\mathrm{d}\nu$. Suppose that g is summable. Then $\hat g$ is continuous, and (a), applied to $\beta = \sum_v c_v 1_{\sigma+v}$, becomes

$$(1 + c_* q^2)^{-1} \int |\hat c|^2 \varpi \le \int |\hat c|^2 \hat g \le \int |\hat c|^2 \varpi.$$

Step 10 (conclusion for (b)). Fejér kernels for $|\hat c|^2$ then give $(1+c_*q^2)^{-1}\varpi \le \hat g \le \varpi$ on $[-\pi,\pi]^3\backslash\{0\}$. However, $\varpi = 0$ on the p_1-axis and $\varpi = 1$ on the plane $p_1 = 0$, so that $\hat g$ cannot have a limit at 0.

Step 11 (set-up for (c)). It suffices to treat $\mu = 1$, the other directions following by the invariance under W. For integers $r \ge 1$ and s, let β^s be equal to 1 on the plaquettes $((s,y);23)$ with $y \in \{0,\dots,r-1\}^2$ and to 0 elsewhere, and put

$$\Phi_s := \langle \beta^s, F\rangle, \qquad g(t) := \int \Phi_0 \Phi_t \,\mathrm{d}\nu.$$

The reflection θ_1 of Lemma E.8 sends Φ_s to Φ_{-1-s}, and the translation τ_{e_1} sends Φ_s to Φ_{s+1}.

Step 12 (spectral representation). By Lemma E.8 the class map extends to the $L^2(\nu)$-closure of $\mathcal{O}_1^+$, to which Φ_0 belongs. By Lemma E.8,

$$g(t) = \langle [\Phi_0], T_1^{|t|-1}[\Phi_0]\rangle = \int \lambda^{|t|-1}\,\mathrm{d}\mathfrak{e}(\lambda) \ge 0 \qquad (|t| \ge 1),$$

where $\mathfrak{e}$ is the spectral measure of $[\Phi_0]$ on $[0,1]$; its mass is $g(1) \le g(0)$.

Step 13 (the sum of g). For $N \ge 1$,

$$\sum_{|t|<N} \Big(1 - \frac{|t|}{N}\Big)\, g(t) = \frac{1}{N}\int \Big\langle \sum_{s=0}^{N-1} \beta^s, F\Big\rangle^2 \,\mathrm{d}\nu.$$

By (a) this quantity lies between $(1 + c_* q^2)^{-1} J_N$ and J_N, where

$$J_N := \frac{1}{N}\Big\| P_{\mathrm{ex}} \sum_{s<N} \beta^s \Big\|^2 \le r^2,$$

and $J_N \to r^2$ by Step 8 and Fejér's theorem in the variable p_1. As $g \ge 0$, it follows that $(1 + c_* q^2)^{-1} r^2 \le \sum_t g(t) \le r^2$; in particular $\mathfrak{e}(\{1\}) = 0$.

Step 14 (the value $g(0)$). Write $k=(p_2,p_3)$, and let 1_r be the indicator function of $\{0,\dots,r-1\}^2$. By (a) and Step 8,

$$g(0)\le\|P_{\mathrm{ex}}\beta^0\|^2=\int\frac{\mathrm{d}^2k}{(2\pi)^2}\,|\hat1_r(k)|^2\,|\mathsf{s}(k)|\,(4+|\mathsf{s}(k)|^2)^{-1/2}$$
$$\le\frac12\|1_r\|\,\|\nabla1_r\|=r^{3/2}.$$

Step 15 (conclusion for (c)). Suppose that the spectral projection of $(1-\varepsilon,1)$ vanishes. Then

$$\sum_t g(t)=g(0)+2\int(1-\lambda)^{-1}\,\mathrm{d}\mathfrak{e}(\lambda)\le\Big(1+\frac{2}{\varepsilon}\Big)\,g(0)\le\Big(1+\frac{2}{\varepsilon}\Big)\,r^{3/2},$$

which contradicts Step 13 as soon as $r>(1+c_*q^2)^2(1+\frac{2}{\varepsilon})^2$. □

5.4. **The flavoured meson.**

Proposition 5.4. Let $\rho,M\in\mathbb{R}$, $w_0,\dots,w_3>0$ and $q>0$, and let $\mathbb{T}=(\Gamma,\chi)$ be any torus. Write $\psi(x)=(\psi^{(1)}(x),\psi^{(2)}(x))$ and $\bar\psi(x)=(\bar\psi^{(1)}(x),\bar\psi^{(2)}(x))$ according to the decomposition $\mathbb{C}^4=\mathbb{C}^2\oplus\mathbb{C}^2$ in which $\gamma_\mu=\mathrm{diag}(\sigma_\mu,-\sigma_\mu)$ (Definition E.1), and define

$$\mathfrak{M}(x):=\bar\psi^{(1)}(x)\,\psi^{(2)}(x),\qquad\mathfrak{M}^\star(x):=-\,\bar\psi^{(2)}(x)\,\psi^{(1)}(x).$$

The Grassmann polynomials $\mathfrak{M}(x),\mathfrak{M}^\star(x)\in\mathrm{Gr}_{\mathbb{T}}$ are even, Γ-periodic, and gauge invariant, with the conventions of Definition E.1.

(a) For every mirror Π and every site x one has $\Theta_\Pi\,\mathfrak{M}(x)=\mathfrak{M}^\star(\theta_\Pi x)$ in $\mathrm{Gr}(\mathbb{X})$, with Θ_Π as in Proposition E.6. On the torus, Θ_Π maps $\mathrm{Gr}_{\mathbb{T}}$ onto $\mathrm{Gr}_{\theta_\Pi\mathbb{T}}$ (Proposition E.6), and the identity holds for the classes, its right side being formed in $\mathrm{Gr}_{\theta_\Pi\mathbb{T}}$. If $\theta_\Pi\mathbb{T}=\mathbb{T}$, as for a torus adapted to the family of Π [4, Section 2.6] and for the cubic torus $C(n)$ of side n, $n\ge4$ an integer, both sides lie in $\mathrm{Gr}_{\mathbb{T}}$.

(b) Let adj denote the adjugate matrix and (y,x) its 2×2 block. For all sites x,y,

$$Z_{\mathbb{T}}\big[\mathfrak{M}^\star(y)\,\mathfrak{M}(x)\big]=\int_{\mathrm{Conf}(\Gamma)}\mathrm{d}(A,z)\,e^{-\frac12\|F\|^2}\,\big\|\mathrm{adj}(\check D_A^{\mathbb{T}})(y,x)\big\|_{\mathrm{HS}}^2\ge0.$$

(c) If $\rho\ge0$ and $\mathbf{m}>0$, then, with ϑ as in Lemma 5.1,

$$\mathbb{E}_{\mathbb{T}}\big[\mathfrak{M}^\star(y)\,\mathfrak{M}(x)\big]=\mathbb{E}_{\mathbb{T}}\big\|(\check D_A^{\mathbb{T}})^{-1}(y,x)\big\|_{\mathrm{HS}}^2\le\frac{8}{\mathbf{m}^2}\,e^{-2\vartheta\,d_{\mathbb{T}}(x,y)},$$
$$\frac{2}{\big(|M|+\frac{1+\rho}{2}\,\mathrm{sw}\big)^2}\le\sum_{y\in\mathbb{X}/\Gamma}\mathbb{E}_{\mathbb{T}}\big[\mathfrak{M}^\star(y)\,\mathfrak{M}(x)\big]\le\frac{2}{\mathbf{m}^2}.$$

The constants do not depend on $\mathbb{T}$ or on q. ■

Proof. Step 1 (proof of (a)). One has $\mathfrak{M}(x)=\bar\psi(x)\,Y\psi(x)$, where the only non-zero block of Y is $Y_{12}=1$. By Proposition E.6,

$$\Theta_\Pi\mathfrak{M}(x)=\bar\psi(\theta_\Pi x)\,\gamma_n Y^*\gamma_n\,\psi(\theta_\Pi x),$$

and $\gamma_nY^*\gamma_n=-Y^*$, since $\gamma_n=\mathrm{diag}(\sigma_n,-\sigma_n)$ and $\sigma_n^2=1$. On the torus, Proposition E.6 gives $\Theta_\Pi[\mathfrak{M}(x)]_{\mathbb{T}}=[\Theta_\Pi\mathfrak{M}(x)]_{\theta_\Pi\mathbb{T}}$, and the identity passes to the classes.

Step 2 (proof of (b)). Write $\check D=\check D_A^{\mathbb{T}}$. By Lemma E.2,

$$S_{\mathbb{T}}(A)=\bar\psi^{(1)}\check D\,\psi^{(1)}+\bar\psi^{(2)}\check D^*\psi^{(2)}.$$

Rearranging even brackets, one obtains

$$\mathfrak{M}^\star(y)\,\mathfrak{M}(x)=\sum_{a,b}\big[\psi_b^{(1)}(y)\,\bar\psi_a^{(1)}(x)\big]\,\big[\psi_a^{(2)}(x)\,\bar\psi_b^{(2)}(y)\big].$$

Step 3. Since the Berezin integral picks out the coefficient of a product of even pairs (Definition E.1), it factorises on the product of an even element of the first flavour with an even element of the second. For every square matrix B, differentiating $\int \exp(-\bar\psi B\psi) = \det B$ with respect to B_{ji} gives

$$\int \exp(-\bar\psi B\psi)\,\psi_i\bar\psi_j = \frac{\partial \det B}{\partial B_{ji}} = \operatorname{adj}(B)_{ij}.$$

Step 4. Applying this with $B = \check D$ and with $B = \check D^*$, and using $\operatorname{adj}(\check D^*) = \operatorname{adj}(\check D)^*$, one finds

$$\int_{\mathbb{T}} e^{-S_{\mathbb{T}}(A)}\,\mathfrak{M}^\star(y)\,\mathfrak{M}(x) = \sum_{a,b} \bigl|\operatorname{adj}(\check D)_{(y,b),(x,a)}\bigr|^2.$$

Convergence of the integral over $\operatorname{Conf}(\Gamma)$ follows from Proposition E.5.

Step 5 (proof of (c)). Lemma 5.1 shows that $\check D$ is invertible. Since $\operatorname{adj}(\check D) = \det \check D \cdot \check D^{-1}$ and $|\det \check D|^2 = \det D_A^{\mathbb{T}}$ (Lemma E.2), the identity follows. The upper bound is a consequence of Lemma 5.1 and of the inequality $\|B\|_{\mathrm{HS}}^2 \le 2\|B\|^2$ for 2×2 blocks B.

Step 6. Finally,

$$\sum_y \|\check D^{-1}(y,x)\|_{\mathrm{HS}}^2 = \operatorname{tr}_{\mathbb{C}^2}\bigl[(\check D\check D^*)^{-1}(x,x)\bigr],$$

and $\|\check D\|^{-2} \le (\check D\check D^*)^{-1} \le \mathbf{m}^{-2}$ by Lemma E.2 and Lemma 5.1. □

For the continuum the notation fixed above is used: β denotes a probe, $a = d^*\beta$ and $f = dd^*\beta$, the charge is $e > 0$, and $\|\cdot\|$, $\langle\cdot,\cdot\rangle$ stand, on forms, for the norm and the inner product of $L^2(\mathbb{R}^3;\Lambda^2)$. The exponential function is written exp.

5.5. **The variance identity.**

Lemma 5.5. In this lemma only, D_A denotes the massless chamber operator of Definition E.1, with standard weights, and $E_n = \mathbb{E}_{C(n)}$ is formed at lattice mass $\boldsymbol{\mu}$. Thus $D_A + \boldsymbol{\mu}$ is the massive operator denoted by D_A elsewhere in this section.

Suppose that $-1 \le \rho \le 1$, $q > 0$, $\boldsymbol{\mu} \in \mathbb{R}$ and $n \ge 4$, and that u is a real one-cochain on $\mathbb{Z}^3/n\mathbb{Z}^3$ with $du \neq 0$. Write ∂_u for the derivative along $A \mapsto A + tu$, and define

$$\ell(A) := \log\det(D_A + \boldsymbol{\mu}), \qquad x := \frac{E_n\langle du, F\rangle^2}{\|du\|^2}, \qquad j := \frac{E_n(\partial_u \ell)^2}{\|du\|^2}.$$

Then $j < \infty$, $x \le 1$, and

$$E_n\bigl(\langle du, F\rangle - \partial_u\ell\bigr)^2 = (2 - x + j)\,\|du\|^2, \qquad (1-x)^2 \le j\,x, \qquad x \ge 1 - \sqrt{j}.$$

■

Proof. 1. Let $\check D_A$ be the upper block of the operator, mass included; then $\det(D_A + \boldsymbol{\mu}) = |\det \check D_A|^2$ (Lemma E.2). This is a trigonometric polynomial in qA, which does not vanish identically, since $Z_n[1] > 0$ (Proposition E.5). Hence ℓ is smooth off a null set, and

$$\frac{\bigl(\partial_u \det(D_A + \boldsymbol{\mu})\bigr)^2}{\det(D_A + \boldsymbol{\mu})} \le 4\,|\partial_u \det \check D_A|^2$$

is bounded; therefore $j < \infty$.

2. The density $\varrho := \exp(-\tfrac12\|F\|^2)\det(D_A + \boldsymbol{\mu})$ is smooth, and almost everywhere one has $\partial_u\varrho = -\varrho\, s$, where $s := \langle du, F\rangle - \partial_u\ell$. The argument of Steps 1 and 2 of the proof of Proposition 5.3, with ϱ substituted for $\exp(-V)$, yields $E_n[\partial_u\Phi] = E_n[\Phi\, s]$ for every Φ of polynomial growth in F; it makes no use of the sign of $\boldsymbol{\mu}$.
3. For $\Phi = \langle du, F\rangle$ this reads $E_n[\langle du, F\rangle\, s] = \|du\|^2$, that is, $E_n[\langle du, F\rangle\,\partial_u\ell] = (x-1)\|du\|^2$. Expanding s^2, one obtains the first identity. By the Schwarz inequality, $\|du\|^4 \le x\|du\|^2 \cdot E_n s^2$, that is, $1 \le x(2 - x + j)$, which is the inequality $(1-x)^2 \le jx$.
4. The bound $x \le 1$ is (13); hence $1 - x \le \sqrt{jx} \le \sqrt{j}$. □

5.6. **Scores and the continuum limit.**

Proposition 5.6. Let $-1 \le \rho \le 1$, $w_0, \dots, w_3 > 0$, $q > 0$ and $M \in \mathbb{R}$; no assumption is made on the sign of $\mathbf{m}$. Let $n \ge 4$ be an integer and $\Gamma = n\mathbb{Z}^3$. Denote by

$$\varrho := \exp(-\tfrac{1}{2}\|F\|^2)\, \det D_A^{C(n)} \ge 0$$

the density of $Z_{C(n)}$ with respect to $\mathrm{d}(A,z)$. Given $u \in C^1(\Gamma)$, define $s_u := \partial_u \varrho/\varrho$ on the set where $\varrho > 0$, and $s_u := 0$ on the set where $\varrho = 0$. The Fisher information of the shift along u is

$$I_n(u) := \mathbb{E}_{C(n)}[s_u^2].$$

(a) *One torus.* One has $I_n(u) < \infty$, $\mathbb{E}_{C(n)}[s_u] = 0$,

$$\|du\|^4 \le \mathbb{E}_{C(n)}\langle du, F\rangle^2 \cdot I_n(u) \qquad \text{and} \qquad I_n(u) \ge \|du\|^2.$$

On the set where $\varrho > 0$ put $\ell_n := \log\det D_A^{C(n)}$; the set where $\varrho = 0$ is a null set for $\mathrm{d}(A,z)$. Whatever $u, u' \in C^1(\Gamma)$, the function $\partial_u \partial_{u'} \ell_n$ is integrable with respect to $Z_{C(n)}[1]^{-1}\varrho\,\mathrm{d}(A,z)$, a probability measure, and the formula

$$K_n(u,u') := -\,\mathbb{E}_{C(n)}[\partial_u \partial_{u'} \ell_n]$$

defines on $C^1(\Gamma)$ a bilinear form that is symmetric and positive semidefinite; for $\rho \ge 0$ and $\mathbf{m} > 0$ it coincides with the form of Lemma 5.2. For every real M, that is, whatever the real lattice mass $\mathbf{m}$, and for all $u, u' \in C^1(\Gamma)$, those with $du = 0$ included,

$$\mathbb{E}_{C(n)}[s_u\, s_{u'}] = \langle du, du'\rangle + K_n(u,u'), \qquad I_n(u) = \|du\|^2 + K_n(u,u),$$
$$K_n(u,u) = \|du\|^2 - \mathbb{E}_{C(n)}\langle du, F\rangle^2 + \mathbb{E}_{C(n)}(\partial_u \ell_n)^2 \ge \mathbb{E}_{C(n)}(\partial_u \ell_n)^2.$$

Suppose in addition that the weights are standard and that $du \neq 0$, and let x and j be the numbers that Lemma 5.5 attaches to u at $\boldsymbol{\mu} = \mathbf{m}$. Then

$$I_n(u) = (2 - x + j)\,\|du\|^2, \qquad K_n(u,u) = (1 - x + j)\,\|du\|^2,$$

and consequently $j \le K_n(u,u)/\|du\|^2 \le j + \min\{1, \sqrt{j}\}$.

For (b), (c) and (d), assume $0 < \rho < 1$ and standard weights, fix $\bar m \in (0,1]$, $e > 0$ and an arbitrary mass assignment $N \mapsto m_{0,N}$, and let μ be a limit point of $(P_{N,T})$ along a defining sequence (N_i, T_i). Set $\eta_i = L^{-N_i}$, $n_i = 8hL^{T_i+N_i}$ and $R_i = 8hL^{T_i}$, and form $E_i := \mathbb{E}_{C(n_i)}$ and I_{n_i} with $q = e\,\eta_i^{1/2}$ and $\mathbf{m} = \boldsymbol{\mu}_{N_i} = \bar m \eta_i + m_{0,N_i}$. Let β be a probe. For all sufficiently large i, let $\hat{\mathfrak{a}}_i$ be the periodized cochain of Lemma E.10 on the torus with n_i sites per side. Put

$$\bar I(\beta) := \liminf_{i\to\infty} I_{n_i}(\hat{\mathfrak{a}}_i) \in [\,\|f\|^2, \infty\,].$$

(b) *Limit points.* $S_2(f,f) \ge \|f\|^4/\bar I(\beta)$.

(c) *Integration by parts.* Suppose that $\bar I(\beta) < \infty$. Then some $\mathsf{s}_\beta \in L^2(\mu)$ with $\|\mathsf{s}_\beta\|_2^2 \le \bar I(\beta)$ satisfies $\mu(\partial_f \Phi) = \mu(\Phi\,\mathsf{s}_\beta)$ for every $\Phi = \phi(X(g_1), \dots, X(g_k))$ in which ϕ is smooth and bounded together with all its derivatives and the $g_j \in L^2(\mathbb{R}^3; \Lambda^2)$ are real; here

$$\partial_f \Phi := \sum_j \langle g_j, f\rangle\, (\partial_j \phi)(X(g_1), \dots, X(g_k)).$$

If $J(a)$ is defined by $J(a) := e^{-1}(X(f) - \mathsf{s}_\beta)$, this identity takes the form $\mu(\partial_f\Phi) = \mu(\Phi\cdot(X(f) - e\,J(a)))$. Moreover, the image μ_t of μ under $\omega \mapsto \omega + t\,f$ satisfies $|\mu_t(\Phi) - \mu(\Phi)| \le |t|\,\bar I(\beta)^{1/2}$ for every real t and every Σ-measurable Φ with $|\Phi| \le 1$.

(d) *The atom at zero.* Assume that μ is volume-first and that $\bar I(\beta) \le (1+c)\,\|f\|^2$ for all probes β, with a constant $c < \infty$. Then $\mathrm{kl}_\mu(\{0\}) \ge (1+c)^{-1}$.

■

Proof. Step 1 (finiteness in (a)). Let $\check{D} := \check{D}_A^{C(n)}$. Then $\det D_A^{C(n)} = |\det \check{D}|^2$ (Lemma E.2), and $\det \check{D}$ is a trigonometric polynomial in the variables $qA(b)$. In particular ϱ is smooth, and both $\det \check{D}$ and $\partial_u \det \check{D}$ are bounded. At a point where $\varrho = 0$ the non-negative function ϱ attains a minimum, so that $\partial_u \varrho = \varrho\, s_u$ everywhere. Where $\varrho > 0$ one has $s_u = -\langle du, F\rangle + 2\,\mathrm{Re}(\partial_u \det \check{D}/\det \check{D})$, and consequently

$$\varrho\, s_u^2 \le 2\exp(-\tfrac{1}{2}\|F\|^2)\big(\langle du, F\rangle^2\,|\det \check{D}|^2 + 4\,|\partial_u \det \check{D}|^2\big).$$

The right side is integrable (Proposition E.5), and $Z_{C(n)}[1] > 0$ (Proposition E.5).

Step 2 (the inequalities in (a)). The argument of Steps 1 and 2 in the proof of Proposition 5.3, carried out with ϱ in place of $\exp(-V)$, gives

$$\mathbb{E}[\partial_u \Phi] = -\mathbb{E}[\Phi\, s_u]$$

for every Φ smooth in A such that Φ and $\partial_u \Phi$ grow at most polynomially in F; the sign of $\mathbf{m}$ plays no part here. Taking $\Phi = 1$ one finds $\mathbb{E}[s_u] = 0$. With $\Phi = \langle du, F\rangle$ one obtains $\|du\|^2 = -\mathbb{E}[\langle du, F\rangle\, s_u]$, and the Schwarz inequality yields the first inequality. The second follows from it together with $\mathbb{E}\langle du, F\rangle^2 \le \|du\|^2$ (see (13)). The identities of (a) in which K_n appears are established in Step 2a below. For the identity $I_n(u) = (2 - x + j)\,\|du\|^2$, let the weights be standard and $du \neq 0$. By Definition E.1, the operator $D_A^{C(n)}$ is then the operator $D_A + \boldsymbol{\mu}$ of Lemma 5.5 at $\boldsymbol{\mu} = \mathbf{m}$, and $\mathbb{E}_{C(n)}$ is the expectation E_n of that Lemma. Where $\varrho > 0$ one has $s_u = -(\langle du, F\rangle - \partial_u \ell)$, with the function ℓ of Lemma 5.5, so that $I_n(u) = \mathbb{E}_{C(n)}(\langle du, F\rangle - \partial_u \ell)^2$; the identity $I_n(u) = (2 - x + j)\,\|du\|^2$ is therefore the first identity in the conclusion of Lemma 5.5.

Step 2a (the identities of (a) that involve K_n; every real M). Write $D := D_A^{C(n)}$, $X_u := \langle du, F\rangle$, $Z := Z_{C(n)}[1]$ and $\mathbb{E} := \mathbb{E}_{C(n)}$.

(i) (The null set.) The A for which $\det D = 0$ form a closed set of Lebesgue measure zero (Proposition E.5), and there are countably many closed z (Definition E.1). Therefore $\{\varrho = 0\}$ is a null set for $\mathrm{d}(A, z)$, and for each closed z the A with $\varrho(A, z) > 0$ form an open set. On that open set ℓ_n depends smoothly on A, and

$$s_u = -X_u + \partial_u \ell_n, \qquad \partial_u s_{u'} = -\langle du, du'\rangle + \partial_u \partial_{u'} \ell_n.$$

(ii) (Integrability.) At the points where $\det D > 0$,

$$\det D \cdot \partial_u \partial_{u'} \ell_n = \partial_u \partial_{u'} \det D - \frac{(\partial_u \det D)(\partial_{u'} \det D)}{\det D},$$

and, because $\det D = |\det \check{D}|^2$ (Lemma E.2),

$$|\partial_u \det D| = 2\,\big|\mathrm{Re}\big(\overline{\det \check{D}}\;\partial_u \det \check{D}\big)\big| \le 2\,(\det D)^{1/2}\,|\partial_u \det \check{D}|.$$

It follows that

$$|\det D \cdot \partial_u \partial_{u'} \ell_n| \le \sup|\partial_u \partial_{u'} \det D| + 4 \sup|\partial_u \det \check{D}|\,\sup|\partial_{u'} \det \check{D}|,$$
$$\det D \cdot (\partial_u \ell_n)^2 \le 4 \sup|\partial_u \det \check{D}|^2,$$

and these suprema are finite, since $\det D$ and $\det \check{D}$ are trigonometric polynomials in the variables $qA(b)$ (Step 1). The function $\exp(-\frac{1}{2}\|F\|^2)$ being integrable (Proposition E.5), both $\partial_u \partial_{u'} \ell_n$ and $(\partial_u \ell_n)^2$ are integrable for the probability measure of density ϱ/Z (Definition E.1). Thus K_n is well defined, bilinear and symmetric.

(iii) (The integral of $\partial_u \partial_{u'} \varrho$ is zero.) Each of the functions $\partial_{u'} \varrho$ and $\partial_u \partial_{u'} \varrho$ is $\exp(-\frac{1}{2}\|F\|^2)$ times a polynomial in F of degree at most two, whose coefficients are trigonometric polynomials in the $qA(b)$. By Step 1 of the proof of Proposition 5.3 and Definition E.1, the integral $\int \mathrm{d}(A, z)\,(\partial_{u'} \varrho)(A + tu, z)$ does not depend on t. When $|t| \le 1$, the derivative in t of the integrand, namely $(\partial_u \partial_{u'} \varrho)(A + tu, z)$, is bounded in modulus by $C \prod_\sigma \exp(-\frac{1}{4} F_\sigma^2)$ for some

constant C, and this function is integrable, as in Step 2 of that proof (Proposition E.5). Differentiating at $t = 0$, one obtains

$$\int \mathrm{d}(A, z)\, \partial_u \partial_{u'} \varrho = 0.$$

(iv) (The identity for $K_n(u, u')$.) By Step 1, $\partial_{u'} \varrho = \varrho\, s_{u'}$. On $\{\varrho > 0\}$, part (i) then gives

$$\partial_u \partial_{u'} \varrho = \varrho\, s_u s_{u'} - \varrho\, \langle du, du' \rangle + \varrho\, \partial_u \partial_{u'} \ell_n.$$

These four terms are integrable, by Step 1 combined with the Schwarz inequality, by (ii) and by (iii). One integrates over $\mathrm{Conf}(\Gamma)$, applies (iii) and divides by $Z > 0$ (Proposition E.5); the result is $\mathbb{E}[s_u s_{u'}] = \langle du, du' \rangle + K_n(u, u')$.

(v) (The expression for $K_n(u, u)$.) Step 2 gives $\mathbb{E}[X_u s_u] = -\|du\|^2$, which is the same as $\mathbb{E}[X_u\, \partial_u \ell_n] = \mathbb{E}X_u^2 - \|du\|^2$. On expanding $s_u^2 = (X_u - \partial_u \ell_n)^2$ one finds, for arbitrary weights,

$$I_n(u) = 2\|du\|^2 - \mathbb{E}X_u^2 + \mathbb{E}(\partial_u \ell_n)^2.$$

Combined with (iv), this is the expression for $K_n(u, u)$ stated in (a); the inequality $\mathbb{E}X_u^2 \leq \|du\|^2$ (see (13)) then yields $K_n(u, u) \geq \mathbb{E}(\partial_u \ell_n)^2$ and, in particular, $K_n(u, u) \geq 0$.

(vi) (Two special cases; the range of ρ.) If $\rho \geq 0$ and $\mathbf{m} > 0$, Lemma 5.1 gives $\varrho > 0$ everywhere, and $s_u = -\partial_u V$; hence K_n is the form of Lemma 5.2, and (iv) with $u' = u$ is the second identity of Step 3 of the proof of Proposition 5.3. For standard weights and $du \neq 0$, (v) takes the form $K_n(u, u) = (1 - x + j)\|du\|^2$, and $0 \leq 1 - x \leq \min\{1, \sqrt{j}\}$, because Lemma 5.5 gives $0 < x \leq 1$ and $x \geq 1 - \sqrt{j}$. Parts (i)–(iv) and the identities of (v) are valid for every real ρ, the zero set of $\det D$ in (i) being described for $\rho^2 \neq 1$ by Proposition E.5; the assumption $-1 \leq \rho \leq 1$ enters the two inequalities of (v).

Step 3 (lattice probes). Let $X_i := \langle d\hat{\mathfrak{a}}_i, F \rangle$ and $\boldsymbol{\sigma}_i := \|d\hat{\mathfrak{a}}_i\|$. By Lemma E.10, $X_i = X_{\eta_i, R_i}(f^{(\eta_i)})$ with $\|f^{(\eta)} - f\| \leq \eta^2 \|\beta\|_{H^4}$, and $\boldsymbol{\sigma}_i \to \|f\|$. The map $\beta \mapsto \hat{\mathfrak{a}}_i$ is linear, so that Lemma E.10 applied to $\beta \pm \beta'$ gives $\langle d\hat{\mathfrak{a}}_i, d\hat{\mathfrak{a}}'_i \rangle \to \langle f, f' \rangle$ for any two probes β, β', the primes referring to β'. For all $e > 0$ and all mass assignments, Lemma E.10 gives

$$E_i\big(X_i - X_{\eta_i, R_i}(f)\big)^2 \leq \|f^{(\eta_i)} - f\|^2 \to 0$$

as soon as $(-\frac{R_i}{2}, \frac{R_i}{2})^3$ contains $\operatorname{supp} \beta + [-1, 1]^3$.

Step 4 (convergence). Under E_i, the family $X_{\eta_i, R_i}(g)$, $g \in \mathcal{S}$, has the same joint laws as the family $\omega(g)$ under P_{N_i, T_i}. By Lemma E.10, using [4, Section 6.2], these laws converge weakly, their generating functions converge uniformly on compact sets, and all their moments converge. Combined with Step 3, this yields $E_i X_i^2 \to S_2(f, f)$. Moreover, let $\beta_1, \dots, \beta_k$ be probes and $g_j = dd^* \beta_j$; then the variables $X_i^{(j)} := \langle d\hat{\mathfrak{a}}_i^{(j)}, F \rangle$ converge jointly in law to the variables $X(g_j)$ under μ.

Step 5 (proof of (b)). For $u = \hat{\mathfrak{a}}_i$, part (a) reads

$$\boldsymbol{\sigma}_i^4 \leq E_i X_i^2 \cdot I_{n_i}(\hat{\mathfrak{a}}_i), \qquad I_{n_i}(\hat{\mathfrak{a}}_i) \geq \boldsymbol{\sigma}_i^2.$$

Letting $i \to \infty$ along a subsequence that realises $\bar{I}(\beta)$ gives (b).

Step 6 (proof of (c) for test forms of probes). Let $g_j = dd^* \beta_j$ and $\Phi_i := \phi(X_i^{(1)}, \dots, X_i^{(k)})$. Then

$$\partial_{\hat{\mathfrak{a}}_i} \Phi_i = \sum_j \langle d\hat{\mathfrak{a}}_i^{(j)}, d\hat{\mathfrak{a}}_i \rangle\, (\partial_j \phi)(X_i^{(1)}, \dots, X_i^{(k)}),$$

and Step 2 together with the Schwarz inequality yields

$$|E_i[\partial_{\hat{\mathfrak{a}}_i} \Phi_i]| \leq (E_i \Phi_i^2)^{1/2}\, I_{n_i}(\hat{\mathfrak{a}}_i)^{1/2}.$$

Along the subsequence chosen in Step 5, Steps 3 and 4 give in the limit

$$|\mu(\partial_f \Phi)| \leq \bar{I}(\beta)^{1/2}\, \|\Phi\|_2.$$

Hence the linear functional $\Phi \mapsto \mu(\partial_f \Phi)$ is well defined on classes in $L^2(\mu)$ and bounded there, and the Riesz representation theorem furnishes s_β.

Step 7 (proof of (c) for all test forms). Every closed form in L^2 is a limit in L^2 of forms $dd^*\beta$. Indeed, the dual fields of the forms $dd^*\beta$ have the Fourier transforms $|p|^2 P^\perp(p)\,\widehat{*\beta}(p)$, and a field $w = P^\perp w \in L^2$ orthogonal to all of them satisfies $|p|^2 w = 0$. Write Π_{c} for the projection of L^2 onto the subspace of closed forms. By Lemma E.11, every real $g \in L^2$ satisfies $X(g) = X(\Pi_{\mathrm{c}} g)$ almost surely and $\|X(g)\|_2 \le \|g\|$; moreover $\langle g, f\rangle = \langle \Pi_{\mathrm{c}} g, f\rangle$. The function ϕ and its partial derivatives $\partial_j \phi$ are bounded and Lipschitz continuous; hence each side of the identity in (c) is a continuous function of $(g_1, \dots, g_k) \in (L^2)^k$, and the identity extends from the case of Step 6 to the general case.

Step 8 (the shift). For $g_j \in \mathcal{S}$ put

$$\Phi_t(\omega) := \Phi(\omega + tf) = \phi(\omega(g_1) + t\langle g_1, f\rangle, \dots).$$

Then $\frac{\mathrm{d}}{\mathrm{d}t}\mu(\Phi_t) = \mu(\partial_f \Phi_t) = \mu(\Phi_t\, \mathsf{s}_\beta)$, whose modulus is at most $\sup|\phi| \cdot \|\mathsf{s}_\beta\|_2$. Integrating in t one obtains the bound for these Φ; this is enough, because the maps $\omega \mapsto \omega(g)$, $g \in \mathcal{S}$, generate Σ.

Step 9 (proof of (d)). A volume-first limit point is, in particular, a limit point [4, Section 6.2]. Part (b) therefore gives $S_2(f,f) \ge (1+c)^{-1}\|f\|^2$ for every probe, and Step 7, combined with the continuity of S_2 on L^2, extends this to $S_2(g,g) \ge (1+c)^{-1}\|g\|^2$ for every real closed $g \in L^2$. Fix such a $g \ne 0$, and set $b = *g$ and $g_l(x) := g(x/l)$. Proposition D.1 gives

$$\frac{S_2(g_l, g_l)}{\|g_l\|^2} = \frac{\int |\hat b(k)|^2\, \mathsf{D}(|k|^2/l^2)\,\mathrm{d}k}{\int |\hat b(k)|^2\,\mathrm{d}k}, \qquad \mathsf{D}(r) := \int \mathrm{d}\,\mathrm{kl}_\mu(t)\,\frac{r}{r+t^2},$$

and $\mathsf{D}(r)$ decreases to $\mathrm{kl}_\mu(\{0\})$ as $r \downarrow 0$. As $l \to \infty$, dominated convergence shows that the left side converges to $\mathrm{kl}_\mu(\{0\})$. □

Given a charge $e > 0$ and a mass assignment, $P_{N,T}$ stands in this section for the law $P^{\mathrm{ch}}_{N,T}$ of [4, Section 6.1] formed with them. For a volume-first limit point μ of these laws, write $\mathrm{kl} = \mathrm{kl}_\mu$ for the measure of Proposition D.1 used in Proposition 5.6 (d). If $0 < e \le e_2$ and the mass assignment is $N \mapsto m_{0,N}(e)$, these laws and this measure coincide with those of the notation fixed above, by Definition E.1 and by the uniqueness of the measure in Proposition D.1. For the remaining fixed-spacing statements of this section, $n \ge 4$ is an integer, $\Gamma = n\mathbb{Z}^3$ and $\mathbb{X}_n := \mathbb{X}/\Gamma$; moreover $D := D_A^{C(n)}$ and $E_n := \mathbb{E}_{C(n)}$, and P_n is the probability that belongs to E_n. Put

$$\mathsf{M}_n(x,y) := E_n \|\check D_A^{-1}(x,y)\|^2_{\mathrm{HS}}, \qquad \mathsf{N}(x,y) := E_n \|D^{-1}(x,y)\|^2_{\mathrm{HS}};$$

the quantity M_n is the expectation of Proposition 5.4 (c). Let $\mathsf{a} := 2\mathbf{m}/\mathsf{h}_3$ and $\varepsilon_* := \frac12 \log(1+\mathsf{a})$. For real a write $|a|_n := \mathrm{dist}(a, n\mathbb{Z})$; for a class s of $(\frac12\mathbb{Z})^3$ modulo Γ write $|s|^2 := \sum_j |s_j|_n^2$ and $|s|_\infty := \max_j |s_j|_n$. For a link b, let $V(b)$ be the set of the 16 end points of the eight bonds over b. The *cell* of $k \in \mathbb{Z}^3/\Gamma$ is the set $\pi^{-1}(k)$, which has 48 sites. Further,

$$L_t := \{y \in \mathbb{X}_n : \pi(y)_1 \equiv t\}, \qquad v(x,y) := \pi(y) - \pi(x), \qquad k(b,b') := K_n(\delta_b, \delta_{b'});$$

$s(b,b')$ denotes the midpoint of b' minus the midpoint of b, and $\mathrm{sv}_j(\cdot)$ are the singular values of an operator.

Define

$$\mathsf{c}_n := \max_b \sum_{b'} |s(b,b')|^2\,|k(b,b')|, \qquad G_n(t) := \sum_{\pi(x)=0}\ \sum_{y \in L_t} \mathsf{M}_n(x,y), \qquad \mathsf{S}_n := \sum_{t \in \mathbb{Z}/n} |t|_n^2\, G_n(t),$$

$$N_A(\lambda) := \#\{j : \mathrm{sv}_j(D_A)^2 \le \lambda\}, \qquad \mathsf{T}_n := n^{-3} E_n\, \mathrm{Tr}\big[(D^*D)^{-2}\big].$$

Accordingly, $G_n(t)$ is the meson correlator summed over the sites of the cell 0 and over the time slice L_t. With $\mathsf{k}_2 := \max_{\mu,\nu} \sum_s |s|^2 |k_{\mu\nu}(s)|$ one has $\mathsf{k}_2 \le \mathsf{c}_n \le 3\mathsf{k}_2$. Condition **(FI)** asserts that there is a

constant C, independent of N, T and u, such that

$$I_n(u) \le (1 + C\,e^2/\bar m)\|du\|^2 \qquad \text{for all } u \text{ with } \sum_x u(x) = 0.$$

The following classical facts enter the proofs.

(K1) (Min-max principle.) Let h be a Hermitian form on an inner product space of finite dimension, and let $h(\psi,\psi) \le \mu\|\psi\|^2$ for all ψ in a subspace of dimension d. Then h has at least d eigenvalues $\le \mu$.

(K2) (Hölder's inequality for Schatten norms.) For complex square matrices,

$$|\operatorname{tr}(XBYB')| \le \|X\|_{\mathrm{HS}}\|B\|\,\|Y\|_{\mathrm{HS}}\|B'\|, \qquad \|XYZ\|_{\mathrm{HS}}^2 \le \|Y\|^2(\operatorname{Tr}[(X^*X)^2]\,\operatorname{Tr}[(Z^*Z)^2])^{1/2}.$$

(K3) Every bounded invertible operator T on a Hilbert space admits a polar decomposition $T = \Omega|T|$ with Ω unitary; moreover $|T^*| = \Omega|T|\Omega^*$.

(K4) On $\mathbb{Z}^3/n\mathbb{Z}^3$ the operator $\Delta := dd^* + d^*d$ is, componentwise, the scalar Laplacian; it commutes with d and with d^*, and its kernel consists of the constant cochains. On a box $\prod_\mu [a_\mu, b_\mu] \cap \mathbb{Z}^3$ every closed one-cochain or two-cochain is exact, and an integer valued one has an integer valued primitive.

(K5) Every sequence of positive measures of bounded mass on a compact interval has a weakly convergent subsequence (Helly), and each such measure is determined by its moments (Weierstrass).

(K6) On a separable metric space, weak convergence of Borel probability measures is metrisable (Lévy, Prokhorov).

5.7. **The Hessian, the Ward identity and the meson moment.**

Theorem 5.7. *Bounds on the Fisher information; an atom at zero when the fermion is heavy.* Let $0 < \rho < 1$, let the weights be standard, and let $q > 0$, $\mathbf{m} > 0$ and $n \ge 4$. For every $u \in C^1(\Gamma)$ with $\sum_x u(x) = 0$,

$$I_n(u) \le \Big(1 + \frac{\pi^2}{4}\,\mathsf{c}_n\Big)\|du\|^2,$$

and

$$\mathsf{c}_n \le 24\,\mathsf{h}_3^2 q^2\,\mathsf{S}_n \le 3\pi^2\mathsf{h}_3^4 q^2\,\mathsf{T}_n.$$

There exist constants $C_A, C_B, C_* < \infty$, depending on ρ alone, with the following properties.

(i) $\mathsf{c}_n \le 576\,\pi^2\mathsf{h}_3^4\,q^2\,\mathbf{m}^{-4}$; this holds also for every $\rho \ge 0$ and for arbitrary weights $w_i > 0$.

(ii) $\mathsf{c}_n \le C_A\,q^2(\mathbf{m}^{-5/2} + n^{-3}\mathbf{m}^{-4})$.

(iii) For every $\mathbf{m}' \in [\mathbf{m}, 1]$ such that $n \ge 20/\mathbf{m}'$ and $\mathbf{m}' \ge C_* q^2 \log^2 \frac{2}{\mathbf{m}}$,

$$\mathsf{c}_n \le C_B\,\frac{q^2}{\mathbf{m}}\Big(\frac{\mathbf{m}'}{\mathbf{m}}\Big)^3.$$

The bounds (i) and (ii) rest on estimates valid in every real field, and (iii) on Gaussian domination. For the continuum scaling, fix an arbitrary charge $e > 0$ and a mass assignment with $m_{0,N} \ge 0$; then $\mathbf{m} = \boldsymbol{\mu}_N \ge \bar m\eta$.

(a) Let $m_{0,N} = 0$. For all T and all N with

$$\max\Big\{\tfrac{5}{2}L\bar m\eta,\ C_* e^2\eta\log^2\frac{2}{\bar m\eta}\Big\} \le 1$$

one has

$$\mathsf{c}_n \le C_B\,\frac{e^2}{\bar m}\,\max\Big\{\tfrac{5}{2}L,\ C_*\frac{e^2}{\bar m}\log^2\frac{2}{\bar m\eta}\Big\}^3.$$

Consequently $\mathsf{k}_2 \le C_\kappa \frac{q^2}{\bar m \eta}\,(\bar m\eta)^{-\kappa}$ for every $\kappa > 0$; the bound (i) corresponds to $\kappa = 3$ and the bound (ii) to $\kappa = \frac{3}{2}$. The bound on k_2 holds for all T and the values of N just specified, with

$$C_\kappa := C_B \max\Big\{(\tfrac{5}{2}L)^3,\ C_*^3\Big(\frac{e^2}{\bar m}\Big)^3\, 2^\kappa\Big(\frac{6}{\kappa \exp 1}\Big)^6\Big\},$$

a constant that depends only on $(\kappa, \rho, L, e^2/\bar m)$. Let $\mathsf{x}_0 = \mathsf{x}_0(\rho, L, e^2/\bar m) \in (0,1]$ be the largest number such that

$$\max\Big\{\tfrac{5}{2}L\mathsf{x},\ C_*\frac{e^2}{\bar m}\,\mathsf{x}\log^2\frac{2}{\mathsf{x}}\Big\} \le 1 \qquad \text{for all } \mathsf{x} \in (0, \mathsf{x}_0].$$

The other values of N are finitely many and satisfy $\bar m\eta > \mathsf{x}_0$; for them (i) gives

$$\mathsf{c}_n \le 576\,\pi^2 \mathsf{h}_3^4\, \mathsf{x}_0^{-3}\,\frac{q^2}{\bar m\eta}.$$

Hence the bound on k_2 holds for all N and T once $576\,\pi^2\mathsf{h}_3^4\,\mathsf{x}_0^{-3}$ is added to C_κ.

(b) Let $\varkappa > 0$, and let $m_{0,N} \ge \varkappa\, e^2\eta \log^{3/2}\frac{2}{\eta}$ for all large N. Then

$$\limsup_{N\to\infty}\ \sup_T\ \mathsf{c}_n \le c_\varkappa := C_B C_*^3 \varkappa^{-4},$$

and $\mathrm{kl}(\{0\}) \ge (1 + \frac{\pi^2}{4}c_\varkappa)^{-1} > 0$ for every volume-first limit point.

(c) Suppose that $m_{0,N}/(\eta \log^{3/2}\frac{2}{\eta}) \to \infty$. Then $P_{N,T}$ converges to the free Maxwell law γ as $N, T \to \infty$ in an arbitrary manner, and $\mathrm{kl}_\gamma = \delta_0$.

■□

In (b) and (c) the bare mass of the fermion in physical units, $\boldsymbol{\mu}_N/\eta$, tends to infinity, so that the fermion is not retained in the limit. The proof of Theorem 5.7 is given after Proposition 5.11.

Lemma 5.8. *From the kernel to the form.* Let $\rho \ge 0$, $w_0, \dots, w_3 > 0$, $q > 0$, $\mathbf{m} > 0$ and $n \ge 4$. The matrix k is real and symmetric. The entry $k((x,\mu),(y,\nu)) = k_{\mu\nu}(s)$ depends only on μ, ν and the class $s = y - x + \frac{1}{2}(e_\nu - e_\mu)$, and

$$k_{\mu\nu}(-s) = k_{\mu\nu}(s) = k_{\nu\mu}(s).$$

For every link b and every $\lambda \in C^0(\Gamma)$ one has $\sum_{b'} k(b,b')\,(d\lambda)(b') = 0$. Finally, for all $u \in C^1(\Gamma)$,

$$|K_n(u,u)| \le \mathsf{c}_n\Big(\frac{\pi^2}{4}\|du\|^2 + 2\pi^2 n^{-5}\Big|\sum_x u(x)\Big|^2\Big).$$

Combined with Theorem 5.7 (i), this yields Lemma 5.2 with

$$c_* = 1152\,\pi^4 \mathsf{h}_3^4\,\mathbf{m}^{-4};$$

the assertions of Proposition 5.3 are therefore valid with this value of c_*. ■

Proof. Step 1 (the kernel). Lemma 5.1 gives $\|D^{-1}\| \le \mathbf{m}^{-1}$, and $\det D = |\det \check D|^2 > 0$ by Lemma E.2; hence ℓ_n is smooth. A phase is carried only by the bonds of type 3; each of them lies over one link, and every site is an end point of exactly one of them [4, Proposition 2.1]. It follows that $\partial_b\partial_{b'} D = \delta_{bb'} D_{bb}$, with $D_b := \partial_b D$ and $D_{bb} := \partial_b^2 D$ with derivatives taken along the indicated links, and that the sets $V(b)$ are pairwise disjoint and cover $\mathbb{X}_n$. Moreover the non-zero blocks of D_b are the sixteen blocks

$$D_b(x_1,x_2) = \pm iq\, D(x_1,x_2), \qquad x_1 \in V(b), \quad x_2 = x_1^{(3)},$$

each of norm $q\mathsf{h}_3$. Jacobi's formula gives

$$-\partial_b\partial_{b'}\ell_n = \mathrm{Tr}(D^{-1}D_bD^{-1}D_{b'}) - \delta_{bb'}\,\mathrm{Tr}(D^{-1}D_{bb});$$

this is a bounded function of the phases of the links.

Step 2 (symmetries). Let $g \in W$. Since $\vec{g}$ is a signed permutation matrix and $\chi(nv) = (-1)^{v_1+v_2+v_3}$, one has $gC(n) = C(n)$. Further $\ell_n((g^{-1})^*A) = \ell_n(A)$ (Lemma E.2), and $(A, z) \mapsto (g^*A, g^*z)$ leaves $\mathrm{d}(A, z)$ and $\|F\|$ unchanged (Definition E.1). The law on $\mathrm{Conf}(\Gamma)$ is therefore invariant under W, and $K_n(g^*u, g^*u') = K_n(u, u')$. The identity $\ell_n(A - d\lambda) = \ell_n(A)$ (Lemma E.2) gives $\partial_u \partial_{d\lambda} \ell_n = 0$ in every field.

Step 3 (evenness). The translations yield the form $k_{\mu\nu}(s)$ of the kernel. For $g = -1$ one has $(g^*u)_\mu(x) = -u_\mu(-x - e_\mu)$ (Definition E.1); the two signs cancel and the midpoints of the links are carried to their negatives, whence $k_{\mu\nu}(-s) = k_{\mu\nu}(s)$. In terms of $k_{\mu\nu}$, the symmetry of k reads $k_{\mu\nu}(s) = k_{\nu\mu}(-s)$.

Step 4 (Parseval's identity with midpoints). Let $\mathcal{B} := \frac{2\pi}{n}\mathbb{Z}^3 \cap (-\pi, \pi]^3$, and for $p \in \mathcal{B}$ let

$$\hat{u}_\mu(p) := \sum_x u_\mu(x) \exp(-ip \cdot (x + \tfrac{1}{2}e_\mu)), \qquad \mathsf{s}_\mu := 2 \sin \frac{p_\mu}{2}, \qquad \hat{k}_{\mu\nu}(p) := \sum_s k_{\mu\nu}(s) \cos(p \cdot s);$$

the cosine is a function of the class s, because $\exp(ip \cdot ne_j) = 1$. By Fourier inversion on $(\mathbb{Z}/n)^3$ and by Step 3,

$$K_n(u, u) = n^{-3} \sum_{p \in \mathcal{B}} \hat{u}^* \hat{k} \hat{u},$$

where each $\hat{k}(p)$ is a real symmetric matrix. If Fourier transforms of two-cochains are taken with respect to the centres of the plaquettes, then $\widehat{du}_{\mu\nu} = i(\mathsf{s}_\mu \hat{u}_\nu - \mathsf{s}_\nu \hat{u}_\mu)$; moreover $\widehat{d\lambda}_\mu = i\mathsf{s}_\mu \hat{\lambda}$ (Definition E.1). Lagrange's identity therefore gives $\|du\|^2 = n^{-3} \sum_p |\mathsf{s}|^2 |\hat{u}^\perp|^2$, where $\hat{u}^\perp$ denotes the component of $\hat{u}$ orthogonal to s.

Step 5 (transversality; the mode $p = 0$). In Fourier variables, Step 2 reads $\hat{k}(p)\mathsf{s}(p) = 0$. Put $m_{\mu\nu} := \sum_s |s|^2 |k_{\mu\nu}(s)|$. Choosing for each class s a shortest representative, one obtains

$$|\hat{k}_{\mu\nu}(p) - \hat{k}_{\mu\nu}(0)| \le \tfrac{1}{2}|p|^2 m_{\mu\nu};$$

by the evenness of the kernel there is no term linear in p. At $p = \frac{2\pi}{n} e_\nu$ transversality gives $\hat{k}_{\mu\nu}(p) = 0$. Consequently $|\hat{k}_{\mu\nu}(0)| \le 2\pi^2 n^{-2} m_{\mu\nu}$, and $|\hat{k}_{\mu\nu}(p)| \le |p|^2 m_{\mu\nu}$ for every $p \in \mathcal{B} \setminus \{0\}$.

Step 6 (conclusion). The norm of a real symmetric matrix is bounded by the largest of the sums of the absolute values of the entries in a row, and $\max_\mu \sum_\nu m_{\mu\nu} = \mathsf{c}_n$. For $p \neq 0$ only $\hat{u}^\perp$ contributes, and $|p| \le \frac{\pi}{2}|\mathsf{s}(p)|$ on $\mathcal{B}$; finally $\hat{u}(0) = \sum_x u(x)$. □

Lemma 5.9. *From the meson to the kernel.* Let the hypotheses of Lemma 5.8 be satisfied.

(a) For every $g \in W$,

$$\mathsf{N}(x, y) = \mathsf{M}_n(x, y) + \mathsf{M}_n(y, x) = \mathsf{N}(y, x) = \mathsf{N}(gx, gy),$$

and $\mathsf{M}_n(gx, gy) = \mathsf{M}_n(x, y)$ for every even $g \in W$. Consequently $G_n(-t) = G_n(t)$, the function G_n is the same for each of the three coordinate directions,

$$\sum_{\pi(x)=0} \sum_{y \in L_t} \mathsf{M}_n(y, x) = G_n(-t),$$

and $\sum_y |v(x, y)|^2 \mathsf{N}(x, y) = \frac{1}{8}\mathsf{S}_n$ for every site x. In addition $\mathsf{M}_n(x, y) = \mathsf{M}_n(y, x)$ (charge conjugation), so that $\mathsf{N} = 2\mathsf{M}_n$ and M_n is invariant under the whole of W.

(b) For all $b \neq b'$, that is, for all $s \neq 0$,

$$|k(b, b')| \le \mathsf{h}_3^2 q^2 \sum_{x' \in V(b)} \sum_{y' \in V(b')} \mathsf{N}(x', y') = 2\mathsf{h}_3^2 q^2 \sum_{x' \in V(b)} \sum_{y' \in V(b')} \mathsf{M}_n(x', y'),$$

a sum over 256 pairs. The term $\mathrm{Tr}(D^{-1} D_{bb})$ occurs only at $b = b'$, where the weight $|s|^2$ is 0; it does not contribute to c_n. For $b \neq b'$ the vertices $V(b)$ and $V(b')$ never overlap.

(c) (Ward identity.) One has $\mathsf{c}_n \le 24\,\mathsf{h}_3^2 q^2\,\mathsf{S}_n$. Links that share an end point enter c_n through pairs of sites in a single cell; no $G_n(t)$ with $t \neq 0$ sees these pairs, and such links are dominated by the other links.

■

Proof. Step 1 (proof of (a)). By Lemma E.2, $D^{-1} = \operatorname{diag}(\check{D}^{-1}, (\check{D}^{-1})^*)$, so that in every field

$$\|D^{-1}(x,y)\|_{\mathrm{HS}}^2 = \|\check{D}^{-1}(x,y)\|_{\mathrm{HS}}^2 + \|\check{D}^{-1}(y,x)\|_{\mathrm{HS}}^2.$$

According to Lemma E.2,

$$D^{-1}_{(g^{-1})^* A}(gx, gy) = \tilde{g}\, D_A^{-1}(x,y)\, \tilde{g}^{-1},$$

where $\tilde{g}$ is a product of unitary matrices $\gamma_n\gamma_5$, each of which interchanges the two summands $\mathbb{C}^2$ (Definition E.1); for even g the matrix $\tilde{g}$ is block diagonal. Combined with Step 2 of the proof of Lemma 5.8 and with $\pi(gx) = g\pi(x)$ [4, Proposition 2.1], this proves the invariances. For the consequences, apply them to $\operatorname{diag}(-1,-1,1)$, to the cyclic permutations and to the translations; the group W acts transitively on $\mathbb{X}$, and summing $\sum_y |v|^2\mathsf{N}$ over the sites of a cell gives $6\mathsf{S}_n$. For the last assertion of (a), the identity of Lemma E.2 is block diagonal with respect to $\mathbb{C}^4 = \mathbb{C}^2 \oplus \mathbb{C}^2$, and on every torus its upper block is

$$\sigma_2 (\check{D}_A)^\top \sigma_2^{-1} = \check{D}_{-A};$$

hence $\|\check{D}^{-1}_{-A}(x,y)\|_{\mathrm{HS}} = \|\check{D}_A^{-1}(y,x)\|_{\mathrm{HS}}$. The map $(A,z) \mapsto (-A,-z)$ is compatible with the equivalence relation and with the product of Lebesgue measure and counting measure, and therefore preserves $\mathrm{d}(A,z)$ (Definition E.1); it preserves $\|F\|$ and $\det D_A$ too.

Step 2 (proof of (b)). Step 1 of the proof of Lemma 5.8 gives

$$\operatorname{Tr}(D^{-1} D_b D^{-1} D_{b'}) = \sum \operatorname{tr}\big[D^{-1}(y_2,x_1)\, D_b(x_1,x_2)\, D^{-1}(x_2,y_1)\, D_{b'}(y_1,y_2)\big],$$

the sum being taken over $x_1 \in V(b)$ and $y_1 \in V(b')$, with $x_2 = x_1^{(3)}$ and $y_2 = y_1^{(3)}$. The assertion follows from (K2), the inequality $2\xi\zeta \le \xi^2 + \zeta^2$, part (a), and the fact that $x_1 \mapsto x_2$ permutes $V(b)$.

Step 3 (proof of (c): the bad links). Fix a link b, parallel to e_μ. A link $b' \neq b$ is called *bad* if it has an end point in common with b, and *good* otherwise. There are 10 bad links, and $|s|^2 \le 1$ for each of them; for good links $|s|^2 \ge 1$, since the value $|s|^2 = \frac{1}{2}$ occurs only at corners. For $\lambda = \delta_y$ the identity of Lemma 5.8 becomes

$$\sum_\nu \big[k(b,(y-e_\nu,\nu)) - k(b,(y,\nu))\big] = 0.$$

Let b' be bad, and let p' be the end point of b' that is not on b. The vector from the end point of b not shared with b' to p' is $\pm 2e_\mu$ or $\pm e_{\mu'} \pm e_\mu$ with $\mu' \neq \mu$; since $n \ge 4$, it is neither 0 nor a unit vector modulo n. Hence the five links at p' other than b' are good, and $|k(b,b')|$ does not exceed the sum of $|k(b,\cdot)|$ over these five links. The map $b' \mapsto p'$ is injective, and every good link arises at most twice in this way. Therefore

$$\sum_{b' \neq b} |s|^2 |k| \le 3 \sum_{\text{good}} |s|^2 |k|.$$

Step 4 (proof of (c): the good links). For a good link b', parallel to e_ν, and sites $x' \in V(b)$, $y' \in V(b')$ one has $v(x',y') \neq 0$ and $s = v(x',y') \pm \frac{1}{2}e_\mu \pm \frac{1}{2}e_\nu$, whence $|s|^2 \le 4|v(x',y')|^2$. From (b), from the disjointness of the sets $V(b')$ and from (a) it follows that

$$\sum_{\text{good}} |s|^2 |k(b,b')| \le 4\mathsf{h}_3^2 q^2 \sum_{x' \in V(b)} \sum_{y'} |v(x',y')|^2 \mathsf{N}(x',y') = 8\mathsf{h}_3^2 q^2 \mathsf{S}_n.$$

□

Lemma 5.10. Let $\rho \ge 0$, $w_i > 0$, $q > 0$, $\mathbf{m} > 0$ and $n \ge 4$, and let A be real and Γ-periodic.

(a) One has

$$\sum_{x,y\in\mathbb{X}_n} |v(x,y)|^2 \,\|D^{-1}(x,y)\|_{\mathrm{HS}}^2 \le \frac{3\pi^2\mathsf{h}_3^2}{4}\ \mathrm{Tr}\big[(D^*D)^{-2}\big],$$

and consequently $\mathsf{S}_n \le \frac{\pi^2\mathsf{h}_3^2}{8}\,\mathsf{T}_n$.

(b) If $0<\rho<1$ and the weights are standard, then

$$\mathrm{Tr}\big[(D^*D)^{-2}\big] \le \mathrm{Tr}\big[(\mathbf{m}+\tfrac{\rho}{2}\nabla_A^*\nabla_A)^{-4}\big] \le C_\rho(\mathbf{m}^{-4}+n^3\mathbf{m}^{-5/2}).$$

(c) *Energy inequality.* Given a site x and $\alpha\in\{1,2\}$, set

$$\psi := \check{D}_A^{-1}\delta_{x,\alpha}, \qquad f_A(t) := \sum_{|v(x,y)_1|_n=t} |\psi(y)|^2, \qquad F_A(t) := \sum_{t'\ge t} f_A(t').$$

Then, for every integer $t\ge 1$,

$$\mathbf{m}\,F_A(t) \le \mathsf{h}_3\sqrt{f_A(t)\,f_A(t-1)}.$$

Let

$$\mathsf{f}(t) := \begin{cases} G_n(t)+G_n(-t) & \text{for } 0<t<\frac{n}{2},\\ G_n(t) & \text{for } t\in\{0,\frac{n}{2}\},\\ 0 & \text{for } t>\lfloor n/2\rfloor,\end{cases} \qquad \mathsf{F}(t) := \sum_{t'\ge t}\mathsf{f}(t').$$

As a consequence,

$$\begin{aligned} \mathsf{a}\,\mathsf{F}(t) &\le \mathsf{f}(t)+\mathsf{f}(t-1) = \mathsf{F}(t-1)-\mathsf{F}(t+1) \qquad (t\ge 1),\\ \mathsf{F}(t+1) &\le \exp(-2\varepsilon_*)\,\mathsf{F}(t-1) \qquad (t\ge 1),\\ \mathsf{F}(0) &\le 96\,\mathbf{m}^{-2},\end{aligned} \tag{E1}$$

and

$$\sum_{t\ge t_1} t^2\,\mathsf{f}(t) \le \mathsf{F}(t_1-1)\big[t_1^2+2(2t_1+1)\,\mathsf{a}^{-1}+8\,\mathsf{a}^{-2}\big] \qquad (t_1\ge 1). \tag{E2}$$

The inequalities (E1) and (E2) carry no multiplicative constant and impose no further condition on n: the periodicity of the torus is incorporated in their statement.

■

Proof. Step 1 (proof of (a)). Let $\phi_j := \exp(2\pi i\,\pi(\cdot)_j/n)$. By [4, Proposition 2.1] the commutator $[D,\phi_j]$ involves hops of type 3 only, each of its blocks has norm at most $2\sin\frac{\pi}{n}\,\mathsf{h}_3$, and $x\mapsto x^{(3)}$ is a bijection. Hence

$$\|[D,\phi_j]\| \le \frac{2\pi}{n}\,\mathsf{h}_3.$$

The blocks of $[\phi_j,D^{-1}] = D^{-1}[D,\phi_j]D^{-1}$ are the matrices $(\phi_j(x)-\phi_j(y))\,D^{-1}(x,y)$, and

$$|\phi_j(x)-\phi_j(y)| \ge \frac{4}{n}\,|v(x,y)_j|_n.$$

The first inequality of (a) is obtained from (K2) on summing over j. By Lemma 5.9 (a), the expectation of its left side equals $6n^3\mathsf{S}_n$, which gives the bound on S_n.

Step 2 (proof of (b)). Put $R_A := \frac{\rho}{2}\nabla_A^*\nabla_A$. For ψ in the span of the eigenvectors of D^*D with eigenvalues $\le\lambda$, Lemma E.2 gives

$$\langle\psi,(\mathbf{m}+R_A)\psi\rangle = \mathrm{Re}\langle\psi,D\psi\rangle \le \sqrt{\lambda}\,\|\psi\|^2.$$

According to (K1), at least $N_A(\lambda)$ eigenvalues of $\mathbf{m}+R_A$ are therefore $\le\sqrt{\lambda}$. This implies

$$\mathrm{Tr}\big[(D^*D)^{-2}\big] \le \mathrm{Tr}\big[(\mathbf{m}+R_A)^{-4}\big] = \frac{1}{6}\int_0^\infty t^3\exp(-\mathbf{m}t)\ \mathrm{Tr}\exp(-tR_A)\,\mathrm{d}t.$$

Step 3 (diamagnetic comparison for (b)). One has $R_A = \frac{\rho}{2}(\mathrm{sw} - H_A)$, and every block of H_A is the product of w_i, a complex number of modulus one and 1_4 (Lemma E.2). Expanding $\exp(\frac{1}{2}t\rho H_A)$ as a power series yields

$$\operatorname{Tr}\exp(-tR_A) \le 4\operatorname{Tr}\exp\bigl(-\tfrac{1}{2}t\rho L_0\bigr),$$

where L_0 is the Laplacian of the weighted graph $\mathbb{X}_n$, acting on periodic functions with scalar values. This graph is connected and invariant under $\mathbb{Z}^3$ [4, Proposition 2.1], so that x and $x+e_\mu$ are joined, covariantly, by at most ℓ_0 bonds. Consequently

$$\sum_{\mu,x} |\psi(x+e_\mu) - \psi(x)|^2 \le 24\,\ell_0^2\,\langle\psi, L_0\psi\rangle.$$

In view of (K1) and of $4\sin^2\frac{\pi j}{n} \ge 16 j^2/n^2$, it follows that $\operatorname{Tr}\exp(-sL_0) \le 48\,(1 + c\,n s^{-1/2})^3$.

Step 4 (proof of (c)). Let $Y := \{y : |v(x,y)_1|_n \ge t\}$ and $P := 1_Y$. Since $x \notin Y$, one has $P\check{D}\psi = 0$; moreover $\operatorname{Re}\check{D}_A \ge \mathbf{m}$ by Lemma E.2. Therefore

$$\mathbf{m}\|P\psi\|^2 \le \operatorname{Re}\langle P\psi, \check{D}P\psi\rangle \le |\langle P\psi, \check{D}(1-P)\psi\rangle|.$$

By [4, Proposition 2.1], the only bonds leaving Y are bonds of type 3; each of them connects distance t with distance $t-1$, at most one of them meets a given site, and their blocks have norm h_3. The Schwarz inequality bounds the right side by $\mathsf{h}_3\sqrt{f_A(t)f_A(t-1)}$, and $\|P\psi\|^2 = F_A(t)$.

Step 5 (proof of (E1) and (E2)). Bounding the right side of the inequality of (c) by means of $2\sqrt{\xi\zeta} \le \xi + \zeta$, summing over α and over the 48 sites of the cell 0, and taking the expectation E_n, one obtains the first inequality of (E1) from Lemma 5.9 (a). The second inequality of (E1) follows from the first together with $\mathsf{F}(t) \ge \mathsf{F}(t+1)$. The bound $\mathsf{F}(0) \le 48 \cdot 2\,\mathbf{m}^{-2}$ is a consequence of Proposition 5.4 (c). Summing the first inequality of (E1) over $t \ge t_1$, first without weight and then with the weight t, one finds

$$\mathsf{a}\sum_{t\ge t_1}\mathsf{F}(t) \le 2\,\mathsf{F}(t_1-1), \qquad \mathsf{a}\sum_{t\ge t_1} t\,\mathsf{F}(t) \le (2t_1+1)\,\mathsf{F}(t_1-1) + 2\sum_{t\ge t_1}\mathsf{F}(t).$$

Together with Abel's summation

$$\sum_{t\ge t_1} t^2\,\mathsf{f}(t) = t_1^2\,\mathsf{F}(t_1) + \sum_{t>t_1}(2t-1)\,\mathsf{F}(t),$$

these two inequalities give (E2). □

5.8. **Localization and the first singular-value estimate.**

Proposition 5.11. Let $0 < \rho < 1$, and let the weights be standard. There exist constants $C_*, C_B' < \infty$, depending only on ρ, with the following property. If $q > 0$, $0 < \mathbf{m} \le \mathbf{m}' \le 1$, $n \ge 20/\mathbf{m}'$ and $\mathbf{m}' \ge C_* q^2 \log^2\frac{2}{\mathbf{m}}$, then

$$n^{-3}E_n N_A(\lambda) \le C_B'\bigl(\sqrt{\lambda} + \mathbf{m}'\bigr)^3 \quad (\lambda > 0), \qquad \mathsf{T}_n \le 20\,C_B'\,\mathbf{m}'^3\,\mathbf{m}^{-4}.$$

■

Proof. Every constant denoted by c with a subscript depends on ρ alone. Let $\ell := \lceil 1/\mathbf{m}'\rceil$. Then $\ell^{-1} \le \mathbf{m}'$, and $10\ell < n$ because $\ell < 1/\mathbf{m}' + 1 \le 2/\mathbf{m}'$. Put $J := \lceil \log_2(4/\mathbf{m}')\rceil$, and let $r \ge 1$.

Step 1 (localisation). Divide $\mathbb{Z}/n$ into arcs $[t_i, t_{i+1}]$ whose lengths lie in $[\ell, 2\ell)$, and define

$$\mathsf{j}_i(t) := \begin{cases} \sin\dfrac{\pi(t-t_{i-1})}{2(t_i - t_{i-1})} & \text{on } [t_{i-1}, t_i], \\ \cos\dfrac{\pi(t-t_i)}{2(t_{i+1}-t_i)} & \text{on } [t_i, t_{i+1}], \\ 0 & \text{elsewhere,} \end{cases} \qquad \mathsf{j}_a(k) := \prod_\mu \mathsf{j}_{a_\mu}(k_\mu).$$

These functions satisfy $\sum_a \mathsf{j}_a^2 = 1$. Each support $B_a := \operatorname{supp} \mathsf{j}_a$ is a box with at most 4ℓ sites per side, the number of these boxes does not exceed $(n/\ell)^3$, and for neighbouring k, k'

$$\sum_a |\mathsf{j}_a(k) - \mathsf{j}_a(k')|^2 \le \frac{\pi^2}{4\ell^2}.$$

The argument of Step 1 of the proof of Lemma 5.10 gives

$$\sum_a \|D\,\mathsf{j}_a\psi\|^2 \le 2\|D\psi\|^2 + c_J\,\ell^{-2}\|\psi\|^2, \qquad c_J = \frac{\pi^2}{2}\,\mathsf{h}_3^2.$$

Since the map $\psi \mapsto (\mathsf{j}_a\psi)_a$ is isometric, (K1) yields

$$N_A(\lambda) \le \sum_a N_a(\lambda'), \qquad \lambda' := 2\lambda + c_J\,\ell^{-2},$$

where $N_a(\lambda')$ counts the eigenvalues not exceeding λ' of the quadratic form $\phi \mapsto \|D\phi\|^2$, restricted to spinors with support over B_a.

Step 2 (a gauge in the box, linear in F). Fix a and write $B = B_a$. Let B^+ be the 1-neighbourhood of B, and B^{++} the $(\ell+1)$-neighbourhood of B^+. These boxes have at most $6\ell + 4 \le 10\ell < n$ sites per side. Hence they, the cochains on them and the spinors over B lift to $\mathbb{Z}^3$ and to $\mathbb{X}$, and the character χ does not enter. As $dF = 0$, (K4) yields

$$F = dA^{\mathrm{C}} + F_{\mathrm{h}}, \qquad A^{\mathrm{C}} := d^*\mathcal{G}F, \qquad \mathcal{G} := \int_0^\infty \exp(-t\Delta)\,P^\perp\,\mathrm{d}t,$$

where $P^\perp$ is the projection onto the orthogonal complement of the constants and F_{h} is constant. On B^{++} define

$$A^{\mathrm{h}}_\nu(k) := \frac{1}{2}\sum_\mu (F_{\mathrm{h}})_{\mu\nu}\,(k - k_B)_\mu, \qquad A' := A^{\mathrm{C}} + A^{\mathrm{h}},$$

with k_B the centre of the box. Then $d(A - A') = -\frac{2\pi}{q}z$, and (K4) furnishes an integer cochain $\mathfrak{k}$ such that $A - A' + \frac{2\pi}{q}\mathfrak{k} = d\lambda$. Put

$$\zeta(k) := \max\{0, 1 - \operatorname{dist}_\infty(k, B^+)/\ell\}, \qquad A''_\mu := \zeta A'_\mu.$$

For ϕ over B, Lemma E.2 gives $\|D_A\phi\| = \|D_{A''}(\exp(iq\,\lambda\circ\pi)\,\phi)\|$.

Step 3 (a test against the polar part of the free operator). Denote by D'_0 the free operator with the mass $\mathbf{m}'$ on $\ell^2(\mathbb{X};\mathbb{C}^4)$, and write $D'_0 = \Omega|D'_0|$ with Ω unitary, according to (K3). Let $\Lambda := (-\Delta_{\mathbb{Z}^3} + \mathbf{m}'^2)^{1/2}$ act in the variable $k = \pi(x)$, for fixed $\vec{g}$. On functions of the form $\psi(k + \vec{g}v_0) = \exp(ip\cdot k)\,u(\vec{g})$ the operator D'_0 acts as the symbol $D^{(p)}$, as a comparison of Lemma E.2 with Definition E.15 shows. By Lemma E.16,

$$|(D^{(p)})^{-1}|_2 \le c_E^{-1}\,(|p|_2 + \mathbf{m}')^{-1} \quad \text{on } [-\pi,\pi]^3, \qquad c_E := (\sqrt{2} + 7C_E)^{-1}.$$

The symbol of Λ is a scalar not larger than $|p|_2 + \mathbf{m}'$; therefore $|D'_0| \ge c_E\Lambda$ and $|D'^*_0| \ge c_E\Lambda$. With $V := D_{A''} - D_0$, suppose that for all ϕ_1, ϕ_2

$$|\langle\phi_1, V\phi_2\rangle| \le \frac{c_E}{2}\,\|\Lambda^{1/2}\phi_1\|\,\|\Lambda^{1/2}\phi_2\|. \tag{5.1}$$

One has $D_{A''} = D'_0 + V - (\mathbf{m}' - \mathbf{m})$ and $c_E\|\Lambda^{1/2}\Omega\phi\|^2 \le \langle\phi, |D'_0|\phi\rangle$. It follows that

$$\|D_{A''}\phi\|\,\|\phi\| \ge \operatorname{Re}\langle\Omega\phi, D_{A''}\phi\rangle \ge \frac{1}{2}\langle\phi, |D'_0|\phi\rangle - \mathbf{m}'\|\phi\|^2 \ge \frac{c_E}{2}\langle\phi,\Lambda\phi\rangle - \mathbf{m}'\|\phi\|^2.$$

Step 4 (the count, given (5.1)). Let $\mathcal{W}$ be a space of spinors ϕ over B on which $\|D_{A''}\phi\|^2 \le \lambda'\|\phi\|^2$. By Step 3, every $\phi \in \mathcal{W}$ satisfies

$$\langle\phi,\Lambda\phi\rangle \le \mathsf{e}\,\|\phi\|^2, \qquad \mathsf{e} := \frac{2}{c_E}(\sqrt{\lambda'} + \mathbf{m}'),$$

and hence $\langle \phi, 1[\Lambda \le 2\mathsf{e}]\phi\rangle \ge \frac{1}{2}\|\phi\|^2$. Consequently

$$\tfrac{1}{2}\dim \mathcal{W} \le \mathrm{Tr}(1_B\, 1[\Lambda \le 2\mathsf{e}]\, 1_B) \le 192\,|B|\,\frac{\pi \mathsf{e}^3}{6},$$

and so $N_a(\lambda') \le 64\pi(4\ell)^3\mathsf{e}^3$.

Step 5 ((5.1) from scalar multipliers). With a matrix w of norm $\|w\| = \mathsf{h}_3$ (Definition E.1) one has

$$(V\psi)(k,\vec{g}) = -w\,(\exp(iqA''(k,k+\vec{g}e_1)) - 1)\,\psi(k+\vec{g}e_1, \vec{g}\,\mathrm{diag}(-1,1,1)).$$

The translations commute with Λ, and the multipliers attached to the directions $-e_\mu$ are translates of the complex conjugates of those attached to $+e_\mu$. It is therefore enough for (5.1) that

$$\mathsf{y} := \max_\mu \|\Lambda^{-1/2}(\exp(iq\zeta A'_\mu) - 1)\Lambda^{-1/2}\| \le \frac{c_E}{2\mathsf{h}_3} \qquad \text{on } \ell^2(\mathbb{Z}^3).$$

From $|\exp(i\theta) - 1 - i\theta| \le \frac{1}{2}\theta^2$ and $\Lambda \ge \mathbf{m}'$ one obtains

$$\mathsf{y} \le q\,\|\Lambda^{-1/2}\zeta A^{\mathrm{C}}_\mu \Lambda^{-1/2}\| + \frac{q}{\mathbf{m}'}\,\|\zeta A^{\mathrm{h}}\|_\infty + \frac{q^2}{2\mathbf{m}'}\,\|\zeta A'\|^2_\infty.$$

Step 6 (scales; the good event). For $\tau > 0$ one has $A^{\mathrm{C}} = d^* c^\tau + A^{>\tau}$, where

$$c^\tau := \int_0^\tau \exp(-t\Delta)\,P^\perp F\,\mathrm{d}t, \qquad A^{>\tau} := d^*\int_\tau^\infty \exp(-t\Delta)\,P^\perp F\,\mathrm{d}t.$$

Every value of these cochains has the form $\langle \beta, F\rangle$ for some $\beta \in C^2(\Gamma)$. The diagonal of the scalar heat kernel obeys, uniformly in n,

$$p_s(0) - n^{-3} \le c_h \min\{1, s^{-3/2}\},$$

as follows from $4\sin^2\frac{\pi j}{n} \ge 16j^2/n^2$. Together with (K4) and $d^*d \le \Delta$, this bounds $\|\beta\|^2$ by $4c_h\sqrt{\tau}$ for $c^\tau_{\mu\nu}(k)$, by $\sqrt{2}\,c_h\tau^{-1/2}$ for $A^{>\tau}_\mu(k)$ and by $3c_h$ for $A^{\mathrm{C}}_\mu(k)$, while $\|\beta\|^2 = n^{-3}$ for $(F_{\mathrm{h}})_{\mu\nu}$. Let $\mathcal{G}_a$ be the event on which $|\langle\beta, F\rangle| \le r\|\beta\|$ holds for all these β, the cells ranging over B^{++} and τ over $\{4^i : 0 \le i \le J\}$. Applying Proposition E.6 to one β at a time, one finds

$$P_n(\mathcal{G}_a^c) \le 20\,(J+1)\,(10\ell)^3\exp(-\tfrac{1}{2}r^2).$$

On $\mathcal{G}_a$ the following bounds hold in B^{++}:

$$|A^{>4^i}| \le c_1 r\,2^{-i/2}, \qquad |c^{4^i}| \le c_2 r\,2^{i/2}, \qquad |A^{\mathrm{C}}| \le c_0 r, \qquad |A^{\mathrm{h}}| \le 5\ell r n^{-3/2} \le r\,\mathbf{m}'^{1/2}.$$

Step 7 (dyadic sum). The spectrum of Λ is contained in $[\mathbf{m}', \sqrt{13}]$. For $0 \le j \le J$ let $\Pi_j := 1[\Lambda \in (2^{1-j}, 2^{2-j}]]$, and for $f \in \ell^2(\mathbb{Z}^3)$ put $f_j := \Pi_j f$ and $x_j := \|\Lambda^{1/2} f_j\|$; then

$$\|f_j\| \le 2^{(j-1)/2}x_j, \qquad \|\nabla f_j\| \le 2^{2-j}\|f_j\|.$$

The quantities g_j and y_j are defined in the same way for a second function g. For $i \le j$ the cochain A^{C} is split at $\tau = 4^i$. The part $A^{>\tau}$ is estimated by its supremum. In the part d^*c^τ a summation by parts shifts the difference onto $\bar{f}_i g_j \zeta$, and the contribution is bounded by

$$2\sup|c^\tau|\,(\|\nabla f_i\|\,\|g_j\| + \|f_i\|\,\|\nabla g_j\| + \ell^{-1}\|f_i\|\,\|g_j\|).$$

Since $\ell^{-1} < 2^{3-J}$, this gives

$$|\langle f_i, \zeta A^{\mathrm{C}}_\mu g_j\rangle| \le (c_1 + 32c_2)\,r\,2^{-i/2}\,\|f_i\|\,\|g_j\| \le \tfrac{1}{2}c_3\,r\,2^{j/2}x_i y_j,$$

and the case $i > j$ is symmetric. The Schwarz inequality yields

$$\sum_{i\le j} 2^{j/2}x_i y_j \le \left((J+1)\,2^{J+1}\right)^{1/2}\|x\|\,\|y\|,$$

and $2^{J+1} < 16/\mathbf{m}'$. Consequently, on $\mathcal{G}_a$,

$$\|\Lambda^{-1/2}\zeta A_\mu^{\mathrm{C}}\Lambda^{-1/2}\| \le 4c_3\, r\sqrt{(J+1)/\mathbf{m}'}.$$

Step 8 (assembly). Put $\mathsf{t} := qr\sqrt{(J+1)/\mathbf{m}'}$. Steps 5 to 7 give $\mathsf{y} \le c_4(\mathsf{t}+\mathsf{t}^2)$; hence there is $\mathsf{t}_0(\rho) > 0$ such that (5.1) holds on $\mathcal{G}_a$ whenever $\mathbf{m}' \ge \mathsf{t}_0^{-2} q^2 r^2 (J+1)$. Choose

$$r^2 := 2\log\bigl(20\,(J+1)\,(10\ell)^3\,\mathbf{m}^{-3}\bigr).$$

With this choice $P_n(\mathcal{G}_a^c) \le \mathbf{m}^3$ and $r^2(J+1) \le c_5 \log^2 \frac{2}{\mathbf{m}}$, and one sets $C_* := c_5 \mathsf{t}_0^{-2}$. On $\mathcal{G}_a^c$ one uses $N_a \le 192(4\ell)^3$. Steps 1 and 4 then yield

$$n^{-3} E_n N_A(\lambda) \le 4096\,\pi\,\mathsf{e}^3 + 12288\,\mathbf{m}^3.$$

By Lemma 5.1, N_A vanishes below $\mathbf{m}^2$, so that

$$\mathrm{Tr}\bigl[(D^*D)^{-2}\bigr] = 2\int_{\mathbf{m}^2}^{\infty} \lambda^{-3} N_A(\lambda)\,\mathrm{d}\lambda;$$

combined with $(\xi+\zeta)^3 \le 4(\xi^3+\zeta^3)$, this gives the bound on T_n. □

Proof of Theorem 5.7. Step 1. By Proposition 5.6 (a), $I_n(u) = \|du\|^2 + K_n(u,u)$; combined with Lemma 5.8, Lemma 5.9 (c) and Lemma 5.10 (a), this identity gives the first assertion of the theorem. Assertion (i) follows from $\|D^{-1}\| \le \mathbf{m}^{-1}$, since D acts on a space of dimension $192n^3$. Assertion (ii) is a consequence of Lemma 5.10 (b), and (iii) of Proposition 5.11.

Step 2 (proof of (a)). From [4, Section 6.1] and the notation fixed above one has $n\bar{m}\eta = 8\bar{m}hL^T > 8/L$. Hence the number

$$\mathbf{m}' := \max\Bigl\{\tfrac{5}{2}L\mathbf{m},\ C_* q^2 \log^2\frac{2}{\mathbf{m}}\Bigr\}$$

satisfies $n \ge 20/\mathbf{m}'$. Inserting $q^2 = e^2\eta$ and $\mathbf{m} = \bar{m}\eta$ into (iii), one arrives at the bound of (a). The constant C_κ results from $\frac{q^2}{\bar{m}\eta} = \frac{e^2}{\bar{m}}$, from $\bar{m}\eta \le 1$ and from

$$\sup_{0<\mathsf{x}<2} \mathsf{x}^\kappa \log^6\frac{2}{\mathsf{x}} = 2^\kappa\Bigl(\frac{6}{\kappa\exp 1}\Bigr)^6,$$

the supremum being attained at $\mathsf{x} = 2\exp(-6/\kappa)$.

Step 3 (the bound in (b)). Let $\mathbf{m} = \boldsymbol{\mu}_N$. Suppose first that $\mathbf{m} \le \eta^{1/3}$ and that N is so large that $\frac{5}{2}L\eta^{1/3} \le 1$ and $C_* e^2 \eta \log^2\frac{2}{\bar{m}\eta} \le 1$; then the $\mathbf{m}'$ of Step 2 satisfies $\mathbf{m}' \le 1$. Part (iii) with this $\mathbf{m}'$, combined with $\frac{q^2}{\mathbf{m}} \le \varkappa^{-1}\log^{-3/2}\frac{2}{\eta}$ and with $\log\frac{2}{\mathbf{m}} \le \log\frac{2}{\bar{m}\eta}$, gives

$$\mathsf{c}_n \le C_B \max\Bigl\{\bigl(\tfrac{5}{2}L\bigr)^3 \varkappa^{-1}\log^{-3/2}\frac{2}{\eta},\ C_*^3\varkappa^{-4}\Bigl(\log\frac{2}{\bar{m}\eta}\Big/\log\frac{2}{\eta}\Bigr)^6\Bigr\}.$$

Suppose next that $\mathbf{m} > \eta^{1/3}$. In this case (ii), together with $n \ge 8/\eta$, gives $\mathsf{c}_n \le C_A e^2(\eta^{1/6} + \eta^{8/3})$.

Step 4 (the atom in (b)). By [4, Section 6.1], every real sequence qualifies as a mass assignment. By Lemma E.10, each lattice probe is orthogonal to the constant cochains, and $\boldsymbol{\sigma}_i \to \|f\|$. Along every defining sequence one therefore has $\bar{I}(\beta) \le (1 + \frac{\pi^2}{4}c_\varkappa)\|f\|^2$ for all probes β, and Proposition 5.6 (d) applies.

Step 5 (proof of (c)). Part (b), applied for every $\varkappa$, gives $\bar{I}(\beta) = \|f\|^2$ for every limit point μ. Let s_β be as in Proposition 5.6 (c). Taking $\Phi = r\tanh(X(f)/r)$ and letting $r \to \infty$, one obtains

$$\|f\|^2 = \mu(X(f)\,\mathsf{s}_\beta) \le \|X(f)\|_2\,\|\mathsf{s}_\beta\|_2 \le \|f\|^2,$$

whence $\mathsf{s}_\beta = X(f)$. The choices $\Phi = \cos tX(f)$ and $\Phi = \sin tX(f)$ then lead to

$$\frac{\mathrm{d}}{\mathrm{d}t}\,\mu(\exp(itX(f))) = -t\,\|f\|^2\,\mu(\exp(itX(f))).$$

By Step 7 of the proof of Proposition 5.6 it follows that $\mu = \gamma$, and the tightness of the laws [4, Section 6.2] yields the convergence. □

5.9. **A finite-torus moment representation.**

Proposition 5.12. Let $0 \le \rho \le 1$, $w_i > 0$, $q > 0$, $\mathbf{m} > 0$ and $n \ge 4$, and set $k_A := \lfloor \frac{n}{2} \rfloor$.

(a) As matrices,
$$\big(G_n(i+j+1)\big)_{0\le i,j\le k_A-1} \ge 0 \qquad \text{and} \qquad \big(G_n(i+j+2)\big)_{0\le i,j\le k_A-2} \ge 0,$$
the first inequality for the link plane $x_1 = \frac{1}{2}$ and the second for the site plane $x_1 = 0$.

(b) One can find a positive measure σ_n on $[0,\infty)$, having at most $k_A - 1$ atoms, for which
$$G_n(t) = \int \lambda^{t-1}\, \mathrm{d}\sigma_n(\lambda)$$
holds for every integer t with $1 \le t \le 2k_A - 2$; this range contains every integer t with $1 \le t \le \frac{n}{2}$. One writes $\lambda = \exp(-E)$, with $E \in (-\infty, \infty]$. This represents every integer step in the stated range. The value $t = 0$ is excluded, since the site plane cuts through the slice L_0.

(c) Let $0 < E_0 < \varepsilon_*$. For $1 \le t \le k_A$ one has
$$G_n(t) = \int_{[E_0,\infty]} \exp(-E(t-1))\, \mathrm{d}\varrho_n(E) + R_n(t),$$
with a measure $\varrho_n \ge 0$ and a remainder subject to
$$0 \le R_n(t) \le \frac{96}{\mathbf{m}^2} \exp\big(\varepsilon_* - (\varepsilon_* - E_0)k_A - E_0 t\big).$$
If the weights are standard and $\mathbf{m} \le (1+\rho)\,\mathrm{sw}$, then $2\vartheta < \varepsilon_*$, and $E_0 = 2\vartheta$ is an admissible choice. If $n \ge 6$ is even and $G_n(2) \ne 0$, then no positive measure carried by $\{E > 0\}$ represents G_n on the range of (b).

(d) Any sequence $n_i \to \infty$ possesses a subsequence along which $G_{n_i}(t)$ converges, for every $t \ge 1$, to a limit $G_\infty(t)$, and there is one positive measure ϱ_∞ such that
$$G_\infty(t) = \int_{[\varepsilon_*,\infty]} \exp(-E(t-1))\, \mathrm{d}\varrho_\infty(E) \qquad (t \ge 1).$$

■

The proof is given in Appendix A.

The finite-torus representation in Proposition 5.12(c) has a remainder. The local criterion below uses (E1) and (E2) directly and does not require an estimate for that remainder.

5.10. **A local meson criterion.**

Proposition 5.13. Let $\rho \ge 0$, $w_i > 0$, $q > 0$, $\mathbf{m} > 0$ and $n \ge 8$. Let t_0 be an integer with $4 \le t_0 \le \frac{n}{2}$, and define
$$\varsigma := \mathsf{a}\, t_0 = \frac{2\mathbf{m}\, t_0}{\mathsf{h}_3}, \qquad c(\varsigma) := \frac{45}{\varsigma} + \frac{157}{\varsigma^2} + \frac{228}{\varsigma^3}.$$

(a) *The far part is dominated by the near part.* One has
$$\mathsf{S}_n \le \big(1 + c(\varsigma)\big) \sum_{0<|t|_n \le t_0} |t|_n^2\, G_n(t).$$

(b) *One box.* If $\varsigma \ge 190$, then $\mathsf{S}_n \le \frac{5}{6} B_n(t_0)$, where
$$B_n(t_0) := \sum_{\pi(x)=0} \sum_{|\pi(y)|_\infty \le t_0} |\pi(y)|^2\, \mathsf{M}_n(x,y).$$

(c) *Mirror pairs.* Assume moreover $\rho \le 1$. For $1 \le d \le n-1$ denote by $\mathsf{p}_n(d)$ the largest value of

$$\mathsf{M}_n(\theta_{x_j,c}x, x) = \mathbb{E}\big[\Theta_{x_j,c}(\mathfrak{M}(x))\,\mathfrak{M}(x)\big]$$

over the directions j, the sites $x \in \mathbb{X}$ and the numbers $c \in \frac{1}{2}\mathbb{Z}$ with $2(\pi(x)_j - c) = d$, and set $\bar{\mathsf{p}}_n(r) := \max\{\mathsf{p}_n(r), \mathsf{p}_n(n-r)\}$. Then $\mathsf{M}_n(x,y) \le \bar{\mathsf{p}}_n(|v(x,y)|_\infty)$ whenever $v(x,y) \ne 0$, and

$$B_n(t_0) \le 1.8 \cdot 10^5 \sum_{r=1}^{t_0} r^4\, \bar{\mathsf{p}}_n(r).$$

In addition $\mathsf{p}_n(n-r) = \mathsf{p}_n(r)$, so that $\bar{\mathsf{p}}_n = \mathsf{p}_n$.

(d) Suppose that $G_n(t) \le C_0 t^{-2}$ for $1 \le t \le t_0$ and that $c_0 \le \mathbf{m} t_0 \le c_0'$. Then $\mathsf{c}_n \le C_1 q^2/\mathbf{m}$, where $C_1 = 48\,\mathsf{h}_3^2 C_0 c_0'(1 + c(2c_0/\mathsf{h}_3))$.

(e) *Continuum.* Let $0 < \rho < 1$, let the weights be standard, and let $e > 0$ be arbitrary. Let $c_{\boldsymbol{\mu}} > 0$, and assume that $\boldsymbol{\mu}_N \ge c_{\boldsymbol{\mu}}\, \bar{m}\eta$ and $\limsup_{T\to\infty} \boldsymbol{\mu}_N \mathsf{S}_n \le C_M$ for all large N. Then

$$\limsup_T \sup_u \frac{I_n(u)}{\|du\|^2} \le 1 + \frac{Ce^2}{\bar{m}},$$

the supremum being taken over the u with $\sum_x u(x) = 0$ and $du \ne 0$, with the constant $C = 6\pi^2 \mathsf{h}_3^2 C_M / c_{\boldsymbol{\mu}}$. Moreover every volume-first limit point satisfies $\mathrm{kl}(\{0\}) \ge (1 + Ce^2/\bar{m})^{-1}$. The ratio $e^2/\bar{m}$ need not be small.

■

Proof. Step 1 (proof of (a)). Let $N_0 := \sum_{1 \le t \le t_0} t^2 \mathsf{f}(t)$; this is the sum that appears on the right of (a). The set $\mathcal{I} := \{t : \frac{t_0}{2} < t \le t_0\}$ has at least $\frac{t_0}{2}$ elements, and each $t \in \mathcal{I}$ satisfies $t - 1 \ge \frac{3}{8} t_0$. Hence

$$\sum_{t \in \mathcal{I}} (\mathsf{f}(t) + \mathsf{f}(t-1)) \le \frac{128}{9 t_0^2}\, N_0,$$

and (E1) provides some $t_1 \in \mathcal{I}$ with $\mathsf{a}\mathsf{F}(t_1) \le \frac{256}{9 t_0^3} N_0$. Since $t_1 + 1 \le \frac{5}{4} t_0$, the inequality (E2) at the point $t_1 + 1$ gives

$$\sum_{t > t_0} t^2 \mathsf{f}(t) \le \frac{256\, N_0}{9\,\mathsf{a}\, t_0^3} \Big[\frac{25}{16}\, t_0^2 + \frac{11\, t_0}{2\mathsf{a}} + \frac{8}{\mathsf{a}^2}\Big] \le c(\varsigma)\, N_0.$$

Step 2 (proof of (b)). For each direction j, split

$$\sum_{\pi(x)=0} \sum_y |\pi(y)_j|_n^2\, \mathsf{M}_n(x,y) = \mathrm{near}_j + \mathrm{tail}_j$$

at the value $|\pi(y)_j|_n = t_0$, the terms with $|\pi(y)_j|_n \le t_0$ forming near_j. The sum over j of the left sides is $3\mathsf{S}_n$, and $\mathrm{tail}_j \le c(\varsigma)\,\mathrm{near}_j$ by (a) and Lemma 5.9 (a). Within near_1, the sites y with $\max(|\pi(y)_2|_n, |\pi(y)_3|_n) > t_0$ contribute no more than $\mathrm{tail}_2 + \mathrm{tail}_3$. Therefore

$$\sum_j \mathrm{near}_j \le B_n(t_0) + 2c(\varsigma) \sum_j \mathrm{near}_j,$$

and, since $c(\varsigma) \le c(190) < \frac{1}{4}$, one arrives at $3\mathsf{S}_n \le (1 + c(\varsigma))\, 2B_n(t_0)$.

Step 3 (proof of (c)). Let x, y be sites with $v(x,y) \ne 0$, and let $\pi(y)_j - \pi(x)_j \equiv d \in [1, n-1]$. Choose lifts of x and y for which this congruence is an equality, and put $c := \frac{1}{2}(\pi(x)_j + \pi(y)_j)$ and $y' := \theta_c x$. Then y and y' both satisfy $2(\pi(\cdot)_j - c) = d$, and both lie in the open slab, since $y_j - c \in (\frac{d}{2} - \frac{1}{2}, \frac{d}{2} + \frac{1}{2})$. In the direction j, the Schwarz inequality for (RP) gives

$$\mathsf{M}_n(x,y)^2 = \mathsf{M}_n(\theta_c y', y)^2 \le \mathsf{M}_n(\theta_c y', y')\, \mathsf{M}_n(\theta_c y, y) \le \mathsf{p}_n(d)^2.$$

At most $24r^2 + 2 \le 26r^2$ cells k satisfy $|k|_\infty = r$; each has $|k|^2 \le 3r^2$, and to each of them correspond 48^2 pairs (x,y) with $\pi(x) = 0$ and $\pi(y) = k$, since a cell has 48 sites. To prove the final claim of (c), assume $2(\pi(x)_j - c) = n - r$ and set $c' := c + \frac{n}{2} \in \frac{1}{2}\mathbb{Z}$. Since $\theta_{c'}$ is θ_c followed by the translation by

ne_j, the two reflections induce one and the same map of $\mathbb{X}_n$. For $x' := \theta_{c'}x$ one has $2(\pi(x')_j - c') = r$ and $\mathsf{M}_n(\theta_c x, x) = \mathsf{M}_n(x', \theta_{c'}x')$. Take $i \neq j$ and $c_i \in \frac{1}{2}\mathbb{Z}$. Then $g := \theta_{x_j,c'}\theta_{x_i,c_i}$ is an even element of W, and $x'' := \theta_{x_i,c_i}x'$ satisfies

$$gx' = \theta_{c'}x'', \qquad g\theta_{c'}x' = x'', \qquad \pi(x'')_j = \pi(x')_j.$$

Lemma 5.9 (a) therefore yields $\mathsf{M}_n(x', \theta_{c'}x') = \mathsf{M}_n(\theta_{c'}x'', x'') \leq \mathsf{p}_n(r)$. This proves $\mathsf{p}_n(n-r) \leq \mathsf{p}_n(r)$, and the reverse inequality is the same statement with $n-r$ put for r.

Step 4 (proof of (d)). Part (a) gives $\mathsf{S}_n \leq (1 + c(\varsigma))\, 2C_0 t_0$, and the function $c(\cdot)$ is decreasing. Lemma 5.9 (c) completes the proof of (d).

Step 5 (proof of (e)). Theorem 5.7 gives $I_n(u) \leq (1 + 6\pi^2 \mathsf{h}_3^2 q^2 \mathsf{S}_n)\, \|du\|^2$, and $q^2/\boldsymbol{\mu}_N \leq e^2/(c_{\boldsymbol{\mu}}\bar{m})$. Let $\mu = \lim \mu^{(N_i)}$, where $\mu^{(N_i)} = \lim_j P_{N_i,T_{ij}}$. By (K6), applied on the separable Hilbert space H_{-4}, the limit point μ admits defining sequences whose T_i can be taken arbitrarily large. Along such a sequence Lemma E.10 yields $\bar{I}(\beta) \leq (1 + Ce^2/\bar{m})\|f\|^2$ for all probes. The assertion on $\mathrm{kl}(\{0\})$ then follows from Proposition 5.6 (d). □

Conventions. Put $g := q^2/\mathbf{m}$. A mass assignment is called *physical* if there are numbers $0 < c \leq C < \infty$ such that $c\,\eta \leq \boldsymbol{\mu}_N \leq C\eta$ for all large N. In Theorem 5.14 and its proof, the letters C, $C_\bullet$ and $c_\bullet$ stand for numbers that depend on ρ alone; the exceptions are the datum $c_{\boldsymbol{\mu}}$ and the numbers C_M, C_F and g_*, which are defined in Theorem 5.14. The norm of $L^{\mathsf{p}}(P_n)$ is written $\|\cdot\|_{L^{\mathsf{p}}}$. The maps J, J^*, the free symbol $D^{(p)}$, its inverse $\mathrm{Rsy}(p)$, and the vertices Vx_ν and $\mathrm{Ehp}_{\pm e_\nu}$ are defined in Definition E.15. The notations $\mathrm{Ehp}_\nu^\pm$ and $\mathrm{Ehp}_{\pm e_\nu}$ agree. In Theorem 5.14 the symbol uses the auxiliary mass $\mathbf{m}'$ of Setting (S). The bounded remainder is

$$\mathrm{Rsy}(p) = J(\mathbf{m}' + i\gamma\cdot p)^{-1}J^* + \mathrm{Er}(p, \mathbf{m}'), \qquad \|\mathrm{Er}(p, \mathbf{m}')\| \leq C_E(\rho),$$

by Lemma E.16; write $\mathrm{Er}(p) = \mathrm{Er}(p, \mathbf{m}')$. In Theorem 5.16, J, J^* act by the same injection and average at each cluster. These maps differ from the integer J in Proposition 5.11; the shells Π_j in those two localization arguments also differ. The vector $\mathsf{s}(p)$ is the one of Step 4 of the proof of Lemma 5.8, and in the proof of Theorem 5.14 the letter s stands for its length $|\mathsf{s}(p)|$.

Classical facts.

(K7) Let g be a standard Gaussian vector in $\mathbb{R}^N$, let $s \in \mathbb{R}^N$ and $\theta \geq 0$, and let $G \geq 0$ be a real symmetric matrix such that $2\theta\|G\| < 1$. Then

$$E\exp(s\cdot\mathsf{g}) = \exp\tfrac{1}{2}|s|^2, \qquad E\exp\bigl(\theta\langle\mathsf{g}, G\mathsf{g}\rangle\bigr) = \det(1-2\theta G)^{-1/2}.$$

(K8) (Poisson summation.) Let $K \in \ell^1(\mathbb{Z}^3)$ be scalar or matrix valued, and let $N \geq 1$ be an integer. A function ψ on $\mathbb{Z}^3/N\mathbb{Z}^3$ is read as an $N\mathbb{Z}^3$-periodic function on $\mathbb{Z}^3$, and the momenta p run through $\frac{2\pi}{N}\mathbb{Z}^3$. Then, on $\mathbb{Z}^3/N\mathbb{Z}^3$, the Fourier multiplier whose symbol is $\sum_z K(z)\exp(ip\cdot z)$ acts as $\psi \mapsto \sum_z K(z)\,\psi(\cdot + z)$. (Plancherel's formula on $\mathbb{Z}^3$.) Every $K \in \ell^1(\mathbb{Z}^3)$ satisfies

$$\sum_z |K(z)|^2 = (2\pi)^{-3}\int_{[-\pi,\pi]^3}\Bigl|\sum_z K(z)\exp(ip\cdot z)\Bigr|^2\,\mathrm{d}p.$$

(K9) The inequalities of Hölder, Minkowski and Markov, Tonelli's theorem, and Lyapunov's inequality

$$\|Z\|_{L^{\mathsf{p}}} \leq \|Z\|_{L^2}^{\vartheta}\,\|Z\|_{L^{2\mathsf{p}}}^{1-\vartheta}, \qquad \vartheta = \frac{1}{\mathsf{p}-1}, \quad \mathsf{p} \geq 2.$$

(K10) Let all spaces be of finite dimension, let $\beta \neq 0$, and let X and $X + \beta R^*R$ be invertible. Then

$$(\beta - \beta^2 R(X+\beta R^*R)^{-1}R^*)(\beta^{-1} + RX^{-1}R^*) = 1 \qquad \text{(Woodbury)}.$$

In finite dimension one has, moreover, $\det(1-XY) = \det(1-YX)$ (Sylvester).

(K11) If a real analytic function on $\mathbb{R}^d$ does not vanish identically, its zero set has Lebesgue measure zero. Two rational functions of one variable which coincide on an interval coincide at every point where both of them are regular.

The earlier localized estimate loses a factor $\log^2(2/\mathbf{m})$ in the auxiliary mass. The next theorem removes this loss. We first factor the localized hopping perturbation so that its high–low terms are absorbed in the coercive free part. The remaining quadratic block term is decomposed into its eight coordinate-reflection sectors. In the sector even under all three reflections, the odd singular part of the free resolvent cancels. Each of the other seven sectors is odd under a reflection, and Lemma 5.15 controls its spatial average. These two estimates are summable over frequency scales. The last three proof steps pass through auxiliary masses on the same box and then to the global eigenvalue count and the continuum Fisher bound.

5.11. **The uniform singular-value estimate.**

Theorem 5.14. *Proposition 5.11 with no logarithmic factor; conditions (M_{loc}) and (FI), and an atom at zero for every charge.* Let $0 < \rho < 1$, and let the weights be standard. There is a constant $C_\sharp = C_\sharp(\rho) < \infty$ with the following property. Whenever $q > 0$, $0 < \mathbf{m} \le \mathbf{m}' \le 1$, $\mathbf{m}' \ge q^2$ and $n \ge 20/\mathbf{m}'$, one has

$$n^{-3} E_n N_A(\lambda) \le C_\sharp \big(\sqrt{\lambda} + \mathbf{m}'\big)^3 \quad (\lambda > 0), \qquad \mathsf{T}_n \le 20\, C_\sharp\, \mathbf{m}'^3\, \mathbf{m}^{-4},$$
$$\mathsf{c}_n \le C_B^\sharp\, g\, \Big(\frac{\mathbf{m}'}{\mathbf{m}}\Big)^3, \qquad C_B^\sharp := 60\, \pi^2 \mathsf{h}_3^4\, C_\sharp.$$

For the continuum scaling, let $e > 0$ and $\bar m \in (0, 1]$ be arbitrary, let $c_{\boldsymbol\mu} > 0$, and let the mass assignment satisfy $c_{\boldsymbol\mu} \bar m \eta \le \boldsymbol\mu_N \le 1$ for all large N; every physical mass assignment is of this kind, for instance $m_{0,N} = 0$, where no mass shift is subtracted. Put $\mathsf{g}_* := \max\{1, e^2/(c_{\boldsymbol\mu}\bar m)\}$. Then the following assertions hold for all large N and for all T such that $n\boldsymbol\mu_N \ge 20$.

(a) For every integer $t_0 \ge 0$,

$$\boldsymbol\mu_N B_n(t_0) \le 3\boldsymbol\mu_N \mathsf{S}_n \le \frac{3\pi^2 \mathsf{h}_3^2}{8}\, \boldsymbol\mu_N \mathsf{T}_n \le C_M := \frac{15}{2}\, \pi^2 \mathsf{h}_3^2\, C_\sharp\, \mathsf{g}_*^3,$$

whereas in Proposition 5.13 the integer t_0 satisfies $4 \le t_0 \le \frac{n}{2}$. This is $(\mathbf{M}_{\mathrm{loc}})$.

(b) For these T condition **(FI)** holds in the form

$$K_n(u,u) \le C_F\, \frac{e^2}{\bar m}\, \|du\|^2 \quad \text{whenever } \sum_x u(x) = 0, \qquad C_F := \frac{15\, \pi^4 \mathsf{h}_3^4\, C_\sharp\, \mathsf{g}_*^3}{c_{\boldsymbol\mu}}.$$

If $e^2 \ge c_{\boldsymbol\mu}\bar m$, then

$$C_F\, \frac{e^2}{\bar m} = 15\, \pi^4 \mathsf{h}_3^4\, C_\sharp \Big(\frac{e^2}{c_{\boldsymbol\mu}\bar m}\Big)^4,$$

so that the bound is of the fourth order in $e^2/\bar m$; a bound linear in $e^2/\bar m$ is not proved. Since $n\boldsymbol\mu_N \ge 8 c_{\boldsymbol\mu} \bar m h L^T$, the T with $n\boldsymbol\mu_N \ge 20$ include every T with $8 c_{\boldsymbol\mu} \bar m h L^T \ge 20$.

(c) Every volume-first limit point satisfies $\mathrm{kl}(\{0\}) \ge (1 + C_F e^2/\bar m)^{-1} > 0$, that is, it has a massless photon.

■□

The ratio $e^2/\bar m$ need not be small, and if $e^2 \le c_{\boldsymbol\mu}\bar m$, the constants depend on ρ and $c_{\boldsymbol\mu}$ alone. Here E_n denotes expectation for the cutoff law formed with the mass assignment under consideration. In Theorem 5.14 the law $P_{N,T}$ is the law $P_{N,T}^{\mathrm{ch}}$ of [4, Section 6.1], and $\mathrm{kl} = \mathrm{kl}_\mu$ is the measure of Proposition D.1 used in Proposition 5.6 (d). Parts (b) to (d) of Proposition 5.6 hold for an arbitrary mass assignment and every $e > 0$. The hypotheses of Theorem 5.14 exclude the mass assignments with $\boldsymbol\mu_N \le 0$ and those with $\boldsymbol\mu_N = o(\eta)$.

Lemma 5.15. *Local functionals odd under mirrors: a law of large numbers from reflection positivity.* Let $-1 \le \rho \le 1$, $w_i > 0$, $M \in \mathbb{R}$ and $q > 0$. Let $\ell \ge 2$, $n > 10\ell$ and $1 \le \mathsf{R} \le \ell$ be integers, and put $a_0 := 2\mathsf{R} + 4$; let $\mathsf{Z} \subset \mathbb{Z}^3$ be a box with at most 7ℓ sites per side, and let $\lambda \in \{1, 2, 3\}$. For each $y \in \mathsf{Z}$ let o_y be a real polynomial in the variables F_σ subject to the following conditions.

- **(L)** The polynomial o_y depends only on the F_σ with $\sigma \subset y + [-\mathsf{R} - \frac{1}{2}, \mathsf{R} + \frac{1}{2}]^3$.
- **(O)** One has $\mathsf{o}_{\theta y}(\theta^* F) = -\mathsf{o}_y(F)$ for every reflection $\theta = \theta_{x_\lambda, c}$ with $c \in \frac{1}{2}\mathbb{Z}$ and every y with $y, \theta y \in \mathsf{Z}$.
- **(S)** With a number t, one has $\| \sum_y v(y) \mathsf{o}_y \|_{L^2} \le \mathsf{t} \|v\|_1$ for every real v on Z.

Let w be a real function carried by Z, extended by zero to $\mathbb{Z}^3$, so that a jump of w at the faces of Z enters ∂w; let x^* be an integer in the projection of Z on the λ-th coordinate, and define

$$\partial w(y) := \begin{cases} w(y) - w(y + e_\lambda) & \text{if } y_\lambda \ge x^*, \\ w(y) - w(y - e_\lambda) & \text{if } y_\lambda < x^*. \end{cases}$$

Then

$$\Big\| \sum_y w(y)\, \mathsf{o}_y \Big\|_{L^2} \le 3\, \mathsf{t}\, a_0^{1/2} \sum_y \big(|y_\lambda - x^*| + 1 + a_0 \big)^{1/2} |\partial w(y)|.$$

■

Proof for $\lambda = 1$. Points of $\mathbb{Z}^3$ are written $y = (y_1, y_\perp)$.

Step 1 (the mirror). For $\alpha, t \in \mathbb{Z}$ put $I_t := [\alpha + t a_0, \alpha + (t+1) a_0 - 1]$, and consider only those t for which I_t lies in the projection of Z. For a real function ω of $y_\perp$ put

$$\Psi_t := \sum_{y_1 \in I_t} \sum_{y_\perp} \omega(y_\perp)\, \mathsf{o}_y.$$

Let $t' \ge t + 2$, and set $c := \alpha + \frac{1}{2}\big((t + t' + 1) a_0 - 1\big) \in \frac{1}{2}\mathbb{Z}$; this is an admissible coordinate reflection plane by Definition E.1. The reflection $\theta := \theta_{x_1, c}$ carries $I_{t'} \times \{y_\perp\}$ onto $I_t \times \{y_\perp\}$. By (L), the plaquettes on which $\Psi_{t'}$ depends satisfy

$$2 \le x_1 - c \le \tfrac{7}{2}\ell + \mathsf{R} + \tfrac{1}{2} < \tfrac{n}{2},$$

which places them in the open slab.

Step 2 (the sign). Consequently the polynomial $\Psi_{t'}$ is an upper-half polynomial admitted by Proposition E.6. The scalar-reflection formula in Proposition E.6 says that $\Theta_{x_1, c}$ acts on functions of F as complex conjugation composed with θ^*; on the real polynomial $\Psi_{t'}$ it therefore acts by $\Theta_{x_1, c} \Psi_{t'} = \Psi_{t'} \circ \theta^*$, and (O) gives $\Psi_{t'} \circ \theta^* = -\Psi_t$. Hence Proposition E.6 yields

$$E_n[\Psi_t \Psi_{t'}] = -E_n\big[\Theta_{x_1, c}(\Psi_{t'})\, \Psi_{t'}\big] \le 0.$$

Step 3 (the intervals). Divide an interval I of the projection of Z into $N_I \le |I|/a_0$ intervals I_t and a remainder of length less than a_0. Step 2 and (S) give

$$E_n\Big(\sum_{t \text{ even}} \Psi_t \Big)^2 \le \sum_{t \text{ even}} E_n \Psi_t^2 \le \frac{N_I + 1}{2} \big(\mathsf{t}\, a_0 \|\omega\|_1 \big)^2,$$

and the odd t obey the same bound; by (S), the sum over the remainder has norm at most $\mathsf{t}\, a_0 \|\omega\|_1$. As $a_0 (N_I + 1)^{1/2} \le (a_0 (|I| + a_0))^{1/2}$ and $\sqrt{2} + 1 \le 3$, one obtains

$$\Big\| \sum_{y_1 \in I} \sum_{y_\perp} \omega(y_\perp)\, \mathsf{o}_y \Big\|_{L^2} \le 3\, \mathsf{t}\, \|\omega\|_1 \big(a_0 (|I| + a_0) \big)^{1/2}.$$

Step 4 (the weights). Let $y_1 \ge x^*$. The sum $\sum_{s \ge y_1} \partial w(s, y_\perp)$ telescopes, and its boundary term vanishes because w is extended by zero; hence it equals $w(y)$, and

$$\sum_{y_1 \ge x^*} w(y)\, \mathsf{o}_y = \sum_{s \ge x^*} \sum_{y_1 \in [x^*, s]} \sum_{y_\perp} \partial w(s, y_\perp)\, \mathsf{o}_y.$$

This sum is bounded by Minkowski's inequality over s together with Step 3 for $\omega = \partial w(s,\cdot)$. The part with $y_1 < x^*$ is bounded by the same argument. □

Proof of Theorem 5.14. **Setting (S) of Steps 1 to 27.** Let $\ell \geq 4$ and $n > 10\ell$ be integers, and let $0 < \mathbf{m} \leq \mathbf{m}' \leq 1$ satisfy $\ell\mathbf{m}' \geq 1$. No upper bound is imposed on the product $\ell\mathbf{m}'$; in particular, the pair $(\ell, 2^k\mathbf{m}')$ fulfils these conditions for every integer $k \geq 0$ with $2^k\mathbf{m}' \leq 1$. Let B be a box with at most 4ℓ sites per side. The boxes B^+ and B^{++}, the second of which has at most 7ℓ sites per side, the cut-off ζ, and the cochains A^{C}, A^{h}, A' and $A'' = \zeta A'$ are formed as in Step 2 of the proof of Proposition 5.11, with the present value of ℓ; that construction is valid for every integer ℓ with $6\ell + 4 < n$. Put

$$\mathsf{n} := 10\ell, \qquad \mathbb{T}' := \mathbb{Z}^3/\mathsf{n}\mathbb{Z}^3 \supset B^{++}, \qquad \mathcal{V} := \mathbb{C}^{48} \otimes \mathbb{C}^4, \qquad \mathcal{H}' := \ell^2(\mathbb{T}') \otimes \mathcal{V},$$

the site $k + \vec{g}v_0$ being identified with the pair $(k, \vec{g})$ (Definition E.1). The geometric and cluster coordinate formulas agree by Definition E.1. The normalized and counting inner products on the fibre differ by the scalar 48, so endomorphism norms, adjoints and polar parts agree; the injection and normalized average remain the explicit maps of Definition E.15. Write $\mathsf{q}(p) := \sum_\mu 4\sin^2\frac{p_\mu}{2}$ and $\mathsf{s} := \mathsf{q}^{1/2}$. Let D_0' denote the free operator with the mass $\mathbf{m}'$ on $\mathcal{H}'$, which is the Fourier multiplier with the symbol $D^{(p)}$, and set

$$S := D_0'^{-1}, \qquad \Lambda := \bigl(\mathsf{q} + \mathbf{m}'^2\bigr)^{1/2}, \qquad \|f\|_\sigma := \|\Lambda^\sigma f\|.$$

As in Step 3 of the proof of Proposition 5.11, Lemma E.16 gives $|D_0'| \geq c_E\Lambda$ and $|D_0'^*| \geq c_E\Lambda$; moreover $D_0' = \Omega|D_0'|$ with a unitary multiplier Ω, and Ω commutes with Λ.

Shells. Put $\kappa_j := 2^j\mathbf{m}'$ and

$$\Pi_0 := 1[\Lambda \leq \kappa_1], \qquad \Pi_j := 1[\kappa_j < \Lambda \leq \kappa_{j+1}] \quad (j \geq 1),$$

and let J be the largest j such that $\kappa_j \leq \sqrt{13}$. Since $\Lambda \leq \sqrt{13} < \kappa_{\mathsf{J}+1}$, one has $\sum_{j=0}^{\mathsf{J}} \Pi_j = 1$. On $\operatorname{ran}\Pi_j$ the inequalities $\Lambda \geq \kappa_j$ and $\mathsf{s} \leq \kappa_{j+1}$ hold, and so does $\mathsf{s} > \frac{\sqrt{3}}{2}\kappa_j$ when $j \geq 1$.

Filters. Fix $\varpi \in C^\infty(\mathbb{R}; [0,1])$ with $\varpi = 0$ on $(-\infty, \frac{1}{4}]$ and $\varpi = 1$ on $[\frac{9}{16}, \infty)$, and set $\varpi_k := \varpi(\mathsf{q}/\kappa_k^2)$; thus $\varpi_k = 0$ for $k \geq \mathsf{J} + 2$. Define

$$\varphi_k := \varpi_k - \varpi_{k+1}, \qquad \Phi_k := 1 - \varpi_{k+1}, \qquad \tilde{\varphi}_k := \varpi_{k-2} - \varpi_{k+3}.$$

Here φ_k is carried by $\frac{1}{2}\kappa_k \leq \mathsf{s} \leq \frac{3}{2}\kappa_k$; next, Φ_k equals 1 where $\mathsf{s} \leq \kappa_k$ and is carried by $\mathsf{s} \leq \frac{3}{2}\kappa_k$; finally, $\tilde{\varphi}_k$ equals 1 where $\frac{3}{16}\kappa_k \leq \mathsf{s} \leq 4\kappa_k$ and is carried by $\frac{1}{8}\kappa_k \leq \mathsf{s} \leq 6\kappa_k$. All these functions are real, even and invariant under O_{cub}. Their convolution kernels on $\mathbb{Z}^3$ are denoted by the same letters, and so are the kernels periodised according to (K8) on $\mathbb{T}'$ and on $\mathbb{Z}^3/n\mathbb{Z}^3$.

Dependence on ℓ. The number e of Step 3 satisfies $\mathsf{ne} \geq 40\ell\mathbf{m}'/c_E$, and the numbers L_j of (BL) satisfy $L_j \leq \ell$; the bounds of Steps 8, 11 and 18 contain the factor $(\ell\mathbf{m}')^3$. Moreover $\ell^{-1} \leq \mathbf{m}'$, one has $(\kappa_k\ell)^{-1} \leq 2^{-k}$ for every integer $k \geq 0$, the integer R of Step 24 satisfies $\mathsf{R} \leq \ell$, and $\frac{2}{\ell} \leq 2\kappa_m$ for every integer $m \geq 0$. The periods obey $\mathsf{n} \geq 1/\kappa_k$ and $n \geq 1/\kappa_k$ for every $k \geq 0$, which are the conditions on the period in (Ker) and in (Per) at $\kappa = \kappa_k$. Apart from the factor $(\ell\mathbf{m}')^3$, each of these relations either follows from $\ell\mathbf{m}' \geq 1$ alone or becomes stronger as $\ell\mathbf{m}'$ increases.

Step 1 (factorisation). Let $V = V^{\mathrm{hl}} + V^{\mathrm{lh}} + V^{\mathrm{d}}$ be a splitting of an operator V on $\mathcal{H}'$, and put

$$\mathsf{A} := SV^{\mathrm{hl}}, \qquad \mathsf{B} := V^{\mathrm{lh}}S, \qquad \mathcal{T} := V^{\mathrm{lh}}SV^{\mathrm{hl}}.$$

Expanding the product $(1 + \mathsf{B})D_0'(1 + \mathsf{A})$, one obtains the identity **(F)**

$$D_0' + V = (1 + \mathsf{B})D_0'(1 + \mathsf{A}) + V^{\mathrm{d}} - \mathcal{T}, \tag{F}$$

in which no term of higher order occurs.

Step 2 (an auxiliary vector). Let $0 \le s < \frac12$ and $D := D_0' + V - (\mathbf{m}' - \mathbf{m})$; throughout Steps 1 to 27 the letter D stands for this operator on $\mathcal{H}'$, and not for $D_A^{C(n)}$. Suppose that the following two conditions hold.

(C1) With respect to $\|\cdot\|_0$, and also with respect to $\|\cdot\|_s$, the operators A and B^* have norm at most $\frac18$, where B^* denotes the adjoint of B in $\mathcal{H}'$.

(C2) For all $f, g \in \mathcal{H}'$,

$$|\langle f, (V^{\mathrm{d}} - \mathcal{T})g\rangle| \le \frac{c_E}{4}\, \mathbf{m}'^{1-2s}\, \|f\|_s\, \|g\|_s.$$

To each $\phi \in \mathcal{H}'$ associate

$$\phi^\sharp := (1+\mathsf{A})\phi, \qquad \phi_1 := (1+\mathsf{B}^*)^{-1}\Omega\phi^\sharp.$$

Then (F) yields

$$\langle \phi_1, D\phi\rangle = \langle \phi^\sharp, |D_0'|\phi^\sharp\rangle + \langle \phi_1, (V^{\mathrm{d}} - \mathcal{T})\phi\rangle - (\mathbf{m}' - \mathbf{m})\langle \phi_1, \phi\rangle.$$

From (C1) and $[\Omega, \Lambda] = 0$ one gets $\|\phi\|_\sigma \le \frac87 \|\phi^\sharp\|_\sigma$ and $\|\phi_1\|_\sigma \le \frac87 \|\phi^\sharp\|_\sigma$ for $\sigma = 0$ and for $\sigma = s$. Taking the real part and using $\mathbf{m}'^{1-2s}\Lambda^{2s} \le \Lambda$ and $0 \le \mathbf{m}' - \mathbf{m} \le \mathbf{m}'$, one arrives at

$$\frac87\, \|\phi^\sharp\|\, \|D\phi\| \ge \frac{c_E}{2}\, \langle \phi^\sharp, \Lambda\phi^\sharp\rangle - 2\mathbf{m}'\|\phi^\sharp\|^2.$$

Step 3 (the count). Assume (C1) and (C2), and put $C_3 := 768\,(12/c_E)^3$. Then the number of eigenvalues not exceeding λ' of the quadratic form $\|D\phi\|^2$, restricted to the spinors over B, is at most $C_3|B|(\sqrt{\lambda'} + \mathbf{m}')^3$. To prove this, let $\mathcal{W}$ be a space of spinors ϕ over B on which $\|D\phi\|^2 \le \lambda'\|\phi\|^2$. For $\phi \in \mathcal{W}$, Step 2 gives

$$\langle \phi^\sharp, \Lambda\phi^\sharp\rangle \le \mathsf{e}\, \|\phi^\sharp\|^2, \qquad \mathsf{e} := \frac{4}{c_E}(\sqrt{\lambda'} + \mathbf{m}').$$

Consequently the projection $\mathsf{E} := 1[\Lambda \le 4\mathsf{e}]$ satisfies

$$\|\mathsf{E}\phi\| \ge \|\mathsf{E}\phi^\sharp\| - \|\mathsf{A}\phi\| \ge \Big(\frac{\sqrt3}{2}\cdot\frac78 - \frac18\Big)\|\phi\| \ge \frac12\, \|\phi\|.$$

It follows that

$$\frac14 \dim \mathcal{W} \le \mathrm{Tr}(1_B \mathsf{E} 1_B) = 192\, |B|\, \mathsf{n}^{-3}\, \#\{p : \Lambda(p) \le 4\mathsf{e}\} \le 192\, |B|\, (3\mathsf{e})^3,$$

where the last inequality rests on $|p| \le \frac{\pi}{2}\mathsf{s}$ and on $\mathsf{ne} \ge 40\ell\mathbf{m}'/c_E \ge 40$, with $c_E = (\sqrt2 + 7C_E)^{-1} \le 1/\sqrt2$. The assertion results on choosing for $\mathcal{W}$ the span of the eigenvectors whose eigenvalues do not exceed λ'.

Step 4 (condition (G) for quadratic forms). **(G1)** Let $\beta_i \in C^2(\Gamma)$ be finitely many real cochains, let $w_i \ge 0$ and $Q := \sum_i w_i \langle \beta_i, F\rangle^2$, let $\mathsf{t} > 0$ be such that $\mathsf{t} \ge \sum_i w_i \|\beta_i\|^2$, and let $\mathsf{p} \ge 1$. Then $E_n \exp(Q/4\mathsf{t}) \le \exp\frac12$, whence

$$P_n(Q \ge 4\mathsf{tr}) \le \exp(\tfrac12 - \mathsf{r}), \qquad \|Q\|_{L^{\mathsf{p}}} \le 7\mathsf{pt}.$$

Indeed, let g_i be the components of the Gaussian vector of (K7). Applying (K7), Tonelli's theorem and (12) at $\beta = \sqrt{2\theta}\sum_i \sqrt{w_i}\, \mathsf{g}_i\beta_i$, one finds

$$E_n \exp\theta Q \le \det(1 - 2\theta G)^{-1/2}, \qquad G_{ii'} = \sqrt{w_i w_{i'}}\, \langle \beta_i, \beta_{i'}\rangle, \qquad \mathrm{tr}\, G \le \mathsf{t},$$

and the inequality $-\log(1-\xi) \le \frac{\xi}{1-\xi}$, used with $\theta = \frac{1}{4\mathsf{t}}$, gives the assertion.

(G2) For real numbers w_i,

$$\Big\| \sum_i w_i \langle \beta_i, F\rangle \langle \beta_i', F\rangle \Big\|_{L^{\mathsf{p}}} \le 7\mathsf{p} \sum_i |w_i|\, \|\beta_i\|\, \|\beta_i'\|.$$

The terms with $\|\beta_i\|\,\|\beta_i'\| = 0$ vanish. For the other terms, $|\xi\zeta| \le \frac12(\vartheta_i\xi^2 + \vartheta_i^{-1}\zeta^2)$ with $\vartheta_i := \|\beta_i'\|/\|\beta_i\|$ produces two positive forms, each of which admits $\mathsf{t} = \sum_i |w_i|\,\|\beta_i\|\,\|\beta_i'\|$; by (G1), the norm is at most $\frac12(7\mathsf{pt} + 7\mathsf{pt})$. Complex cochains are treated with the bilinear pairing and with $\|\beta\|^2 = \|\operatorname{Re}\beta\|^2 + \|\operatorname{Im}\beta\|^2$. The product of two complex linear forms is then a combination of four real products, and

$$(\|\operatorname{Re}\beta\| + \|\operatorname{Im}\beta\|)(\|\operatorname{Re}\beta'\| + \|\operatorname{Im}\beta'\|) \le 2\,\|\beta\|\,\|\beta'\|;$$

hence (G2) is valid for complex cochains with $14\mathsf{p}$ in place of $7\mathsf{p}$.

Step 5 (functionals of A^{C}). To a function k on $\mathbb{Z}^3/n\mathbb{Z}^3$ associate

$$A^{\mathrm{C}}_\nu(\mathsf{k}) := \sum_y \mathsf{k}(y)\,A^{\mathrm{C}}_\nu(y) = \langle \beta_{\mathsf{k}}, F\rangle, \qquad \beta_{\mathsf{k}} = \mathcal{G}d(\mathsf{k}e_\nu).$$

Since $d^*d \le \Delta$ by (K4), one has the bound **(B0)**

$$\|\beta_{\mathsf{k}}\|^2 \le \langle \mathsf{k}, \mathcal{G}\mathsf{k}\rangle = n^{-3}\sum_{p\neq 0} \frac{|\hat{\mathsf{k}}(p)|^2}{\mathsf{q}(p)}. \tag{B0}$$

The bound on the heat kernel in Step 6 of the proof of Proposition 5.11 gives **(B1)**

$$\langle \mathsf{k}, \mathcal{G}\mathsf{k}\rangle \le \int_0^\infty \min\{\|\mathsf{k}\|_2^2,\ c_h\|\mathsf{k}\|_1^2\,t^{-3/2}\}\,\mathrm{d}t = 3\,c_h^{2/3}\,\|\mathsf{k}\|_1^{4/3}\,\|\mathsf{k}\|_2^{2/3}. \tag{B1}$$

In particular $\|\mathsf{k}\|_1 \le c$ and $\|\mathsf{k}\|_2^2 \le c\kappa^3$ imply $\|\beta_{\mathsf{k}}\|^2 \le c'\kappa$.

(Per) For $\mathsf{k} \in \ell^1(\mathbb{Z}^3)$ let

$$A^{\mathrm{C}}_\nu(\mathsf{k}) := \sum_{y\in\mathbb{Z}^3} \mathsf{k}(y)\,A^{\mathrm{C}}_\nu(y) = A^{\mathrm{C}}_\nu(\operatorname{per}_n \mathsf{k}), \qquad \operatorname{per}_n \mathsf{k} := \sum_{\mathsf{m}\in\mathbb{Z}^3} \mathsf{k}(\cdot + n\mathsf{m}).$$

Then $\|\operatorname{per}_n \mathsf{k}\|_1 \le \|\mathsf{k}\|_1$. Assume further that $|\mathsf{k}(y')| \le \mathsf{k}_0(1+\kappa|y'-y|_\infty)^{-8}$, with $n\kappa \ge 1$. On the fundamental domain $y'-y \in (-\frac{n}{2}, \frac{n}{2}]^3$ the image of index $\mathsf{m} \neq 0$ does not exceed $\mathsf{k}_0(\frac12\kappa n|\mathsf{m}|_\infty)^{-8}$, and Minkowski's inequality over the images yields

$$\|\operatorname{per}_n \mathsf{k}\|_2 \le \|\mathsf{k}\|_{\ell^2(\mathbb{Z}^3)} + C\mathsf{k}_0\,\kappa^{-3/2}(\kappa n)^{-13/2}.$$

Step 6 (frequencies add). On $(\mathbb{R}/2\pi)^3$ one has $\mathsf{s}(p+p') \le \mathsf{s}(p)+\mathsf{s}(p')$, by $|\sin\frac{a+b}{2}| \le |\sin\frac{a}{2}|+|\sin\frac{b}{2}|$ and Minkowski's inequality. Therefore, if $\hat g$ is carried by $\mathsf{s} \le a$, the component of ug with $b_1 < \mathsf{s} \le b_2$ involves only those modes of the multiplier u that have $b_1 - a < \mathsf{s} \le b_2 + a$; this is exact, and no aliasing occurs. The mode of u which takes p to P is $p' = P - p$, and s is even; hence

$$|\mathsf{s}(p) - \mathsf{s}(P)| \le \mathsf{s}(p') \le \mathsf{s}(p) + \mathsf{s}(P).$$

For an input with $a_1 \le \mathsf{s} \le a_2$ and an output with $\mathsf{s} \le b < a_1$ it follows that only the modes with $a_1 - b \le \mathsf{s} \le a_2 + b$ are selected. In the left factor of Step 16 the inner momenta form the input and the outer momenta the output, and the selected modes are those with $\frac14\kappa_k \le \mathsf{s} \le \frac74\kappa_k$, on which $\tilde\varphi_k = 1$.

Step 7. **(Ker)** Let $0 < \kappa \le 64$, and let f be a smooth, periodic, matrix valued function carried by $\mathsf{s} \le 6\kappa$ such that $|\partial^\gamma \mathsf{f}| \le \mathsf{f}_0\kappa^{-|\gamma|}$ for $|\gamma| \le 8$. The kernel K of f obeys

$$|K(z)| \le C\mathsf{f}_0\,\kappa^3\,(1+\kappa|z|_\infty)^{-8},$$

as one sees from the volume of the support and from eight integrations by parts; the same bound holds after periodisation with a period $\ge 1/\kappa$, $|z|_\infty$ being then the distance on the torus. At $\kappa = \kappa_k$ this applies to the filters φ_k, Φ_k and $\tilde\varphi_k$ with $\mathsf{f}_0 = C$, because $|\partial_\mu \mathsf{q}| \le 2\mathsf{s}$, the higher derivatives of q are at most 2, and $\kappa \le 64$; to $\tilde\varphi_k\,\mathsf{q}^{-1}(1-\exp(\mp ip_\mu))$ and to $\varphi_k\mathrm{Rsy}$ with $\mathsf{f}_0 = C\kappa_k^{-1}$; to the product of these two with $\mathsf{f}_0 = C\kappa_k^{-2}$; and to $\Phi_k(1-\exp(\mp ip_\mu))$ with $\mathsf{f}_0 = C\kappa_k$, since $|1-\exp(\mp ip_\mu)| \le \mathsf{s}$ and the derivatives of $1-\exp(\mp ip_\mu)$ have modulus at most 1. For Rsy the required bounds are those of (19), which applies because $\operatorname{dist}_2(p, 2\pi\mathbb{Z}^3) \ge \mathsf{s} \ge \frac12\kappa_k$ on the support of φ_k.

Step 8. **(BL)** Let $0 \le j \le \mathsf{J}+1$ and $L_j := \lceil 1/\kappa_j \rceil$, and denote by $\mathcal{C}_j$ a partition of $\mathbb{T}'$ of product type, into boxes all of whose sides lie in $[L_j, 2L_j)$. Such a partition exists, since $L_j \le \lceil 1/\mathbf{m}' \rceil \le \ell < \mathsf{n}$, and $\#\mathcal{C}_j \le (\mathsf{n}/L_j)^3 \le (\mathsf{n}\kappa_j)^3$. For a box c put $\langle W \rangle_c := |c|^{-1} \sum_{y \in c} W(y)$. If $W \ge 0$ and $\hat{g}$ is carried by $\mathsf{s} \le \kappa_j$, then

$$\langle g, Wg \rangle \le C_{\mathrm{BL}} \max_{c \in \mathcal{C}_j} \langle W \rangle_c \, \|g\|^2 .$$

In fact $g = \Phi_j * g$, so that the Schwarz inequality gives $|g(x)|^2 \le \|\Phi_j\|_1 \sum_y |\Phi_j(x-y)|\,|g(y)|^2$; and (Ker), applied with the period $\mathsf{n} \ge 10/\kappa_j$, gives $\|\Phi_j\|_1 \le C$ and $\sum_c |c| \max_{x \in c} |\Phi_j(y-x)| \le C$.

Step 9 (the potential). Let e range over $\{\pm e_\nu\}$, and write $(\tau_e f)(k) := f(k+e)$ and $\mathrm{Ehp}_{\pm e_\nu} := \mathrm{Ehp}_\nu^{\pm}$. The operator $V := D_{A''} - D_0$ takes the form

$$V = -\sum_e u_e \, \tau_e \otimes \mathrm{Ehp}_e, \qquad u_e(k) := \exp\bigl(iqA''(k, k+e)\bigr) - 1$$

by Definition E.1; the hop points in the direction $\vec{g}e_1$, because $x^{(3)} - x = (1 - 2v_1)\vec{g}e_1$ and $0 < v_1 < \frac{1}{2}$. Further $\mathrm{Ehp}_e = 1_e \mathrm{Ehp}_e 1_{-e}$, where 1_e is the projection onto $\{\vec{g}e_1 = e\}$, and $\|\mathrm{Ehp}_e\| = \mathsf{h}_3$. Consequently every family of multipliers v_e satisfies

$$\Bigl\| \sum_e v_e \, \tau_e \otimes \mathrm{Ehp}_e \, g \Bigr\|^2 \le \mathsf{h}_3^2 \sum_e \|v_e \, \tau_e 1_{-e} g\|^2 .$$

Extend A'' by zero. As $D_0' - D_0 = \mathbf{m}' - \mathbf{m}$, the operator D of Step 2 is the chamber operator on $\mathcal{H}'$ with the mass $\mathbf{m}$ and the field A''. For ϕ over B the value $(D\phi)(k, \vec{g})$ involves ϕ only at $(k, \vec{g})$, at $(k, \vec{g}s_i)$ with $i \le 2$, and at $(k + \vec{g}e_1, \vec{g}\,\mathrm{diag}(-1,1,1))$; thus $D\phi$ lives over B^+. Since $7\ell < 10\ell$, the quotient map $\mathbb{Z}^3 \to \mathbb{T}'$ is injective on B^{++}. Hence $\|D\phi\|_{\mathcal{H}'} = \|D_{A''}\phi\|$, and by Step 2 of the proof of Proposition 5.11 this equals $\|D_A \exp(-iq\,\lambda \circ \pi)\phi\|$. The phase factor is unitary on the spinors over B, so that the forms $\|D\phi\|^2$ and $\|D_A\phi\|^2$ over B have the same eigenvalues.

Step 10 (the splitting). Let $\mathsf{u}_\nu := \zeta A_\nu^{\mathrm{C}}$, and let V_1 be the operator obtained by putting

$$u^{(1)}_{e_\nu} := iq\,\mathsf{u}_\nu, \qquad u^{(1)}_{-e_\nu} := -iq\,\tau_{-e_\nu}\mathsf{u}_\nu$$

in the places of the u_e; the adjoint V_1^* has the same multipliers, with Ehp^*_{-e} in place of Ehp_e. Set $V_{\mathrm{r}} := V - V_1$ and

$$V^{\mathrm{hl}} := \sum_{i \ge j+3} \Pi_i V_1 \Pi_j, \qquad V^{\mathrm{lh}} := \sum_{j \ge i+3} \Pi_i V_1 \Pi_j, \qquad V^{\mathrm{d}} := \sum_{|i-j| \le 2} \Pi_i V_1 \Pi_j + V_{\mathrm{r}}.$$

With this splitting, $\mathcal{T}$ is exactly quadratic in F.

Step 11 (the linear event). For $r \ge 1$ put $\mathsf{r}_{kj} := r + 3j + 2\log(2 + k - j)$, and let $\mathcal{G}^{\mathrm{lin}}(r)$ be the event on which the following inequalities hold for all μ, ν, all $0 \le j \le \mathsf{J}+1$ and all $c \in \mathcal{C}_j$:

(a) $\langle |\tilde{\varphi}_k * \mathsf{u}_\nu|^2 \rangle_c \le c_1 \kappa_k \mathsf{r}_{kj}$ for $j \le k \le \mathsf{J}+1$;
(b) $\langle |\Phi_{j+3} * \mathsf{u}_\nu|^2 \rangle_c \le c_1 \kappa_j \mathsf{r}_{jj}$;
(c) $\langle |\mathsf{u}_\nu|^2 \rangle_c \le c_1 \mathsf{r}_{jj}$;
(d) $|(F_{\mathrm{h}})_{\mu\nu}| \le (2r)^{1/2} n^{-3/2}$.

In (a), (b) and (c) each value of $\tilde{\varphi}_k * \mathsf{u}_\nu$, of $\Phi_{j+3} * \mathsf{u}_\nu$ and of u_ν is of the form $A_\nu^{\mathrm{C}}(\mathsf{k})$, with k a kernel multiplied by ζ. By (Ker) and (B1) the three quadratic forms admit $\mathsf{t} \le c\kappa_k$, $\mathsf{t} \le c\kappa_j$ and $\mathsf{t} \le c$ respectively, c being the largest of the three constants; in (b) this constant contains the ratio $\kappa_{j+3}/\kappa_j = 8$, and (Ker) is applicable there because $\kappa_{j+3} \le 16\sqrt{13} < 64$. Put $c_1 := 4c$; the thresholds in (a), (b) and (c) are then the numbers $4\mathsf{t}\mathsf{r}$ of (G1). Combining (G1), Proposition E.6 for a single β, and

$$\sum_j (10\ell\kappa_j)^3 \exp(-3j) \sum_k (2 + k - j)^{-2} \le C(\ell\mathbf{m}')^3,$$

one obtains

$$P_n\bigl(\mathcal{G}^{\mathrm{lin}}(r)^c\bigr) \le C(\ell\mathbf{m}')^3 \exp(-r).$$

Step 12 (blocks on the event $\mathcal{G}^{\rm lin}(r)$).

(i) Let $\hat g$ be carried by $\mathsf{s} \le \kappa_j$, and let $j \le k$. By (BL),

$$\|(\tilde\varphi_k * \mathsf{u}_\nu)g\|^2 \le C_{\rm BL}\, c_1 \kappa_k \mathsf{r}_{kj}\, \|g\|^2.$$

This extends to translates of u_ν. Indeed, for $w \in \mathbb{Z}^3$ and a multiplier U one has $\|(\tau_w U)g\| = \|U\tau_{-w}g\|$, and $\tau_{-w}g$ has the same norm and the same Fourier support as g; consequently the inequality of (i) is valid with $\tilde\varphi_k * \mathsf{u}_\nu$ replaced by any of its translates.

(ii) Let $i \ge j+3$. The input of $\Pi_i V_1 \Pi_j$ has $\mathsf{s} \le \frac14 \kappa_i$, and its output has $\frac{\sqrt3}{2}\kappa_i < \mathsf{s} \le 2\kappa_i$. According to Step 6, only the modes of u_ν with $0.6\,\kappa_i < \mathsf{s} \le 2.25\,\kappa_i$ contribute, and on them $\tilde\varphi_i = 1$. Step 9 therefore gives

$$\|\Pi_i V_1 \Pi_j\|,\ \|\Pi_i V_1^* \Pi_j\| \le c_2\, q\, (\kappa_i \mathsf{r}_{i,j+1})^{1/2}, \qquad c_2 := \mathsf{h}_3\, (2C_{\rm BL}c_1)^{1/2}.$$

(iii) Let $|i-j| \le 2$. Only the modes with $\mathsf{s} \le \kappa_{j+4}$ contribute, and (b) at the index $j+1$ gives

$$\|\Pi_i V_1 \Pi_j\| \le c_2\, q\, (\kappa_j \mathsf{r}_{j+1,j+1})^{1/2};$$

the factor 2 in c_2 comes from $\kappa_{j+1} = 2\kappa_j$.

Step 13 (the condition (C1)). For an operator X one has $\|X\|^2 \le \sum_{i,j} \|\Pi_i X \Pi_j\|^2$, and $\|S\Pi_i\| \le (c_E\kappa_i)^{-1}$. Let $0 \le \sigma < \frac12$. From (ii),

$$\begin{aligned}\|\Lambda^\sigma \mathsf{A} \Lambda^{-\sigma}\|^2 &\le \sum_{i \ge j+3} \kappa_{i+1}^{2\sigma}\, (c_E\kappa_i)^{-2}\, c_2^2 q^2\, \kappa_i \mathsf{r}_{i,j+1}\, \kappa_j^{-2\sigma} \\ &\le C_\sigma q^2 \sum_j \kappa_j^{-1}(r+3j+3) \le C'_\sigma\, q^2 (r+1)/\mathbf{m}'.\end{aligned}$$

Since $\sigma < \frac12$, the summation over i is that of a geometric series; in $\mathsf{r}_{i,j+1} = r+3j+3+2\log(1+i-j)$ the term $2\log(1+i-j)$ is absorbed into C_σ, as $r+3j+3 \ge 4$. The operator $\mathsf{B}^* = S^*(V^{\rm lh})^*$ satisfies the same bound.

Step 14 (the operator $V^{\rm d}$). Henceforth $s = \frac38$, $s' = \frac{5}{16}$ and $\mathsf{p} = 16$. By Step 12 (iii), the near-diagonal part of $V^{\rm d}$ contributes to $\langle f, V^{\rm d} g\rangle$ a term of modulus at most

$$Cq \sup_j \kappa_j^{-1/4} \mathsf{r}_{j+1,j+1}^{1/2}\, \|f\|_s \|g\|_s \le Cq(r+1)^{1/2} \mathbf{m}'^{-1/4} \|f\|_s \|g\|_s.$$

Moreover

$$|u_e - u_e^{(1)}| \le q^2 |\zeta A^{\rm C}|^2 + q^2 |\zeta A^{\rm h}|^2 + q|\zeta A^{\rm h}|,$$

where for $e = -e_\nu$ the right side is to be replaced by its translate by $-e_\nu$; the argument of Step 12 (i) applies to such translates. From (BL) and Step 11 (c) one obtains

$$\big\| |\mathsf{u}_\nu| f \big\| \le \sum_j C \mathsf{r}_{j+1,j+1}^{1/2} \kappa_j^{-s} \|\Pi_j f\|_s \le C(r+1)^{1/2} \mathbf{m}'^{-s} \|f\|_s.$$

Step 11 (d), together with $n > 10\ell$ and $\ell \mathbf{m}' \ge 1$, gives $|\zeta A^{\rm h}| \le (r\mathbf{m}')^{1/2}$, and therefore $q^2|\zeta A^{\rm h}|^2 \le q^2 r$, because $\mathbf{m}' \le 1$. The summation over the six directions e is controlled by

$$\sum_e \|1_e f\|_s \|1_{-e} g\|_s \le \|f\|_s \|g\|_s,$$

a consequence of Step 9 and the Schwarz inequality. It follows that there is a number $c_6 > 0$ with the following property: on $\mathcal{G}^{\rm lin}(r)$, whenever $q^2(r+1) \le c_6 \mathbf{m}'$, condition (C1) is satisfied and

$$|\langle f, V^{\rm d} g\rangle| \le \frac{c_E}{8}\, \mathbf{m}'^{1-2s} \|f\|_s \|g\|_s.$$

Step 15 (blocks of $\mathcal{T}$). Let $m := \max(i,j)+3$. Then

$$\Pi_i \mathcal{T} \Pi_j = \Pi_i V_1 S\, 1[\Lambda > \kappa_m]\, V_1 \Pi_j,$$

and this block is zero unless $m \le \mathsf{J}$. One has $1[\Lambda > \kappa_m] = \varpi_m - \varpi_m \Pi_{m-1}$, for $\Lambda > \kappa_m$ implies $\mathsf{q} \ge \frac{9}{16}\kappa_m^2$, while $\varpi_m \neq 0$ implies $\Lambda > \kappa_{m-1}$. By Step 12 (ii), (iii), the second term contributes at most $Cq^2 \mathsf{r}_{mm}$ to $\|\Pi_i \mathcal{T} \Pi_j\|$. Indeed, the input index of the left factor $\Pi_i V_1 \Pi_{m-1}$ is $m-1$. If $i = m-3$, the bound (iii) applies, Step 11 (b) being used at the index $m \le \mathsf{J}+1$, and this yields r_{mm}; if $i < m-3$, the left factor is the adjoint of $\Pi_{m-1} V_1^* \Pi_i$, a block covered by (ii). Finally, $\varpi_m = \sum_{k=m}^{\mathsf{J}+1} \varphi_k$.

Step 16 (filtered fields). Let $\tilde{S}^{(k)} := \varphi_k S$; thus

$$(\tilde{S}^{(k)}\psi)(y) = \sum_z \tilde{S}^{(k)}(z)\, \psi(y+z),$$

where z runs through $\mathbb{Z}^3$ and $\tilde{S}^{(k)}(z)$ is the kernel on $\mathbb{Z}^3$ of (K8), the same kernel as in the definition of $\mathsf{b}_{e'}$ in Step 20. According to (Ker), $|\tilde{S}^{(k)}(z)| \le C\kappa_k^2 (1+\kappa_k |z|_\infty)^{-8}$. The inner momenta satisfy $\frac{1}{2}\kappa_k \le \mathsf{s} \le \frac{3}{2}\kappa_k$, and the outer ones $\mathsf{s} \le \frac{1}{4}\kappa_k$. By Step 6, the right factor involves only those modes of its multiplier for which $\frac{1}{4}\kappa_k \le \mathsf{s} \le \frac{7}{4}\kappa_k$. The left factor, whose input is the inner momentum p and whose output is the outer momentum P, has the same property: a mode p' carrying p to P equals $P-p$, so that $\mathsf{s}(p) - \mathsf{s}(P) \le \mathsf{s}(p') \le \mathsf{s}(p) + \mathsf{s}(P)$. On these modes $\tilde{\varphi}_k = 1$. Consequently $\Pi_i V_1 \tilde{S}^{(k)} V_1 \Pi_j = \Pi_i \mathcal{X}_k \Pi_j$, where $\mathcal{X}_k$ is built from the multipliers $u_{e[k]}$, in which $\tilde{\varphi}_k * \mathsf{u}_\nu$ stands in place of u_ν:

$$(\mathcal{X}_k g)(y) = \sum_{e,e',z} u_{e[k]}(y)\, u_{e'[k]}(y+e+z)\, \mathrm{Ehp}_e \tilde{S}^{(k)}(z) \mathrm{Ehp}_{e'}\, g(y+e+z+e').$$

Step 17 ($\mathcal{T}$ is a multiplication operator modulo convergent errors). Denote by $\Xi_k(y) \in \operatorname{End}\mathcal{V}$ the sum of Step 16 with the factor $g(y+e+z+e')$ deleted:

$$\Xi_k(y) := \sum_{e,e',z} u_{e[k]}(y)\, u_{e'[k]}(y+e+z)\, \mathrm{Ehp}_e \tilde{S}^{(k)}(z) \mathrm{Ehp}_{e'}.$$

On $\operatorname{ran}\Pi_j$ one has $\|(\tau_v - 1)g\| \le |v|_2 \kappa_{j+1} \|g\|$. Together with the Schwarz inequality and Step 12 (i), this gives

$$\|\Pi_i (\mathcal{X}_k - \Xi_k) \Pi_j\| \le Cq^2 \kappa_k \mathsf{r}_{km} \kappa_{j+1} \sum_z |\tilde{S}^{(k)}(z)|\, (|z|_2 + 2) \le Cq^2 \mathsf{r}_{km}\, \frac{\kappa_j}{\kappa_k},$$

because $\sum_z |\tilde{S}^{(k)}(z)|(|z|_2+2) \le C(\kappa_k^{-2} + \kappa_k^{-1}) \le C'\kappa_k^{-2}$, by $\kappa_k \le 2\sqrt{13}$. Summed over $k \ge m$, these bounds amount to at most $Cq^2 \mathsf{r}_{mm}$. Here Step 12 (i) bounds $\|\overline{u_{e[k]}} f\|$, where $f \in \operatorname{ran}\Pi_i$ has $\mathsf{s} \le \kappa_{i+1}$, and, for every z, $\|(\tau_{e+z} u_{e'[k]})(\tau_v - 1)g\|$ with $v = e+z+e'$, whose multiplier is a translate. As $i+1, j+1 \le m \le k$ and r_{kj} is an increasing function of j, both bounds carry the factor r_{km}. By Step 6, the block $\Pi_i \Xi_k \Pi_j$ involves only the modes of Ξ_k with $\mathsf{s} \le \kappa_{m-1}$. Put

$$Y_m := \Phi_m * \sum_{k=m}^{\mathsf{J}+1} \Xi_k, \qquad \|Y_m\|_\infty := \sup_{x \in \mathbb{T}'} \|Y_m(x)\|,$$

the norm of $Y_m(x)$ being the operator norm of $\operatorname{End}\mathcal{V}$. Then

$$\|\Pi_i \mathcal{T} \Pi_j\| \le Cq^2 \mathsf{r}_{mm} + \|Y_m\|_\infty.$$

Step 18 (the quadratic event). Let $\epsilon_3 > 0$ be a number depending on ρ alone, whose value is chosen in Step 27. This positive number is distinct from the component ε_3 of the sign vectors $\varepsilon \in \{\pm 1\}^3$ of Step 21. Define

$$\mathcal{G}^{\mathrm{qu}} := \{\|Y_m\|_\infty \le \epsilon_3 \mathbf{m}' 2^{s'm} \text{ for } 3 \le m \le \mathsf{J}\}.$$

On $\mathcal{G}^{\mathrm{lin}}(r) \cap \mathcal{G}^{\mathrm{qu}}$ one has

$$|\langle f, \mathcal{T} g\rangle| \le C\big[q^2(r+1) + \epsilon_3 \mathbf{m}'\big]\, \mathbf{m}'^{-2s} \|f\|_s \|g\|_s,$$

since $\sum_{i,j} 4^{-s(i+j)}(1+m)^2 4^{s'm} < \infty$; the number C in this bound does not depend on ϵ_3. If $q^2(r+1) = c_*\mathbf{m}'$ with $0 < c_* \le c_6$, then on $\mathcal{G}^{\mathrm{lin}}(r) \cap \mathcal{G}^{\mathrm{qu}}$ Step 14 yields (C1), and Step 14 together with this bound yields

$$|\langle f, (V^{\mathrm{d}} - \mathcal{T})g\rangle| \le \Big[\frac{c_E}{8} + C(c_* + \epsilon_3)\Big]\, \mathbf{m}'^{1-2s}\|f\|_s\|g\|_s,$$

with the same number C. Consequently (C2) holds on this event whenever $C(c_*+\epsilon_3) \le \frac{c_E}{8}$, because $\frac{c_E}{8} + \frac{c_E}{8} = \frac{c_E}{4}$. As $Y_m = \Phi_{m+1} * Y_m$, Hölder's inequality and (Ker) yield $\|Y_m\|_\infty \le C\kappa_m^{3/\mathsf{p}}\|Y_m\|_{\ell^{\mathsf{p}}(\mathbb{T}')}$, where $\|Y_m\|_{\ell^{\mathsf{p}}(\mathbb{T}')}$ is the ℓ^{p} norm of the function $x \mapsto \|Y_m(x)\|$. Suppose that condition **(H)** is satisfied:

$$\big\|\, \|Y_m(x)\| \,\big\|_{L^{\mathsf{p}}} \le \mathsf{C}_{\mathsf{p}} q^2 \qquad \text{for all } x \in \mathbb{T}' \text{ and } 3 \le m \le \mathsf{J}.$$

Markov's inequality then gives

$$P_n\big((\mathcal{G}^{\mathrm{qu}})^c\big) \le \sum_m C(\ell\mathbf{m}')^3\, 2^{(3-\mathsf{p}s')m}\Big(\frac{\mathsf{C}_{\mathsf{p}} q^2}{\epsilon_3 \mathbf{m}'}\Big)^{\mathsf{p}} \le C(\ell\mathbf{m}')^3\Big(\frac{\mathsf{C}_{\mathsf{p}} q^2}{\epsilon_3\mathbf{m}'}\Big)^{16},$$

as $\mathsf{p}s' = 5$. Step 27 establishes (H) with $\mathsf{C}_{\mathsf{p}} = C\mathsf{p}^2$, and ϵ_3 depends on ρ alone; the factor $(\mathsf{C}_{\mathsf{p}}/\epsilon_3)^{16}$ of the last bound is therefore a number depending on ρ alone, and it is absorbed into the constant C_{11} of Step 27.

Step 19 (covariance). Let $g \in W$. Then $U_g = T_g \otimes \mathsf{U}_{\vec g}$, where $(T_g f)(k) := f(g^{-1}k)$ and where the unitary operator $\mathsf{U}_{\vec g}$ acts by $(\mathsf{U}_{\vec g} u)(\vec h) := \tilde g\, u(\vec g^{-1}\vec h)$; furthermore $U_g D_A U_g^{-1} = D_{g_*A}$ with $g_* := (g^{-1})^*$, and $\tilde g \gamma_v \tilde g^{-1} = \gamma_{\vec g v}$ (Lemma E.2). The matrix $\tilde g$ is formed with a word $\vec g = r_{n_1}\cdots r_{n_k}$. For a different word it changes at most by a sign (an even word gives $\mathrm{diag}(X, X)$ with $X \in SU(2)$), and hence $\mathrm{Ad}\,\mathsf{U}_{\vec g}$ depends on $\vec g$ alone. Since

$$(\mathrm{Ehp}_e u)(\vec h) = 1[\vec h e_1 = e]\, 3(\rho - \gamma\cdot e)\, u(\vec h s_3),$$

with the linear map $s_3 = \mathrm{diag}(-1,1,1)$ of Definition E.15, one finds $\mathsf{U}_{\vec g}\mathrm{Ehp}_e\mathsf{U}_{\vec g}^{-1} = \mathrm{Ehp}_{\vec g e}$. At $A = 0$ the relation $\mathsf{U}_{\vec g} D^{(p)} \mathsf{U}_{\vec g}^{-1} = D^{(\vec g p)}$ holds, whence $\mathsf{U}_{\vec g}\tilde S^{(k)}(z)\mathsf{U}_{\vec g}^{-1} = \tilde S^{(k)}(\vec g z)$.

Step 20 (the local functional). Fix an integer R with $1 \le \mathsf{R} \le \ell$; its value is chosen in Step 24 and depends on (k, m). Write ξ_k for $\xi_k^{(\mathsf{R})}$. On $\mathbb{Z}^3$, with the periodic field, define the one-cochain $a^{[k]} := \tilde\varphi_k * A^{\mathrm{C}}$ and

$$\mathsf{b}_{e'}(x) := \sum_z \tilde S^{(k)}(z)\, a^{[k]}(x+z, x+z+e'),$$
$$\xi_k^\infty(y; F) := \sum_{e,e'} a^{[k]}(y, y+e)\, \mathrm{Ehp}_e\, \mathsf{b}_{e'}(y+e)\, \mathrm{Ehp}_{e'}.$$

Each of the two factors is a sum $\sum_\sigma \mathsf{K}(\sigma - y)F_\sigma$ over plaquettes, and ξ_k is obtained from ξ_k^∞ by restricting both sums to the plaquettes whose centres satisfy $|\sigma - y|_\infty \le \mathsf{R}$. The operators d^* and $\mathcal{G}$, as well as the O_{cub}-invariant filters, commute with g_* (Definition E.1). Therefore $a^{[k]}[g_*F](gx, gx + \vec g e) = a^{[k]}[F](x, x+e)$. The centre of $g\sigma$ is the image under g of the centre of σ, and $|\vec g v|_\infty = |v|_\infty$. Accordingly, the identity is valid also when the sums are cut off at $|\sigma - y|_\infty \le \mathsf{R}$. In each of the two factors this cut-off is centred at y, which gives the property (L) of Lemma 5.15 for this R. In detail, (K8) gives on $\mathbb{Z}^3$ the representation $a^{[k]}(b) = \langle \mathsf{K}_b, F\rangle$ for a bond b, with $\mathsf{K}_b := d(\mathsf{G}_k * \delta_b) \in \ell^1$ and $\langle \delta_b, a\rangle := a(b)$, where G_k denotes the kernel of the symbol $\tilde\varphi_k/\mathsf{q}$. This symbol is O_{cub}-invariant, and it is smooth because $\tilde\varphi_k$ vanishes near $p = 0$. Moreover $g_*\delta_b = \delta_{gb}$, and g_* commutes with d and with this convolution. Hence $g_*\mathsf{K}_b = \mathsf{K}_{gb}$ and $g_*(\mathsf{K}_b 1_{y,\mathsf{R}}) = \mathsf{K}_{gb} 1_{gy,\mathsf{R}}$, where $1_{y,\mathsf{R}}$ is the indicator of the plaquettes with $|\sigma - y|_\infty \le \mathsf{R}$. Since g_* is an orthogonal map, $\langle \mathsf{K}_{gb} 1_{gy,\mathsf{R}}, g_* F\rangle$ coincides with $\langle \mathsf{K}_b 1_{y,\mathsf{R}}, F\rangle$ for every bounded F. Change the summation variables in

the sums defining $\xi_k^\infty(gy; g_*F)$ and $\xi_k(gy; g_*F)$, which converge absolutely, to $\tilde e$, $\tilde z$, $\tilde e'$, where $e = \vec g\tilde e$, $z = \vec g\tilde z$, $e' = \vec g\tilde e'$, and apply both identities of Step 19. The outcome is

$$\xi_k(gy; g_*F) = \mathsf{U}_{\vec g}\, \xi_k(y; F)\, \mathsf{U}_{\vec g}^{-1},$$

together with the same identity for ξ_k^∞.

Step 21 (sectors). Let r_λ be the reflection $x_\lambda \mapsto -x_\lambda$; then $\tilde r_\lambda = \gamma_\lambda\gamma_5$ (formed with the word r_{e_λ}), and one puts $\mathrm{Ad}_\lambda := \mathrm{Ad}\, \mathsf{U}_{r_\lambda}$. These maps are commuting isometric involutions of $\mathrm{End}\, \mathcal{V}$, because $(\gamma_\lambda\gamma_5)^2 = -1$ and $\gamma_\lambda\gamma_5$ anticommutes with $\gamma_\mu\gamma_5$ for $\mu \neq \lambda$ (Definition E.1). For $\varepsilon \in \{\pm 1\}^3$ set

$$\mathsf{P}^\varepsilon := \prod_\lambda \tfrac12(1 + \varepsilon_\lambda\, \mathrm{Ad}_\lambda),$$

so that $\sum_\varepsilon \mathsf{P}^\varepsilon = 1$. For $\theta = \theta_{x_\lambda, c}$ with $c \in \frac12\mathbb{Z}$, Step 20 implies

$$(\mathsf{P}^\varepsilon \xi_k)(\theta y; \theta^* F) = \varepsilon_\lambda\, (\mathsf{P}^\varepsilon \xi_k)(y; F).$$

Step 22 (sizes). A matrix valued function $\mathsf{z}(y)$ is said to have *size* N if every entry of $\mathsf{z}(y)$ can be written as a sum of no more than C products $\langle \beta, F\rangle\langle \beta', F\rangle$, where β, β' are complex cochains such that $\|\beta\|\|\beta'\| \le \mathsf{N}$. Let $\mathsf{p}' \ge 1$ be an arbitrary real number. This exponent is independent of the number $\mathsf{p} = 16$ fixed in Step 14, and its prime does not mark a conjugate exponent. Let further v be a real function of y with finite support, as in condition (S) of Lemma 5.15 (a weight, not a vector), and put $\|v\|_1 := \sum_y |v(y)|$. For a function z of size N, the inequality (G2), with p' in the place of p, gives

$$\Big\| \sum_y v(y)\mathsf{z}(y) \Big\|_{L^{\mathsf{p}'}} \le C\mathsf{p}'\mathsf{N}\|v\|_1,$$

because the product of two complex linear forms is a linear combination of four products of real ones, built from the real and the imaginary parts of β and β'. In this bound the norm of $L^{\mathsf{p}'}$ is applied to each real component of the matrix on the left, that is, to the real part and to the imaginary part of each of its entries; for C one may then take the factor 14 of (G2) for complex cochains times the number of products admitted in the definition of size. With another C, the bound holds as well when the matrix is measured by its operator norm in $\mathrm{End}\, \mathcal{V}$. Steps 26 and 27 use this bound at $\mathsf{p}' = 2$, $2\mathsf{p}$ and p. The size is preserved by P^ε. Indeed, $\gamma_\lambda\gamma_5$ is a monomial matrix (Definition E.1), so that U_{r_λ} is monomial too, and each entry of $\mathrm{Ad}_\lambda \mathsf{z}$ equals an entry of z times a unimodular number. If the symbol of a kernel is supported in $\frac18\kappa_k \le \mathsf{s} \le 6\kappa_k$ and has modulus at most f_0, then (B0) gives $\|\beta\|^2 \le C\mathsf{f}_0^2\kappa_k$. For the factors of ξ_k^∞ this yields $\|\beta\| \le C\kappa_k^{1/2}$ and, as $\tilde\varphi_k\varphi_k = \varphi_k$ and $|\mathrm{Rsy}| \le 2/(c_E\kappa_k)$, also $\|\beta'\| \le C\kappa_k^{-1/2}$; so ξ_k^∞ has size C for every k. Denote by K the kernel on $\mathbb{Z}^3$ of either factor, and by $\hat{\mathsf{k}}$ the symbol in front of $d^*\mathcal{G}$; it is supported in $\frac18\kappa_k \le \mathsf{s} \le 6\kappa_k$ and obeys $|\hat{\mathsf{k}}| \le \mathsf{f}_0$. The truncated factor has the cochain $\mathsf{K}(\cdot - y)1[|\cdot - y|_\infty \le \mathsf{R}]$. Because $2\mathsf{R} + 2 \le 2\ell + 2 < n$, no two of its plaquettes are congruent modulo Γ, so that its norm in $C^2(\Gamma)$ does not exceed $\|\mathsf{K}\|_{\ell^2(\mathbb{Z}^3)}$. Plancherel's formula on $\mathbb{Z}^3$ (K8) and $d^*d \le \Delta$ give

$$\|\mathsf{K}\|^2_{\ell^2(\mathbb{Z}^3)} \le (2\pi)^{-3} \int_{[-\pi,\pi]^3} |\hat{\mathsf{k}}|^2\, \mathsf{q}^{-1}\, \mathrm{d}p \le C\mathsf{f}_0^2\kappa_k.$$

This is (B0), with the integral replacing the Riemann sum; the factors of ξ_k are therefore subject to the same two bounds. For a lattice difference of either untruncated factor of ξ_k^∞, the corresponding cochain norm bound acquires an additional factor $C\kappa_k$.

Step 23 (cut-off and period). One has

$$(\tilde\varphi_k * \mathsf{u}_\nu)(y) - \zeta(y)\, a_\nu^{[k]}(y) = A_\nu^{\mathrm{C}}(\mathsf{k}_1 + \mathsf{k}_2),$$

where

$$\mathsf{k}_1 := \tilde\varphi_k(y - \cdot)\, (\zeta - \zeta(y)), \qquad \mathsf{k}_2 := \sum_{\mathsf{m}\neq 0} \tilde\varphi_k(y + \mathsf{nm} - \cdot)\, \zeta.$$

In this identity $y \in \mathbb{T}'$ is lifted to a point of $\mathbb{Z}^3$ with $|y - k_B|_\infty \le 5\ell$, where k_B is the centre of B^{++} (Step 2 of the proof of Proposition 5.11). On B^{++} one has $|y' - k_B|_\infty \le \frac{7}{2}\ell$, and so the arguments of $\tilde\varphi_k$ in k_2 have $|\cdot|_\infty \ge 10\ell - 5\ell - \frac{7}{2}\ell = \frac{3}{2}\ell$. The function k_1 lives on $\mathbb{Z}^3$ and is read through (Per), with $\mathsf{k}_0 = C\kappa_k^3$ for the first factor and $\mathsf{k}_0 = C\kappa_k^2$ for the second, whose kernel is $\tilde S^{(k)}$ since $\tilde\varphi_k\varphi_k = \varphi_k$. From $|\zeta(y') - \zeta(y)| \le \min\{1, |y - y'|_\infty/\ell\}$, $(\kappa_k n)^{-13/2} \le (\kappa_k\ell)^{-1}$ and (Ker) one obtains for the periodisation of k_1

$$\begin{aligned} &\|\cdot\|_1 \le C(\kappa_k\ell)^{-1}, && \|\cdot\|_2^2 \le C\kappa_k^3(\kappa_k\ell)^{-2} && \text{in the first factor,}\\ &\|\cdot\|_1 \le C\kappa_k^{-1}(\kappa_k\ell)^{-1}, && \|\cdot\|_2^2 \le C\kappa_k(\kappa_k\ell)^{-2} && \text{in the second factor;}\end{aligned}$$

the function k_2 has finite support, and the same bounds hold for it. If $e = -e_\nu$, as well as in the second factor, the centre of the kernel is a point y_0 such that $|y_0 - y|_\infty \le 2$, namely $y - e_\nu$, respectively $y + e$ or $y + e - e_\nu$, while ζ is evaluated at y. In these cases $|\zeta(y') - \zeta(y)| \le \min\{1, (|y' - y_0|_\infty + 2)/\ell\}$, and the arguments in k_2 are at least $\frac{3}{2}\ell - 2 \ge \ell$; since $\kappa_k \le 2\sqrt{13}$, only C is altered. Accordingly (B1) gives $\|\beta\| \le C\kappa_k^{1/2}(\kappa_k\ell)^{-1}$, and in the same manner the bound $C\kappa_k^{-1/2}(\kappa_k\ell)^{-1}$ for the second factor, which is formed with $\tilde S^{(k)}$. Consequently

$$\Xi_k = -q^2\big[\zeta^2\xi_k^\infty + \mathrm{err}_k\big],$$

where err_k has size $C(\kappa_k\ell)^{-1}$, and $\sum_k(\kappa_k\ell)^{-1} \le 2$.

Step 24 (truncation; the leading part). Fix $m \le k$, and let $\mathsf{R} := \lceil \kappa_k^{-1} 2^{(k-m)/2}\rceil$; then $\mathsf{R} \le \ell$. A far part has as its cochain the n-periodisation of $\mathsf{K}(\cdot - y)1[|\cdot - y|_\infty > \mathsf{R}]$. The bound (Ker) gives $|\mathsf{K}(z)| \le C\mathsf{f}_1\kappa_k^3(1 + \kappa_k|z|_\infty)^{-8}$, where $\mathsf{f}_1 = \kappa_k^{-1}$ for the first kernel and $\mathsf{f}_1 = \kappa_k^{-2}$ for the second. Applying Minkowski's inequality to the sum over the images $n\mathsf{m}$, $\mathsf{m} \in \mathbb{Z}^3$, one bounds the norm of this cochain in $C^2(\Gamma)$ by

$$\begin{aligned}\|\mathsf{K}1_{>\mathsf{R}}\|_{\ell^2(\mathbb{Z}^3)} + \sum_{\mathsf{m}\neq 0} Cn^{3/2}\mathsf{f}_1\kappa_k^3\big(\tfrac12\kappa_k n|\mathsf{m}|_\infty\big)^{-8}&\\ \le C\mathsf{f}_1\kappa_k^{3/2}\big[(\kappa_k\mathsf{R})^{-13/2} + (\kappa_k n)^{-13/2}\big] &\le C\kappa_k^{\pm 1/2}(\kappa_k\mathsf{R})^{-6},\end{aligned}$$

the last inequality resting on $n > 10\mathsf{R}$ and $\kappa_k\mathsf{R} \ge 1$. The second kernel decays around $y + e$ rather than around y, and from $|\sigma - y|_\infty > \mathsf{R}$ one gets $|\sigma - y - e|_\infty > \mathsf{R} - 1$. Since $1 + \kappa_k|z - e|_\infty \ge (1 + \kappa_k)^{-1}(1 + \kappa_k|z|_\infty)$ and $\kappa_k \le 2\sqrt{13}$, this affects only C. Likewise for the first kernel at $e = -e_\nu$, whose centre of decay is $y - e_\nu$, and for the second kernel at $e' = -e_\nu$, whose centre of decay is $y + e - e_\nu$: here $1 + \kappa_k|z - v|_\infty \ge (1 + 2\kappa_k)^{-1}(1 + \kappa_k|z|_\infty)$ for $|v|_\infty \le 2$. The far parts of the two kernels thus have $\|\beta\| \le C\kappa_k^{\pm 1/2}(\kappa_k\mathsf{R})^{-6}$, the bounds (G1) and (G2) being valid for every $\beta \in C^2(\Gamma)$, closed or not. It follows that $\xi_k^\infty - \xi_k$ has size $C2^{-3(k-m)}$. Let $\mathsf{X}_\nu := \mathrm{Vx}_\nu(0) = -i(\mathrm{Ehp}_{e_\nu} - \mathrm{Ehp}_{-e_\nu})$ (see (17)). Up to a lattice difference, $\sum_e a^{[k]}(y, y+e)\mathrm{Ehp}_e$ equals $i\sum_\nu a_\nu^{[k]}(y)\mathsf{X}_\nu$. Hence $\xi_k^\infty = \mathsf{L}_k + \varrho_k$, with

$$\mathsf{L}_k(y) := -\sum_{\mu,\nu,z} a_\mu^{[k]}(y)\, a_\nu^{[k]}(y+z)\, \mathsf{X}_\mu \tilde S^{(k)}(z)\mathsf{X}_\nu$$

and with ϱ_k of size $C\kappa_k$.

Step 25 (the even sector). Put $\mathrm{Ad}_\iota := \mathrm{Ad}_1\,\mathrm{Ad}_2\,\mathrm{Ad}_3$. Then $\mathsf{P}^{+++} = \mathsf{P}^{+++}\,\frac12(1 + \mathrm{Ad}_\iota)$, and by Step 19 the map Ad_ι carries X_ν into $-\mathsf{X}_\nu$ and $\tilde S^{(k)}(z)$ into $\tilde S^{(k)}(-z)$. Hence $\frac12(1 + \mathrm{Ad}_\iota)\mathsf{L}_k$ is the expression L_k in which the kernel $\tilde S^{(k)}$ receives the symbol

$$\varphi_k\,\tfrac12\big(\mathrm{Rsy}(p) + \mathrm{Rsy}(-p)\big) = \varphi_k\Big[J\,\frac{\mathbf{m}'}{\mathbf{m}'^2 + |p|^2}\,J^* + \tfrac12\big(\mathrm{Er}(p) + \mathrm{Er}(-p)\big)\Big]$$

by Lemma E.16. The norm of this symbol does not exceed $4\mathbf{m}'\kappa_k^{-2} + C_E$. Being odd in p, the singular part $-iJ\frac{\gamma\cdot p}{\mathbf{m}'^2 + |p|^2}J^*$ cancels. It follows that $\mathsf{P}^{+++}\xi_k^\infty$ has size $C(\kappa_k + \mathbf{m}'/\kappa_k)$. Since $\varphi_k = 0$

for $k \ge \mathsf{J}+2$ and $\kappa_{\mathsf{J}+1} \le 2\sqrt{13}$, one has

$$\sum_{k=3}^{\mathsf{J}+1} \Big(\kappa_k + \frac{\mathbf{m}'}{\kappa_k} \Big) \le 4\sqrt{13} + \frac{1}{4}.$$

Step 26 (the seven odd sectors). Let $\varepsilon_\lambda = -1$, and let o_y denote the real part or the imaginary part of an entry of $(\mathsf{P}^\varepsilon \xi_k)(y;\cdot)$. By Steps 20 to 22, these functionals satisfy (L), (O) and (S) of Lemma 5.15 with $\mathsf{Z} = B^{++}$ and $\mathsf{t} = C$. Here $m \le k$ are the integers of Step 24, with $3 \le m \le \mathsf{J}$, as in (H), and $k \le \mathsf{J}+1$, as in the definition of Y_m in Step 17. Let $x \in \mathbb{T}'$ be the point that occurs in condition (H) of Step 18; the bounds of this step hold uniformly in x. Denote by the same letter a representative of x in $\mathbb{Z}^3$, chosen arbitrarily, so that the argument $x + \mathsf{nm} - y$ of the kernel of Φ_m on $\mathbb{Z}^3$ is a point of $\mathbb{Z}^3$. Another representative has the form $x + \mathsf{nm}_0$ with $\mathsf{m}_0 \in \mathbb{Z}^3$; the summand of index m formed with it equals the summand of index $\mathsf{m} + \mathsf{m}_0$ formed with x, so that the sums over $\mathsf{m} \in \mathbb{Z}^3$ in this step do not depend on the choice of the representative. For $\mathsf{m} \in \mathbb{Z}^3$ set $w = w_\mathsf{m} := \Phi_m(x + \mathsf{nm} - \cdot)\, \zeta^2$, and let $x^* = x^*_\mathsf{m}$ be the point of the projection of Z that is nearest to $(x + \mathsf{nm})_\lambda$. This point is unique, because $(x + \mathsf{nm})_\lambda$ is an integer, and it is an integer, as Lemma 5.15 requires, because $\mathsf{Z} \subset \mathbb{Z}^3$. As the symbol of Φ_m is real and even, its kernel is real, and so is ζ; moreover ζ vanishes off B^{++}, a box of $\mathbb{Z}^3$ on which the quotient map $\mathbb{Z}^3 \to \mathbb{T}'$ is injective (Step 9), and the kernel of Φ_m on $\mathbb{T}'$ is the periodisation of its kernel on $\mathbb{Z}^3$ ((K8) and the paragraph on filters). Hence $\sum_\mathsf{m} \sum_y w\, \mathsf{o}_y$ is the real part or the imaginary part of an entry of $\big[\Phi_m * (\zeta^2 \mathsf{P}^\varepsilon \xi_k)\big](x)$, namely of the entry and the part that define o_y. Lemma 5.15 is applied to each m separately, and Minkowski's inequality over m combines the resulting bounds. The cut-off ζ vanishes wherever $\operatorname{dist}_\infty(\cdot, B^+) \ge \ell$, that is, on the two outermost layers of B^{++} and off B^{++}. Extended by zero, w therefore has no jump across the faces of Z, and

$$|\partial w| \le |\nabla_\lambda \Phi_m| + \frac{2}{\ell}\, |\Phi_m|,$$

the kernel of the second term being taken at the shifted point; as in Step 24, the shift affects only the constant C. Apply (Ker) at $\kappa = \kappa_m$ to Φ_m and, with $\mathsf{f}_0 = C\kappa_m$, to the symbol $\Phi_m(p)(1 - \exp(\mp i p_\lambda))$ of $\nabla_\lambda \Phi_m$. Together with $\ell^{-1} \le \mathbf{m}' \le \kappa_m$ this yields

$$|\partial w(y)| \le C\kappa_m^4 (1 + \kappa_m |x + \mathsf{nm} - y|_\infty)^{-8}.$$

Moreover $|y_\lambda - x^*| \le |y_\lambda - (x + \mathsf{nm})_\lambda|$, because x^* is the point nearest to $(x + \mathsf{nm})_\lambda$ of an interval to which y_λ belongs. The sets $x + \mathsf{nm} - \mathsf{Z}$, $\mathsf{m} \in \mathbb{Z}^3$, are mutually disjoint, as $7\ell < \mathsf{n}$. Consequently

$$\begin{aligned} \sum_\mathsf{m} \sum_y (|y_\lambda - x^*| + 1 + a_0)^{1/2} |\partial w| &\le \sum_{z \in \mathbb{Z}^3} (|z|_\infty + 1 + a_0)^{1/2} C\kappa_m^4 (1 + \kappa_m |z|_\infty)^{-8} \\ &\le C(\kappa_m^{-1} + a_0)^{1/2} \kappa_m, \end{aligned}$$

where, owing to $\kappa_k \le 2\sqrt{13}$,

$$a_0 \kappa_m \le 2 \cdot 2^{-(k-m)/2} + 6\kappa_k 2^{-(k-m)} \le 45.3 \cdot 2^{-(k-m)/2}.$$

Lemma 5.15 then gives

$$\Big\| \sum_\mathsf{m} \sum_y w\, \mathsf{o}_y \Big\|_{L^2} \le C\, 2^{-(k-m)/4},$$

while (G2) bounds the norm of the same sum in $L^{2\mathsf{p}}$ by $C\mathsf{p}$. Lyapunov's inequality (K9) shows that its norm in L^p is at most $C\mathsf{p}\, 2^{-(k-m)/(4(\mathsf{p}-1))}$, and the sum over k of these numbers does not exceed $C\mathsf{p}^2$.

Step 27 ((H) for one box and one level). In ξ_k take $\mathsf{R} = \mathsf{R}(k,m)$, the number of Step 24. From Steps 17 and 23 and from $\sum_\varepsilon \mathsf{P}^\varepsilon = 1$ one obtains

$$Y_m = -q^2 \sum_{k=m}^{\mathsf{J}+1} \Big[\Phi_m * (\zeta^2 \mathsf{P}^{+++} \xi_k^\infty) + \sum_{\varepsilon \neq +++} \Phi_m * (\zeta^2 \mathsf{P}^\varepsilon \xi_k) + \sum_{\varepsilon \neq +++} \Phi_m * \big(\zeta^2 \mathsf{P}^\varepsilon (\xi_k^\infty - \xi_k)\big) + \Phi_m * \mathrm{err}_k \Big];$$

here the even sector enters without truncation, and the odd sectors enter truncated. For each real component of the four terms, Steps 25, 26, 24 and 23 give, in this order, the bounds

$$C\mathsf{p}\Big(\kappa_k + \frac{\mathbf{m}'}{\kappa_k}\Big), \qquad C\mathsf{p}\, 2^{-(k-m)/60}, \qquad C\mathsf{p}\, 2^{-3(k-m)}, \qquad C\mathsf{p}\, (\kappa_k \ell)^{-1}$$

on the L^{p}-norm. Summed over k, these bounds are at most $C\mathsf{p}$ times $4\sqrt{13} + \frac{1}{4}$ (because $m \geq 3$), $(1 - 2^{-1/60})^{-1} < 87.07$, $\frac{8}{7}$ and 2, respectively. Minkowski's inequality over k, over the sectors and over the $2 \cdot 192^2$ components, combined with Steps 23 to 26 and with $\|\Phi_m\|_1 \leq C$, consequently yields (H), the constant being $\mathsf{C}_{\mathsf{p}} = C\mathsf{p}^2$. Fix $\epsilon_3 > 0$ and $c_* \in (0, c_6]$, sufficiently small for Steps 14 and 18 to imply (C2), and let $r := c_* \mathbf{m}'/q^2 - 1$. Define

$$\mathcal{G} = \mathcal{G}(B, \mathbf{m}') := \begin{cases} \mathcal{G}^{\mathrm{lin}}(r) \cap \mathcal{G}^{\mathrm{qu}} & \text{if } r \geq 1, \\ \emptyset & \text{otherwise.} \end{cases}$$

If $r \geq 1$, Steps 11 and 18 and the inequality

$$\exp(-r) \leq \exp(1)\, 16! \Big(\frac{q^2}{c_* \mathbf{m}'}\Big)^{16}$$

lead to

$$P_n(\mathcal{G}^c) \leq C_{11} (\ell \mathbf{m}')^3 \Big(\frac{q^2}{\mathbf{m}'}\Big)^{16},$$

with a constant $C_{11} \geq (2/c_*)^{16}$ depending on ρ alone. If $r < 1$, then $q^2/\mathbf{m}' > c_*/2$; as $\ell \mathbf{m}' \geq 1$, it follows that $C_{11}(\ell \mathbf{m}')^3 (q^2/\mathbf{m}')^{16} \geq 1 \geq P_n(\mathcal{G}^c)$, so that the same bound is valid. On $\mathcal{G}$, it follows from Step 3 above and from Step 2 of the proof of Proposition 5.11 that the number $N_B(\lambda')$ of eigenvalues not exceeding λ' of the form $\|D_A \phi\|^2$ on the spinors over B satisfies

$$N_B(\lambda') \leq C_3 |B| \big(\sqrt{\lambda'} + \mathbf{m}'\big)^3.$$

The gauge transformation and unitary identification of Step 9 identify this count with the count for the original field on B.

Step 28 (levels). Let $\mathbf{m}$, $\mathbf{m}'$ and n be as in the theorem, and set $\ell := \lceil 1/\mathbf{m}' \rceil$. In the case $\ell < 4$ one has $N_A \leq 192 n^3 \leq 192 \cdot 27\, \mathbf{m}'^3 n^3$. In the case $\ell \geq 4$ one has $10\ell < n$, and Step 1 of the proof of Proposition 5.11 gives $N_A(\lambda) \leq \sum_a N_a(\lambda')$, where $\lambda' = 2\lambda + c_J \ell^{-2}$ and the number of boxes B_a is at most $(n/\ell)^3$. For each integer $k \geq 0$ with $\mathbf{m}'_k := 2^k \mathbf{m}' \leq 1$ let $\mathcal{G}_{a,k} := \mathcal{G}(B_a, \mathbf{m}'_k)$, formed with the same ℓ. As $\ell \mathbf{m}' \leq 2$ and $q^2 \leq \mathbf{m}'$, Step 27 yields, also when $\mathcal{G}_{a,k} = \emptyset$,

$$P_n(\mathcal{G}_{a,k}^c) \leq C_{11} (\ell \mathbf{m}'_k)^3 \Big(\frac{q^2}{\mathbf{m}'_k}\Big)^{16} = C_{11} (\ell \mathbf{m}')^3 \Big(\frac{q^2}{\mathbf{m}'}\Big)^{16} 2^{-13k} \leq 8 C_{11} 2^{-13k}.$$

Here Steps 1 to 27 are applied to the pair $(\ell, \mathbf{m}'_k)$. This pair satisfies Setting (S), because $\mathbf{m} \leq \mathbf{m}'_k \leq 1$ and $\ell \mathbf{m}'_k \geq 1$. The conditions of Setting (S) bound $\ell \mathbf{m}'$ from below only, and for the largest admissible k one has $\ell \mathbf{m}'_k > \ell/2$. Every bound of Steps 1 to 27 in which ℓ and $\mathbf{m}'$ occur together (Setting (S) enumerates these bounds, in the paragraph on the dependence on ℓ) holds as soon as $\ell \mathbf{m}' \geq 1$, or becomes better as $\ell \mathbf{m}'$ increases, with the exception of the factor $(\ell \mathbf{m}')^3$ of Steps 8, 11 and 18. This factor is retained; in the estimate above it appears as $(\ell \mathbf{m}'_k)^3 = (\ell \mathbf{m}')^3 2^{3k}$.

Step 29 (summation over the levels). Denote by $k_{\max}$ the largest integer k for which $2^k \mathbf{m}' \leq 1$. On the event that $\mathcal{G}_{a,k}$ is the first of $\mathcal{G}_{a,0}, \ldots, \mathcal{G}_{a,k_{\max}}$ to occur, $N_a(\lambda')$ is estimated by Step 27

at the level k. On the event that none of them occurs, $N_a \le 192(4\ell)^3(2^{k_{\max}+1}\mathbf{m}')^3$. With $C_{12} := \max\{C_3, 192\}\max\{1, 8C_{11}\}$ it follows that

$$E_n N_a(\lambda') \le C_{12}(4\ell)^3(\sqrt{\lambda'} + \mathbf{m}')^3\Big[1 + \sum_{k\ge 1} 2^{3k}2^{-13(k-1)}\Big],$$

which gives the first bound in the display of Theorem 5.14. The second bound is a consequence of the formula

$$\mathsf{T}_n = 2\int_{\mathbf{m}^2}^{\infty} \lambda^{-3}n^{-3}E_n N_A(\lambda)\,\mathrm{d}\lambda$$

and of the inequality $(\xi+\zeta)^3 \le 4(\xi^3+\zeta^3)$, as in Step 8 of the proof of Proposition 5.11. The third bound results from $\mathsf{c}_n \le 3\pi^2\mathsf{h}_3^4 q^2 \mathsf{T}_n$, which is part of Theorem 5.7.

Step 30 (continuum). Choose $\mathbf{m} := \boldsymbol{\mu}_N$ and $\mathbf{m}' := \max\{\boldsymbol{\mu}_N, e^2\eta\} \le 1$. Then $\mathbf{m}'/\mathbf{m} \le \mathsf{g}_*$ and $g \le e^2/(c_{\boldsymbol{\mu}}\bar{m})$. Assertion (a) follows from Lemma 5.10 (a) together with $B_n(t_0) \le 3\mathsf{S}_n$; the latter inequality holds since the terms of $B_n(t_0)$ are non-negative and since, by Lemma 5.9 (a), the corresponding sum over all y equals $3\mathsf{S}_n$. For (b), Proposition 5.6 (a) and Theorem 5.7 give

$$K_n(u,u) = I_n(u) - \|du\|^2 \le \frac{\pi^2}{4}\,\mathsf{c}_n\|du\|^2.$$

Assertion (c) is Proposition 5.13 (e), applied with $C_M/3$ in place of C_M. □

5.12. **The tuned-mass sum rule.**

Theorem 5.16. *The tuned masses: a bound from above, a sum rule, a bound from below.* Let $0<\rho<1$, let the weights be standard, let $\bar{m} \in (0,1]$, and let $e>0$ be such that $e_K = e\,h^{1/2} \le e_1$; the data of Section F.2 satisfy this condition. Let $N \ge N_{\min}(R_{\mathrm{top}}(e), e)$, and denote by $(\varepsilon_k^{\mathrm{v},t}, m_k, E_k)$ the trajectory of the tuned pair $P_N(e)$ on $R_{\mathrm{top}}(e)$ [4, Section 16.4, Theorems 16.3–16.4]; in particular $m_{k_\star} = 0$, where $k_\star := k_{\mathrm{top}}$. Use the construction constants of [4, Proposition 24.1], the RG notation of Definitions F.1–F.6, and the free-coercivity constant c_b of [4, Proposition 9.4]. Put $K_{\mathrm{pol}} = 72^{54}/36$, the constant in its equation (A.3). The index $\star$ refers to the level $k_\star$: the numbers $a_\star \in \{h, h/L\}$, $e_\star = e\,a_\star^{1/2} \le e_K$, $\mu_\star = \bar{m}\,a_\star \in (L^{-2}, 1]$ and $\ell_\star$ are independent of N, and $L^{-k_\star} = \eta/a_\star$ [4, Proposition 16.1 and Section 16.1]. For $0 \le j \le k_\star$ put $a_j := L^{j-N}$; the running charge of level j is $e_j := e\,a_j^{1/2} = q\,L^{j/2}$ [4, Proposition 16.1 and Sections 15–16]. In formulas with a varying index j, e_j always denotes the running charge, including $e_0 = q$, $e_1 = L^{1/2}q$ and $e_2 = Lq$. In the condition $e_K \le e_1$, the symbol e_1 denotes the fixed charge ceiling of [4, Proposition 24.1]. Put

$$d_\star := \big(b_\star^{-1} + \mu_\star^{-1}\big)^{-1}, \qquad Z_0 := b_\star^2\,(b_\star+\mu_\star)^{-2}, \qquad c_\star := d_\star - \frac{d_\star^2}{b_\star} = \mu_\star Z_0.$$

Here $b_\star = b_{k_\star} = b_-\,(1-\eta/a_\star)^{-1}$, with $b_- = b(1-L^{-1})$, as in Definition F.5. As $a_\star$ is independent of N and $\eta = L^{-N}$, the number $b_\star$ decreases to b_- as $N\to\infty$; thus $b_\star$, $d_\star$, Z_0 and $c_\star$ depend on N. With the number $Z_0^\infty := b_-^2\,(b_- + \mu_\star)^{-2} > 0$, which does not depend on N, one has

$$Z_0^\infty \le Z_0 \le 1, \qquad \mu_\star Z_0^\infty \le c_\star \le \mu_\star,$$

and Z_0 decreases to Z_0^∞ as $N\to\infty$. In this theorem and its proof, $n := R_{\mathrm{top}}(e)/\eta = L^{k_\star}\ell_\star = \ell_0$ is odd by [4, Proposition 16.1]; this n differs from the number $n = 8hL^{T+N}$ of the notation fixed above and of Theorem 5.14, whereas η, q and $\boldsymbol{\mu}_N$ have the same meaning as there.

Let Ω be the constant bond function on $T^0_{k_\star}$ with $e_\star\Omega_\mu\ell_\star = \pi$. For $\zeta = (\zeta_j)_{j<k_\star} \in \prod_j \mathrm{Box}_j$ define $A_{k_\star} := \Omega$ and, for $j < k_\star$, $A_j := A_j^{\min}(A_{j+1}) + \mathrm{Sq}_j\zeta_j$ [4, equation (22.1)]. Throughout this theorem and its proof, D_A is the massless operator of Definition F.5, and $\nabla_A^*\nabla_A$ has the meaning given there. The operators D_{A_0}, $D_0 := D_A|_{A=0}$ and D_{Ω_0} act at level zero, with mesh 1 and charge q, on $\mathfrak{S}^{\mathrm{ch}}(1,n)$; the connection Ω_0 is defined in Step 13 of the proof. the operator D_Ω is that of level $k_\star$, with mesh $L^{-k_\star}$ and charge $e_\star$. At level zero, $D_A + \boldsymbol{\mu}$ is the operator of Definition E.1 with scalar

lattice mass $\mathbf{m} = \boldsymbol{\mu}$. On $\mathcal{H}_0 := \mathfrak{S}^{\mathrm{ch}}(1,n)$ let $\mathcal{D}_0 := D_{A_0} + \boldsymbol{\mu}_N$. With $\mathcal{H}_j := \mathfrak{S}(1,\ell_j)$, the map $\mathcal{R} = \mathcal{R}(\zeta)\colon \mathcal{H}_0 \to \mathcal{H}_{k_\star}$ is the product of the averaging maps, defined in Step 1 of the proof. Set

$$B := L^{k_\star}\mathcal{D}_0 + b_\star\,\mathcal{R}^*\mathcal{R}, \qquad \mathcal{D} := b_\star - b_\star^2\,\mathcal{R}B^{-1}\mathcal{R}^*,$$

let P_N be the measure with

$$\mathrm{d}\mathsf{P}_N \propto \det B \prod_j \mathbf{1}_{\mathrm{Box}_j}\,\mathrm{d}\mu_I(\zeta_j),$$

and let E_N denote its mean. Finally, with $u := (1,0,0,0)$, let

$$\Psi_u(y) := \ell_\star^{-3/2}\exp(-ie_\star\Omega\cdot y)\,u, \qquad \psi := b_\star B^{-1}\mathcal{R}^*\Psi_u, \qquad \Phi := \mathcal{D}\Psi_u,$$
$$Z_N := \mathsf{E}_N\|\psi\|^2, \qquad \mathsf{W}_N := \mathsf{E}_N\,\frac{\rho}{2}\,(\psi,\nabla_{A_0}^*\nabla_{A_0}\psi)^{\mathrm{ch}} \geq 0;$$

the number W_N is the Wilson energy of the dressed zero mode.

(a) *The flow density as a mixture of Gaussians with a non-negative weight.* Whatever the real value of $\boldsymbol{\mu}_N$, one has $\det B \geq 0$, the measure P_N is a probability measure, and

$$\frac{\mathscr{F}_{k_\star}(\Omega;\Psi)}{\mathscr{F}_{k_\star}(\Omega;0)} = \mathsf{E}_N\exp(-\langle\bar\Psi,\mathcal{D}\Psi\rangle), \qquad \mathscr{F}_{k_\star} := \mathscr{F}_{k_\star}[\varepsilon_{k_\star}^{\mathrm{v},t},0,E_{k_\star}].$$

If $\boldsymbol{\mu}_N > 0$, then $\mathrm{Re}\,\mathcal{D}(\zeta) \geq \big(b_\star^{-1} + (L^{k_\star}\boldsymbol{\mu}_N)^{-1}\big)^{-1}$ for every ζ.

(b) *From above.* For all $N \geq 0$,

$$m_{0,N}(e) \leq 10^{-4}\,e_\star^{3/4-10\varepsilon}\,\bar m\,\eta.$$

Consequently $\limsup_N \boldsymbol{\mu}_N/\eta \leq \bar m(1+10^{-4}e_K^{3/4-10\varepsilon})$; the tuned assignment is therefore physical from above, that is, $\boldsymbol{\mu}_N \leq C\eta$ for large N.

(c) *A sum rule, for every real value of* $\boldsymbol{\mu}_N$. At every ζ with $\det B \neq 0$ the identity **(SR)**

$$\mathrm{Re}(\Psi_u,\mathcal{D}\Psi_u) = b_\star^{-1}\|\Phi\|^2 + L^{k_\star}\Big[\boldsymbol{\mu}_N\|\psi\|^2 + \frac{\rho}{2}\,(\psi,\nabla_{A_0}^*\nabla_{A_0}\psi)^{\mathrm{ch}}\Big] \tag{SR}$$

holds. In the mean,

$$\boldsymbol{\mu}_N Z_N + \mathsf{W}_N = \frac{\eta}{a_\star}\,c_\star\,(1+\vartheta_N), \qquad |\vartheta_N| \leq e_\star^{3/4-10\varepsilon}, \qquad Z_N \geq \tfrac12 Z_0.$$

(d) *From below.* For all large N,

$$\boldsymbol{\mu}_N \geq -2\rho\,C_5\,q^{2-\varepsilon}, \qquad C_5 := 108\,c_b^{-1} + 6.$$

For comparison, the bound $|m_{0,N}(e)| \leq \frac18 q^{3/4-8\varepsilon}$ of [4, Theorem 16.4, equation (16.19)] gives $\boldsymbol{\mu}_N \geq -\frac18 q^{3/4-8\varepsilon}$.

■□

Proof of Theorem 5.16. Step 1 (averaging maps). By [4, Proposition 10.1 and equation (10.29)] the functions $\mathscr{B}_j := H_j A_j^{\min}(A_{j+1})$ and $\omega_j := \omega_j(A_{j+1})$ are real. The map $(\mathsf{S}_j v)(y') := L^{-3/2}v(y'/L)$ is unitary from $\mathcal{H}_{j+1}$ onto $\mathfrak{S}(L,\ell_j)$, with its weighted inner product $\langle\cdot,\cdot\rangle_L$. The dilation in Definition F.6 is $v_L = L^{1/2}\mathsf{S}_j v$. Let

$$R_j := \mathsf{S}_j^{-1}\exp(-ie_j\omega_j)\,Q(\mathscr{B}_j) \quad (0<j<k_\star), \qquad R_0 := \mathsf{S}_0^{-1}\exp(-iq\omega_0)\,Q(\mathscr{B}_0)J^*,$$

and $\mathcal{R} := R_{k_\star-1}\cdots R_0$. It follows from [4, Section 11.2, equation (11.8)] that

$$R_j^* = Q^T(-\mathscr{B}_j)\exp(ie_j\omega_j)\,\mathsf{S}_j \quad (0<j<k_\star), \qquad R_0^* = JQ^T(-\mathscr{B}_0)\exp(iq\omega_0)\,\mathsf{S}_0,$$

that $R_jR_j^* = 1$, and that every R_j commutes with the spin matrices.

Step 2 (a single Gaussian integration). Suppose that $\mathcal{D}_j$ acts on $\mathcal{H}_j$ and that $G_j := \mathcal{D}_j + b_L R_j^* R_j$ is invertible. In the block transformation $\mathbf{T}_j$ of Definition F.6, the two averaging maps obey $\langle Q(-\mathscr{B})\bar\Psi, v'\rangle_L = \langle \bar\Psi, Q^T(-\mathscr{B})v'\rangle$. Hence [4, Section 11.5, equation (11.28)] yields

$$\int \mathrm{D}\Psi_j \, \exp\{-b_L\langle\cdots\rangle_L - \langle\bar\Psi_j, \mathcal{D}_j\Psi_j\rangle\} = \det G_j \, \exp\bigl(-\langle\bar\Psi_{j+1}, \mathcal{D}_{j+1}\Psi_{j+1}\rangle\bigr),$$

where $\mathcal{D}_{j+1} := L\bigl[b_L - b_L^2 R_j G_j^{-1} R_j^*\bigr]$. When $\mathcal{D}_j$ is invertible, (K10) gives $\mathcal{D}_{j+1}^{-1} = b^{-1} + L^{-1} R_j \mathcal{D}_j^{-1} R_j^*$.

Step 3 (accretivity). If $\operatorname{Re}\mathcal{D}_j \geq d > 0$, then $\operatorname{Re}\mathcal{D}_{j+1} \geq \bigl(b^{-1} + (Ld)^{-1}\bigr)^{-1}$. Indeed, let $\Phi' = \mathcal{D}_{j+1}\Psi'$ and $\phi = \mathcal{D}_j^{-1} R_j^* \Phi'$. Then $\Psi' = b^{-1}\Phi' + L^{-1} R_j \phi$ and

$$\operatorname{Re}(\Psi', \Phi') = b^{-1}\|\Phi'\|^2 + L^{-1}\operatorname{Re}(\phi, \mathcal{D}_j\phi) \geq b^{-1}\|\Phi'\|^2 + L^{-1} d\,\|\phi\|^2,$$

while the Schwarz inequality gives

$$\|\Psi'\|^2 \leq \bigl(b^{-1} + (Ld)^{-1}\bigr)\bigl(b^{-1}\|\Phi'\|^2 + L^{-1} d\,\|\phi\|^2\bigr).$$

Definition F.5 gives $\operatorname{Re}\mathcal{D}_0 \geq \boldsymbol{\mu}_N$ and $b_1 = b$, $b_{j+1}^{-1} = b^{-1} + L^{-1} b_j^{-1}$ for $j \geq 1$. For $\boldsymbol{\mu}_N > 0$ the implication above, used at $j = 0$ with $d = \boldsymbol{\mu}_N$ and then by induction on $j \geq 1$, therefore gives for every ζ and $1 \leq k \leq k_\star$

$$\operatorname{Re}\mathcal{D}_k \geq \bigl(b_k^{-1} + (L^k \boldsymbol{\mu}_N)^{-1}\bigr)^{-1}.$$

Applying Step 2 with the same start, at $j = 0$ with $b_1 = b$, then by induction on $j \geq 1$, and using $R_j R_j^* = 1$, one finds $\mathcal{D}_{k_\star}^{-1} = b_\star^{-1} + L^{-k_\star}\mathcal{R}\mathcal{D}_0^{-1}\mathcal{R}^*$; by (K10) this means $\mathcal{D}_{k_\star} = \mathcal{D}$.

Step 4 (the identity of a step, iterated). At every level $j < k_\star$, the actual real data obey $|m_j| \leq w_j^{\mathrm{m}}$, $\|\mathscr{E}_j\|_j^x \leq w_j^{\mathrm{E}}$, $\mathscr{E}_j|_0 = E_j$ and $\mathfrak{P}_j(\cdot, 0) = 0$ [4, Proposition 16.2, Theorems 16.3–16.4]. Constant connections belong to Dom_k. Proposition 10.4 of [4] therefore gives $A_j \in \mathrm{Dom}_j$ at each preceding level. Starting from

$$\mathscr{F}_0 = \det(D_0 + \bar m\eta)^{-1} \exp\{-\langle\bar\Psi_0, \mathcal{D}_0\Psi_0\rangle^{\mathrm{ch}} - \varepsilon_{\mathrm{in}}^{\mathrm{v}} n^3\}$$

Iterating [4, Theorem 14.8, equations (14.47)–(14.48)] at zero external slot, with the block marginal (13.20) there, gives

$$\mathscr{F}_{k_\star}(\Omega; \Psi) = \int \prod_j \mathbf{1}_{\mathrm{Box}_j} \, \mathrm{d}\mu_I(\zeta_j) \, \mathcal{G}(\zeta; \Psi);$$

the integrand $\mathcal{G}$ arises by iterating the Berezin integration of Step 2, the factors c_j included.

Step 5 (the weight). Let $\boldsymbol{\mu}_N > 0$. By Steps 2 and 3, $\mathcal{G} = \mathsf{w}\exp(-\langle\bar\Psi, \mathcal{D}\Psi\rangle)$ with $\mathsf{w} \propto \prod_j \det G_j$. Integration over Ψ, together with $k_\star$ applications of [4, equation (13.20)], gives

$$\mathsf{w}\det\mathcal{D} = \exp\bigl(-\varepsilon_{\mathrm{in}}^{\mathrm{v}} n^3\bigr)\,\frac{\det\mathcal{D}_0}{\det(D_0 + \bar m\eta)},$$

and by (K10), $\det\mathcal{D}\,\det B = b_\star^{4n_{k_\star}}\det\bigl(L^{k_\star}\mathcal{D}_0\bigr)$. Hence $\mathsf{w} = C_N \det B$ with a constant $C_N > 0$.

Step 6 (all real masses; proof of (a)). At a given ζ, each coefficient of $\mathcal{G}$ is a polynomial function of the mass, whereas each coefficient of $C_N \det B \exp(-\langle\bar\Psi, \mathcal{D}\Psi\rangle)$ is a rational function of it; they coincide for $\boldsymbol{\mu}_N > 0$ by Step 5, and so, by (K11), for every real $\boldsymbol{\mu}_N$ at which $\det B \neq 0$. By Definition F.5 and Step 1, each of $X = B$ and $X = \mathcal{D}$ satisfies

$$[X, \Sigma_3] = 0, \tag{P1}$$

$$X^* = \gamma_5 X \gamma_5, \tag{P2}$$

and γ_5 interchanges the eigenspaces of Σ_3; hence $\det B = |\det B^+|^2 \geq 0$, where B^+ is the block of B on the eigenspace of Σ_3 for the eigenvalue 1. The function $\det B$ is real analytic in ζ, and it does not vanish identically, because $\mathscr{F}_{k_\star}(\Omega; 0) \neq 0$; by (K11), $\det B \neq 0$ almost everywhere. Since $\Sigma_3 u = u$, the function

$$\det B\,\|B^{-1}\mathcal{R}^*\Psi_u\|^2 = \|\operatorname{adj}(B^+)\,\mathcal{R}^*\Psi_u\|^2$$

is bounded, and therefore $\|\psi\|^2$, $\|\Phi\|^2$ and $(\psi, \nabla^*\nabla\psi)$ are integrable.

Step 7 (the vector Ψ_u). The photon minimizer preserves constants, so $H_{k_\star}\Omega = \Omega$; the domain statement in [4, Proposition 10.4] gives $\Omega \in \frac{1}{4}\widetilde{R}_{k_\star}$. The function $f(c,g) := \ell_\star^{-3/2}\exp(-ie_\star\Omega\cdot c)\,u$ is antiperiodic and covariantly constant with respect to Ω; The covariantly constant identities in Definition F.5 give $D_\Omega f = 0$, $Q_{k_\star}(\Omega)J^*f = \Psi_u$ and $JQ^T_{k_\star}(-\Omega)\Psi_u = f$. Its formulas for $S_{k_\star}$ and $D_{k_\star}$ then yield $S_{k_\star}(\Omega)f = (\mu_\star + b_\star)^{-1}f$ and $D_{k_\star}(\Omega)\Psi_u = d_\star\Psi_u$.

Step 8 (degrees two and four). Since $m_{k_\star} = 0$, the density of Definition F.6 has the expansion

$$\mathscr{F}_{k_\star}(\Omega;\Psi) = \mathscr{F}_{k_\star}(\Omega;0)\exp\{-\langle\bar\Psi, D_{k_\star}(\Omega)\Psi\rangle + \mathcal{A}_2 + \mathcal{A}_4 + \dots\},$$

where $\mathcal{A}_{2i}$ denotes the part of degree (i,i) of $\sum_X E_{k_\star}(X;\Omega,H^{\#}_{k_\star}(\Omega)\Psi)$. Write $\mathcal{A}_2 =: \langle\bar\Psi, K_E\Psi\rangle$ and $\delta\mathcal{D} := \mathcal{D} - \mathsf{E}_N\mathcal{D}$. A comparison of the logarithms of the two sides of the identity in (a) gives

$$\mathsf{E}_N\mathcal{D} = D_{k_\star}(\Omega) - K_E.$$

Let a, b be indices with $\Sigma_3 = +1$ and c, d indices with $\Sigma_3 = -1$. As $\delta\mathcal{D}_{ad} = \delta\mathcal{D}_{cb} = 0$ by (P1), the coefficient of $\bar\Psi_a\Psi_b\bar\Psi_c\Psi_d$ in $\mathcal{A}_4$ is $\pm\mathsf{E}_N[\delta\mathcal{D}_{ab}\,\delta\mathcal{D}_{cd}]$.

Step 9 (sizes). Apply [4, equation (21.28)], Definition F.1, the polymer bound $c_\kappa \le K_{\rm pol}$ from equation (A.3) there, and the trajectory bound $\|E_{k_\star}\|_{k_\star} \le 4e_\star^{1/4-6\varepsilon}$ from its Theorem 16.4. They give

$$\Big|\sum_X E_{k_\star}(X;\Omega,H^{\#}_{k_\star}(\Omega)\Psi)\Big|_{\rm aw_2} \le 4e_\star^{1/4-6\varepsilon}K_{\rm pol}\Big(\frac{\ell_\star}{M}\Big)^3, \qquad {\rm aw}_2 = C_0^{-1}e_\star^{-1/4+\varepsilon}.$$

Therefore the moduli of the entries of K_E have a sum not larger than $\ell_\star^3\delta_\star$, and $|\mathcal{A}_4|_1 \le \ell_\star^3\delta_2$, with

$$\delta_\star := 4C_0^2K_{\rm pol}M^{-3}e_\star^{3/4-8\varepsilon}, \qquad \delta_2 := 4C_0^4K_{\rm pol}M^{-3}e_\star^{5/4-10\varepsilon}.$$

The terminal inequalities (21) and the identity $c_{\rm fl} = 3294K_{\rm pol}$ in [4, Proposition 24.1] imply

$$\ell_\star^3\delta_\star \le 10^{-5}e_\star^{3/4-10\varepsilon}d_\star, \qquad \delta_2 \le 10^{-6}e_\star^{5/4-12\varepsilon}d_\star^2.$$

With Step 7 it follows that the number $\bar z := (\Psi_u, \mathsf{E}_N\mathcal{D}\Psi_u)$ satisfies $|\bar z - d_\star| \le \delta_\star$, and that $\big|\,\|\mathsf{E}_N\mathcal{D}\Psi_u\| - d_\star\big| \le \ell_\star^3\delta_\star$.

Step 10 (proof of (b)). If $\boldsymbol{\mu}_N \le 0$, there is nothing to prove, and for $N < N_{\min}$ one has $m_{0,N} = 0$ by definition. Let $\boldsymbol{\mu}_N > 0$. Taking the mean in (a), one finds

$$\big(b_\star^{-1} + (L^{k_\star}\boldsymbol{\mu}_N)^{-1}\big)^{-1} \le \operatorname{Re}\bar z \le d_\star(1+t), \qquad t := \delta_\star/d_\star.$$

Hence

$$(L^{k_\star}\boldsymbol{\mu}_N)^{-1} \ge d_\star^{-1}(1-t) - b_\star^{-1} = \mu_\star^{-1}(1 - t(1+\mu_\star/b_\star)) \ge \mu_\star^{-1}(1-3t),$$

so that $L^{k_\star}\boldsymbol{\mu}_N \le \mu_\star(1+6t)$; moreover $L^{-k_\star}\mu_\star = \bar m\eta$.

Step 11 (the identity (SR)). Let B be invertible. Then $\Phi = b_\star(\Psi_u - \mathcal{R}\psi)$, and $\mathcal{R}^*\Phi = B\psi - b_\star\mathcal{R}^*\mathcal{R}\psi = L^{k_\star}\mathcal{D}_0\psi$. Consequently

$$(\Psi_u,\Phi) = b_\star^{-1}\|\Phi\|^2 + (\mathcal{R}\psi,\Phi) = b_\star^{-1}\|\Phi\|^2 + L^{k_\star}(\psi,\mathcal{D}_0\psi),$$

and (SR) results on taking real parts by means of Definition F.5.

Step 12 (proof of (c)). Denote by $a \mapsto \hat a$ the permutation of the indices given by γ_5. By (P2), $\overline{\delta\mathcal{D}_{ab'}} = \delta\mathcal{D}_{\hat b'\hat a}$, and Steps 8 and 9 give

$$\mathsf{E}_N\|\delta\mathcal{D}\Psi_u\|^2 = \sum_{a,b,b'}\mathsf{E}_N[\delta\mathcal{D}_{ab}\,\delta\mathcal{D}_{\hat b'\hat a}]\;\Psi_u(b)\,\overline{\Psi_u(b')} \le \ell_\star^{-3}|\mathcal{A}_4|_1 \le \delta_2.$$

Averaging (SR), one obtains

$$L^{k_\star}(\boldsymbol{\mu}_N Z_N + \mathsf{W}_N) = \operatorname{Re}\bar z - b_\star^{-1}(\|\mathsf{E}_N\mathcal{D}\Psi_u\|^2 + \mathsf{E}_N\|\delta\mathcal{D}\Psi_u\|^2).$$

By Step 9 and the inequalities $b_\star^{-1} \le 2$, $d_\star \le 1$ and $c_\star \ge \frac{1}{3} d_\star$, the right side differs from $c_\star = d_\star - d_\star^2/b_\star$ by at most $e_\star^{3/4-10\varepsilon} c_\star$; in addition $\mathsf{E}_N \|\Phi\|^2 \le 2$. Finally, $\|\psi\| \ge |(\Psi_u, \mathcal{R}\psi)| = |1 - b_\star^{-1}(\Psi_u, \mathcal{D}\Psi_u)|$, whence

$$Z_N \ge |1 - \bar{z}/b_\star|^2 \ge \big(Z_0^{1/2} - 2\delta_\star\big)^2.$$

Step 13 (a gauge with a small potential). Put $\mathscr{A}_j := H_j A_j$. By [4, equation (10.29)],

$$\mathscr{A}_j = (\mathscr{A}_{j+1})_L - d\omega_j + H_j \mathrm{Sq}_j \zeta_j,$$

with $H_0 = 1$ and $\mathscr{A}_{k_\star} = \Omega$. It follows that $A_0 = \Omega_0 + \tilde{A} - d\lambda$ with a real λ, where

$$\Omega_0 := L^{-k_\star/2}\Omega, \qquad \tilde{A} := \sum_{j<k_\star} \big(H_j \mathrm{Sq}_j \zeta_j\big)_{L^j}.$$

The definition $\mathrm{Sq}_j = C C_j^{1/2}$ in Definition F.6, the fluctuation bound of [4, Proposition 10.4], and the dilation $A_L(b) = L^{-1/2} A(b/L)$ give $|\tilde{A}|_\infty \le \sum_j L^{-j/2} \frac{1}{2} e_j^{-\varepsilon/2}$. Since $e_j = q L^{j/2} \ge q$ and $L \ge 13$,

$$|\tilde{A}|_\infty \le \tfrac{1}{2}\big(1 - L^{-1/2}\big)^{-1} q^{-\varepsilon/2} \le q^{-\varepsilon/2}.$$

Therefore, for $x = (c, g)$,

$$\varsigma(x) := \big|\exp\big(iq\tilde{A}(c, c_3)\big) - 1\big| \le q^{1-\varepsilon/2}.$$

Step 14 (the Wilson energy for fixed ζ). By gauge covariance, we may use $A_0 = \Omega_0 + \tilde{A}$ in the Wilson energy. Since $w_3 = 6$ and every site lies on a single bond of type 3, the bond formula of Definition F.5 gives

$$(\psi, \nabla_{A_0}^* \nabla_{A_0} \psi) \le 2\,(\psi, \nabla_{\Omega_0}^* \nabla_{\Omega_0} \psi) + 12\,\|\varsigma\psi\|^2.$$

Apply [4, Proposition 9.4] to the free operator with mesh 1 and antiperiodic character. The perturbation bound following Definition F.5 and $\|\mathcal{D}_0\psi\| = L^{-k_\star}\|\Phi\|$ give

$$(\psi, \nabla_{\Omega_0}^* \nabla_{\Omega_0} \psi) \le c_b^{-1} \|D_{\Omega_0}\psi\|^2 \le 3c_b^{-1}\big(L^{-2k_\star}\|\Phi\|^2 + \boldsymbol{\mu}_N^2 \|\psi\|^2 + 36\,\|\varsigma\psi\|^2\big).$$

Together these give the inequality **(WE)**:

$$\frac{\rho}{2}\,(\psi, \nabla_{A_0}^* \nabla_{A_0} \psi) \le 3\rho c_b^{-1}\big(L^{-2k_\star}\|\Phi\|^2 + \boldsymbol{\mu}_N^2 \|\psi\|^2\big) + \rho\, C_5\, \|\varsigma\psi\|^2. \tag{WE}$$

Step 15 (proof of (d)). Let $\boldsymbol{\mu}_N < 0$. From (c), (WE) and Steps 12 and 13,

$$\begin{aligned} |\boldsymbol{\mu}_N| Z_N = \mathsf{W}_N - \frac{\eta}{a_\star}\, c_\star(1 + \vartheta_N) \\ \le \Big[6\rho c_b^{-1} \frac{\eta^2}{a_\star^2} - \frac{\eta}{2a_\star}\, c_\star\Big] + \big(3\rho c_b^{-1} \boldsymbol{\mu}_N^2 + \rho\, C_5\, q^{2-\varepsilon}\big) Z_N. \end{aligned}$$

Since $c_\star \ge \mu_\star Z_0^\infty$ and $\eta \to 0$, once N is sufficiently large the square bracket takes a negative value and, by [4, Theorem 16.4, equation (16.19)], one has $3\rho c_b^{-1} |\boldsymbol{\mu}_N| \le \frac{1}{2}$. Division by $Z_N > 0$ then yields (d), for every $N \ge N_{\min}(R_{\mathrm{top}}(e), e)$ such that

$$12\rho\,\eta < c_b c_\star a_\star \qquad \text{and} \qquad 6\rho\big(\bar{m}\eta + \tfrac{1}{8} q^{3/4-8\varepsilon}\big) \le c_b;$$

the first of these conditions holds whenever $12\rho\,\eta < c_b\,\mu_\star Z_0^\infty a_\star$, an inequality whose right side does not depend on N. In the estimate above,

$$|\vartheta_N| \le e_\star^{3/4-10\varepsilon} \le e_\star^{3(1/4-9\varepsilon)} \le \tfrac{1}{512},$$

on account of $e_\star \le 1$ and of the third inequality of (21). □

5.13. **A clustering estimate at a fixed spacing.** Here P_λ denotes conditional expectation onto the $\mathbb{Z}^3$-invariant events. The local algebra $\mathcal{O}(Q)$ consists of bounded Borel functions of finitely many curvature plaquettes all of whose corners lie in Q; its closure is taken in $L^2(\lambda)$.

Proposition 5.17. (A clustering rate at a fixed spacing.) Consider a law λ on Ω which is invariant under the group W and which is reflection positive at each mirror belonging to one of the nine families. For example, fix $0<\rho<1$, $w_i>0$, $M\in\mathbb{R}$ and $q>0$, and let $n_i\to\infty$ be a sequence such that $8\mid n_i$ eventually and

$$\{m\in\mathbb{N} : m\mid n_i \text{ for all sufficiently large } i\}$$

is infinite. Every weak limit point of $\mu_{C(n_i)}$ is then such a law by [4, Theorem 4.7]. In particular, this applies to $n_i=8L^i$ for any fixed odd integer $L\geq 3$. Let $\mathsf{r}\geq 1$ be an integer, put

$$Q_{\mathsf{r}} := \{x : |x_j+\tfrac12|<\mathsf{r},\ j=1,2,3\},$$

let G and G' belong to the closure of $\mathcal{O}(Q_{\mathsf{r}})$ in $L^2(\lambda)$, and let $a\in\mathbb{Z}^3$ be such that $\mathsf{a}:=|a|_\infty-2\mathsf{r}\geq 1$. Then, with the scalar product and the norm of $L^2(\lambda)$,

$$\big|\langle G,\tau_a G'\rangle_2-\langle G,P_\lambda G'\rangle_2\big|\leq C_\flat\,\|G\|_2\,\|G'\|_2\Big(\frac{\mathsf{r}}{\mathsf{a}}+\frac{\mathsf{r}^2}{\mathsf{a}^2}\Big),$$

where

$$C_\flat := \frac{128(1+\sqrt2)}{4\pi^2}\,C''<2\cdot 10^4,\qquad C'':=200+64\Big(\frac{32}{\pi}+\frac{256}{\pi^2}\Big).$$

■

Proof. We use the filter from the proof of Proposition C.1 and the lattice cone of Proposition E.13. The discrete summation and its constants are given below. Both sides of the inequality depend continuously on G and G' in $L^2(\lambda)$, so that it suffices to treat $G,G'\in\mathcal{O}(Q_{\mathsf{r}})$.

1. *Pictures.* The reflection $\theta_j:=\theta_{x_j,-1/2}$ maps Q_{r} onto itself, and $\tau_{\mathsf{r}e_j}Q_{\mathsf{r}}\subset\{x_j>-\frac12\}$. Let $\{i,i'\}=\{1,2,3\}\setminus\{j\}$. In the axis picture of Proposition E.13, $0\leq T_j\leq 1$, the transverse operators $U_{j,i}$ and $U_{j,i'}$ are unitary, and all three commute. By the joint spectral theorem they possess a joint spectral measure $\mathfrak{E}_j$, written in the variables

$$t=\exp(-\mathsf{e}),\quad \mathsf{e}\in[0,\infty],\qquad u=(\exp ik_i,\exp ik_{i'}),\quad k\in(-\pi,\pi]^2.$$

The vector $\psi^j_G:=[\tau_{\mathsf{r}e_j}G]_j$ satisfies $\|\psi^j_G\|\leq\|G\|_2$. As $\theta_j\tau_{\mathsf{r}e_j}\theta_j=\tau_{-\mathsf{r}e_j}$, one has for $y_j\geq 2\mathsf{r}$

$$\langle G,\tau_y G'\rangle_2=\langle\psi^j_{\theta_jG},T_j^{y_j-2\mathsf{r}}U^{y_\perp}\psi^j_{G'}\rangle,\qquad U^{y_\perp}:=U^{y_i}_{j,i}U^{y_{i'}}_{j,i'}.$$

2. *The cone.* By Proposition E.13, the measure $\mathfrak{E}_j$ is carried by the set where $(1-t)^2\geq t\,|1-u_i|^2$, and likewise by the corresponding set for i'. Thus $t=1$ forces $u=1$, and $\sinh\frac{\mathsf{e}}{2}\geq|\sin\frac{k_i}{2}|$ whenever $0<t<1$. The function $x\mapsto\sin\frac{x}{2}-\sinh\frac{x}{4}$ is concave on $[0,\pi]$, zero at 0 and positive at π; it follows that $\mathsf{e}\geq\frac12|k|_\infty$.
3. *The kernel.* Suppose that $\theta_jG=\pm G$, and let $\varrho:=\langle\psi^j_G,\mathfrak{E}_j(\cdot)\psi^j_G\rangle$. Use the sequences z_m,b_m, the periodic function B_z, and the function Ψ constructed in the proof of Proposition C.1; write $\Upsilon=\Psi$ here. Define

$$K(y):=\sum_m z_m b_m\,\langle\theta_jG,\tau_{2\mathsf{r}me_j+y}G\rangle_2,\qquad y\in\mathbb{Z}^2.$$

The series converges absolutely because $\sum_m|z_mb_m|=8$. Reflection invariance and $\theta_jG=\pm G$ make its correlation even in the e_j coordinate. Pairing the terms m and $-m$ and applying Step 1 gives

$$K(y)=\int d\varrho\,\Upsilon(t^{2\mathsf{r}})\,u^y,\qquad K(-y)=\overline{K(y)},\qquad |K|\leq 8\|G\|_2^2.$$

4. *Decay.* Let $y_i \geq 2\mathsf{r}$. Apply Step 1 in the picture i before summing over m. Since $\theta_j G = \pm G$, the sum of the transverse translations is $B_z(2\mathsf{r}k_j)$, giving

$$K(y) = \pm\langle \psi^i_{\theta_i G}, T_i^{y_i - 2\mathsf{r}}\, \mathsf{B}\, U^{y_{i'}}_{i,i'} \psi^i_G \rangle, \qquad \mathsf{B} = \int d\mathfrak{E}_i\, B_z(2\mathsf{r}k_j).$$

Since $B_z(u) = 0$ for $\operatorname{dist}(u, 2\pi\mathbb{Z}) \leq \pi/4$, nonvanishing of $B_z(2\mathsf{r}k_j)$ forces $|k_j| > \frac{\pi}{8\mathsf{r}}$. Step 2 gives $\mathsf{e} > \alpha := \frac{\pi}{16\mathsf{r}}$ wherever $B_z(2\mathsf{r}k_j) \neq 0$, and so

$$|K(y)| \leq 8\exp(-\alpha(|y|_\infty - 2\mathsf{r}))\|G\|_2^2 \qquad \text{for } |y|_\infty \geq 2\mathsf{r}.$$

5. *Density.* For $s \geq 1$ there are $8s$ points $y \in \mathbb{Z}^2$ with $|y|_\infty = s$, and $\sum_{m\geq 1}(2\mathsf{r}+m)\exp(-\alpha m) \leq 2\mathsf{r}\alpha^{-1} + \alpha^{-2}$. Consequently

$$\sum_y |K(y)| \leq \big[8(4\mathsf{r}+1)^2 + 64(2\mathsf{r}\alpha^{-1} + \alpha^{-2})\big]\|G\|_2^2 \leq C''\mathsf{r}^2\|G\|_2^2.$$

By Fourier inversion for an absolutely summable sequence, the image of the measure $\Upsilon(t^{2\mathsf{r}})\, d\varrho$ under $(t,u) \mapsto k$ is $h\, dk$, where

$$0 \leq h \leq \frac{C''\mathsf{r}^2\|G\|_2^2}{4\pi^2}.$$

Since $\Upsilon > 0$ on $[0,1)$, this implies $\varrho(\{t < 1,\ k = 0\}) = 0$.

6. *Cube vectors.* The inequality $\Upsilon(\exp(-x)) \geq \tanh x/(1+\sqrt{2})$ from (5), Step 2, the monotonicity of $x \mapsto \exp(-\mathsf{a}x)/\tanh(2\mathsf{r}x)$ and the bound $\coth x \leq 1 + \frac{1}{x}$ yield, since $t^{\mathsf{a}} = 0$ at $t = 0$,

$$\int_{\{t<1\}} t^{\mathsf{a}}\, d\varrho \leq (1+\sqrt{2}) \int dk\, h\, \frac{\exp(-\mathsf{a}|k|_\infty/2)}{\tanh(\mathsf{r}|k|_\infty)}$$
$$\leq (1+\sqrt{2})\, \frac{C''}{4\pi^2}\, \|G\|_2^2 \Big(\frac{32\mathsf{r}^2}{\mathsf{a}^2} + \frac{16\mathsf{r}}{\mathsf{a}}\Big).$$

For a general $G = G_+ + G_-$, with $G_\pm = (G \pm \theta_j G)/2$, use the triangle inequality for the seminorm defined by $T_j^{\mathsf{a}}(1 - \hat{P})$. Since $\|G_\pm\|_2 \leq \|G\|_2$, the bound acquires a factor 4; here $\hat{P} = \mathbf{1}_{\{1\}}(T_j)$.

7. *Conclusion.* Let $a_j = |a|_\infty$; if no coordinate of a equals $|a|_\infty$, one takes complex conjugates. Put $\hat{P} := \mathfrak{E}_j(\{t = 1\})$ and $S := T_j^{\mathsf{a}} - \hat{P}$, so that $T_j^{\mathsf{a}} = \hat{P} + S$. By Step 2, $\hat{P}U^{a_\perp} = \hat{P}$; moreover

$$\langle \psi^j_{\theta_j G}, \hat{P}\psi^j_{G'} \rangle = \langle \tau_{-\mathsf{r}e_j} G, P_j \tau_{\mathsf{r}e_j} G' \rangle_2 = \langle G, P_\lambda G' \rangle_2,$$

where P_j is the projection onto $\{H : \tau_{e_j} H = H\}$. The equality $P_j = P_\lambda$ and the reflected projection identity are proved in Proposition E.13. The term containing $S \geq 0$ is estimated by the Schwarz inequality and Step 6.

□

6. Bounds on the tuned lattice masses

Given a power h of L, one calls *class* the interval $I_h := (\frac{1}{Lh}, \frac{1}{h}]$. For $m > 0$ put $h(m) := L^{\lfloor \log_L(1/m) \rfloor}$; then I_h consists of those $m > 0$ for which $h(m) = h$, so that $m \in I_{h(m)}$, and the number h of the notation fixed above is $h(\bar{m})$. Where h is the index of a class, as here and in Proposition 6.3(c), (d), the starred numbers, ϵ_e and $R_{\mathrm{top}}(e)$ are formed with this power of L; where h enters together with the parameter $\bar{m}$, as in Theorem 6.1 and in the number ϵ_e of that theorem, it is the number $h = h(\bar{m})$ of the notation fixed above, and $\bar{m} \in I_h$. Let $m \in I_h$, $e\, h^{1/2} \leq e_1$ and $N \geq N_{\min}(R_{\mathrm{top}}(e), e)$. Then $m_{0,N}(e; m)$ denotes the tuned shift formed with $\bar{m} = m$ [4, Section 16.4, Theorems 16.3–16.4], and one puts $\varphi_N(m) := m + m_{0,N}(e; m)/\eta$, so that $\boldsymbol{\mu}_N = \varphi_N(\bar{m})\eta$. The index $\star$ refers to the level $k_\star = k_{\mathrm{top}}$ on $R_{\mathrm{top}}(e) = hM\, r_K(e)$, where r_K is the smallest power L^i with $L^i \geq \log^r(1/(e\, h^{1/2}))$.

More generally, for a power $h' \geq 1$ of L with $e\,h'^{1/2} \leq e_1$ let $R_{\mathrm{top}}(e;h') := h'M\,r'$, where r' is the smallest power L^i with $L^i \geq \log^r(1/(e\,h'^{1/2}))$. Thus r' is the number r_k of [4, Proposition 16.1 and Sections 15–16] with $k = N + \log_L h'$, and $R_{\mathrm{top}}(e;h')$ is the side $R_{\mathrm{top}}(e)$ of [4, Proposition 16.1 and Sections 15–16] formed at any parameter $\bar m \in I_{h'}$; for the class I_h at hand, $R_{\mathrm{top}}(e) = R_{\mathrm{top}}(e;h)$. The numbers $a_\star \in \{h, h/L\}$, $e_\star = e\,a_\star^{1/2} \leq e\,h^{1/2}$, $\ell_\star$ and $N_{\min}$ depend on (e,h), but neither on N nor on $m \in I_h$ [4, Proposition 16.1 and Section 16.1]; Indeed $k_\star \in \{K, K-1\}$ and $a_K = h$ at $K = N + \log_L h$, by [4, Proposition 16.1]. On the other hand, $b_\star = b_{k_\star} = b_-/(1-\eta/a_\star)$ depends on N, and $b_- := 1 - \frac{1}{L} \leq b_\star \leq 1$. Further, $\mu_k = m a_k$, and

$$d_k(\mu) := \frac{b_k\mu}{b_k+\mu}, \qquad Z_k(\mu) := \frac{b_k^2}{(b_k+\mu)^2} = d_k'(\mu), \qquad \delta_k^\flat := 4K_{\mathrm{pol}}M^{-3}e_k^{2\varepsilon}w_k.$$

Assume the flow data (FL) of Definition F.1 on R at $\bar m = m$; the set $\mathrm{Tor}(e)$ to which R belongs, the side $R_{\mathrm{top}}(e)$, the number $N_{\min}(R,e)$ and the stop level $k_*(R)$ are those of [4, Proposition 16.1 and Sections 15–16]. The *flow of* $(m;t)$ *on* R is then the trajectory [4, Section 16.2, equations (16.9)–(16.10)] of a real pair with $m_{\mathrm{in}} = (t-m)\eta$, that is, of a real pair with the lattice mass $t\eta$. Its entries $m_k = m_k(t;m)$ and E_k do not depend on $\varepsilon_{\mathrm{in}}^{\mathrm{v}}$. Let h_J be the largest power of L with $e\,h_J^{1/2} \leq e_1$, and $m_\flat := 1/h_J$; then $m_\flat < Le^2/e_1^2$. The operator $\mathcal{K}_j := H_j C C_j^{1/2}$ has the kernel $\mathcal{K}_j(b,\Upsilon)$, where b is a fine bond of T_j. Put

$$\kappa_1 := \sup \sum_{\Upsilon} |\mathcal{K}_j(b,\Upsilon)|, \qquad \kappa_\infty := \sup \eta_j^3 \sum_{b} |\mathcal{K}_j(b,\Upsilon)|,$$

the first supremum being taken over the levels $j \geq 0$, the sides and b, the second over the levels $j \geq 0$, the sides and Υ, where a side is an element of $\mathrm{Cl} = \{\ell : \ell/(LM) \text{ is an odd integer} \geq 13\}$ [4, Proposition 16.1 and Sections 15–16], [4, Proposition 16.1]. Here $\eta_j := L^{-j}$ is the mesh of the level j [4, Proposition 16.1 and Sections 15–16], and at the level k it is written $\eta_k = L^{-k}$. With the place i in a sequence (N_i) or (N_i, T_i) of cutoffs as index, $\eta_i := L^{-N_i}$ is the spacing at the cutoff N_i, as in Proposition 5.6 and Lemma 6.4(a). With the constant $C_5 = 108\,c_b^{-1} + 6$ of Theorem 5.16, let

$$C_{\mathsf{W}} := 1200\,\rho\,C_5\,\kappa_1^2, \qquad C_\natural := C_{\mathsf{W}} + 3Le_1^{-2},$$
$$\epsilon_e := \min\Big\{\frac{1}{12},\, 4\big(L^{1/2}e\,h^{1/2}\big)^{\varepsilon}\Big\}, \qquad e_7 := \Big(\frac{\bar m}{4C_\natural}\Big)^{1/2}.$$

6.1. **Background estimates for the mass comparison.**

(K22) (Šidák 1967 [8].) Let $(Y_1,\dots,Y_d)$ be a centred jointly Gaussian vector, and let $c_1,\dots,c_d > 0$. Then

$$P(|Y_i| \leq c_i \text{ for every } i) \geq \prod_i P(|Y_i| \leq c_i).$$

(K23) (Schur.) For a matrix K,

$$\|K\|^2_{\ell^2\to\ell^2} \leq \sup_i \sum_j |K_{ij}| \cdot \sup_j \sum_i |K_{ij}|.$$

If X is a $d\times d$ matrix with $\|X\| < 1$, then $|\det(1+X)| \geq (1-\|X\|)^d \geq 1 - d\|X\|$.

(K24) Every coclosed one-cochain A of mean zero on $\mathbb{Z}^3/\ell\mathbb{Z}^3$ satisfies

$$\|dA\|^2 \geq 4\sin^2\frac{\pi}{\ell}\,\|A\|^2 \geq 16\ell^{-2}\|A\|^2.$$

If ω is a function on a cube of d_1^3 sites and $\bar\omega$ is its mean, then $\sum|\omega-\bar\omega|^2 \leq \frac{d_1^2}{4}\sum_b |d\omega(b)|^2$. (Kato.) For $a,b\in\mathbb{C}^4$ and $|u|=1$ one has $\big||a|-|b|\big| \leq |ua-b|$.

(K28) (Stone–Weierstrass.) Let X be a locally compact Hausdorff space. A subalgebra of $C_0(X;\mathbb{R})$ which separates the points of X and has no common zero is dense.

Theorem 6.1. *Two-sided bounds on the tuned lattice masses.* Suppose that $\bar m \in (0,1]$ and $e>0$ satisfy $e\,h^{1/2} \le e_1$; the data of Section F.2 satisfy this condition. Then there is a number $N_1(e)<\infty$, one and the same for all such $\bar m$, such that every $N \ge N_1(e)$ satisfies

$$(\bar m(1-\epsilon_e) - C_\natural e^2)\,\eta \le \boldsymbol{\mu}_N = \bar m\eta + m_{0,N}(e) \le \bar m\,\eta\,(1+10^{-4}e_\star^{3/4-10\varepsilon}).$$

(a) Let $0<e\le e_7$. Then $e\,h^{1/2}\le e_1$, and

$$\tfrac12 \bar m\,\eta \le \boldsymbol{\mu}_N \le 2\bar m\,\eta \qquad \text{for } N \ge N_1(e).$$

Hence $\liminf_N \boldsymbol{\mu}_N/\eta \ge \frac12\bar m>0$, and the tuned assignment satisfies the mass hypothesis of Theorem A. Here $e_7 = c\,\bar m^{1/2}$, and the number $c=(4C_\natural)^{-1/2}$ does not depend on $\bar m$.

(b) For $N\ge N_1(e)$ one has $|m_{0,N}(e)| \le (C_\natural e^2 + \epsilon_e \bar m)\,\eta$. Since $\bar m\le 1$ and $e\le 1$, the right side does not exceed $(C_\natural + 4L^{\varepsilon/2})\,e^{\varepsilon}\eta$, uniformly in $\bar m$. This bound does not reach the power e^2, on account of the term $\epsilon_e\bar m$; in Steps 2 and 3 of the proof this term bounds from above the errors of the slope, which come from (M) of Proposition 6.3 (c), together with the errors at the junctions, which come from (J) of Proposition 6.3 (d) combined with (M).

(c) If $e\le e_7$ and $N\ge N_1(e)$, then $Z_N \le 2.1\,Z_0$.

■□

Proposition 6.2. *A bound from below at every sign of the mass, and the second moment that yields it.* Let $m\in(0,1]$ with $e\,h(m)^{1/2}\le e_1$, and form the objects of Theorem 5.16 at $\bar m = m$. Put

$$v_0 := \log\frac{1}{q}, \qquad j_1 := \lceil 2p_0 \log_L v_0\rceil, \qquad \mathsf{l} := L^{2j_1}, \qquad \tilde a_j(b) := L^{-j/2}(\mathcal{K}_j\zeta_j)(L^{-j}b),$$

so that $\tilde A = \sum_{j<k_\star}\tilde a_j$ by Step 13 of the proof of Theorem 5.16. Let $\mathcal{G}$ be the set of those ζ for which

$$\sum_{b\cap \mathsf{Q}\neq\emptyset} \tilde a_j(b)^2 \le 16 L^{-j}\kappa_1^2 \mathsf{l}^3$$

holds for every $j<j_1$ and every block Q of side l of $\mathbb{Z}^3/n\mathbb{Z}^3$. The set $\mathcal{G}$ and the blocks are formed only for the N with $2j_1\le k_\star$, which are the large N; for them the levels $j<j_1$ exist, and l divides $n = L^{k_\star}\ell_\star$.

(a) There is a number $N_A(e,m)$, defined in Step 10 of the proof, which is a function of $(e,h(m),\mu_\star)$ and of constants, such that $\varphi_N(m) \ge -C_{\mathsf{W}}e^2$ for all $N\ge N_A(e,m)$. By comparison, Theorem 5.16 (d) gives the lower bound $-2\rho C_5 e^{2-\varepsilon}\eta^{-\varepsilon/2}$ for $\varphi_N(m)$.

(b) Let $2j_1\le k_\star$. On $\mathcal{G}\cap\prod_j \mathrm{Box}_j$, every block Q satisfies

$$\sum_{b\cap\mathsf{Q}\neq\emptyset}\tilde A(b)^2 \le 200\,\kappa_1^2\mathsf{l}^3;$$

this bounds a second moment by a number of the order one, whereas the bound on the supremum is $q^{-\varepsilon}$. Let $\bar\mu$ denote the zero-flux law of Definition F.2, taken on the torus with $n = R_{\mathrm{top}}/\eta$ sites per side and at the lattice mass $\varphi_N(m)\eta$. Then

$$\bar\mu(\mathcal{G}^c) \le j_1 n^3 \exp\Big(-\frac{2\kappa_1}{\kappa_\infty}v_0^{6p_0}\Big) \longrightarrow 0 \qquad (N\to\infty).$$

■

Proposition 6.3. *Running mass versus bare mass; slope one; junctions.* Assume $e\,h^{1/2}\le e_1$, where $h=h(m)$ in (a) and (b).

(a) *One parameter.* Suppose that (FL) holds on R at $\bar m = m$, that $k\le k_*(R)$, and that the flows of $(m;t)$ and of $(m;t')$ on R satisfy $|m_j|\le w_j$ for $j<k$. Then

$$m_k(t;m) - m_k(t';m) = a_k(t-t')(1+\theta), \qquad |\theta| \le \tfrac{1}{15}e_k^{\varepsilon}.$$

Thus, on the set of those t whose flows stay in these windows, the running mass is a strictly increasing function of the bare mass.

(b) *Reading.* Suppose that (FL) holds on R at $\bar m = m$, that $1 \le k \le k_*(R)$, and that $|m_j(t;m)| \le w_j$ for $j < k$. Then there is a complex number z_k, defined in Step 2 of the proof, which depends on (e, N, R, k, t) alone and not on m: it is one and the same number for every m with the two properties just required, although for each such m the flow of $(m;t)$ is formed with the primitives of (FL) at (m, e, R, N); these primitives depend on m through $\mu_k = m\,a_k$ and through the scheme of $h(m)$, that is, through $R_{\mathrm{top}}(e; h(m))$ and, on a torus $R < R_{\mathrm{top}}(e; h(m))$, through the projections with $\mathrm{Jr} = R_{\mathrm{top}}/R$ [4, Proposition 16.1 and Sections 15–16]; in Step 7 of the proof, for instance, which proves (d) by the argument of Step 5, the flows of $(m_b;t)$ and of $(m_c;t)$ with $t = \varphi_N(m_c)$, where $m_b \in I_{Lh}$ and $m_c \in I_h$, are taken on one torus, $R = R_{\mathrm{top}}(e; h)$, which satisfies $R \le R_{\mathrm{top}}(e; Lh)$ and belongs to the tori of the scheme of Lh as well; they are formed with the schemes of Lh and of h, respectively; apart from (e, N, R, k, t), the number z_k involves only constants and the spinor $u = (1, 0, 0, 0)$ of the statement of Theorem 5.16. This number satisfies

$$\mathsf{z}_k = d_k(\mu_k) + Z_k(\mu_k)\,(m_k(t;m) - \varkappa_k), \qquad |\varkappa_k| \le \delta_k^\flat.$$

(c) **(M)** Let $N \ge N_{\min}(R_{\mathrm{top}}(e), e)$, and let $m' < m''$ be points of I_h. Then

$$|\varphi_N(m'') - \varphi_N(m') - (m'' - m')| \le e_\star^\varepsilon\,[(m'' - m') + w_\star/a_\star].$$

In particular $\varphi_N(m'') > \varphi_N(m')$ whenever $m'' - m' \ge 2e_\star^\varepsilon w_\star/a_\star$.

(d) **(J)** Suppose, in addition, that $e\,(Lh)^{1/2} \le e_1$. Let $N \ge N_{\min}(R_{\mathrm{top}}(e; h), e)$, let $m_b := \frac{1}{Lh}$ be the upper end point of I_{Lh}, and let $m_c \in I_h$ satisfy $m_c - m_b \le \frac{w_\star}{2a_\star}$. Then

$$\varphi_N(m_c) - \varphi_N(m_b) \ge -\frac{w_\star}{6a_\star}.$$

■

Proof of Proposition 6.3. Step 1 (one density under two names). By Definition F.6, the density $\mathscr{F}_0[\varepsilon_{\mathrm{in}}^{\mathrm{v}}, m_{\mathrm{in}}, 0](A_0; \Psi)$ equals a positive number multiplied by

$$\exp\{-\langle\bar\Psi, (D_{A_0} + t\eta)\Psi\rangle^{\mathrm{ch}}\}.$$

Each map $\mathbf{T}_j$ is linear. Its constituents c_j, Box_j, μ_I, b_L, ω_j, $Q(\mathscr{B}_j)$, $A_j^{\min}$ and Sq_j depend only on (e_j, ℓ_j) and the fixed constants, by Definition F.6; thus neither a fermion mass nor a primitive enters $\mathbf{T}_j$. Indeed, H_j (which occurs in $\mathscr{B}_j = H_j A_j^{\min}(A')$), $C_j^{1/2}$, Sq_j, $A_j^{\min}$ and ω_j are the exact photon objects of [4, Section 10.5, equations (10.21)–(10.29), and Section 13.3]; they are formed with (j, ℓ_j, L) and with the basis $\{e_i^0\}$, which is fixed once for each L. Assume that $|m_j| \le w_j$ for $j < k$. At the levels $j < k$, [4, Proposition 16.2] then furnishes the hypotheses of [4, Theorem 14.8, equations (14.47)–(14.48)] at $\mathscr{C}' = 0$; the primitive families are fixed by (m, e, R, N) as specified in Definition F.1, and the trajectory remains real. Combined with [4, Proposition 10.4], this gives

$$\mathscr{F}_k[\varepsilon_k^{\mathrm{v},t}, m_k, E_k] = \mathbf{T}_{k-1}\cdots\mathbf{T}_0\,\mathscr{F}_0 \qquad \text{on } \mathrm{Dom}_k.$$

Consequently $\mathscr{F}_k(A;\cdot)$ is, up to a positive factor, a function of (e, N, R, k, t) and of A only.

Step 2 (the constant mode). Let Ω_k be the constant bond function with $e_k\Omega_{k,\mu}\ell_k = \pi$, let

$$\Psi_u(y) := \ell_k^{-3/2}\exp(-ie_k\Omega_k\cdot y)\,u, \qquad f(c,g) := \ell_k^{-3/2}\exp(-ie_k\Omega_k\cdot c)\,u,$$

and define the number z_k by the condition that the component of degree $(1,1)$ of

$$\log\frac{\mathscr{F}_k(\Omega_k;\Psi)}{\mathscr{F}_k(\Omega_k;0)},$$

evaluated at $(\overline{\Psi_u}, \Psi_u)$, equals $-\mathsf{z}_k$. The calculation in Step 7 of the proof of Theorem 5.16 applies at every level $k \ge 1$, using the operators of Definition F.5. It gives the following identities. The

photon minimizer satisfies $H_k\Omega_k = \Omega_k \in \frac{1}{4}\widetilde{R}_k$ by [4, Propositions 10.1 and 10.4]. Thus Ω_k lies in the supremum defining the activity norm. The map $H_k\Omega_k$ is the photon minimizer; the map $H_k(A)$ below acts on spinors, as in Definition F.5. By Definition F.5,

$$H_k(A) = b_k S_k(A) J Q_k^T(-A), \qquad \bar{H}_k(A) = \mathsf{C}^{-1} H_k(-A)\mathsf{C},$$

where $\mathsf{C} = i\gamma_2$ is the real matrix of Definition F.5. By Definition F.5, the fields inserted into an activity at the connection Ω_k are

$$\psi = H_k(\Omega_k)\Psi, \qquad \chi = \delta_{\alpha,\Omega_k}\psi, \qquad \bar{\psi} = \bar{H}_k(\Omega_k)\bar{\Psi}, \qquad \bar{\chi} = \delta_{\alpha,-\Omega_k}\bar{\psi}.$$

The maps $\delta_{\alpha,\pm\Omega_k}$ are the Hölder quotients of Definition F.5, with fixed exponent $1-\varepsilon < \alpha < 1$. Furthermore $D_{\Omega_k} f = 0$, $D_k(\Omega_k)\Psi_u = d_k(\mu_k)\Psi_u$ and $H_k(\Omega_k)\Psi_u = Z_k^{1/2} f$. The vector $\mathsf{C}\overline{\Psi_u}$ and the function $\mathsf{C}\bar{f}$ are Ψ_u and f formed at $-\Omega_k$ with the spinor $\mathsf{C}\bar{u}$ in the place of u; the same computation at $-\Omega_k$ therefore yields $H_k(-\Omega_k)\mathsf{C}\overline{\Psi_u} = Z_k^{1/2}\mathsf{C}\bar{f}$, that is, $\bar{H}_k(\Omega_k)\overline{\Psi_u} = Z_k^{1/2}\bar{f}$. Since $U_{\Omega_k}(x,x')f(x') = f(x)$ for every pair, the quotient formula gives $\delta_{\alpha,\Omega_k} f = 0 = \delta_{\alpha,-\Omega_k}\bar{f}$. At $(\bar{\Psi},\Psi) = (\overline{\Psi_u},\Psi_u)$ the fields ψ, χ, $\bar{\psi}$, $\bar{\chi}$ therefore take the values

$$\psi = Z_k^{1/2} f, \qquad \chi = Z_k^{1/2}\delta_{\alpha,\Omega_k} f = 0, \qquad \bar{\psi} = Z_k^{1/2}\bar{f}, \qquad \bar{\chi} = Z_k^{1/2}\delta_{\alpha,-\Omega_k}\bar{f} = 0.$$

The definition of the density in Definition F.6, with these fields inserted, then gives

$$\mathsf{z}_k = d_k + Z_k m_k - Z_k \sum_X \iint \bar{f}^T E_{k,20}(X;\Omega_k)\, f.$$

Step 3 (proof of (b)). The weighted coefficient norm in Definition F.1 gives

$$e_k^{-1/2} \iint \sum_{\beta\beta'} |E_{20}(X)_{\beta\beta'}| = \|E^{[2,0]}(X)\|_{h_k} \le \|E(X)\|_{h_k},$$

and $|f|^2 = \ell_k^{-3}$. The norm in Definition F.1 uses $\kappa \ge 7\kappa_0 + 7 \ge \kappa_0$ by [4, Proposition 24.1]; by [4, equation (A.3)], [4, equation (A.3)], applied to the whole torus, which is the union of $(\ell_k/M)^3$ M-cubes,

$$\sum_{X\in D_k} \exp\bigl(-\kappa_0\, d_M(X)\bigr) \le K_{\mathrm{pol}}\Bigl(\frac{\ell_k}{M}\Bigr)^3;$$

and, for the flow of $(m;t)$, $\|E_k\|_k \le 4e_k^{1/4-6\varepsilon}$ by [4, Proposition 16.2]. It follows that the sum in the last term of Step 2,

$$\varkappa_k := \sum_X \iint \bar{f}^T E_{k,20}(X;\Omega_k)\, f,$$

satisfies

$$|\varkappa_k| \le 4K_{\mathrm{pol}} M^{-3} e_k^{3/4-6\varepsilon} = \delta_k^\flat.$$

By Step 1, z_k is determined by (e,N,R,k,t), by the spinor u and by constants, the basis $\{e_i^0\}$ among them, so that it does not depend on m; the identity of Step 2, read with $m_k = m_k(t;m)$ and with $\varkappa_k$ formed from E_k of the flow of $(m;t)$, and the bound on $\varkappa_k$ prove (b). The number $\delta_\star$ of Step 9 of the proof of Theorem 5.16, formed with the bound on $H_{k_\star}$ in the supremum norm, equals $C_0^2 e_\star^{-2\varepsilon}\delta_\star^\flat$; here $H_k(\Omega_k)\Psi_u$ is known exactly, and this makes the accuracy $\delta_k^\flat$ with which z_k determines $m_k(t;m)$ finer than the window w_k.

Step 4 (proof of (a)). Let $t \neq t'$, and let primed letters refer to the flow of $(m;t')$. Put

$$x_j := m_j - m_j', \qquad Y_j := \|E_j - E_j'\|_j;$$

one has $Y_0 = 0$. The recursion of the flow [4, Section 16.2, equations (16.9)–(16.10)], in whose difference the term LE_j^{det} cancels, and Lemma F.7 give

$$x_{j+1} = L(x_j + m(E_j - E_j')), \qquad |m(E_j - E_j')| \le c_{\mathrm{fl}}\, e_j^{1/2}\, Y_j,$$

$$Y_{j+1} \le (c_{\mathrm{fl}} L^{-1/4+2\varepsilon} + e_j^{\varepsilon}) Y_j + e_j^{-1/2+2\varepsilon} |x_j|.$$

From $c_{\mathrm{fl}} L^{-3/8+6\bar{\varepsilon}} \le \frac{1}{16}$ and $L \ge 5$ [4, Proposition 24.1], and from $(c_{\mathrm{fl}}+2)e_j^{\varepsilon} \le \frac{1}{16}$ (20), induction on j shows that $x_j \neq 0$ and

$$u_j := \frac{e_j^{1/2} Y_j}{|x_j|} \le e_j^{2\varepsilon},$$

the inductive step being

$$|x_{j+1}| \ge \tfrac{15}{16} L |x_j|, \qquad u_{j+1} \le \tfrac{16}{15} L^{-3/4} (c_{\mathrm{fl}} L^{-1/4+2\varepsilon} + e_j^{\varepsilon} + 1) e_j^{2\varepsilon}.$$

Hence $x_{j+1} = L x_j (1+\theta_j)$ with $|\theta_j| \le c_{\mathrm{fl}} e_j^{2\varepsilon}$, and by (20)

$$\mathsf{s} := \sum_{j<k} |\theta_j| \le \tfrac{1}{16} e_k^{\varepsilon}.$$

Because $\mathsf{s} \le \frac{1}{16}$, one has

$$\Big| \prod_{j<k} (1+\theta_j) - 1 \Big| \le \exp(\mathsf{s}) - 1 \le \tfrac{16}{15}\, \mathsf{s}.$$

As $x_0 = (t-t')\eta$ and $L^k \eta = a_k$, this proves (a).

Step 5 (one link). Let $m_1 < m_2 = m_1 + s$ be points of I_h with $s \le \frac{w_\star}{2a_\star}$; put $t_2 := \varphi_N(m_2)$, $R = R_{\mathrm{top}}(e)$ and $\mu_k^{(i)} := m_i a_k$. By [4, Theorem 16.3], the flow of $(m_2; t_2)$ satisfies $|m_k| \le \frac{1}{8} w_k$ and $m_{k_\star} = 0$. The flow of $(m_1; t_2)$ is estimated by induction on k. Let $|m_j(t_2; m_1)| \le w_j$ for $j < k$. Both flows have the same $t = t_2$, so that (b) attaches to them one and the same number z_k; with the numbers $\varkappa_k^{(1)}$, $\varkappa_k^{(2)}$ that (b) gives for the two flows, it follows that

$$Z_k(\mu_k^{(1)})(m_k(t_2;m_1) - \varkappa_k^{(1)}) = d_k(\mu_k^{(2)}) - d_k(\mu_k^{(1)}) + Z_k(\mu_k^{(2)})(m_k(t_2;m_2) - \varkappa_k^{(2)}).$$

Since $Z_k = d_k'$ is decreasing, $0 \le d_k(\mu_k^{(2)}) - d_k(\mu_k^{(1)}) \le s\, a_k Z_k(\mu_k^{(1)})$; since a_k/w_k increases with k, one obtains

$$|m_k(t_2; m_1)| \le (\tfrac{1}{2} + \tfrac{1}{8}) w_k + 2\delta_k^{\flat} \le \tfrac{3}{4} w_k.$$

For $k = 0$ the same bound follows from the relation $m_{\mathrm{in}}^{(1)} = m_{\mathrm{in}}^{(2)} + s\eta$ between the initial masses of the two flows. At $k = k_\star$, where $m_{k_\star}(t_2; m_2) = 0$, the identity above gives

$$m_{k_\star}(t_2; m_1) = s\, a_\star \vartheta' + r, \qquad \vartheta' = \frac{b_\star + \mu_\star^{(1)}}{b_\star + \mu_\star^{(2)}} \in [1 - w_\star, 1], \qquad |r| \le 2\delta_\star^{\flat}.$$

Step 6 (proof of (c)). Clause (a), applied to the flow of $(m_1; t_2)$ and to the tuned pair of m_1, gives

$$m_{k_\star}(t_2; m_1) = a_\star(\varphi_N(m_2) - \varphi_N(m_1))(1+\theta).$$

By the third inequality of (21), $w_\star \le e_\star^{\varepsilon} (e_\star^{1/4-9\varepsilon})^3 \le \frac{1}{512} e_\star^{\varepsilon}$. With Step 5 it follows that

$$|\varphi_N(m_2) - \varphi_N(m_1) - s| \le \tfrac{1}{8} e_\star^{\varepsilon} s + 2.2\, \frac{\delta_\star^{\flat}}{a_\star}.$$

Let $[m', m'']$ be divided into $\lceil 2a_\star(m''-m')/w_\star \rceil$ intervals of equal length, each of which is a link as in Step 5. By (20),

$$\frac{\delta_\star^{\flat}}{w_\star} = 4 K_{\mathrm{pol}} M^{-3} e_\star^{2\varepsilon} \le 10^{-4} e_\star^{\varepsilon},$$

and the sum over the links of the inequality for one link yields (c).

Step 7 (proof of (d)). Let $R := R_{\mathrm{top}}(e; h)$; then $R = Lh \cdot L^T$, where $L^T := M r_K / L \ge 1$. Applied to Lh, [4, Proposition 16.1 and Section 16.1] gives $v_K^r < L v_{K+1}^r \le L r_{K+1}$, and $L r_{K+1}$ is a power of L; hence $r_K \le L r_{K+1}$ and $R \le R_{\mathrm{top}}(e; Lh)$. Thus R belongs to the tori of the scheme of Lh, while the numbers $N_{\min}(R, e)$, $k_*(R) = k_\star$, ℓ_k and e_k do not depend on h [4, Proposition 16.1 and

Section 16.1]. By [4, Proposition 16.1 and Section 16.1], $N \geq N_{\min}(R,e) \geq N_{\min}(R_{\mathrm{top}}(e;Lh),e)$, so that the shift entering $\varphi_N(m_b)$ is the tuned one. By [4, Theorem 16.4, equation (16.18)], the tuned pair of m_b has $|m_k| \leq \frac{1}{8} w_k$ and is controlled on R. The argument of Step 5, with m_b and m_c in the places of m_1 and m_2, shows that the flow of $(m_b;\varphi_N(m_c))$ on R is controlled and has $m_{k_\star} \geq -2\delta_\star^\flat$. Clause (a) on R therefore gives

$$a_\star(\varphi_N(m_c)-\varphi_N(m_b))(1+\theta) \geq -2\delta_\star^\flat - \tfrac{1}{8} w_\star,$$

and (d) follows. □

Proof of Proposition 6.2. Throughout the proof put

$$\mathrm{mup} := \prod_{j<k_\star} \mathbf{1}_{\mathrm{Box}_j}\, \mu_I, \qquad g_\star := b_\star^{-1}+\mu_\star^{-1} = d_\star^{-1}, \qquad g_- := b_-^{-1}+\mu_\star^{-1},$$

$$c_- := \mu_\star\, b_-^2\,(b_-+\mu_\star)^{-2}, \qquad \varkappa_D := \bigl(16\,\ell_\star^3\, g_-\bigr)^{-1}, \qquad r_1 := \frac{\mu_\star^2}{24}\min\Bigl\{\varkappa_D, \frac{c_-}{28}\Bigr\}, \qquad \boldsymbol{\mu} := \varphi_N(m)\,\eta.$$

Since $b_- \leq b_\star$ and the map $b \mapsto \mu b^2(b+\mu)^{-2}$ is increasing, one has $g_\star \leq g_-$ and $c_- \leq c_\star$. The numbers g_-, c_-, $\varkappa_D$ and r_1 depend on $(e,h,\mu_\star)$ but not on N, whereas $g_\star$, $d_\star$, $c_\star$ and Z_0 depend on N through $b_\star$.

Step 1 (Theorem 5.16 at a general block field). Denote by $V_{\mathrm{t}}(T^0_{k_\star})$ the space of coclosed mean-zero cochains in Definition F.2. Put $p_\star = v_\star^p$, where $v_\star = -\log e_\star$ and p is the fixed terminal-cutoff exponent of [4, Proposition 24.1]. Let $\theta \in \mathbb{R}^3$, let $A \in V_{\mathrm{t}}(T^0_{k_\star})$ satisfy $|dA|_\infty \leq p_\star$, and put $A' := A + c_\theta$, where c_θ is the constant bond function with the twist θ. Then $A' \in \mathrm{Dom}_{k_\star}$ by [4, Proposition 10.4, Theorem 22.2 and Proposition 22.6]. The objects A_j, $\mathcal{R}$, B, $\mathcal{D}$, $\mathcal{D}_0$, ψ and Φ are formed as in Theorem 5.16, with $A_{k_\star} := A'$ and with the same vector Ψ_u, and $z := (\Psi_u, \Phi)$. Let $\mathsf{P}^{A'}$ and $\mathsf{P}^{0,A'}$ be the measures proportional to $\det B\,\mathrm{mup}$ and to $\det \mathcal{D}_0\,\mathrm{mup}$, respectively, and let $\mathsf{E}^{A'}$ be the mean under $\mathsf{P}^{A'}$. Put $\mathscr{A}' := H_{k_\star} A'$; by [4, Proposition 10.1], $\mathscr{A}' = H_{k_\star} A + c_\theta$. The following parts of the proof of Theorem 5.16 are carried out at $A_{k_\star} = A'$: Steps 1 to 6; Step 8, with $\mathscr{A}'$ in the place of $H_{k_\star}\Omega = \Omega$; in Step 9, the bound on $\sum_X E_{k_\star}$ in the norm $|\cdot|_{\mathrm{aw}_2}$, the bounds on K_E and on $\mathcal{A}_4$ in terms of $\delta_\star$ and δ_2, and the two inequalities for $\ell_\star^3\delta_\star$ and δ_2 derived from (21); Step 11; in Step 12, the permutation $a \mapsto \hat{a}$ of the indices, the bound $\mathsf{E}_N\|\delta\mathcal{D}\Psi_u\|^2 \leq \delta_2$ and the identity obtained by averaging (SR); Step 13, with $\mathscr{A}_{k_\star} = \mathscr{A}'$. These arguments are valid with any real element of $\mathrm{Dom}_{k_\star}$ in the place of Ω ([4, Proposition 10.4 and equation (21.28)]; the bounds of Step 9 of that proof are suprema over $\tilde{R}_{k_\star}$ [4, equation (21.28)]), and with any antiperiodic unit vector Ψ_u such that $\Sigma_3 u = u$ and $|\Psi_u(y)|^2 = \ell_\star^{-3}$. Step 7 of that proof establishes the relation $D_{k_\star}(\Omega)\Psi_u = d_\star\Psi_u$; on this relation rest the bounds of its Step 9 on $\bar{z} = (\Psi_u, \mathsf{E}_N\mathcal{D}\Psi_u)$ and on $\|\mathsf{E}_N\mathcal{D}\Psi_u\|$, its Step 10, and, in its Step 12, the estimate comparing the right side of the averaged identity (SR) with $c_\star$, the bound $\mathsf{E}_N\|\Phi\|^2 \leq 2$ and the inequality $|1-\bar{z}/b_\star|^2 \geq (Z_0^{1/2}-2\delta_\star)^2$. Step 8 below takes the place of these parts and of Step 7 of that proof. Consequently, whatever the sign of $\boldsymbol{\mu}$, the following hold.

(i) One has $\det B \geq 0$; the objects $\mathcal{D}$, ψ, Φ and z are defined off the null set $\{\det B = 0\}$, and the relations below are understood off this set. Both measures can be normalised, as $\mathscr{F}_{k_\star}(A';0) \neq 0$ and $Y_*(A') > 0$ [4, Proposition 10.4, Theorem 22.2 and Proposition 22.6]. Moreover $\det\mathcal{D} \geq 0$, and with the constant $\mathsf{C}_{\det} > 0$ of Step 5 of the proof of Theorem 5.16, which is denoted C_N there,

$$\mathscr{F}_{k_\star}(A';\Psi) = \mathsf{C}_{\det}\int \mathrm{d}\,\mathrm{mup}\ \det B\,\exp\bigl(-\langle\bar{\Psi},\mathcal{D}\Psi\rangle\bigr), \qquad y_0(\mathrm{Af}_0(A',\zeta)) = \mathsf{C}_{\det}\,\det B\,\det\mathcal{D}.$$

In the second identity, $\mathrm{Af}_0(A',\zeta)$ is the last field A_0 of the recursion $A_j = A_j^{\min}(A_{j+1}) + \mathrm{Sq}_j\zeta_j$ begun at $A_{k_\star} = A'$ [4, equation (22.1)], and y_0 is the function

$$y_0(\breve{A}) := \exp(-\varepsilon_{\mathrm{in}}^{\mathrm{v}} n^3)\,\frac{\det(D_{\breve{A}} + \boldsymbol{\mu})}{\det(D_0 + m\eta)}$$

of a real one-cochain $\breve{A}$ on $\mathbb{Z}^3/n\mathbb{Z}^3$; Thus $y_0 = w_N$ of Definition F.2, for the present mass m and incoming pair; integrating the density of Definition F.6 at level zero gives this same function. For $A_0 = \mathrm{Af}_0(A',\zeta)$ one has $\det \mathcal{D}_0 = \det(D_{A_0} + \boldsymbol{\mu})$; the second identity of the display is therefore the relation $\mathsf{C}_{\det} \det B \cdot \det \mathcal{D} = \exp(-\varepsilon_{\mathrm{in}}^{\mathrm{v}} n^3) \det \mathcal{D}_0 / \det(D_0 + m\eta)$ obtained in Steps 5 and 6 of the proof of Theorem 5.16.

(ii) With $\delta\mathcal{D} := \mathcal{D} - \mathsf{E}^{A'}\mathcal{D}$ one has

$$\mathsf{E}^{A'}\mathcal{D} = D_{k_\star}(\mathscr{A}') - K_E,$$

where the moduli of the entries of K_E have a sum not exceeding $\ell_\star^3\delta_\star$; furthermore $\mathsf{E}^{A'}\|\delta\mathcal{D}\Psi_u\|^2 \le \delta_2$, and summation of the identities of Steps 8 and 12 of that proof over all indices yields $\mathsf{E}^{A'}\|\delta\mathcal{D}\|_{\mathrm{HS}}^2 \le 2\,\ell_\star^3\delta_2$.

(iii) The identity (SR) of Theorem 5.16 (c) holds.

(iv) One has $A_0 = H_\theta + \mathbf{a} + \tilde{A} - d\lambda$, where H_θ is the constant one-cochain of the twist θ, as in [4, equation (22.8)], and $\mathbf{a} := (H_{k_\star}A)_{L^{k_\star}}$; moreover $q|\mathbf{a}|_\infty = e_\star L^{-k_\star}|H_{k_\star}A|_\infty$.

Step 2 (the complement of an event costs at most $\frac14$ and carries no weight $\|\psi\|^2$). Put

$$W := \frac{\rho}{2}\,(\psi, \nabla_{A_0}^*\nabla_{A_0}\psi), \qquad c(A') := \mathsf{E}^{A'}\big[\mathrm{Re}\, z - b_\star^{-1}\|\Phi\|^2\big].$$

As $\mathrm{Re}\, z \le \|\Phi\|$ and $x - x^2/b_\star \le b_\star/4 \le \frac14$ for real x, the identity (SR) gives, for every Borel set $\mathcal{B}$,

$$c(A') \le L^{k_\star}\,\mathsf{E}^{A'}\big[\boldsymbol{\mu}\|\psi\|^2 + W; \mathcal{B}\big] + \tfrac14\,\mathsf{P}^{A'}(\mathcal{B}^c).$$

Step 3 (Kato's inequality and a Poincaré inequality). Let Q be a cube of l^3 photon sites, let $f \in \mathcal{H}_0$, and let $\tau \ge 0$ be a function on the sites over Q, whose mean is $\langle\tau\rangle_{\mathsf{Q}}$. In this step $\|\cdot\|_{\mathsf{Q}}$ denotes the norm of $\mathcal{H}_0$ restricted to the $48\,\mathsf{l}^3$ sites over Q; the symbols P_{Q} and $\langle\cdot\rangle_{\mathsf{Q}}$ both stand for averaging over these sites; The restricted energy $(f, \nabla_{A_0}^*\nabla_{A_0}f)_{\mathsf{Q}}$ sums the bond terms of Definition F.5 with both endpoints over Q. The Poincaré inequality in (K24), Katoś inequality on each bond, and [4, Proposition 9.1 and its proof], applied to $(|f|,0,0,0)$, yield

$$\big\|\,|f| - P_{\mathsf{Q}}|f|\big\|_{\mathsf{Q}}^2 \le C_{\mathrm{Po}}\,\mathsf{l}^2\,(f, \nabla_{A_0}^*\nabla_{A_0}f)_{\mathsf{Q}}.$$

Together with $|f|^2 \le 2(P_{\mathsf{Q}}|f|)^2 + 2(|f| - P_{\mathsf{Q}}|f|)^2$, it follows that

$$\|\tau^{1/2}f\|_{\mathsf{Q}}^2 \le 2\,\langle\tau\rangle_{\mathsf{Q}}\,\|f\|_{\mathsf{Q}}^2 + 2\,|\tau|_\infty\,C_{\mathrm{Po}}\,\mathsf{l}^2\,(f, \nabla_{A_0}^*\nabla_{A_0}f)_{\mathsf{Q}}.$$

Step 4 (the good set $\mathcal{G}$; the first half of (b)). In the suprema that define κ_1 and κ_∞, the side ranges, as in [4, Proposition 16.1 and Sections 15–16], over the class Cl of those ℓ for which $\ell/(LM)$ is an odd integer not smaller than 13 [4, Proposition 16.1]. The exponent $1 - \varepsilon < \alpha < 1$ is the fixed Hölder order in Definition F.5. The numbers κ_1 and κ_∞ are finite, and each of them is at most a number that depends on L and α but not on e, m, N, j or the side. Indeed, for $j \ge 0$ and for sides in Cl, $|H_j(b,b')|$ and $|C_j^{1/2}(\Upsilon,\Upsilon')|$ are at most $C_P\exp(-\gamma_P d)$, where C_P and γ_P depend on L and α, and C has range at most $2L$ and row sums at most $c_C = c_C(L)$ [4, equations (10.16), (10.21)–(10.22), (10.26)]; the sides $\ell_j(R_{\mathrm{top}})$, $j < k_\star$, at which these bounds are applied lie in Cl [4, Proposition 16.1 and Section 16.1]; by the locality stated in [4, equations (10.21)–(10.22)], each column of C has at most $C(L)$ non-zero entries, each of modulus at most c_C; and the lattice sums of $\exp(-\gamma_P d)$ over the unit bonds and over the fine bonds with the weight η_j^3 are bounded by numbers depending on γ_P, while the sum over Υ is bounded by a number depending on γ_P and L. For large N one has $2j_1 \le k_\star$, so that the blocks Q tile $\mathbb{Z}^3/n\mathbb{Z}^3$; at most $4\,\mathsf{l}^3$ bonds meet a block Q. On $\prod_j \mathrm{Box}_j$ every coordinate

satisfies $|\zeta_j(\Upsilon)| \le p_{0,j} = v_j^{p_0}$, where $v_j = -\log e_j$ [4, Section 10.5, equations (10.21)–(10.29), and Section 13.3], [4, Proposition 16.1 and Sections 15–16]; since $e_j = qL^{j/2} \ge q$, one has $v_j \le v_0$, whence $|\tilde a_j|_\infty \le \kappa_1 L^{-j/2} v_0^{p_0}$, and therefore

$$|\tilde A|_\infty \le 2\,\kappa_1 v_0^{p_0}, \qquad \sum_{j \ge j_1} |\tilde a_j|_\infty \le 1.81\,\kappa_1.$$

On $\mathcal{G}$, the inequality $\big(\sum_{j<j_1} \tilde a_j\big)^2 \le 3.02 \sum_j L^{j/4} \tilde a_j^2$ leads to

$$\sum_{b \cap \mathsf{Q} \ne \emptyset} \tilde A(b)^2 \le 2\,(3.02 \cdot 16 \cdot 1.43 + 4 \cdot 3.3)\,\kappa_1^2\,\mathsf{I}^3 \le 200\,\kappa_1^2\,\mathsf{I}^3.$$

Hence the function $\varsigma(x) = |\exp(iq\tilde A(c, c_3)) - 1|$ (eight labels for each direction) satisfies $\langle \varsigma^2 \rangle_{\mathsf{Q}} \le 67\,\kappa_1^2 q^2$.

Step 5 (the Wilson energy on $\mathcal{G}$). Step 14 of the proof of Theorem 5.16, with H_θ in the place of Ω_0 ([4, Proposition 9.4] holds for every character), and the function

$$\varsigma'(x) := \big|\exp(iq(\tilde A + \mathbf{a})(c, c_3)) - 1\big| \le \varsigma(x) + q|\mathbf{a}|_\infty$$

give

$$X := (\psi, \nabla_{A_0}^* \nabla_{A_0} \psi) \le 6\,c_b^{-1}\big(L^{-2k_\star}\|\Phi\|^2 + \boldsymbol{\mu}^2 \|\psi\|^2\big) + 2\,C_5\,\|\varsigma'\psi\|^2.$$

The bounds that follow are asserted for the pairs (θ, A) with $q|\mathbf{a}|_\infty \le L^{-k_\star}$, among which, by Step 8, are those of the set N of Step 7. Step 3, with $\tau = \varsigma^2$ and summed over the blocks Q, together with Step 4, gives

$$\|\varsigma\psi\|^2 \le 134\,\kappa_1^2 q^2 \|\psi\|^2 + 2\,|\varsigma|_\infty^2\,C_{\mathrm{Po}}\,\mathsf{I}^2 X,$$

and $|\varsigma|_\infty^2\,\mathsf{I}^2 \le 4\,\kappa_1^2 L^4 q^2 v_0^{10 p_0}$, a quantity which tends to 0 as $N \to \infty$. For large N it follows that on $\mathcal{G}$

$$\boldsymbol{\mu}\|\psi\|^2 + W \le 6\rho\,c_b^{-1} L^{-2k_\star}\|\Phi\|^2 + \beta_N \|\psi\|^2,$$
$$\beta_N := \boldsymbol{\mu} + 6\rho\,c_b^{-1}\boldsymbol{\mu}^2 + 4\rho\,C_5\,L^{-2k_\star} + 540\,\rho\,C_5\,\kappa_1^2 q^2.$$

Step 6 (Gaussian domination and large deviations; the second half of (b)). Here $n = R_{\mathrm{top}}(e)/\eta = L^{k_\star}\ell_\star$ is the number of Theorem 5.16, which differs from the number n of the notation fixed above, and $C^1(n)$, $C^2(n)$ are the spaces of the real one-cochains and two-cochains on $\mathbb{Z}^3/n\mathbb{Z}^3$. Use the counting inner product on $F_N = dC^1(n)$ from Definition F.2. In this step write $|w| = \langle w, w\rangle^{1/2}$ for its norm. One has $\zeta_j(\Upsilon) = \langle \phi_{j,\Upsilon}, F^{\mathrm{u}}\rangle$, the $\phi_{j,\Upsilon}$ forming an orthonormal family in F_N [4, Proposition 22.1]; hence $\bar\mu(\exp\langle v, \zeta\rangle) \le \exp\frac12|v|^2$ at every real mass [4, Proposition 21.2]. From (K7) of Section 5 and Tonelli's theorem one obtains, for every real symmetric $\mathsf{M} \ge 0$ and every $s \ge 0$ with $4s\|\mathsf{M}\| \le 1$,

$$\bar\mu \exp\big(s\langle\zeta, \mathsf{M}\zeta\rangle\big) \le \det(1 - 2s\mathsf{M})^{-1/2} \le \exp(2s\,\mathrm{tr}\,\mathsf{M}).$$

Let M be the matrix with $\sum_{b\cap\mathsf{Q}\ne\emptyset} \tilde a_j(b)^2 = \langle \zeta_j, \mathsf{M}\zeta_j\rangle$; then $\mathrm{tr}\,\mathsf{M} \le 4\,\mathsf{I}^3 L^{-j}\kappa_1^2$, and $\|\mathsf{M}\| \le L^{2j}\kappa_1\kappa_\infty$ by (K23) of this section. Markov's inequality, with $s = (4L^{2j}\kappa_1\kappa_\infty)^{-1}$, bounds the $\bar\mu$-probability that this sum exceeds $16 L^{-j}\kappa_1^2\mathsf{I}^3$ by

$$\exp\Big(-\frac{2\kappa_1}{\kappa_\infty}\,(\mathsf{I}L^{-j})^3\Big);$$

moreover $\mathsf{I}L^{-j} \ge v_0^{2p_0}$ and $\log n \le 2v_0 + C(e)$.

Step 7 (from $\bar\mu$ to $\mathsf{P}^{0,A'}$). Apply Lemma F.8 with $\Phi(A, \zeta) = \mathbf{1}_{\mathcal{B}}(\zeta)$, and average over θ. At fixed (θ, A), the integral of w_N against the fluctuation boxes is $Y_*(A + c_\theta)$; dividing by this positive

number gives the probability $\mathsf{P}^{0,A+c_\theta}(\mathcal{B})$. The finite-dimensional integrals are Borel in (θ,A) by the lemma and Fubini's theorem. Consequently

$$\bar\nu(\{\zeta\in\mathcal{B}\}\cap\mathfrak{G})=\int\frac{d\theta}{(2\pi)^3}\int \mathsf{dm}_*(A)\,\chi_*(A)\,Y_*(A+c_\theta)\,\mathsf{P}^{0,A+c_\theta}(\mathcal{B}).$$

In this step $\bar\nu$ is the average over the twist

$$\bar\nu:=\int_{[-\pi,\pi]^3}\frac{d\theta}{(2\pi)^3}\,\nu^\theta,\qquad \nu^\theta:=\mathrm{dens}_N^{\theta,0}\,\gamma_N,$$

The density $\mathrm{dens}_N^{\theta,0}$ is $w_N(A(F)+H_\theta)$ of Definition F.2, with the tuned pair for m on $R_{\mathrm{top}}(e)$. By Definition F.2, $\bar\mu=\bar\nu/\bar\nu(1)$. Further, A_* is the field of the level $k_\star$ that [4, Section 22.1, Proposition 22.1] attaches to F^{u}, m_* is the law of A_* under γ_N, a measure carried by $V_{\mathrm{t}}(T^0_{k_\star})$, and $\chi_*(A):=\mathbf{1}[|dA|_\infty\le p_\star]$; the set $\mathfrak{G}$ of this step is the small-field set

$$\mathfrak{G}:=\{F^{\mathrm{u}}\in F_N:\ |\zeta_j(\Upsilon)|\le p_{0,j}\text{ for every }\Upsilon\text{ and every }j<k_\star,\ |dA_*|_\infty\le p_\star\}$$

of Lemma F.8. Here $Y_*(A+c_\theta)=\mathrm{Rtw}_\theta\,(1+\mathrm{vrem})$ with $|\mathrm{vrem}|\le\varepsilon_1$ ([4, Proposition 10.4, Theorem 22.2 and Proposition 22.6]; $\varepsilon_\star^{\mathrm{v},t}=0$ by [4, Theorem 16.3]), and

$$|\log\mathrm{Rtw}_\theta|\le c_{\mathrm{tw}}:=2\varepsilon_5(\tfrac1L)+C_\Sigma$$

by [4, Propositions 22.4–22.5] and [4, Proposition 16.1 and Sections 15–16]; moreover $\bar\nu(1)\le\bar\nu(\mathfrak{G})/(1-\varepsilon_3)$ [4, Proposition 22.7, equation (22.45)]. Let N be the set of the pairs (θ,A) with $\theta\in[\pi-r_1,\pi]^3$ and $|dA|_\infty\le r_1$; on N one has $\chi_*=1$, since $r_1\le1\le p_\star$, the second inequality because $v_\star\ge8$ by Definition F.1. By [4, Proposition 22.1], to each plaquette pl of $T^0_{k_\star}$ there corresponds an element $\phi_{\mathrm{pl}}\in F_N$ with $|\phi_{\mathrm{pl}}|\le1$ and $(dA_*)(\mathrm{pl})=\langle\phi_{\mathrm{pl}},F^{\mathrm{u}}\rangle$; as γ_N is the standard Gaussian measure of F_N (Definition F.2), the variables $(dA_*)(\mathrm{pl})$ are centred Gaussian under γ_N, with variance $|\phi_{\mathrm{pl}}|^2\le1$; so (K22) of Section 6 gives

$$\mathsf{m}_*(|dA|_\infty\le r_1)\ge\mathrm{erf}(r_1/\sqrt2)^{3\ell_\star^3}.$$

(A lower bound that does not depend on N holds also without (K22). One has $|dA|_\infty\le|dA|_2\le|dH_{k_\star}A|_2$, the first inequality because each plaquette of $T^0_{k_\star}$ carries the weight one, the second by [4, equation (10.2)]; by [4, Proposition 22.1], m_* has a density proportional to $\exp(-\frac12|dH_{k_\star}A|_2^2)$ on $V_{\mathrm{t}}(T^0_{k_\star})$, whose dimension is $2\ell_\star^3-2$ [4, Section 22.1], so that under m_* the variable $|dH_{k_\star}A|_2^2$ is χ^2-distributed with $2\ell_\star^3-2$ degrees of freedom; hence $\mathsf{m}_*(|dA|_\infty\le r_1)\ge(\frac12r_1^2)^{\ell_\star^3-1}\exp(-\frac12r_1^2)/\Gamma(\ell_\star^3)$.) Taking the mean over N, one finds a pair $(\theta,A)\in\mathsf{N}$ with $\mathsf{P}^{0,A+c_\theta}(\mathcal{G}^c)\le\varpi\,\bar\mu(\mathcal{G}^c)$, where

$$\varpi:=\frac{(1+\varepsilon_1)\exp(2c_{\mathrm{tw}})}{(1-\varepsilon_1)(1-\varepsilon_3)}\Big(\frac{2\pi}{r_1}\Big)^3\,\mathrm{erf}\Big(\frac{r_1}{\sqrt2}\Big)^{-3\ell_\star^3}.$$

The number ϖ depends on $(e,h,\mu_\star)$ and not on N: ε_1 and ε_3 depend on $(e_\star,\ell_\star,p_{0\star},p_\star)$ [4, Section 22.3, equation (22.15)], where $p_{0\star}:=p_{0,k_\star}=v_\star^{p_0}$ and $p_\star=v_\star^p$ are functions of $e_\star$ and of constants, and neither $e_\star$ nor $\ell_\star$ depends on N [4, Proposition 16.1 and Sections 15–16]; the estimate of [4, Propositions 22.4–22.5] holds uniformly with respect to the mesh $L^{-k_\star}$ and to $b_\star\in[b_-,1]$, so that c_{tw} is a constant; finally, r_1 is defined through b_-.

Step 8 (near Ω). For real $\mathscr{A}_1$ and $\mathscr{A}_2$,

$$\|D_k(\mathscr{A}_1)-D_k(\mathscr{A}_2)\|\le12\,\mu_k^{-2}e_k\,|\mathscr{A}_1-\mathscr{A}_2|_\infty.$$

This follows from the definitions of $S_k(\mathscr{A})$ and $D_k(\mathscr{A})$ in Definition F.5 and from the classical resolvent identity

$$S_k(\mathscr{A}_1)-S_k(\mathscr{A}_2)=S_k(\mathscr{A}_1)\Big(D_{\mathscr{A}_2}-D_{\mathscr{A}_1}+b_k\,J\big(P_k(\mathscr{A}_2)-P_k(\mathscr{A}_1)\big)J^*\Big)S_k(\mathscr{A}_2),$$

Here $b_k \le 1$, $\mu_k \le 1$, and accretivity of $D_{\mathscr{A}} + \mu_k + b_k J P_k(\mathscr{A}) J^*$ gives $\|S_k(\mathscr{A})\| \le \mu_k^{-1}$ for real $\mathscr{A}$. The two perturbation bounds following Definition F.5 are

$$\big\|\big(Q_k(\mathscr{A}_1) - Q_k(\mathscr{A}_2)\big) J^*\big\| \le \tfrac{3}{2}\, e_k |\mathscr{A}_1 - \mathscr{A}_2|_\infty, \qquad \|D_{\mathscr{A}_1} - D_{\mathscr{A}_2}\| \le 6\, e_k |\mathscr{A}_1 - \mathscr{A}_2|_\infty;$$

These follow from the explicit phase averages and four-hop formula in Definition F.5: the nested paths satisfy $\eta_k |\tau_k \mathscr{A}| \le \frac{3}{2}|\mathscr{A}|_\infty$, and the Schwarz inequality bounds the average on Δ_y. The terms add up to $b_k^2[3\mu_k^{-1} + (6+3b_k)\mu_k^{-2}] e_k |\mathscr{A}_1 - \mathscr{A}_2|_\infty$. Moreover $\|D_k(\mathscr{A})^{-1}\| \le g_k := b_k^{-1} + \mu_k^{-1}$ by (K10) of Section 5. On N, the fact (K24), the first clause of (21) and the bound $|H_{k_\star} A|_\infty \le C_0 |A|_\infty$, which is the kernel bound quoted in [4, equation (10.16) and the definition of the terminal constant C_0 in Section 21.4] and, being linear, holds for every real A, give

$$e_\star |H_{k_\star} A|_\infty \le e_\star C_0\, \ell_\star^{5/2} r_1 \le r_1.$$

Consequently $q|\mathbf{a}|_\infty \le L^{-k_\star}$ and $e_\star |\mathscr{A}' - \Omega|_\infty \le 2 r_1$. It follows that $D_{k_\star}(\mathscr{A}')$ differs in norm from $D_{k_\star}(\Omega)$ and from $D_{k_\star}(c_\theta)$ by at most

$$12\, \mu_\star^{-2} \cdot 2 r_1 = \min\Big\{\varkappa_D, \frac{c_-}{28}\Big\} \le \min\Big\{\varkappa_D, \frac{c_\star}{28}\Big\}.$$

From $D_{k_\star}(\Omega)\Psi_u = d_\star \Psi_u$, from Step 1 (ii) and from the sizes obtained in Steps 9 and 12 of the proof of Theorem 5.16 one concludes

$$c(A') \ge c_\star - 7\Big(\frac{c_\star}{28} + \ell_\star^3 \delta_\star\Big) - 2\delta_2 \ge \tfrac{1}{2} c_\star, \qquad \mathsf{E}^{A'} \|\Phi\|^2 \le 2.$$

Step 9 (from $\mathsf{P}^{0,A'}$ to $\mathsf{P}^{A'}$). By Step 1 (i), $\mathsf{E}^{A'}[\det \mathcal{D}; \mathcal{B}] = \mathsf{E}^{A'}[\det \mathcal{D}]\, \mathsf{P}^{0,A'}(\mathcal{B})$. Let $E_{\star,00}(X) := E_{k_\star}(X; \mathscr{A}', 0, 0)$ be the component of degree $(0,0)$ of $E_{k_\star}(X; \mathscr{A}', H^{\#}_{k_\star}(\mathscr{A}')\Psi)$. Since $m_{k_\star} = 0 = \varepsilon^{\mathrm{v},t}_{k_\star}$, Definition F.6 gives $\mathscr{F}_{k_\star}(A'; 0) = \det D_{k_\star}(0)^{-1} \exp \sum_X E_{\star,00}(X)$, and therefore

$$\mathsf{E}^{A'} \det \mathcal{D} = \frac{Y_*(A')}{\mathscr{F}_{k_\star}(A'; 0)} = \det D_{k_\star}(c_\theta)\, (1 + \mathrm{vrem}) \exp\Big(-\sum_X E_{\star,00}\Big) \le 1.21\, \det D_{k_\star}(c_\theta),$$

by [4, Proposition 22.5, equation (22.31), and its proof] and because $|\sum_X E_{\star,00}| \le e_\star^{1/4-7\varepsilon} \le \frac{1}{8}$. Here the bound of Step 9 of the proof of Theorem 5.16, taken at $\mathscr{A}'$, gives $|\sum_X E_{\star,00}| \le 4 e_\star^{1/4-6\varepsilon} K_{\mathrm{pol}} (\ell_\star/M)^3$, and the second and third clauses of (21), with $c_{\mathrm{fl}} = 3294 K_{\mathrm{pol}}$, give the two inequalities. On the event $\{\|\mathcal{D} - D_{k_\star}(\mathscr{A}')\| \le \varkappa_D\}$, two applications of (K23), together with $g_\star \varkappa_D \le g_- \varkappa_D = (16 \ell_\star^3)^{-1}$, give $\det \mathcal{D} \ge \frac{9}{16} \det D_{k_\star}(c_\theta)$. By the first clause of (21), which reads $8 C_0^2 \ell_\star^6 g_- \le e_\star^{-\varepsilon/2}$, and by (20), one has $\|K_E\| \le \frac{1}{2}\varkappa_D$; by Chebyshev's inequality,

$$\mathsf{P}^{A'}\big(\|\delta\mathcal{D}\|_{\mathrm{HS}} > \tfrac{1}{2}\varkappa_D\big) \le 8\, \varkappa_D^{-2}\, \ell_\star^3\, \delta_2 \le e_\star^{5/4-12\varepsilon}.$$

Hence $\mathsf{P}^{A'}(\mathcal{B}) \le e_\star^{5/4-12\varepsilon} + 3\, \mathsf{P}^{0,A'}(\mathcal{B})$.

Step 10 (conclusion). Let (θ, A) be the pair of Step 7. Steps 2 (with $\mathcal{B} = \mathcal{G}$), 5, 8 and 9 give

$$\tfrac{1}{2} c_\star \le 12 \rho\, c_b^{-1} L^{-k_\star} + L^{k_\star} \beta_N\, \mathsf{E}^{A'}[\|\psi\|^2; \mathcal{G}] + \tfrac{1}{4}\big(e_\star^{5/4-12\varepsilon} + 3\varpi\, \bar{\mu}(\mathcal{G}^c)\big).$$

By (21), $c_\star \ge \frac{1}{3} d_\star \ge \frac{8}{3} e_\star^{\varepsilon/2}$ and $e_\star^{5/4-12.5\varepsilon} \le e_\star^{1/4-9\varepsilon} \le \frac{1}{8}$, so that $\frac{1}{4} e_\star^{5/4-12\varepsilon} \le \frac{1}{16} c_\star$. For large N the first term on the right is at most $\frac{1}{8} c_\star$, and $\frac{3}{4}\varpi\, \bar{\mu}(\mathcal{G}^c) \le \frac{1}{16} c_\star$, since $c_\star \ge c_-$ and neither c_- nor ϖ depends on N (Steps 6 and 7); thus the first and the last term are at most $\frac{1}{8} c_\star$ each. As $\mathsf{E}^{A'}\|\psi\|^2 < \infty$ by Step 6 of the proof of Theorem 5.16, it follows that $\beta_N > 0$. If $\boldsymbol{\mu} < 0$, then for large N one has $6\rho\, c_b^{-1} |\boldsymbol{\mu}| \le \frac{1}{2}$ [4, Theorem 16.4, equation (16.19)] and $4 L^{-2k_\star} \le 60\, \kappa_1^2 q^2$, whence $\frac{1}{2}\boldsymbol{\mu} + 600\, \rho\, C_5\, \kappa_1^2 q^2 > 0$. Accordingly, $N_A(e, m)$ is the least $N_0 \ge N_{\min}(R_{\mathrm{top}}(e), e)$ such that every $N \ge N_0$ satisfies

$$2 j_1 \le k_\star, \qquad 64\, C_5\, C_{\mathrm{Po}}\, \kappa_1^2 L^4 q^2 v_0^{10 p_0} \le 1 \quad \text{(Step 5)}, \qquad 96 \rho\, \eta \le c_b c_- a_\star,$$

$$12\,\varpi\, j_1 n^3 \exp\Big(-\frac{2\kappa_1}{\kappa_\infty}\, v_0^{6p_0}\Big) \le c_-, \qquad 3\rho\, q^{3/4-8\varepsilon} \le 2c_b, \qquad \eta \le 15\,\kappa_1^2 e^2 a_\star^2.$$

This number is a function of $(e, h, \mu_\star)$ and of constants. □

Proof of Theorem 6.1. Put $h_j := L^j h$ for $0 \le j \le J'$, where $h_{J'} = h_J$; a subscript j marks the numbers with the index $\star$ that belong to the class I_{h_j}. Let $N_1(e)$ be the largest of the numbers $N_A(e, m_\flat)$ and $N_{\min}(R_{\mathrm{top}}(e; h'), e)$, where h' ranges over the finite set of the powers of L in $[1, h_J]$, that is, of the numbers $h(m)$ with $m \in (0, 1]$ and $e\,h(m)^{1/2} \le e_1$.

Step 1. The upper bound is Theorem 5.16 (b). By Proposition 6.2 (a), $\varphi_N(m_\flat) \ge -C_{\mathsf{W}} e^2$. Let $h = h_J$. Then (M) of Proposition 6.3 (c), applied on $[\bar m, m_\flat]$ together with $w_\star/a_\star \le \frac{1}{8} m_\flat$ (the bound on w_j/a_j of Step 3), gives $\varphi_N(\bar m) \ge -C_{\mathsf{W}} e^2 - 2m_\flat$; this is at least $\bar m - C_\natural e^2$, because $\bar m \le m_\flat < L e^2/e_1^2$.

Step 2 (a chain of classes, $h < h_J$). Let $h < h_J$, and for $j = J' - 1, \dots, 0$ put

$$m_b^{(j)} := \frac{1}{L h_j}, \qquad \mathrm{top}_j := \frac{1}{h_j}\ \ (j \ge 1), \qquad \mathrm{top}_0 := \bar m, \qquad m_c^{(j)} := \min\Big\{ m_b^{(j)} + \frac{w_j}{2a_j},\, \mathrm{top}_j \Big\}.$$

Adding the inequalities (J) of Proposition 6.3 (d), from $m_b^{(j)} = \mathrm{top}_{j+1}$ to $m_c^{(j)}$, and (M), from $m_c^{(j)}$ to top_j, one obtains

$$\varphi_N(\bar m) - \varphi_N(m_\flat) \ge (\bar m - m_\flat) - \sum_j e_j^\varepsilon (\mathrm{top}_j - m_b^{(j)}) - \sum_j \frac{w_j}{a_j}\Big(\frac{2}{3} + e_j^\varepsilon\Big).$$

Step 3 (the error terms). One has $\mathrm{top}_0 - m_b^{(0)} \le \bar m$ and $\mathrm{top}_j < L^{1-j}\bar m$; moreover $e_j \le L^{j/2} e\, h^{1/2}$, and $e_j^\varepsilon \le \frac{1}{48}$ by (20). By (21), $\mu_\star^{-1} \le \frac{1}{8} e_\star^{-\varepsilon/2}$ for every parameter of the class; letting the parameter tend to $\frac{1}{L h_j}$, one obtains

$$\frac{w_j}{a_j} \le \frac{1}{8 L h_j}\, e_j^{3/4-9\varepsilon} < \frac{\bar m}{8}\, L^{-j} e_j^{3/4-9\varepsilon},$$

and $e_j^{1/4-9\varepsilon} \le \frac{1}{8}$. Therefore the two sums of Step 2 are together at most $\epsilon_e \bar m$.

Step 4 (proof of (a) to (c)). If $e \le e_7$, then $C_\natural e^2 \le \frac{1}{4}\bar m$ and $e^2 L h \le e_1^2/12$, so that $\varphi_N(\bar m) \ge (\frac{11}{12} - \frac{1}{4})\bar m$. For (b), one has $h^{\varepsilon/2}\bar m \le 1$. For (c): by Theorem 5.16 (c), since $\mathsf{W}_N \ge 0$ and $c_\star = \bar m\, a_\star Z_0$,

$$Z_N \le \frac{\eta}{a_\star}\, c_\star\, \frac{1 + \vartheta_N}{\boldsymbol{\mu}_N} \le 2.1\, Z_0.$$

The numbers Z_0 and $c_\star$ depend on N through $b_\star$. □

6.2. **One law.**

Lemma 6.4. *What the atom needs; the measure* kl *and the components.* Let $e > 0$, let the mass assignment be arbitrary, and let μ be a volume-first limit point with the ergodic decomposition $\mu = \int \mu_\xi\, d\nu(\xi)$.

(a) The bounds of Theorem 5.14 are statements at a single cutoff, and the conclusion of Theorem 5.14 (c) is obtained along one defining sequence, in the following form. Let $c > 0$, let $\mu = \lim_i \mu^{(N_i)}$ with $\mu^{(N_i)} \in \mathcal{V}_{N_i}$, and suppose that $c\,\eta_i \le \boldsymbol{\mu}_{N_i} \le 1$ for all large i, where $\eta_i = L^{-N_i}$. Then, along a defining sequence of μ, every probe β satisfies

$$\bar I(\beta) \le \Big(1 + \frac{C_F' e^2}{c}\Big)\|f\|^2, \qquad C_F' := 15\,\pi^4 \mathsf{h}_3^4\, C_\sharp\, \max\{1, e^2/c\}^3,$$

and

$$\mathrm{kl}_\mu(\{0\}) \ge \Big(1 + \frac{C_F' e^2}{c}\Big)^{-1}.$$

Consequently, if $\limsup_N \boldsymbol{\mu}_N/\eta > 0$, the atom is obtained at the limit points taken along those N_i for which $c\,\eta_i \le \boldsymbol{\mu}_{N_i} \le 1$, and nothing is asserted at the limit points taken along other sequences.

(b) The map $\xi \mapsto \mathrm{kl}_\xi$ is measurable, and $\mathrm{kl}_\mu = \int \mathrm{kl}_\xi \, d\nu(\xi)$; in particular $\mathrm{kl}_\mu(\{0\}) = \int \mathrm{kl}_\xi(\{0\})\, d\nu(\xi)$.

■

Proof. Step 1 (proof of (a)). The first display of Theorem 5.14 is a statement about one quadruple $(q, \mathbf{m}, \mathbf{m}', n)$, and Step 30 of the proof of Theorem 5.14 takes $\mathbf{m} = \boldsymbol{\mu}_N$ and $\mathbf{m}' = \max\{\boldsymbol{\mu}_N, e^2\eta\}$. Theorem 5.14 (b) therefore holds, with $c_{\boldsymbol{\mu}}\bar{m} = c$, at every cutoff at which $c\,\eta \le \boldsymbol{\mu}_N \le 1$, $e^2\eta \le 1$ and $n\boldsymbol{\mu}_N \ge 20$. By Step 5 of the proof of Proposition 5.13, μ has defining sequences (N_i, T_i) whose T_i are as large as one wishes. The assertions of (a) then follow from Lemma E.10 and Proposition 5.6 (d).

Step 2 (proof of (b)). By [5, Theorem A] and Proposition D.1, almost every μ_ξ has a Källén–Lehmann measure of mass at most one. Let D_ξ and D_μ be the functions D of Step 9 of the proof of Proposition 5.6, formed with kl_ξ and with kl_μ respectively. The function $\mathsf{D}_\xi(r)$ is continuous in r, and it arises as the limit of $S_2^\xi(g_n, g_n)$ along forms g_n whose weights accumulate on the sphere $|p|^2 = r$; it is therefore measurable in ξ (Lemma G.2). The identity $S_2^\mu = \int S_2^\xi \, d\nu$ [5, Theorem A and Section 5] and the bounds $0 \le \mathsf{D}_\xi \le 1$ give $\mathsf{D}_\mu(r) = \int \mathsf{D}_\xi(r)\, d\nu(\xi)$. By partial fractions, the closed linear span of the functions $s \mapsto (r+s)^{-1}$, $r > 0$, is an algebra, and by the theorem of Stone and Weierstrass (K28) it is the whole of $C_0([0,\infty))$. Hence $\int \mathrm{kl}_\xi \, d\nu$ is a measure of mass at most one that represents S_2^μ. Such a measure is unique, at every $e > 0$ and for every mass assignment, by Proposition D.1, and also by the density just established. For the atoms one lets $r \downarrow 0$ and applies dominated convergence.

□

7. Scores and ergodic components

The notation (FI_λ) with constant C means $\lambda(\partial_f\Phi)^2 \le C\|f\|^2\lambda(\Phi^2)$ for every probe and every smooth bounded cylinder function with all derivatives bounded. Equivalently $C(\lambda) \le C$.

Conventions. Unless stated otherwise, μ denotes a law that is invariant under $\mathbb{R}^3$ and has the property (G) of the notation fixed above; the σ-algebra I, the kernel $\omega \mapsto \mu_\omega$ and the measure ν are those of [5, Theorem A and Section 5], and "almost every ξ" is understood in the sense of Definition G.1. Put $L^2_{\mathrm{c}} := \Pi_{\mathrm{c}} L^2(\mathbb{R}^3; \Lambda^2)$. For a law lw, the symbol $\mathcal{A}(\mathrm{lw})$ stands for the class of the functions $\Phi = \phi(X(g_1), \dots, X(g_k))$ of Proposition 5.6 (c), regarded as elements of $L^2(\mathrm{lw})$, with $\partial_f\Phi$ defined as there; its subclass $\mathcal{A}_{\mathrm{c}}$ is formed by the Φ with $g_j \in C_c^\infty$, which are functions on $\mathcal{S}'$ and constitute an algebra, and on which $U(a)\Phi = \phi(X(\tau_a \cdot g_1), \dots, X(\tau_a \cdot g_k))$. For a law lw with (G) and for $f \in L^2_{\mathrm{c}}$ set

$$I_{\mathrm{lw}}(f) := \sup_{\Phi \in \mathcal{A}(\mathrm{lw})} \left[2\,\mathrm{lw}(\partial_f\Phi) - \mathrm{lw}(\Phi^2)\right] \in [0,\infty], \qquad C(\mathrm{lw}) := \sup_{\beta:\, dd^*\beta \neq 0} \frac{I_{\mathrm{lw}}(dd^*\beta)}{\|dd^*\beta\|^2}.$$

Replacing Φ by $t\Phi$, with real t, one finds that $I_{\mathrm{lw}}(f) \le \mathsf{c}$ is equivalent to $\mathrm{lw}(\partial_f\Phi)^2 \le \mathsf{c}\,\mathrm{lw}(\Phi^2)$ for all $\Phi \in \mathcal{A}(\mathrm{lw})$; thus ($\mathrm{FI}_{\mathrm{lw}}$) with constant C is equivalent to $C(\mathrm{lw}) \le C$. One writes $I_\xi := I_{\mu_\xi}$, $C_\xi := C(\mu_\xi)$ and $\mathsf{z}_\xi := \mathrm{kl}_\xi(\{0\})$. For the probes considered, $f = dd^*\beta \neq 0$; with $\beta^{(l)} := l^{1/2}\beta(\cdot/l)$ and $f_l := dd^*\beta^{(l)}$ one has $\|f_l\| = \|f\|$. A *law of the class* means a member of $\mathcal{K}$.

Put $\mathcal{K}(C) = \{\lambda \in \mathcal{K} : C(\lambda) \le C\}$. The space $\mathcal{D}_0 \subset C_c^\infty$ is the countable $\mathbb{Q}$-vector space of [5, Section 5]. The Gaussian laws γ^ϱ and the operation $*_\epsilon$ are defined in Section 4. Scores are denoted by s_f; $T_f\omega = \omega + f$. The constants $C_\natural$, e_7 and $N_1(e)$ are those of Section 6, and $C_\flat$ is the lattice clustering constant in Proposition 5.17. *Classical facts.*

(K12) (Riesz.) Let a linear functional be defined on a dense subspace of a Hilbert space and have norm κ. Then it equals $\langle s, \cdot\rangle$ for exactly one vector s, and $\|s\| = \kappa$.

(K13) The theorems of monotone and of dominated convergence, Fatou's lemma and Tonelli's theorem; Jensen's inequality for the function $x \mapsto 1/x$. (Monotone class theorem.) On some set, consider a vector space Mv of bounded real functions which contains the constant functions and is stable under pointwise limits of increasing sequences that are uniformly bounded, and a subset Mu $\subset$ Mv that is closed under products. Then every bounded σ(Mu)-measurable function belongs to Mv.

(K14) If $(Y_0, \dots, Y_k)$ is a centred Gaussian vector and $\phi \in C^1_b$, then

$$E\big[Y_0\, \phi(Y_1, \dots, Y_k)\big] = \sum_j \operatorname{Cov}(Y_0, Y_j)\, E[\partial_j \phi].$$

(K15) (von Neumann.) For a unitary representation of $\mathbb{Z}^3$, the averages over the sets $\{|x|_\infty \le \ell\}$ converge strongly to the projection onto the invariant vectors.

(K16) (Stone, Naimark, Ambrose, Godement.) Every strongly continuous unitary representation of $\mathbb{R}^3$ has the form $\int \exp(i\langle a, \mathsf{p}\rangle)\, dE(\mathsf{p})$ with a unique projection valued measure E. A closed subspace that is invariant under a bounded self-adjoint operator reduces that operator. The identity theorem. (Transitive actions.) Suppose that a locally compact second countable group G, with left Haar measure λ_G, acts continuously and transitively on a second countable, locally compact Hausdorff space Y. If m is a σ-finite Borel measure on Y and $S \subset Y$ is a Borel set with $m(gS \triangle S) = 0$ for all $g \in G$, then either $m(S) = 0$ or $m(Y \setminus S) = 0$.

Proof. The set $N := \{(g, y) : 1_S(gy) \neq 1_S(y)\}$ is Borel, and its section at g is $g^{-1}S \triangle S$, which is m-null. Tonelli's theorem shows that for m-almost every y the set $\{g : 1_S(gy) \neq 1_S(y)\}$ is λ_G-null. Assume that $m(S) > 0$ and $m(Y \setminus S) > 0$. Then there are points $y \in S$ and $y' \notin S$ with this property, and some $g_0 \in G$ with $y' = g_0 y$. It follows that $gy \in S$ for λ_G-almost every g, and that $gg_0y \notin S$ for λ_G-almost every g. The second assertion means that $g'y \notin S$ for every g' in a right translate of a conull set; since a right translate of a λ_G-null set is null, by the modular function, this is incompatible with $\lambda_G \neq 0$. □

Four facts on one law with (G). In (S1)–(S4), lw is a law with (G).

(S1) The functions $\cos X(g)$ and $\sin X(g)$, $g \in \mathcal{D}_0$, span over $\mathbb{R}$ a subspace of $\mathcal{A}_{\mathrm{c}}$ that is dense in the real space $L^2(\mathrm{lw})$ [5, Section 5].

(S2) If $I_{\mathrm{lw}}(f) < \infty$, then $\mathrm{lw}(\partial_f \Phi)$ is a linear functional of the class of Φ, of norm $I_{\mathrm{lw}}(f)^{1/2}$. Hence, by (S1) and (K12), there is a unique $\mathsf{s}_f \in L^2(\mathrm{lw})$, called the *score*, such that $\mathrm{lw}(\partial_f \Phi) = \mathrm{lw}(\Phi\, \mathsf{s}_f)$ for every $\Phi \in \mathcal{A}(\mathrm{lw})$; moreover $\|\mathsf{s}_f\|_2^2 = I_{\mathrm{lw}}(f)$. For the s_β of Proposition 5.6 (c) this gives $\mathsf{s}_\beta = \mathsf{s}_f$ and $I_\mu(f) \le \bar{I}(\beta)$.

(S3) Let $\mathcal{F}_k \subset C^\infty_c(\mathbb{R}^k)$ be a countable set that is dense in $C^\infty_c(\mathbb{R}^k)$ for the C^2_b norm, and let $\mathcal{A}_0 \subset \mathcal{A}_{\mathrm{c}}$ be the countable set of those Φ for which $g_j \in \mathcal{D}_0$ and $\phi \in \mathcal{F}_k$. Each $\Phi \in \mathcal{A}(\mathrm{lw})$ is the limit of a uniformly bounded sequence of $\Phi_n \in \mathcal{A}_0$ such that $\Phi_n \to \Phi$ and $\partial_f \Phi_n \to \partial_f \Phi$ in $L^2(\mathrm{lw})$ for every f. Such a sequence is obtained by cutting ϕ off at the scale n, approximating in C^2_b, choosing $g_j^{(n)} \to g_j$ in L^2, and using the bound $\|X(g)\|_2 \le \|g\|$ [4, Proposition 5.1]. It follows that the supremum defining I_{lw} may be restricted to $\mathcal{A}_0$, that $\xi \mapsto I_\xi(f)$ is measurable by Lemma G.2, and that an identity $\mathrm{lw}(\partial_f \Phi) = \mathrm{lw}(\Phi\, \mathsf{s})$ with $\mathsf{s} \in L^2(\mathrm{lw})$ extends from $\mathcal{A}_0$ to $\mathcal{A}(\mathrm{lw})$.

(S4) On L^2_{c}, the function $f \mapsto I_{\mathrm{lw}}(f)$ is a supremum of continuous affine functions and is therefore lower semicontinuous; moreover the forms $dd^*\beta$, $\beta \in \mathcal{D}_0$, are dense in L^2_{c}, by Step 7 of the proof of Proposition 5.6 and the density of $\mathcal{D}_0$ in $\mathcal{S}$. Hence $C(\mathrm{lw})$ equals the supremum over $\beta \in \mathcal{D}_0$, the map $\xi \mapsto C_\xi$ is measurable, and $I_{\mathrm{lw}} \le C(\mathrm{lw})\, \|\cdot\|^2$ on L^2_{c}.

7.1. **Disintegration of the score.**

Theorem 7.1. *The atom in almost every component.* Assume that $0 < \rho < 1$, that the weights are standard, that $\bar{m} \in (0, 1]$ and that $e > 0$; the mass assignment is arbitrary, every physical one and the tuned ones being admitted.

(a) *One probe; limits in any order.* Fix a limit point μ together with a defining sequence, a probe β with $\bar{I}(\beta) < \infty$, and a Σ-measurable representative s of s_β. Then, for almost every ξ, the function s belongs to $L^2(\mu_\xi)$ and serves as the score of μ_ξ along f; furthermore

$$\int I_\xi(f)\, d\nu = I_\mu(f) \le \bar{I}(\beta), \qquad \|f\|^4 \le S_2^\xi(f,f)\, I_\xi(f), \qquad \int \frac{\|f\|^4}{S_2^\xi(f,f)}\, d\nu(\xi) \le \bar{I}(\beta).$$

Consequently $S_2^\xi(f,f) > 0$, and the components equal to the zero field form a ν-null set.

(b) *The atom.* Let μ be volume-first, and let (N_i, T_i) be a defining sequence of μ as a limit point; such a sequence exists, since by [4, Section 6, Section 7.5, and Theorem 8.6] each iterated limit is a limit point; compare Step 9 of the proof of Proposition 5.6. Let $\bar{I}$ be formed along this sequence, let β be *one* probe, and let $l_k \to \infty$ be *one* sequence. Then

$$\int \mathsf{z}_\xi^{-1}\, d\nu \le \liminf_{k\to\infty} \frac{\bar{I}(\beta^{(l_k)})}{\|f\|^2}.$$

This inequality holds for every defining sequence, and the hypothesis of Proposition 5.6 (d) is understood along one of them, as in Lemma 6.4(a) and Theorem 7.1(b). If this hypothesis is satisfied, then

$$\int \mathrm{kl}_\xi(\{0\})^{-1}\, d\nu(\xi) \le 1 + c.$$

Hence $\mathrm{kl}_\xi(\{0\}) > 0$ **for ν-almost every** ξ, and $\nu\{\mathsf{z}_\xi \le t\} \le ct/(1-t)$ for $0 < t < 1$. Since $\mathrm{kl}_\mu(\{0\}) = \int \mathsf{z}_\xi\, d\nu$, an identity contained in the equality $\mathrm{kl}_\mu = \int \mathrm{kl}_\xi\, d\nu$ of measures in Lemma 6.4 (b), Jensen's inequality gives back the conclusion of Proposition 5.6 (d).

(c) *Laws.* Part (a) extends to every law μ that has (G) and is invariant under $\mathbb{R}^3$, with $I_\mu(f)$ replacing $\bar{I}(\beta)$, and to every $f \in L^2_{\mathrm{c}} \setminus \{0\}$ of compact support such that $I_\mu(f) < \infty$; without the last condition, the identity $\int I_\xi(f)\, d\nu = I_\mu(f)$ holds in $[0, \infty]$. Part (b) extends to all laws of the class, with $I_\mu(f_{l_k})$ replacing $\bar{I}(\beta^{(l_k)})$. Accordingly $\int \mathsf{z}_\xi^{-1}\, d\nu \le C$ for $\mu \in \mathcal{K}(C)$.

■□

Proof. Step 1 (reduction to (c)). Every limit point is invariant under $\mathbb{R}^3$ and has (G), and a volume-first limit point belongs to the class [4, Section 6, Section 7.5, and Theorem 8.6]. Moreover (S2) gives $I_\mu(f) = \|\mathsf{s}_\beta\|_2^2 \le \bar{I}(\beta)$.

Step 2 (an invariant function is a constant for ∂_f). Consider $f \in L^2_{\mathrm{c}}$ of compact support with $I_\mu(f) < \infty$, a function $\Phi \in \mathcal{A}_{\mathrm{c}}$, and $\mathsf{v} \in L^2(\mu)$ such that $U(a)\mathsf{v} = \mathsf{v}$ for all a. It is claimed that

$$\mu(\mathsf{v}\, \partial_f \Phi) = \mu(\mathsf{v}\, \Phi\, \mathsf{s}_f).$$

Given $\epsilon > 0$, (S1) furnishes $\Psi = \psi(X(g_1'), \dots, X(g_{k'}')) \in \mathcal{A}_{\mathrm{c}}$ with $\|\Psi - \mathsf{v}\|_2 < \epsilon$; choose a such that none of the sets $a + \operatorname{supp} g_i'$ meets $\operatorname{supp} f$. The function $\Psi_a := U(a)\Psi$ lies in $\mathcal{A}_{\mathrm{c}}$ and satisfies $\partial_f \Psi_a = 0$; moreover $\|\Psi_a - \mathsf{v}\|_2 = \|U(a)(\Psi - \mathsf{v})\|_2 < \epsilon$, because translation invariance of μ makes $U(a)$ unitary. As $\Phi\Psi_a \in \mathcal{A}_{\mathrm{c}}$, the Leibniz rule and (S2) yield

$$\mu(\Psi_a\, \partial_f \Phi) = \mu(\partial_f(\Phi\Psi_a)) = \mu(\Psi_a\, \Phi\, \mathsf{s}_f).$$

Replacing Ψ_a by v changes the first member of this chain by at most $\epsilon \sup|\partial_f \Phi|$ and the last one by at most $\epsilon \sup|\Phi|\, \|\mathsf{s}_f\|_2$.

Step 3 (components). For $\mathrm{Ie} \in I$, Step 2 with $\mathsf{v} = 1_{\mathrm{Ie}}$ expresses the equality $E_\mu[\partial_f \Phi \,|\, I] = E_\mu[\Phi\, \mathsf{s} \,|\, I]$. Consequently $\mu_\omega(\partial_f \Phi) = \mu_\omega(\Phi\, \mathsf{s})$ outside a μ-null set of ω; moreover $\int \mu_\omega(\mathsf{s}^2)\, d\mu = \mu(\mathsf{s}^2)$ by [5, Theorem A and Section 5]. By Lemma G.2 the set

$$\mathcal{B}_\Phi := \{\xi :\ \xi(\mathsf{s}^2) < \infty,\ \xi(\partial_f \Phi) = \xi(\Phi\, \mathsf{s})\}$$

is measurable, and [5, Theorem A and Section 5] gives $\nu(\mathcal{B}_\Phi) = 1$. By [5, Theorem A and Section 6], read with Definition G.1, some set of the σ-algebra of Definition G.1 has ν-measure 1 and consists of laws with (G); let Ξ_f be the intersection of such a set with the sets $\mathcal{B}_\Phi$, $\Phi \in \mathcal{A}_0$. If $\xi \in \Xi_f$, then by (S3) and (S2) the function s is the score of μ_ξ, and $I_\xi(f) = \xi(\mathsf{s}^2)$; the integral of the latter with respect to ν is $\mu(\mathsf{s}^2) = I_\mu(f)$.

Step 4 (the identity in $[0,\infty]$). Assume that $\int I_\xi(f)\,d\nu < \infty$. For $\Phi \in \mathcal{A}_\mathrm{c}$ the Schwarz inequality gives

$$\mu(\partial_f \Phi)^2 \le \Big(\int I_\xi(f)^{1/2}\, \xi(\Phi^2)^{1/2}\, d\nu \Big)^2 \le \int I_\xi(f)\, d\nu \cdot \mu(\Phi^2).$$

By (S3) it follows that $I_\mu(f) < \infty$, and Step 3 applies.

Step 5 (the Cramér–Rao inequality within a component). For $\xi \in \Xi_f$ put $\Phi_r := r \tanh(X(f)/r)$; then $\|f\|^2\, \xi(\cosh^{-2}(X(f)/r)) = \xi(\Phi_r\, \mathsf{s})$. Since $|\Phi_r| \le |X(f)|$, letting $r \to \infty$ and using (K13) one obtains

$$\|f\|^2 = \xi(X(f)\,\mathsf{s}) \le S_2^\xi(f,f)^{1/2}\, I_\xi(f)^{1/2}.$$

Division by $S_2^\xi(f,f)$ and integration over ξ by means of Step 3 give (a).

Step 6 (scales). Let μ be a law of the class, and let Ξ_1 be a measurable set with $\nu(\Xi_1) = 1$ on which μ_ξ is a law of the class, is ergodic, clusters on $L^2(\mu_\xi)$, and has a two-point function S_2^ξ of Källén–Lehmann form with the measure kl_ξ ([5, Theorem A], Corollary 2.2 and Proposition D.1). For $\xi \in \Xi_1$ and $r > 0$ put $\mathsf{D}_\xi(r) := \int d\mathrm{kl}_\xi(t)\, r/(r + t^2)$, the function D of Step 9 of the proof of Proposition 5.6 formed with kl_ξ, as in Step 2 of the proof of Lemma 6.4; let ϖ be the probability measure on the momenta k with $d\varpi \propto |\hat f|^2\, dk$, which is the Fourier probability measure used in Proposition 4.1. The number

$$x_l(\xi) := \frac{S_2^\xi(f_l, f_l)}{\|f\|^2} = \int \mathsf{D}_\xi(|k|^2/l^2)\, d\varpi(k)$$

is a non-increasing function of $l > 0$ with limit z_ξ, since the computation in Proposition 4.1 is valid for every law whose two-point function has the Källén–Lehmann form. Hence $\xi \mapsto \mathsf{z}_\xi$ is measurable and $x_l^{-1} \uparrow \mathsf{z}_\xi^{-1}$. Choose an increasing subsequence of (l_k) along which $I_\mu(f_{l_k})/\|f\|^2$ tends to its lower limit, which may be taken finite, since otherwise there is nothing to prove. Along it, (K13) and Step 5 at f_{l_k} give

$$\int \mathsf{z}_\xi^{-1}\, d\nu = \lim_k \int x_{l_k}^{-1}\, d\nu \le \liminf_k \frac{I_\mu(f_{l_k})}{\|f\|^2}.$$

Step 7 (consequences). As $\mathsf{z}_\xi \le 1$, the inequality $\int (\mathsf{z}_\xi^{-1} - 1)\, d\nu \le c$ has a non-negative integrand, and the tail bound is Markov's inequality. By [5, Theorem A and Section 5], $S_2^\mu(f_l, f_l) = \int S_2^\xi(f_l, f_l)\, d\nu$, and the limit $l \to \infty$ gives

$$\mathrm{kl}_\mu(\{0\}) = \int \mathsf{z}_\xi\, d\nu \ge \Big(\int \mathsf{z}_\xi^{-1}\, d\nu \Big)^{-1}.$$

For volume-first limit points this equality is the case of the atom in Lemma 6.4 (b).

Step 8 (the order of the choices). First μ, the kernel and ν are fixed; then the set Ξ_1; then a single probe; then a countable family of scales $l = l_k$; for every such l, a representative of s_{f_l}; then the countable class $\mathcal{A}_0$, which for (b) may be replaced by the functions $r \tanh(X(f_l)/r)$, $r \in \mathbb{N}$; and last of all one exceptional null set, namely the complement of $\Xi_1 \cap \bigcap_l \Xi_{f_l}$, which belongs to the σ-algebra of Definition G.1. The function Ψ and the translation a of Step 2 serve an approximation in $L^2(\mu)$ and create no further null set. The limits $r \to \infty$ and then $l \to \infty$ are taken inside a fixed component, and the integration over ξ is performed last. The null set depends on β, whereas the conclusion does not. □

7.2. **Sharpness for Gaussian mixtures.**

Proposition 7.2. *Optimality; what is false.* The equality in (b) concerns the full class $\mathcal{K}(C)$: Theorem 7.1(c) gives the upper-bound inclusion, and the Gaussian mixtures in (a) give the reverse inclusion. The examples below are not asserted to be QED3 cutoff limit points or their components.

(a) Let π be a Borel probability measure on $(0,1]$, and put

$$\mu_\pi := \int \gamma_a \, d\pi(a).$$

The law μ_π belongs to the class, and the map $\omega \mapsto -\omega$ leaves it invariant. The components of μ_π are the laws γ_a; its measure ν is the image of π under $a \mapsto \gamma_a$; the component $\xi = \gamma_a$ has $\mathsf{z}_\xi = a$ and $C_\xi = 1/a$. For every probe,

$$I_{\mu_\pi}(f) = \|f\|^2 \int a^{-1} \, d\pi(a).$$

(b) Let $C \geq 1$. As μ runs through $\mathcal{K}(C)$, the distribution of $\xi \mapsto \mathsf{z}_\xi$ under ν runs *exactly* through the probability measures π on $(0,1]$ such that $\int a^{-1} \, d\pi(a) \leq C$. Consequently, inside $\mathcal{K}(1+c)$, the law of z_ξ is subject to no restriction other than $\int \mathsf{z}_\xi^{-1} \, d\nu \leq 1 + c$; each μ_π satisfies $\int \mathsf{z}_\xi^{-1} \, d\nu = C(\mu_\pi)$; and the tail bound $\nu\{\mathsf{z}_\xi \leq t\} \leq ct/(1-t)$ of Theorem 7.1 (b) is attained for each $0 < t < 1$ such that $ct/(1-t) \leq 1$.

(c) Let $c > 0$. Each of the three statements below is false for some law in $\mathcal{K}(1+c)$. The first is the inequality $\mathsf{z}_\xi \geq (1+c)^{-1}$ for almost every ξ: if $0 < a < a' \leq 1$, the law $\mu = \frac{1}{2}(\gamma_a + \gamma_{a'})$ has

$$C(\mu) = \int \mathsf{z}_\xi^{-1} \, d\nu = \frac{a + a'}{2aa'}, \qquad \mathrm{kl}_\mu(\{0\}) = \frac{a + a'}{2},$$

and $\mathsf{z}_\xi = a < 1/C(\mu)$ on a set of ν-measure $\frac{1}{2}$. Take a and a' subject to $\frac{a+a'}{2aa'} = 1 + c$, for example $a' = 1$ and $a = (1+2c)^{-1}$. Then $C(\mu) = 1 + c$, whence $\mu \in \mathcal{K}(1+c)$ by (a), and $1/C(\mu) = (1+c)^{-1}$, so that $\mathsf{z}_\xi = a < (1+c)^{-1}$ on that set. The second is a bound $\mathsf{z}_\xi \geq \delta(c) > 0$ for almost every ξ, whatever the number $\delta(c)$. The third, for a given $p > 1$, is the integrability $\mathsf{z}_\xi^{-1} \in L^p(\nu)$.

(d) The conclusion of Proposition 5.6 (d), a positive lower bound for $\mathrm{kl}_\mu(\{0\})$, does not imply that $\mathsf{z}_\xi > 0$ for almost every ξ. Take $M > 0$ and $\mu = \frac{1}{2}(\gamma + \gamma^{\delta_M})$; then $\mathrm{kl}_\mu(\{0\}) = \frac{1}{2}$, while the components of μ are γ, for which $\mathsf{z} = 1$, and γ^{δ_M}, for which $\mathsf{z} = 0$. At every probe the information of μ is finite,

$$I_\mu(f_l) = \|f\|^2 + \tfrac{1}{2} M^2 l^2 \|d^*\beta\|^2,$$

and yet $C(\mu) = \int \mathsf{z}_\xi^{-1} \, d\nu = \infty$.

(e) A law lw may satisfy (FI), in the form ($\mathrm{FI}_{\mathrm{lw}}$), without being Gaussian. Examples are the laws $\mathrm{lw} = \mu_\xi^\epsilon = \mu_\xi *_\epsilon \gamma$ of Proposition 4.4, for the tuned masses, $e \leq e_2$, $\xi \in \Xi_\mu$ and $0 < \epsilon \leq \epsilon_0$: they are ergodic laws of the class, they are not Gaussian, and they satisfy $C(\mathrm{lw}) \leq 1/\epsilon$; their measure $\mathrm{kl}_{\mathrm{lw}}$ has at zero the atom $\epsilon + (1-\epsilon)\mathsf{z}_\xi \geq \epsilon$.

■

Proof. Step 1 (Gaussian laws). Let γ^ϱ be one of the laws of Proposition 4.4 (d), let $\Phi = \phi(X(g_1), \ldots, X(g_k))$ be an element of $\mathcal{A}(\gamma^\varrho)$, and let $g \in L^2_{\mathrm{c}}$. Then (K14) yields

$$\gamma^\varrho(X(g)\,\Phi) = \sum_j C_\varrho(g, g_j)\, \gamma^\varrho(\partial_j \phi).$$

Write $\mathsf{D}_\varrho(r) := \int d\varrho(t)\, r/(r + t^2)$; this is the function D of Step 9 of the proof of Proposition 5.6 with ϱ in the place of kl_μ, and by the formula for C_ϱ in Proposition 4.4 (d) the multiplier of

C_ϱ is $\Pi(p)\,\mathsf{D}_\varrho(|p|^2)$. Suppose that the form g given by $\hat g := \hat f/\mathsf{D}_\varrho(|p|^2)$ belongs to L^2. Then $C_\varrho(g,g_j)=\langle f,g_j\rangle$ for every j, so that $X(g)$ is the score of γ^ϱ along f, and

$$I_{\gamma^\varrho}(f)=(2\pi)^{-3}\int |\hat f|^2\,\mathsf{D}_\varrho(|p|^2)^{-1}\,dp.$$

For γ_a this gives $I_{\gamma_a}(f)=\|f\|^2/a$; for γ^{δ_M}, use $\operatorname{div}(d^*\beta)=0$ and $*f=\operatorname{curl}(d^*\beta)$. The Fourier formula in Step 1 then gives $I_{\gamma^{\delta_M}}(f)=\|f\|^2+M^2\|d^*\beta\|^2$. Finally, $\|d^*\beta^{(l)}\|=l\,\|d^*\beta\|$.

Step 2 (the law μ_π). Tonelli's theorem shows that, for each $\mathrm{Ev}\in\Sigma$, the map $a\mapsto\gamma_a(\mathrm{Ev})=\gamma\{\omega: a^{1/2}\omega\in\mathrm{Ev}\}$ is measurable; moreover

$$\int \exp\bigl(\tfrac{a}{2}\|\Pi_{\mathrm c}g\|^2\bigr)\,d\pi(a)\le\exp\bigl(\tfrac12\|g\|^2\bigr).$$

Being linear in the law, condition (dF) of the notation fixed above, the invariances and the inequalities (2) carry over to μ_π from the laws γ_a, which satisfy them by Lemma D.2. The laws γ_a are ergodic, and distinct values of a give distinct laws; the uniqueness statement of [5, Theorem A and Section 5] therefore identifies ν with the image of π. By Proposition 4.4 (d), $\mathrm{kl}_{\gamma_a}=a\delta_0$. The formula for I_{μ_π} is a consequence of Step 1 and Theorem 7.1 (c).

Step 3 (proof of (b) and (c)). Theorem 7.1 (c) gives one of the two inclusions in (b), and (a) gives the other. Let $0<t<1$ be such that $p:=\frac{ct}{1-t}\le 1$, and let $\pi=(1-p)\delta_1+p\,\delta_t$; then $\int a^{-1}\,d\pi(a)=1+c$. For this measure the tail bound is attained, and $\operatorname{ess\,inf}\mathsf{z}_\xi=t$, which can be taken as small as one wishes. Let $p>1$ and $0<\theta\le\min\{1,c(p-1)\}$; the measure $\pi=(1-\theta)\delta_1+\theta\,p\,a^{p-1}\,da$ on $(0,1]$ has

$$\int a^{-1}\,d\pi(a)=1+\frac{\theta}{p-1}\le 1+c,\qquad \int a^{-p}\,d\pi(a)=\infty,$$

and (a) places μ_π in $\mathcal K(1+c)$.

Step 4 (proof of (d) and (e)). Proposition 4.4(d) shows that γ^{δ_M} is an ergodic law of the class whose measure kl is δ_M; with Steps 1 and 2, this proves (d). For (e), take F and G independent, of the laws μ_ξ and γ, as in the definition of $*_\epsilon$ in Section 4, and let $Y:=(1-\epsilon)^{1/2}F+\epsilon^{1/2}G$. For $\Phi\in\mathcal A(\mathrm{lw})$ and $f\in L^2_{\mathrm c}$, (K14), applied to G while F is held fixed, gives

$$\epsilon^{1/2}\,E\bigl[(\partial_f\Phi)(Y)\bigr]=E\bigl[\Phi(Y)\,G(f)\bigr];$$

consequently $\mathrm{lw}(\partial_f\Phi)^2\le\epsilon^{-1}\|f\|^2\,\mathrm{lw}(\Phi^2)$. The other assertions of (e) are contained in Proposition 4.4 (a)–(c). □

7.3. **Weighted Fisher criteria.**

Proposition 7.3. *The bound* (FI) *in the components, deduced from* (FI_μ) *alone.* Let $\mu\in\mathcal K(C)$. For example, at any charge $e>0$ and for any mass assignment, every volume-first limit point μ that satisfies the hypothesis of Proposition 5.6 (d) belongs to $\mathcal K(1+c)$.

(a) *Each probe.* Each $f\in L^2_{\mathrm c}$ satisfies

$$\int I_\xi(f)\,d\nu=I_\mu(f)\le C\|f\|^2.$$

In particular, for almost every ξ, the inequality $\mu_\xi(\partial_f\Phi)^2\le I_\xi(f)\,\mu_\xi(\Phi^2)$ holds for all $\Phi\in\mathcal A(\mu_\xi)$; and $\nu\{I_\xi(f)>t\|f\|^2\}\le C/t$ for $t>0$. For a law lw with (G), a probe β, $a=d^*\beta$ and $f=da$ with $I_{\mathrm{lw}}(f)<\infty$, the current $J(a)$ is defined by

$$e\,J(a):=X(f)-\mathsf{s}_f,$$

where s_f is the score of (S2) and $e>0$ is a fixed number; the product $e\,J(a)$ does not depend on e. If lw is a limit point at the charge e and $\bar I(\beta)<\infty$, then (S2) yields $\mathsf{s}_\beta=\mathsf{s}_f$, and this current is the function $J(a)$ of Proposition 5.6 (c). For $f=da$ as above, let $J(a)$ be formed

for μ with a Σ-measurable representative s of s_f, as in Theorem 7.1 (a). Then, for almost every ξ, the field equation of Proposition 5.6 (c) is valid in μ_ξ with this $J(a)$, and

$$\int \|e\,J(a)\|^2_{L^2(\mu_\xi)}\,d\nu(\xi) \le (C-1)\,\|f\|^2.$$

For an arbitrary $f \in L^2_{\mathrm{c}}$, the field equation and this bound hold with $X(f) - \mathsf{s}$ in the place of $e\,J(a)$, where s is a Σ-measurable representative of s_f. Proposition 7.2 shows that integrability in $L^1(\nu)$ is the best possible here.

(b) *Large scales.* Fix a probe β and define

$$c_\xi(\beta) := \liminf_{\mathbb{N}\ni l\to\infty} \frac{I_\xi(f_l)}{\|f\|^2}.$$

Then $\int c_\xi(\beta)\,d\nu \le C$ and $\mathsf{z}_\xi \ge 1/c_\xi(\beta)$. Hence $c_\xi(\beta) < \infty$ for almost every ξ, and almost every component satisfies the hypothesis of Proposition 5.6 (d) in the form in which it enters Theorem 7.1 (b).

(c) *One null set for all probes; a stronger norm.* Let $s > \frac{3}{2}$, and put $\mathrm{ZH}_{2s} := \sum_\kappa \lambda_\kappa^{-2s}$, with the Hermite eigenvalues of (3). This sum is finite: the eigenvalue $2n+4$ has multiplicity $3\binom{n+2}{2}$ and $2s > 3$. One can choose a measurable function $\xi \mapsto T_\xi \ge 0$, with $\int T_\xi\,d\nu \le C\,\mathrm{ZH}_{2s}$, such that

$$I_\xi(f) \le T_\xi \inf_a \|\tau_a f\|_s^2$$

holds for almost every ξ and *all* $f \in L^2_{\mathrm{c}} \cap H_s$ simultaneously; these f include $dd^*\beta$ for every probe β. As a consequence, for almost every ξ the law μ_ξ possesses, at every probe, a score depending linearly on f, the current $J(a) \in L^2(\mu_\xi)$ of (a), formed with $\mathrm{lw} = \mu_\xi$, and the bound on shifts of Proposition 5.6 (c) with $I_\xi(f)$ instead of $\bar{I}(\beta)$. Since $\|\cdot\|_s$ is not invariant under the scaling $\beta \mapsto \beta^{(l)}$, the bound in (c) gives no lower bound for z_ξ.

(d) *One constant: a criterion.* One has $C_\xi \ge \mathsf{z}_\xi^{-1}$. Moreover $\int C_\xi\,d\nu$ is the smallest constant C' for which

$$(\mathrm{FI}^I_\mu) \qquad \mu\Big(\sum_{k\le m} 1_{B_k}\,\partial_{f_k}\Phi\Big)^2 \le C'\,\mu(\Phi^2)\max_k \|f_k\|^2$$

holds for every finite partition $(B_k) \subset I$, all probes β_k and every $\Phi \in \mathcal{A}_{\mathrm{c}}$; for $m = 1$ this inequality is (FI_μ). If $\nu(\{\xi_0\}) > 0$, then $C_{\xi_0} \le C/\nu(\{\xi_0\})$. If the components of μ are laws of the form γ^ϱ, then $C_\xi = \mathsf{z}_\xi^{-1}$ and $\int C_\xi\,d\nu = C(\mu)$.

(e) It is not decided whether every $\mu \in \mathcal{K}(C)$ has $C_\xi < \infty$ for almost every ξ. Invariance under all Euclidean maps, (G) of the notation fixed above, (dF), mixing of the components and (FI_μ) do not imply it: some mixture μ of Gaussian laws has all these properties, is *not of the class, as it is reflection positive at no plane*, and satisfies $C(\mu) \le 2$ and $C_\xi = \infty$ for every ξ. ■

Proof. Step 1 (proof of (a)). By (S4), the score s_f depends linearly and boundedly on $f \in L^2_{\mathrm{c}}$, as an element of $L^2(\mu)$. In Step 2 of the proof of Theorem 7.1 both members of the identity depend continuously on $f \in L^2_{\mathrm{c}}$; consequently that step holds for every $f \in L^2_{\mathrm{c}}$, and so do Steps 3 to 5 of the same proof. Read for μ, Step 5 gives $\mu(X(f)\,\mathsf{s}_f) = \|f\|^2$; expanding the square with the help of (S2), one obtains the equality in

$$\mu\big((X(f) - \mathsf{s}_f)^2\big) = S_2(f,f) - 2\|f\|^2 + I_\mu(f) \le (C-1)\,\|f\|^2.$$

The inequality follows from $I_\mu(f) \le C\|f\|^2$, by (S4), and from $S_2(f,f) = \mu(X(f)^2) \le \|f\|^2$, a consequence of (G) [4, Proposition 5.1].

Step 2 (proof of (b)). Fatou's lemma, combined with (a), bounds $\int c_\xi(\beta)\,d\nu$. Steps 5 and 6 of the proof of Theorem 7.1 give $x_l^{-1} \le I_\xi(f_l)/\|f\|^2$ and $x_l^{-1} \uparrow \mathsf{z}_\xi^{-1}$, whence $\mathsf{z}_\xi^{-1} \le c_\xi(\beta)$.

Step 3 (proof of (c)). With the Hermite functions h_κ and the eigenvalues λ_κ of (3), put $e_\kappa := \Pi_{\mathrm{c}} h_\kappa$, choose a representative ς_κ of s_{e_κ}, and define

$$T_\xi := \sum_\kappa \lambda_\kappa^{-2s}\, \xi(\varsigma_\kappa^2).$$

From (a) one obtains $\int T_\xi \, d\nu \le C \sum_\kappa \lambda_\kappa^{-2s} = C\,\mathrm{ZH}_{2s} < \infty$, the sum being finite because $2s > 3$. Take $\xi \in \bigcap_\kappa \Xi_{e_\kappa}$ with $T_\xi < \infty$, where Ξ_f denotes the set of Step 3 of the proof of Theorem 7.1. Every $f \in L^2_{\mathrm{c}} \cap H_s$ has the expansion $f = \sum_\kappa \langle h_\kappa, f\rangle\, e_\kappa$ in L^2, and Cauchy–Schwarz in the Hermite expansion (3) yields

$$\sum_\kappa |\langle h_\kappa, f\rangle|\, \|\varsigma_\kappa\|_{L^2(\xi)} \le \|f\|_s\, T_\xi^{1/2}.$$

The map $f \mapsto \xi(\partial_f \Phi)$ is linear and continuous for the norm of L^2; hence $\sum_\kappa \langle h_\kappa, f\rangle\, \varsigma_\kappa$ is the score of ξ along f. Translation invariance of ξ gives $I_\xi(\tau_a f) = I_\xi(f)$. For the bound on shifts, Step 8 of the proof of Proposition 5.6 is read in μ_ξ.

Step 4 (proof of (d), from (FI^I_μ)). Step 2 of the proof of Theorem 7.1 turns the left side of (FI^I_μ) into $\mu\big(\Phi \sum_k 1_{B_k} \mathsf{s}_{f_k}\big)^2$. Taking the supremum over the Φ with $\mu(\Phi^2) = 1$ and using (S1), [5, Theorem A and Section 5] and Step 3 of the proof of Theorem 7.1, one finds the value

$$\sum_k \mu(1_{B_k} \mathsf{s}_{f_k}^2) = \sum_k \int_{B_k} I_{\mu_\omega}(f_k)\, d\mu(\omega).$$

Let $f_1, f_2, \dots$ be an enumeration of the normalised forms $dd^*\beta/\|dd^*\beta\|$ with $\beta \in \mathcal{D}_0$. Let $B_k \in I$ be the set of the ω for which k is the smallest index $j \le m$ at which $I_{\mu_\omega}(f_j)$ is maximal. With this partition, $\int \max_{j\le m} I_\xi(f_j)\, d\nu \le C'$. As $m \to \infty$, (K13) and (S4) yield $\int C_\xi \, d\nu \le C'$.

Step 5 (proof of the other assertions of (d)). Conversely, $\int_{B_k} I_{\mu_\omega}(f_k)\, d\mu \le \|f_k\|^2 \int_{B_k} C_{\mu_\omega}\, d\mu$ for each k. The inequality $C_\xi \ge \mathsf{z}_\xi^{-1}$ follows from (b). Since an atom of ν is contained in no null set, (a) gives $\nu(\{\xi_0\})\, I_{\xi_0}(f) \le C\|f\|^2$ for every f. Let $\varrho(\{0\}) > 0$. As $\mathsf{D}_\varrho \ge \varrho(\{0\})$, Step 1 of the proof of Proposition 7.2 shows $C(\gamma^\varrho) \le \varrho(\{0\})^{-1}$; Theorem 7.1 (c) and Proposition 7.3 (a) then give

$$\int C_\xi \, d\nu = \int \mathsf{z}_\xi^{-1}\, d\nu \le C(\mu) \le \int C_\xi \, d\nu.$$

Step 6 (proof of (e): the law μ). For $\xi \in [0,1)$ define

$$m_\xi(p) := 1 + \sum_{n\ge 0} 2^n\, 1\big[2^j \le |p| < 2^{j+1},\ j = 2^n + \lfloor 2^n \xi \rfloor\big],$$

denote by $\gamma_{[\xi]}$ the centred Gaussian law whose covariance has the multiplier $\Pi(p)/m_\xi(p)$, and set

$$\mu := \int_0^1 \gamma_{[\xi]}\, d\xi.$$

By Lemma D.2, every $\gamma_{[\xi]}$ exists, satisfies (G) and (dF), is invariant under all Euclidean maps, and is mixing; distinct values of ξ give distinct laws. On $[0,1) \times \mathbb{R}^3$ the function $(\xi, p) \mapsto m_\xi(p)$ is Borel. Indeed, $m_\xi(p) - 1$ is the countable sum, taken over the pairs (n, i) with $n \ge 0$ and $0 \le i < 2^n$, of 2^n multiplied by the indicator function of the set

$$[i2^{-n}, (i+1)2^{-n}) \times \{p : 2^{2^n+i} \le |p| < 2^{2^n+i+1}\}.$$

For each $g \in \mathcal{S}$, Tonelli's theorem therefore makes the function

$$\xi \mapsto \mathrm{Cov}_\xi(g,g) := (2\pi)^{-3} \int |\Pi(p)\, \widehat{*g}(p)|^2\, m_\xi(p)^{-1}\, dp$$

Borel, and with it $\xi \mapsto \gamma_{[\xi]}(\exp iX(g)) = \exp(-\tfrac{1}{2}\,\mathrm{Cov}_\xi(g,g))$. Denote by Mv the class of the bounded real Σ-measurable functions Φ such that $\xi \mapsto \gamma_{[\xi]}(\Phi)$ is Borel. This class is a linear space containing the constants, and it contains the limit of every bounded increasing sequence of its elements. A

finite product of functions $\cos X(g)$ and $\sin X(g)$, $g \in \mathcal{S}$, is a linear combination of functions of the same kind; hence Mv contains the set Mu of all such products. Moreover $\sigma(\mathrm{Mu}) = \Sigma$, because $X(g) = \lim_n n \sin X(g/n)$. The monotone class theorem shows that every bounded Σ-measurable function lies in Mv. Consequently $\mu(\Phi) = \int_0^1 \gamma_{[\xi]}(\Phi)\, d\xi$ for every bounded measurable Φ, and the map $\xi \mapsto \gamma_{[\xi]}$ is measurable with respect to the σ-algebra of Definition G.1. Under this map, Lebesgue measure on $[0,1)$ has an image that is concentrated on ergodic laws and represents μ; by [5, Theorem A and Section 5] this image is ν, and the laws $\gamma_{[\xi]}$ are the components of μ.

The law μ does not belong to the class. The multiplier of S_2^μ is $\Pi(p)M(p)$, with

$$M(p) := \int_0^1 \frac{d\xi}{m_\xi(p)} = \begin{cases} 1 & \text{if } |p| < 2, \\ \dfrac{2^n}{1+2^n} & \text{if } 2^j \le |p| < 2^{j+1} \text{ and } 2^n \le j < 2^{n+1}. \end{cases}$$

If μ were a law of the class, Proposition D.1 would yield $M(p) = \mathsf{D}(|p|^2)$ for Lebesgue-almost all p, where $\mathsf{D}(r) = \int d\mathrm{kl}_\mu(t)\, r/(r+t^2)$ is a non-decreasing function of r; whereas $M = 1$ on $\{|p| < 2\}$ and $M = \frac{1}{2}$ on $\{2 \le |p| < 4\}$. The law μ is invariant under all Euclidean maps and satisfies (G) and (dF); among the conditions of (H) and of $\mathcal{C}_9$, reflection positivity is therefore the only one that μ can fail. Let $r_{\hat{n},\mathrm{po}}$ be the reflection in the plane $\{\hat{n} \cdot y = \mathrm{po}\}$, and let Gi be a Euclidean map that carries this plane onto the plane $\{\hat{n}' \cdot y = \mathrm{po}'\}$. Because $r_{\hat{n}',\mathrm{po}'} = \mathrm{Gi}\, r_{\hat{n},\mathrm{po}}\, \mathrm{Gi}^{-1}$ and μ is invariant, reflection positivity of μ at a single plane would entail reflection positivity at every plane. Therefore μ is reflection positive at no plane.

Step 7 (proof of (e): the constants). The computation of Step 1 of the proof of Proposition 7.2 gives $I_{\gamma_{[\xi]}}(f) = (2\pi)^{-3} \int m_\xi\, |\hat{f}|^2\, dp$ whenever $m_\xi \hat{f} \in L^2$. For every p one has $\int_0^1 m_\xi(p)\, d\xi \le 2$, and Theorem 7.1 (c) then gives $C(\mu) \le 2$. Choosing forms f whose $\hat{f}$ is carried by the shell $\{2^j \le |p| < 2^{j+1}\}$ with $j = 2^n + \lfloor 2^n \xi \rfloor$, and applying (S4), one obtains $C(\gamma_{[\xi]}) \ge 1 + 2^n$ for every n. □

7.4. **One constant for all probes.**

Theorem 7.4. *A single constant valid at every probe, for the volume-first limit points of the model.* Let $0 < \rho < 1$, let the weights be standard, let $\bar{m} \in (0,1]$ and $e > 0$ be arbitrary, and let the mass assignment be such that $\boldsymbol{\mu}_N > 0$ for all large N. Let μ be a volume-first limit point,

$$\mu = \lim_i \mu^{(N_i)}, \qquad \mu^{(N_i)} = \lim_j P_{N_i,T_{ij}},$$

and define

$$c_W := \liminf_{i\to\infty} 3\pi^2 \mathsf{h}_3^2\, q_i^2 \liminf_{j\to\infty} \mathsf{S}_{n_{ij}},$$

where $\eta_i = L^{-N_i}$, $q_i = e\,\eta_i^{1/2}$ and n_{ij} are the spacing, the lattice charge and the number of sites per side at the cutoff (N_i, T_{ij}) (the notation fixed above), and $\mathsf{S}_{n_{ij}}$ is the number S_n of Section 5 formed with $q = q_i$ and $\mathbf{m} = \boldsymbol{\mu}_{N_i}$. Then

$$\int \mathsf{z}_\xi^{-1}\, d\nu \le \int C_\xi\, d\nu(\xi) \le 1 + c_W.$$

In other words, if $c_W < \infty$, then for ν-almost every ξ the component μ_ξ fulfils the condition (FI_{μ_ξ}) at every probe and for every $\Phi \in \mathcal{A}(\mu_\xi)$ with one and the same constant $C_\xi < \infty$, and $C_\xi \in L^1(\nu)$. Among all laws of the class, the condition (FI^I_μ) of Proposition 7.3 (d) implies no integrability of C_ξ beyond this one: for each $p > 1$ there is a law of the class which satisfies (FI^I_μ) and for which $C_\xi \notin L^p(\nu)$ (Proposition 7.2 (c), Proposition 7.3 (d)). For the volume-first limit points formed with a physical mass assignment, or with the tuned masses of Section F.2, it is not decided whether C_ξ belongs to $L^p(\nu)$ for some $p > 1$. If the hypothesis of Proposition 5.13 (e) is satisfied, then $c_W \le \frac{1}{2} C\, e^2/\bar{m}$, with the constant C of that proposition. By Theorem 5.14 (a), for every charge

$e > 0$ and every mass assignment such that $c_{\boldsymbol{\mu}} \bar{m} \eta \le \boldsymbol{\mu}_N \le 1$ for all large N and some $c_{\boldsymbol{\mu}} > 0$ (every physical mass assignment is of this kind),

$$c_W \le \tfrac{1}{2} C_F \, \frac{e^2}{\bar{m}};$$

here C_F is the number of Theorem 5.14 (b), whose dependence on $e^2/(c_{\boldsymbol{\mu}} \bar{m})$ is described in Theorem A. ■□

Proof. Put $\lambda_i := (J_{\eta_i})_* \mu^{(N_i)}$. By Definition E.9 and [4, Section 6.3], the laws $\mu_{C(n_{ij})}$ converge weakly on Ω to λ_i as $j \to \infty$, the law $\mu^{(N_i)} = (\sharp_{\eta_i})_* \lambda_i$ has the property (G) of the notation fixed above, and λ_i is invariant under $W = \mathbb{Z}^3 \rtimes O_{\mathrm{cub}}$. Throughout Steps 1 to 6 the number $N = N_i$ is held fixed, with $\mathbf{m} = \boldsymbol{\mu}_N > 0$; one writes $\lambda = \lambda_i$, and $n = n_{ij}$ runs through a subsequence along which $\mathsf{S}_{n_{ij}}$ converges to $\liminf_j \mathsf{S}_{n_{ij}}$. For a real field A on $C(n)$, one-cochains u, u', links b, b' and sites x, y set

$$k_A(u,u') := -\partial_u \partial_{u'} \ell_n(A), \qquad k_A(b,b') := k_A(\delta_b, \delta_{b'}), \qquad \mathsf{n}_A(x,y) := \|D_A^{-1}(x,y)\|_{\mathrm{HS}}^2.$$

A function is called *local* if it depends smoothly on finitely many of the F_σ and is bounded together with all its derivatives. On $C(n)$ with n large, local functions and the one-cochains u, u_α of finite support on $\mathbb{Z}^3$ are read in the manner of Step 4 of the proof of Proposition 5.3. For $g \in W$ and a function G one writes $(gG)(F) := G(g^* F)$.

Step 1 (field by field). In each real field A on $C(n)$ the following three statements hold.

(i) There is a bound on $k_A(b,b')$ and on $\mathsf{n}_A(x,y)$ that does not depend on n (Lemma 5.1; Step 1 of the proof of Lemma 5.8).

(ii) One has $\mathsf{n}_A(x,y) = \mathsf{n}_A(y,x)$, and whenever $b \ne b'$,

$$|k_A(b,b')| \le \mathsf{h}_3^2 q^2 \sum_{x' \in V(b),\, y' \in V(b')} \mathsf{n}_A(x',y'),$$

which is the inequality of Steps 1 and 2 of the proof of Lemma 5.9 before the expectation is formed.

(iii) Whenever ϑ is a zero-cochain of finite support, $k_A(u, d\vartheta) = 0$; moreover, for each $g \in W$,

$$k_{g^*A}(g^*u, g^*u') = k_A(u,u'), \qquad \mathsf{n}_{(g^{-1})^* A}(gx, gy) = \mathsf{n}_A(x,y)$$

(Step 2 of the proof of Lemma 5.8; Step 1 of the proof of Lemma 5.9).

Step 2 (a weighted identity). For every local h,

$$E_n[h\, s_u^2] = E_n[h]\, \|du\|^2 + E_n[h\, k_A(u,u)] + E_n[\partial_u^2 h].$$

To see this, take the relation $E_n[\partial_u \Phi] = -E_n[\Phi\, s_u]$ of Step 2 of the proof of Proposition 5.6 once with $\Phi = h\, s_u$, using $\partial_u s_u = -\|du\|^2 - k_A(u,u)$ (Step 2a (i) of that proof), and once with $\Phi = \partial_u h$.

Step 3 (weights). Consider local functions $\Psi_1, \dots, \Psi_m \ge 0$ with $\sum_\alpha \Psi_\alpha \le 1$, let $\Lambda_\ell := \{x \in \mathbb{Z}^3 : |x|_\infty \le \ell\}$, and define

$$h_\alpha = h_{\alpha,\ell} := \frac{1}{48\,|\Lambda_\ell|} \sum_{g_0 \in O_{\mathrm{cub}}} \sum_{x \in \Lambda_\ell} (\tau_x g_0) \Psi_\alpha.$$

These functions satisfy $h_\alpha \ge 0$ and $\sum_\alpha h_\alpha \le 1$; for each $g \in W$ the number $\sup |g h_\alpha - h_\alpha|$ tends to 0 as $\ell \to \infty$; and, the number of x that contribute being bounded,

$$\sup |\partial_u h_\alpha| + \sup |\partial_u^2 h_\alpha| \le \frac{\mathsf{c}}{|\Lambda_\ell|}.$$

Let G be local. Applying Step 2 of the proof of Proposition 5.6 with $\Phi = h_\alpha G$, and then the Schwarz inequality together with $(\sum_\alpha h_\alpha s_{u_\alpha})^2 \le \sum_\alpha h_\alpha s_{u_\alpha}^2$, one finds

$$\Big| E_n \sum_\alpha h_\alpha\, \partial_{u_\alpha} G \Big| \le (E_n G^2)^{1/2} \Big(\sum_\alpha E_n[h_\alpha s_{u_\alpha}^2] \Big)^{1/2} + \frac{\mathsf{c}'}{|\Lambda_\ell|}.$$

Step 4 (limit kernels). On account of (i) and of a diagonal argument, letting first $j \to \infty$ and then $\ell = \ell_r \to \infty$, one obtains for all links b, b' of $\mathbb{Z}^3$ and all sites x, y of $\mathbb{X}$ limits

$$E_n[h_\alpha k_A(b,b')] \to k^{(\alpha)}(b,b'), \qquad E_n[h_\alpha \mathsf{n}_A(x,y)] \to \mathsf{n}^{(\alpha)}(x,y) \geq 0.$$

The expectation E_n is invariant under W (Step 2 of the proof of Lemma 5.8), so that (iii) yields $E_n[h\, k_A(g^*u, g^*u')] = E_n[(gh)\, k_A(u,u')]$; in view of Step 3, the kernels $k^{(\alpha)}$ and $\mathsf{n}^{(\alpha)}$ are therefore invariant under W. From (ii) and (iii) it follows that $k^{(\alpha)}$ is real and symmetric, that it annihilates each $d\delta_y$, and that

$$|k^{(\alpha)}(b,b')| \leq \mathsf{h}_3^2 q^2 \sum_{x' \in V(b)} \sum_{y' \in V(b')} \mathsf{n}^{(\alpha)}(x',y') \qquad (b \neq b').$$

The group W acts transitively on the sites, and hence

$$\sigma^{(\alpha)} := \sum_y |v(x,y)|^2\, \mathsf{n}^{(\alpha)}(x,y)$$

is the same number for every site x. By Fatou's lemma, by $\sum_\alpha h_\alpha \leq 1$ and by Lemma 5.9 (a),

$$\sum_\alpha \sigma^{(\alpha)} \leq \frac{1}{8} \liminf_{j\to\infty} \mathsf{S}_{n_{ij}}.$$

Step 5 (Lemma 5.9 (c) and Lemma 5.8 on $\mathbb{Z}^3$). With the properties of $k^{(\alpha)}$ and $\mathsf{n}^{(\alpha)}$ obtained in Step 4 and the geometry of the links, the argument of Steps 3 and 4 of the proof of Lemma 5.9 applies to $k^{(\alpha)}$ and gives

$$\mathsf{c}^{(\alpha)} := \max_b \sum_{b'} |s(b,b')|^2\, |k^{(\alpha)}(b,b')| \leq 192\, \mathsf{h}_3^2 q^2 \sigma^{(\alpha)}.$$

Form $\hat{k}_{\mu\nu} = \hat{k}^{(\alpha)}_{\mu\nu}$ and $m_{\mu\nu}$ from $k^{(\alpha)}$ as in Steps 4 and 5 of the proof of Lemma 5.8. Plancherel's formula on $\mathbb{Z}^3$, contained in (K8) of Section 5, takes in polarised form the shape

$$k^{(\alpha)}(u,u) = (2\pi)^{-3} \int_{[-\pi,\pi]^3} \hat{u}^*\, \hat{k}^{(\alpha)}\, \hat{u}\, dp$$

for every u of finite support. With this formula, Steps 3 to 6 of the proof of Lemma 5.8 give four facts: the matrix $\hat{k}^{(\alpha)}(p)$ is real and symmetric; $\hat{k}^{(\alpha)}\mathsf{s} = 0$; $|\hat{k}_{\mu\nu}(p) - \hat{k}_{\mu\nu}(0)| \leq \frac{1}{2}|p|^2 m_{\mu\nu}$; and $\hat{k}_{\mu\nu}(te_\nu) = 0$ whenever $0 < t \leq \pi$. Continuity gives $\hat{k}^{(\alpha)}(0) = 0$, whence $|\hat{k}_{\mu\nu}(p)| \leq \frac{1}{2}|p|^2 m_{\mu\nu}$ at every p; on a torus, where the mode $p = 0$ contributes, Step 5 of the proof of Lemma 5.8 yields at $p \neq 0$ the bound $|p|^2 m_{\mu\nu}$, twice as large. As $|p| \leq \frac{\pi}{2}|\mathsf{s}(p)|$, it follows that

$$|k^{(\alpha)}(u,u)| \leq \frac{\pi^2}{8}\, \mathsf{c}^{(\alpha)} \|du\|^2 \leq 24\pi^2 \mathsf{h}_3^2 q^2 \sigma^{(\alpha)} \|du\|^2.$$

Step 6 (fixed N). In the inequality of Step 3 let $j \to \infty$, local functions being bounded and continuous on Ω, and afterwards let $\ell = \ell_r \to \infty$. With

$$\bar{\Psi}_\alpha := \frac{1}{48} \sum_{g_0 \in O_{\mathrm{cub}}} g_0 \Psi_\alpha,$$

(K15) shows that $h_{\alpha,\ell} \to P_\lambda \bar{\Psi}_\alpha$ in $L^2(\lambda)$, where P_λ stands, as in the Conventions of Section 5 and in Proposition 5.17, for the projection of $L^2(\lambda)$ onto the vectors invariant under $\mathbb{Z}^3$; as τ_x corresponds to $U(\eta x)$ under $\Psi \mapsto \Psi \circ \sharp_\eta$ (Step 7), (K15), used in $L^2(\lambda)$ and in $L^2(\mu^{(N_i)})$, gives $P_\lambda(\Psi \circ \sharp_\eta) = (P_i \Psi) \circ \sharp_\eta$ for $\Psi \in \mathcal{A}_{\mathrm{c}}$, with the projection P_i of Step 8. Steps 2, 4 and 5 thus lead to

$$\begin{aligned} \lambda\Big(\sum_\alpha (P_\lambda \bar{\Psi}_\alpha)\, \partial_{u_\alpha} G\Big)^2 &\leq \lambda(G^2) \sum_\alpha \big[\lambda(\Psi_\alpha) + 24\pi^2 \mathsf{h}_3^2 q^2 \sigma^{(\alpha)}\big]\, \|du_\alpha\|^2 \\ &\leq \lambda(G^2) \Big(1 + 3\pi^2 \mathsf{h}_3^2 q^2 \liminf_{j\to\infty} \mathsf{S}_{n_{ij}}\Big) \max_\alpha \|du_\alpha\|^2. \end{aligned} \tag{7.1}$$

Step 7 (data for $N \to \infty$). Let $\Psi_\alpha = \psi_\alpha(X(g'_1), \dots)$ and $\Phi = \phi(X(g_1), \dots, X(g_k))$ be elements of $\mathcal{A}_{\mathrm{c}}$, so that ψ_α, like ϕ, is smooth and bounded together with all its derivatives, and suppose that $\psi_\alpha \geq 0$ and $\sum_\alpha \psi_\alpha \leq 1$, all the forms g'_c and g_j lying in $C_c^\infty((-\mathrm{vp}, \mathrm{vp})^3; \Lambda^2)$; let further β_α be probes, and $f_\alpha := dd^*\beta_\alpha$. Since $X(g) \circ \sharp_\eta = \langle g_{[\eta]}, \cdot \rangle$, the functions $\Psi_\alpha \circ \sharp_\eta$ and $G := \Phi \circ \sharp_\eta$ are local. The operators $\hat{w}$ by which the elements $w \in O_{\mathrm{cub}}$ act on functions on $\mathcal{S}'$ correspond to w, and the operator $U(\eta x)$ corresponds to τ_x ([4, equation (6.9)]). Let u_α be the finitely supported cochain $\widehat{\mathfrak{a}}_\eta$ of Lemma E.10, formed with β_α. Then $du_\alpha = (f_\alpha^{(\eta)})_{[\eta]}$. Consequently, as $N \to \infty$, one has $\|du_\alpha\| \leq \|f_\alpha^{(\eta)}\| \to \|f_\alpha\|$ and

$$\langle g_{[\eta]}, du_\alpha \rangle = \langle \mathsf{P}_\eta g, f_\alpha^{(\eta)} \rangle \to \langle g, f_\alpha \rangle,$$

with P_η the projection onto the forms which are constant on each cube $Q_\eta(\sigma)$ (Lemma E.10). Hence, uniformly,

$$\partial_{u_\alpha} G = (\partial_{f_\alpha} \Phi) \circ \sharp_\eta + o(1).$$

Step 8 (an ergodic theorem uniform in the spacing). Denote by P_i the projection of $L^2(\mu^{(N_i)})$ onto the vectors invariant under $\eta_i \mathbb{Z}^3$, and, for $\mathsf{L} \in \mathbb{N}$, by A_{L}^η the average of the operators $U(a)$ over the points $a \in \eta\mathbb{Z}^3$ with $|a|_\infty \leq \mathsf{L}$. Then $\|A_{\mathsf{L}}^\eta \bar{\Psi}_\alpha - P_i \bar{\Psi}_\alpha\|_2^2$ equals the average, over the pairs (a, a') of such points, of the numbers

$$\langle \bar{\Psi}_\alpha, U(a' - a) \bar{\Psi}_\alpha \rangle - \langle \bar{\Psi}_\alpha, P_i \bar{\Psi}_\alpha \rangle.$$

Proposition 5.17, taken with $\mathsf{r} = \lceil \mathrm{vp}/\eta \rceil + 2$, bounds each of these numbers by

$$\min\Big\{2,\ \frac{3C_\flat(\mathrm{vp}+3)}{|a-a'|_\infty}\Big\}.$$

For $\mathsf{L} \geq \mathrm{vp} + 3$ the average is therefore at most $\mathsf{c}\,(\mathrm{vp}+3)/\mathsf{L}$, where the constant c is absolute; the bound holds *uniformly in* i.

Step 9 (the limit). Let A_{L} be the average of $U(a)$ over $a \in [-\mathsf{L}, \mathsf{L}]^3$. Condition (G) for $\mu^{(N_i)}$ gives

$$\|(A_{\mathsf{L}}^\eta - A_{\mathsf{L}}) \bar{\Psi}_\alpha\|_2 \leq \mathrm{Lip}(\psi_\alpha) \sum_c \sup_{|y|_\infty \leq \eta} \|\tau_y g'_c - g'_c\| + \frac{\mathsf{c}\,\eta}{\mathsf{L}},$$

a quantity which tends to 0 as $i \to \infty$; moreover the function $(A_{\mathsf{L}} \bar{\Psi}_\alpha)\, \partial_{f_\alpha} \Phi$ is bounded and continuous on H_{-4}, since $|\omega(g)| \leq \|\omega\|_{-4}\, \|g\|_4$ whenever $\omega \in H_{-4}$ and $g \in \mathcal{S}$ (see (3)). Write $P := P_\mu^{123}$. By [5, Lemma 7.2] one has $P = E_\mu[\,\cdot\,|I]$ and $A_{\mathsf{L}} \bar{\Psi}_\alpha \to P \bar{\Psi}_\alpha$ in $L^2(\mu)$; moreover $P\bar{\Psi}_\alpha = P\Psi_\alpha$, because ν-almost every component is invariant under O_{cub} [4, Section 6, Section 7.5, and Theorem 8.6]. As to the convergence, let $A_{2\mathsf{L}}^{123}$ be the average of $U(a)$ over $a \in [0, 2\mathsf{L}]^3$ which enters that lemma; then

$$A_{\mathsf{L}} = U(-\mathsf{L}(e_1 + e_2 + e_3))\, A_{2\mathsf{L}}^{123}, \qquad U(y)P = P,$$

and consequently $\|A_{\mathsf{L}} \Psi - P\Psi\|_2 = \|A_{2\mathsf{L}}^{123} \Psi - P\Psi\|_2 \to 0$ for each $\Psi \in L^2(\mu)$. Together with Steps 7 and 8, an argument with three epsilons shows that the left side of (7.1) at N_i converges to $\mu\big(\sum_\alpha (P\Psi_\alpha)\, \partial_{f_\alpha} \Phi\big)^2$, and $\lambda_i(G^2) \to \mu(\Phi^2)$. Letting $i \to \infty$ along a subsequence that realises c_W, one arrives at

$$\mu\Big(\sum_\alpha (P\Psi_\alpha)\, \partial_{f_\alpha} \Phi\Big)^2 \leq (1 + c_W)\, \mu(\Phi^2) \max_\alpha \|f_\alpha\|^2. \tag{7.2}$$

Step 10 (partitions). As a function of $(\Psi_\alpha) \in L^2(\mu)^m$, the left side of (7.2) is continuous. Let $(B_\alpha) \subset I$ be a finite partition; by (S1) one may choose $\Psi'_\alpha \in \mathcal{A}_{\mathrm{c}}$ converging to 1_{B_α} in $L^2(\mu)$. Fix smooth functions $\sigma_\delta \colon \mathbb{R} \to [0,1]$ with $|\sigma_\delta(t) - t| \leq \delta$ on $[0,1]$ and ζ_δ with $\max(1,t) \leq \zeta_\delta(t) \leq \max(1,t) + \delta$, and set

$$\Psi_\alpha := \frac{\sigma_\delta(\Psi'_\alpha)}{\zeta_\delta\big(\sum_{\alpha'} \sigma_\delta(\Psi'_{\alpha'})\big)}.$$

These functions are admissible in Step 7, and, the map $(t_{\alpha'})_{\alpha'} \mapsto \sigma_\delta(t_\alpha)/\zeta_\delta(\sum_{\alpha'} \sigma_\delta(t_{\alpha'}))$ being Lipschitz, $\limsup \|\Psi_\alpha - 1_{B_\alpha}\|_2 \le \mathsf{c}\, m\, \delta$. Since $P 1_{B_\alpha} = 1_{B_\alpha}$, the inequality (7.2) yields in the limit (FI^I_μ) with $C' = 1 + c_W$. Suppose first that $c_W < \infty$. For $m = 1$ and $B_1 = \mathcal{S}'$, the condition (FI^I_μ) is the inequality

$$\mu(\partial_f \Phi)^2 \le (1 + c_W)\, \|f\|^2\, \mu(\Phi^2),$$

valid for every probe and every $\Phi \in \mathcal{A}_\mathrm{c}$. According to (S3), in the supremum which defines I_μ the function Φ may be restricted to $\mathcal{A}_0 \subset \mathcal{A}_\mathrm{c}$; thus $C(\mu) \le 1 + c_W$. By Step 1 of the proof of Theorem 7.1, μ is a law of the class. So $\mu \in \mathcal{K}(1 + c_W)$, and Proposition 7.3 (d) is applicable. Suppose, finally, that $c_W = \infty$. The right-hand inequality of the theorem then holds trivially. The left-hand one is proved by means of Theorem 7.1 (c), which is applicable since μ is a law of the class also when $c_W = \infty$: for a probe β and a sequence $l_k \to \infty$ it gives $\int \mathsf{z}_\xi^{-1}\, d\nu \le \liminf_k I_\mu(f_{l_k})/\|f\|^2$, and in $[0, \infty]$

$$I_\mu(f_{l_k}) = \int I_\xi(f_{l_k})\, d\nu \le \|f\|^2 \int C_\xi\, d\nu,$$

because $\|f_{l_k}\| = \|f\|$ and $I_\xi(f_{l_k}) \le C_\xi \|f_{l_k}\|^2$ by the definition of C_ξ; the two bounds together give $\int \mathsf{z}_\xi^{-1}\, d\nu \le \int C_\xi\, d\nu$.

Step 11 (the constants). Under the lower bound $\boldsymbol{\mu}_N \ge c_{\boldsymbol{\mu}} \bar{m} \eta$, which is part of the hypotheses of Proposition 5.13 (e) and of Theorem 5.14,

$$q^2 \mathsf{S}_n \le \frac{e^2}{c_{\boldsymbol{\mu}} \bar{m}}\, \boldsymbol{\mu}_N \mathsf{S}_n.$$

Under the hypothesis of Proposition 5.13 (e), let C_M be the number of that hypothesis, so that $\limsup_T \boldsymbol{\mu}_N \mathsf{S}_n \le C_M$ for all large N. Since $T_{ij} \to \infty$ as $j \to \infty$, one obtains for all large i

$$3\pi^2 \mathsf{h}_3^2\, q_i^2 \liminf_{j \to \infty} \mathsf{S}_{n_{ij}} \le \frac{3\pi^2 \mathsf{h}_3^2 C_M\, e^2}{c_{\boldsymbol{\mu}} \bar{m}} = \frac{1}{2}\, C\, \frac{e^2}{\bar{m}}, \qquad C = \frac{6\pi^2 \mathsf{h}_3^2 C_M}{c_{\boldsymbol{\mu}}},$$

and C is the constant of Proposition 5.13 (e); hence $c_W \le \frac{1}{2} C e^2/\bar{m}$. For the mass assignments of Theorem 5.14, part (a) of that theorem gives $\boldsymbol{\mu}_N \mathsf{S}_n \le \frac{1}{3} C_M$ for all large N and all large T, where C_M denotes the number defined in that part and not the datum of Proposition 5.13 (e); finally, $2\pi^2 \mathsf{h}_3^2 C_M / c_{\boldsymbol{\mu}} = C_F$. □

8. Proof of Theorem A and Corollary B

Proof of Theorem A. Step 1 (hypotheses). The limit point μ has defining sequences whose T_i are as large as one wishes (Step 5 of the proof of Proposition 5.13); the lattice probes satisfy $\sum_x u(x) = 0$ (Lemma E.10); and $\bar{I}(\beta) = \|f\|^2 + \liminf_i K_{n_i}(\hat{\mathfrak{a}}_i, \hat{\mathfrak{a}}_i)$ (Proposition 5.6(a) and Step 3 of its proof). Theorem 5.14 (b) therefore yields the hypothesis of Proposition 5.6 (d) with $c = c_F$, and Theorem 7.1 (b) then gives the atom. The bound $\bar{I}(\beta) \le (1 + c_F)\|f\|^2$ at all probes, which is that hypothesis, is contained in Lemma 6.4 (a), with $c_{\boldsymbol{\mu}} \bar{m}$ in the place of the constant c of that statement. Theorem 5.14 (a) gives $c_W \le \frac{1}{2} c_F$ in Theorem 7.4. Put $\Xi^\gamma_\mu := \Xi_1 \cap \{C_\xi < \infty\}$, with the set Ξ_1 of Step 6 of the proof of Theorem 7.1. By Theorem 7.4 this set has full ν-measure. Fix $\xi \in \Xi^\gamma_\mu$ and put $\mathrm{lw} = \mu_\xi$ for the remaining steps. *Step 2* ((i) and (ii)). Assertion (i) follows from the Gaussian moment and analytic bounds in [4, Section 5], the rotation and reflection conclusion of [4, Theorem 8.6], and the inheritance and clustering of ergodic components in [5, Theorem A]. Since $C_\xi < \infty$, Proposition 7.3(a) and Step 1 of its proof give $\|f\|^4 \le S_2^\xi(f, f) I_\xi(f)$ for every real $f \in L^2_\mathrm{c}$. Applying this inequality to one probe gives $x_l(\xi)^{-1} \le I_\xi(f_l)/\|f\|^2 \le C_\xi$ for all l, while $x_l^{-1} \uparrow \mathsf{z}_\xi^{-1}$ (Step 6 of the proof of Theorem 7.1); hence $\mathsf{z}_\xi^{-1} \le C_\xi$. The tail bound is Markov's inequality, applied to $\mathsf{z}_\xi^{-1} - 1$. The description of z_ξ as a best constant is Proposition 4.1, which holds for every law whose second moment has the Källén–Lehmann form.

Step 3 (infrared limit). For each ξ in this full-measure set, the law μ_ξ is ergodic and belongs to $\mathcal{K}$. Apply Theorem 4.3(b), with $c_0 = \mathrm{kl}_\xi(\{0\}) = z_\xi$, to obtain exactly the characteristic-functional convergence in Theorem A(iv).

Put

$$\mathcal{H}_{\mathrm{E}}^{(1)} = \overline{\{[X(g)] : g \in C_c^\infty(\{y_1 > 0\}; \Lambda^2\mathbb{C}^3)\}}, \qquad \mathcal{H}_{\mathrm{E},\gamma} = E(\partial V)\mathcal{H}_{\mathrm{E}}^{(1)}.$$

Step 11 identifies these Euclidean spaces with the spaces $\mathcal{H}^{(1)}$ and $\mathcal{H}_\gamma$ in the statement.

Step 4 (shell and group). The law lw has the properties listed in Proposition 3.1, and Theorems 3.2 and 3.3 hold for such laws (Remark after Proposition 3.1). From $\mathcal{U}(0,\Lambda)\mathcal{U}(a,1)\mathcal{U}(0,\Lambda)^{-1} = \mathcal{U}(\Lambda a, 1)$, from Theorem 3.2 (a) and from (K16) one obtains

$$\mathcal{U}(0,\Lambda)\, E(\Delta)\, \mathcal{U}(0,\Lambda)^{-1} = E((\Lambda^T)^{-1}\Delta),$$

and $\Lambda^T \in SO_0(1,2)$ preserves ∂V and $\{0\}$. By the property (Sp) of Proposition 3.1, E is carried by $\{\omega \geq |\vec{p}|\}$; hence $E(\{0\}) = 1_{\{0\}}(H) = |\Omega\rangle\langle\Omega|$ by the property (Vac) of Proposition 3.1.

Step 5 (the vectors ψ_h). Proposition 4.2 (a)–(c) is stated for every probability measure with (G) and (dF), the conditions (T_{123}), (R_1), (RP_1) of Section 2, and the Källén–Lehmann form in Proposition D.1; the identities obtained in item 4 of its proof hold for every such measure whose spectral measure is carried by the cone. Consequently $\langle\psi_h, E(\cdot)\psi_h\rangle$ is the image of the measure

$$\mathrm{kl}(ds)\, \frac{|\hat{h}(k)|^2\, |k|^2}{2\,\omega_s(k)}\, \frac{d^2k}{(2\pi)^2}$$

under $(s,k) \mapsto (\omega_s(k), k)$, and $\omega_s(k) = |k|$ holds for $s = 0$ alone. This gives the norm of $E(\partial V)\psi_h$; moreover $\|\psi_h\|^2 = \int c_h\, d\mathrm{kl} \leq c_h(0) = \frac{1}{2}\|h\|_{1/2}^2$, and the measure has no atom at $k = 0$.

Step 6 ($\mathcal{H}_{\mathrm{E}}^{(1)}$ is invariant). The operators $\exp(-tH)$ with $t \geq 0$ and $V(y) = \exp i(y_2P_2 + y_3P_3)$, and the rotations about e_1, map $[X(g)]$ to $[X(\mathrm{Gi}\cdot g)]$, with the corresponding Euclidean map Gi (the property (Rec) of Proposition 3.1, Theorem 3.2 (b)); by (K16), $\mathcal{H}_{\mathrm{E}}^{(1)}$ reduces H. Let g have compact support in $\{y_1 > 0\}$, and let $t_0 > 0$ be small. Then $\zeta \mapsto \exp(\zeta L_j)[X(g)]$ is holomorphic on $|\operatorname{Re}\zeta| < t_0$, and at real ζ its value is $[X(R_\zeta^{(j)} \cdot g)] \in \mathcal{H}_{\mathrm{E}}^{(1)}$ (Theorem 3.2 (c)). Pairing with a vector $\phi \perp \mathcal{H}_{\mathrm{E}}^{(1)}$ and using the identity theorem, one finds $\exp(isL_j)[X(g)] \in \mathcal{H}_{\mathrm{E}}^{(1)}$. Together with Step 4, it follows that $\mathcal{H}_{\mathrm{E},\gamma}$ is closed and invariant under $\mathcal{U}$.

Step 7 (spectral measure of the vectors $[X(g)]$). Take g as in Step 6, but allow complex values; the first entry of C_ϱ is the conjugate linear one. The reconstruction identity (Rec) and the Källén–Lehmann representation give

$$\langle [X(g)], \exp(-tH)\, V(y)[X(g)]\rangle = \int d\mathrm{kl}(s)\, C_{\delta_s}(\theta\cdot g, \tau_{te_1+y}\cdot g).$$

Repeating the computation in Step 11 of the proof of Proposition 4.4, with $\tau_{te_1+y}\cdot g$ in the second entry and with $E_s = (|k|^2 + s^2)^{1/2}$, one obtains

$$C_{\delta_s}(\theta\cdot g, \tau_{te_1+y}\cdot g) = \int \frac{d^2k}{(2\pi)^2}\, \frac{|v_2|^2 + |v_3|^2 - |v_1|^2}{2E_s}\, \exp(-tE_s - ik\cdot y),$$

whose density is non-negative. Here v is the vector in Step 11 of the proof of Proposition 4.4, with its mass parameter replaced by s. For $s = 0$ the density is $|a_g|^2/(2|k|)$, where

$$a_g(k) = \int_0^\infty dt \int_{\mathbb{R}^2} d^2y\, e^{-|k|t - ik\cdot y}\, \mathrm{div}(*g)(t,y).$$

This is the linear amplitude of Lemma D.4, extended complex linearly.

Step 8 (a single particle). According to Lemma D.3, the measures of Step 7, carried over by the map $(s,k) \mapsto (E_s(k), -k)$, give the spectral measure of $[X(g)]$; on ∂V only $s = 0$ contributes, and the apex carries no mass. Therefore $\iota\colon E(\partial V)[X(g)] \mapsto \mathsf{z}_\xi^{1/2} a_g$ is isometric, and $\iota\, \mathsf{F}(H, \vec{P}) = \mathsf{F}(|k|, -k)\, \iota$ for every bounded Borel function F.

Step 9 (ι is onto). As $\varepsilon \to 0$, and then $t \downarrow 0$, one has $X(u_{\varepsilon,t}) \to B_t(h) \to B_0(h)$ in $L^2(\mathrm{lw})$ (Proposition 4.2 (a) and item 2 of its proof). Hence $\psi_h \in \mathcal{H}_{\mathrm{E}}^{(1)}$, and the integration by parts of Lemma D.4 gives $\iota E(\partial V)\psi_h = \mathsf{z}_\xi^{1/2}|k|\hat{h}$. The span of these functions is dense, and ι is therefore onto.

Step 10 (irreducibility; spectra). On $\mathcal{H}_{\mathrm{E},\gamma}$ the operators $\mathsf{F}(H,\vec{P})$ act as the maximal abelian algebra $L^\infty(\mathbb{R}^2)$, so that every closed $\mathcal{U}$-invariant subspace equals $\iota^{-1}L^2(S)$ for some Borel set $S \subset \mathbb{R}^2 \setminus \{0\}$. Denote by m the measure on ∂V obtained from d^2k by the map $k \mapsto (|k|,-k)$. Steps 8 and 9 show that $E(\Delta)$ vanishes on $\mathcal{H}_{\mathrm{E},\gamma}$ if and only if $m(\Delta) = 0$. Suppose that $E(\tilde{S})\mathcal{H}_{\mathrm{E},\gamma} = \iota^{-1}L^2(S)$, where $\tilde{S} := \{(|k|,-k) : k \in S\}$, is invariant under $\mathcal{U}$. Step 4 and $\mathcal{U}(0,\Lambda)\mathcal{H}_{\mathrm{E},\gamma} = \mathcal{H}_{\mathrm{E},\gamma}$ give

$$E((\Lambda^T)^{-1}\tilde{S})\mathcal{H}_{\mathrm{E},\gamma} = \mathcal{U}(0,\Lambda)\,E(\tilde{S})\mathcal{H}_{\mathrm{E},\gamma} = E(\tilde{S})\mathcal{H}_{\mathrm{E},\gamma},$$

so that $m((\Lambda^T)^{-1}\tilde{S} \triangle \tilde{S}) = 0$ for all Λ. The group $SO_0(1,2)$ acts on $\partial V \setminus \{0\}$ by $\mathsf{p} \mapsto (\Lambda^T)^{-1}\mathsf{p}$; this action is continuous, and the rotations about the axis together with the boosts make it transitive. By (K16), either $\tilde{S}$ or its complement is an m-null set. The assertions on the spectra follow from $H \cong |k|$ on $\mathcal{H}_{\mathrm{E},\gamma}$ and from Step 4.

Step 11 (the field F and the subspace $\mathcal{H}_{\mathrm{E}}^{(1)}$). For $z \in \mathcal{T}$ and $b \in \Lambda^2\mathbb{C}^3$, denote by $\Psi_1(z;b)$ the vectors introduced in Step 6 of the proof of Theorems 3.2 and 3.3, and by $T(\zeta)$ the semigroup of Step 1 of that proof. As functions of z on the convex set $\mathcal{T}$, the $\Psi_1(z;b)$ are holomorphic in norm, and they depend linearly on b; their closed span is denoted $\mathcal{H}^{\Psi_1}$. (a) Step 7 of that proof, at $n = 1$, gives $[X(g)] = \int dy\,\Psi_1(Jy;Jg(y)) \in \mathcal{H}^{\Psi_1}$. If $\phi \perp \mathcal{H}_{\mathrm{E}}^{(1)}$, then $\int dy\,\langle\phi,\Psi_1(Jy;Jg(y))\rangle = 0$ for every $g \in C_c^\infty(\{y_1 > 0\};\Lambda^2)$. For each b, the continuous function $y \mapsto \langle\phi,\Psi_1(Jy;Jb)\rangle$ therefore vanishes on $\{y_1 > 0\}$; Fact 2 of that proof (with $L = J$) then shows $\phi \perp \mathcal{H}^{\Psi_1}$, and therefore $\mathcal{H}_{\mathrm{E}}^{(1)} = \mathcal{H}^{\Psi_1}$. (b) Write $\mathcal{H}^F$ for the closure of $\{F(h')\Omega : h' \in \mathrm{Sw}_1\}$. The rule (e4) of Step 18 of that proof, at $n = 1$, gives $\exp(-tH)F(h')\Omega = \int dx\,\Psi_1(x + ite_0;h'(x)) \in \mathcal{H}^{\Psi_1}$ for $t > 0$, and the limit $t \downarrow 0$ gives $\mathcal{H}^F \subset \mathcal{H}^{\Psi_1}$. The rule (e3) of the same step and $T(-a)\Omega = \Omega$ give $T(a)F(h')\Omega = F(\tau_a h')\Omega$ for real a. Thus $\mathcal{H}^F$ and its orthogonal complement are invariant under the unitary group $T(a) = \exp i(a^0H + \vec{a}\cdot\vec{P})$, $a \in \mathbb{R}^3$. Conjugation by the unitary operator $2Q - 1$, where the projection Q onto $\mathcal{H}^F$ commutes with this group, turns E into a projection valued measure for the same group; by the uniqueness in (K16) it is E itself. Consequently $\mathcal{H}^F$ reduces H, and $\exp(-tH)$ leaves it invariant. For $\phi \perp \mathcal{H}^F$ one obtains $\int dx\,\langle\phi,\Psi_1(x + ite_0;h'(x))\rangle = 0$ for every $h' \in \mathrm{Sw}_1$, whence $\langle\phi,\Psi_1(x + ite_0;b)\rangle$ vanishes for every real x and every b. The function $z \mapsto \langle\phi,\Psi_1(z + ite_0;b)\rangle$ is holomorphic on $\mathcal{T} - ite_0$, which contains $\mathbb{R}^3$, and Fact 2 with $L = 1$ yields $\phi \perp \mathcal{H}^{\Psi_1}$. Thus the three spaces $\mathcal{H}^F$, $\mathcal{H}^{\Psi_1}$ and $\mathcal{H}_{\mathrm{E}}^{(1)}$ coincide, and $\{E(\partial V)F(h')\Omega\}$ is dense in $\mathcal{H}_{\mathrm{E},\gamma}$. By the definition of $\mathcal{H}^{(1)}$ in Theorem A, $\mathcal{H}_{\mathrm{E}}^{(1)} = \mathcal{H}^{(1)}$ and $\mathcal{H}_{\mathrm{E},\gamma} = \mathcal{H}_\gamma$. That the boosts leave $\mathcal{H}_{\mathrm{E}}^{(1)}$ invariant, shown in Step 6, can also be read off from $\mathcal{U}(0,\Lambda)\Psi_1(z;b) = \Psi_1(\Lambda z;\Lambda b)$ (Step 12 of the proof of Theorems 3.2 and 3.3).

□

Proof of Corollary B. The physical mass bounds are Theorem 6.1(a). The fourth-cumulant bounds in the parent law and the components are [4, Proposition 26.2 and the proof of Theorem 1.1]. Theorem A gives $z_\xi > 0$. If $\mathrm{kl}_\xi((0,\infty)) = 0$, the rigidity theorem in Section 4 gives $\mu_\xi = \gamma_{z_\xi}$, contradicting the positive fourth cumulant. Since the total spectral mass is at most one, $z_\xi < 1$. The Maxwell limit follows from Theorem A(iv). □

The construction gives limit points. Convergence to a unique law and ergodicity of the parent limit point remain open. The mass hypothesis in Theorem A is essential to the Fisher estimate supplied here; at nonpositive lattice mass or at mass $o(\eta)$ this argument supplies no positive photon weight. At arbitrary charge the theorem does not decide Gaussianity. It also gives neither a scattering theory nor a description of possible additional massless sectors.

Appendix A. The finite-torus moment representation

We prove Proposition 5.12 using reflection positivity and a truncated Stieltjes moment representation. The lattice quantities and the facts (K1)–(K11) are those of Section 5.

Proof of Proposition 5.12. Step 1 (positivity). The torus $C(n)$ is adapted to x_1, x_2 and x_3, with height period n [4, Section 2.6], and $\mathrm{Adm}(x_1) = \frac{1}{2}\mathbb{Z}$ (Definition E.1). Fix $c \in \frac{1}{2}\mathbb{Z}$, and consider the sites x of the open slab $\{c < x_1 < c + \frac{n}{2}\}$. For these sites $\mathfrak{M}(x) \in \mathcal{B}^+$ and $\Theta_c\mathfrak{M}(x) = \mathfrak{M}^\star(\theta_c x)$, while $\mathbb{E}[\mathfrak{M}^\star(y)\mathfrak{M}(x)] = \mathsf{M}_n(y,x)$ (Proposition 5.4 (a), (c)). Consequently Proposition E.6 shows that, on these sites, the kernel $(x,x') \mapsto \mathsf{M}_n(\theta_c x, x')$ is Hermitian and positive semidefinite. This property is denoted **(RP)**.

Step 2 (proof of (a)). Every site of the layer L_τ has $\tau - \frac{1}{2} < x_1 < \tau + \frac{1}{2}$ (Definition E.1), and $\theta_c L_\tau = L_{2c-\tau}$ [4, Proposition 2.1]. Evaluated on the indicator functions of the layers that lie in the slab, (RP) together with translation invariance yields the positive semidefinite matrix $\big(n^2 G_n(\tau + \tau' - 2c)\big)_{\tau,\tau'}$. When $c = \frac{1}{2}$, the slab contains the layers L_τ with $1 \le \tau \le k_A$; when $c = 0$, it contains those with $1 \le \tau \le \lfloor \frac{n-1}{2} \rfloor$, and $\lfloor \frac{n-1}{2} \rfloor \ge k_A - 1$.

Step 3 (proof of (b), a truncated Stieltjes problem). Put $\mathsf{g}_j := G_n(j+1)$ and $k := k_A - 1$, write $\mathcal{P}_d$ for the space of real polynomials of degree at most d, and let L be the linear form on $\mathcal{P}_{2k}$ with $\mathsf{L}(\lambda^j) := \mathsf{g}_j$. Part (a) shows that $(f,h) \mapsto \mathsf{L}(fh)$ is positive semidefinite on $\mathcal{P}_k$, and that $\mathsf{b}(f,h) := \mathsf{L}(\lambda f h)$ defines a positive semidefinite form on $\mathcal{P}_{k-1}$. Whenever $h \in \mathcal{P}_{k-1}$ and $\mathsf{L}(h^2) = 0$,

$$\mathsf{b}(f,h)^2 \le \mathsf{L}(\lambda^2 f^2)\, \mathsf{L}(h^2) = 0.$$

Hence b descends to the quotient $\mathcal{P}_{k-1}/\{h : \mathsf{L}(h^2) = 0\}$, equipped with the inner product induced by L. There it takes the form $\mathsf{b}(f,h) = \langle [f], \mathsf{J}[h] \rangle$ with an operator $\mathsf{J} = \mathsf{J}^* \ge 0$ satisfying $\mathsf{J}^i[1] = [\lambda^i]$ for $i \le k-1$. Let σ_n be the spectral measure of J at the vector $[1]$; then

$$\int \lambda^{i+i'}\, \mathrm{d}\sigma_n = \mathsf{g}_{i+i'} \quad (i, i' \le k-1), \qquad \int \lambda^{2k-1}\, \mathrm{d}\sigma_n = \mathsf{b}(\lambda^{k-1}, \lambda^{k-1}) = \mathsf{g}_{2k-1}.$$

Step 4 (proof of (c)). Decompose σ_n at $\lambda_0 := \exp(-E_0)$: its part on $\{\lambda \le \lambda_0\}$, expressed in the variable E, is ϱ_n, and its part on $\{\lambda > \lambda_0\}$ produces $R_n(t)$. If $\lambda > \lambda_0$ and $t \le k_A$, then $\lambda^{t-1} \le \lambda_0^{t-k_A} \lambda^{k_A - 1}$, and therefore $R_n(t) \le \exp(E_0(k_A - t))\, G_n(k_A)$. From (E1) one obtains

$$G_n(k_A) \le \mathsf{F}(k_A) \le \exp\!\big(-2\varepsilon_* \lfloor k_A/2 \rfloor\big)\, \mathsf{F}(0) \le 96\, \mathbf{m}^{-2} \exp\!\big(-\varepsilon_*(k_A - 1)\big).$$

Next, for standard weights and $\mathbf{m} \le (1+\rho)\,\mathrm{sw}$, put $\xi := \frac{\mathbf{m}}{(1+\rho)\mathrm{sw}} \le 1$; then

$$\exp(4\vartheta) = (1+\xi)^4 \le 1 + 15\xi < 1 + 4\,\frac{\mathrm{SW}}{w_3}\,\xi = \exp(2\varepsilon_*).$$

Finally, let $n \ge 6$ be even. By Lemma 5.9 (a), $G_n(2) = G_n(n-2)$, and both arguments lie in the range of (b). For a measure $\sigma \ge 0$ on $[0,1)$, on the other hand, $\int \lambda^{n-3}\, \mathrm{d}\sigma < \int \lambda\, \mathrm{d}\sigma$ unless $\sigma((0,1)) = 0$.

Step 5 (proof of (d)). The total mass of σ_n is $G_n(1) \le 96\, \mathbf{m}^{-2}$, and $R_n(t) \to 0$ as $n \to \infty$ for every $E_0 < \varepsilon_*$. By (K5) there is a subsequence along which the restrictions of σ_{n_i} to $[0, \exp(-E_0)]$ converge. Their limit represents G_∞, is independent of E_0, and is carried by $\bigcap_{E_0 < \varepsilon_*} [0, \exp(-E_0)]$. □

Appendix B. Further reconstruction identities

B.1. **Plate vectors and unitary boosts.** The point plates and their integral formula are given in [4, Proposition 8.2]. To make the vector notation explicit, put $\Psi(\varnothing) = \Omega_1$. For $x \in \mathcal{W}^{(n)}$, $n \ge 1$, put

$$\Psi(x; \lambda) = \mathcal{B}_{\lambda_1}(x_1, (x_2 - x_1)/2) \cdots \mathcal{B}_{\lambda_n}((x_n - x_{n-1})/2, \kappa)\Omega_1, \qquad \kappa \in \mathcal{W}.$$

For $n = 1$ this means $\mathcal{B}_{\lambda_1}(x_1, \kappa)\Omega_1$. At each intermediate factor the two arguments are half the adjacent gaps. The choice of κ does not matter. Indeed, the point-plate translation law and $\mathsf{T}(\kappa)\Omega_1 = \Omega_1$ show that an argument beside the vacuum can be changed arbitrarily; between two

plates only the sum of their adjacent arguments matters. These identities, the holomorphy, the Gram formula, and the time-ordered Bochner integral are proved in [4, Section 8.3 and Proposition 8.3].

Lemma B.1. Under the one-plane hypotheses of Theorem 2.1, the closed span $\mathcal{H}_{12}^{\Psi}$ reduces $\mathsf{T}(\kappa)$ for $\kappa \in \overline{\mathcal{W}}$ and is invariant under every point plate. It contains every time-ordered product class occurring in Theorem 2.1. There is a unique strongly continuous unitary group on this span satisfying

$$\mathfrak{L}(\beta)\Psi(x;\lambda) = e^{w(\lambda)\beta}\Psi(\Lambda_\beta x;\lambda).$$

On this span it fixes the vacuum and obeys

$$\mathfrak{L}(\beta)\mathsf{T}(\kappa)\mathfrak{L}(\beta)^{-1} = \mathsf{T}(\Lambda_\beta\kappa),$$
$$\mathfrak{L}(\beta)\mathcal{B}_{\lambda_0}(\xi,\xi')\mathfrak{L}(\beta)^{-1} = e^{w(\lambda_0)\beta}\mathcal{B}_{\lambda_0}(\Lambda_\beta\xi,\Lambda_\beta\xi').$$

■

Proof. The time-ordered Bochner formula places each product class in the closed span. The translation law is $\mathsf{T}(\kappa)\Psi(x;\lambda) = \Psi(x+\kappa;\lambda)$. It applies also to $\mathsf{T}(\kappa)^* = \mathsf{T}(\kappa^*)$, since $\kappa^* = (\overline{\kappa^0}, -\overline{\kappa^1})$ belongs to the same closed tube. Hence this span reduces $\mathsf{T}(\kappa)$. Combining adjacent plate arguments gives

$$\mathcal{B}_{\lambda_0}(\xi,\xi')\Psi(x;\lambda) = \Psi((\xi,\xi+\xi'+x_1,\dots,\xi+\xi'+x_n);\lambda_0,\lambda_1,\dots,\lambda_n).$$

All consecutive differences lie in $\mathcal{W}$, including $\xi'+x_1$, so this proves invariance under point plates.

The Gram formula writes the inner product of two plate vectors as $s(\lambda)G_{n+n'}$ evaluated at differences, sums, and starred arguments. These operations commute with Λ_β. The proof of the angular identity in [4, Section 8.5, equation (8.41)] uses cubic invariance only through the quarter turn in the chosen plane. The one-plane hypotheses therefore suffice for that identity. On the ordered tube it gives

$$G_{n+n'}(\Lambda_\beta\zeta;\lambda^{\mathrm{rev}},\lambda') = e^{-(w(\lambda)+w(\lambda'))\beta}G_{n+n'}(\zeta;\lambda^{\mathrm{rev}},\lambda').$$

Here λ^{rev} reverses the order of the tuple. It follows that the proposed map on plate vectors preserves every Gram matrix. For total degree zero this is the vacuum norm identity; for total degree one both sides vanish since $G_1 = 0$. It therefore extends linearly and continuously to an isometry on the closed span. The relations $\Lambda_\beta\Lambda_{\beta'} = \Lambda_{\beta+\beta'}$ and $w(\lambda) \in \mathbb{R}$ give the group law; $-\beta$ supplies its inverse. Holomorphy of Ψ gives continuity on the spanning vectors, and the unit operator norm gives strong continuity on their closure. Uniqueness follows from density. The two intertwining identities now follow from the translation and product formulas above, because Λ_β commutes with sums and the new tuple has weight $w(\lambda_0)+w(\lambda)$. All the operators are bounded, so the identities extend from the span. □

B.2. **Spatial symmetries and the round cone.** Let the law also be invariant under $O(2)_1 = \{g \in O(3) : ge_1 = e_1\}$. Then $[\Phi]_1 \mapsto [\hat{g}\Phi]_1$ is unitary: g preserves the positive half-space, commutes with its reflection, and leaves the law invariant. Strong continuity holds first on character classes by continuity of the action on tests and bounded convergence, and then on the whole space by their density. This unitary fixes Ω_1, commutes with $e^{-t\mathsf{H}_1}$, and carries $V_1(y)$ to $V_1(gy)$. The joint spectral theorem consequently gives

$$\mathcal{U}(g)\mathfrak{E}_1(A)\mathcal{U}(g)^{-1} = \mathfrak{E}_1(gA), \qquad gA = \{(\omega, gp_\perp) : (\omega,p_\perp) \in A\}.$$

If (C_{12}) holds, rotate its null set $\{\omega < |p_2|\}$ by a countable dense set of spatial rotations. Their union is $\{\omega < |p_\perp|\}$, so the spectral measure is carried by the round forward cone. Spectral integration then gives $\|\mathsf{P}_2\psi\|^2 + \|\mathsf{P}_3\psi\|^2 \le \|\mathsf{H}_1\psi\|^2$ on $D(\mathsf{H}_1)$. The same quotient argument transports one coordinate reconstruction to another: for an orthogonal coordinate permutation g_{ji}, use the image law $(g_{ji}^{-1})_*\mu$ and the map $[\Phi]_1 \mapsto [\hat{g}_{ji}\Phi]_j$. It conjugates $H_1, V_1(y)$ and the joint spectral measure to $H_j, V_j(g_{ji}y)$ and the image spectral measure. This establishes the coordinate-change identities used in Corollary 2.2.

Appendix C. A uniform continuum clustering estimate

Let $\mu \in \mathcal{C}_9$: the law is translation invariant and is invariant and reflection positive at the nine coordinate and diagonal planes through the origin. No Gaussian bound or coupling restriction is needed in this appendix. For an open set O, let Σ_O be the sigma algebra generated by $X(f)$ with real $f \in C_c^\infty(O;\Lambda^2)$. The reflection quotients, their joint spectral measures, and the identification of invariant projections are those of [4, Propositions 7.1–7.2 and Section 7.3]. Thus the joint spectral measure $\mathfrak{E}_j$ of $(\mathsf{H}_j, \mathsf{P}_i, \mathsf{P}_{i'})$ is supported on $\lambda \geq |k|_\infty$, and its zero-energy projection $P_{0,j}$ corresponds to conditional expectation P onto all translation-invariant events. The linear reflection on observables is denoted by $R_j\Phi = \overline{\Theta_j\Phi}$.

Proposition C.1. Put

$$C_2 = \frac{128 + 64(16/\pi + 64/\pi^2)}{4\pi^2}, \qquad C_3 = 32(1+\sqrt{2})C_2, \qquad C_0 = 3C_3.$$

For $r > 0$, $\Phi, \Phi' \in L^2(\Sigma_{(-r,r)^3}, \mu)$, and $a \in \mathbb{R}^3$ with $t = |a|_\infty - 2r > 0$,

$$\left|\langle \Phi, U(a)\Phi'\rangle - \langle \Phi, P\Phi'\rangle\right| \leq C_3 \|\Phi\|_2 \|\Phi'\|_2 \left(\frac{r}{t} + \frac{r^2}{t^2}\right).$$

In particular, for $|a|_\infty \geq 4r$ the right side can be replaced by $C_0\|\Phi\|_2\|\Phi'\|_2 r/|a|_\infty$. If μ is translation ergodic, the subtracted term is $\overline{\mu(\Phi)}\mu(\Phi')$. ∎

Proof. The cube vector $\psi^j_\Phi = [U(re_j)\Phi]_j$ is well defined because $(-r,r)^3 + re_j \subset \{x_j > 0\}$, and $\|\psi^j_\Phi\| \leq \|\Phi\|_2$. The reflection form and translation law give, for $y_j \geq 2r$,

$$\langle \Phi, U(y)\Phi'\rangle = \langle \psi^j_{R_j\Phi}, e^{-(y_j - 2r)\mathsf{H}_j} V_j(y - y_j e_j)\psi^j_{\Phi'}\rangle. \tag{4}$$

Indeed, reflecting the first translated observable changes re_j to $-re_j$, and then translating both factors by re_j gives the left side. Also $\langle \Phi, U(-y)\Phi'\rangle = \overline{\langle \Phi', U(y)\Phi\rangle}$.

A filter excluding small transverse momentum. Let z_m be even and four-periodic, with values $(0, 1, -2, 1)$, and let B be the 2π-periodic function $B(u) = 8(1 - 4|u|/\pi)_+$ on $[-\pi, \pi]$. Its Fourier coefficients are

$$b_0 = 1, \qquad b_m = \left(\frac{8\sin(m\pi/8)}{m\pi}\right)^2 \quad (m \neq 0).$$

Two integrations give these formulas. Their absolute summability and Fourier uniqueness give $B(u) = \sum_m b_m e^{imu}$. Define

$$B_z(u) = \sum_m z_m b_m e^{imu}, \qquad \Psi(v) = 2\sum_{m \geq 1} z_m b_m v^{m-1} \quad (0 \leq v \leq 1).$$

The identities $z_m = \frac{1}{2}i^m - (-1)^m + \frac{1}{2}(-i)^m$ and $|z_m| = 1 - \frac{1}{2}i^m - \frac{1}{2}(-i)^m$ imply

$$\sum_m |z_m| b_m = 8, \quad B_z(u) = \tfrac{1}{2}B(u + \pi/2) - B(u + \pi) + \tfrac{1}{2}B(u - \pi/2).$$

Consequently $|B_z| \leq 8$ and $B_z(u) = 0$ if $\operatorname{dist}(u, 2\pi\mathbb{Z}) \leq \pi/4$; also $|\Psi| \leq 8$ and $\Psi(1) = 0$. The Poisson kernel $P_x(u) = \sinh x/(\cosh x - \cos u) = \sum_m e^{-|m|x}e^{imu}$ gives

$$\Psi(e^{-x}) = \frac{e^x}{2\pi}\int_{-\pi}^{\pi} B(v)\left\{\tfrac{1}{2}P_x(v - \pi/2) - P_x(v - \pi) + \tfrac{1}{2}P_x(v + \pi/2)\right\} dv.$$

For $|v| \leq \pi/4$, the expression in braces is

$$\sinh x\left(\frac{\cosh x}{\cosh^2 x - \sin^2 v} - \frac{1}{\cosh x + \cos v}\right) \geq \frac{\tanh x}{\sqrt{2}\cosh x + 1}.$$

Since $B \geq 0$, its mean is one, and $e^x/(\sqrt{2}\cosh x + 1) \geq (1+\sqrt{2})^{-1}$,

$$\Psi(e^{-x}) \geq \frac{\tanh x}{1+\sqrt{2}} \quad (x > 0). \tag{5}$$

Thus $\Psi \geq 0$ on $[0,1]$ and is positive on $[0,1)$, including zero because $\Psi(0) = 2b_1 > 0$.

The weighted momentum density. First suppose $R_j\Phi = \varepsilon\Phi$, $\varepsilon \in \{1,-1\}$. Let $\varrho = \langle \psi^j_\Phi, \mathfrak{E}_j(\cdot)\psi^j_\Phi\rangle$ and, for $y \perp e_j$, set

$$K(y) = \sum_m z_m b_m \langle R_j\Phi, U(2rme_j + y)\Phi\rangle.$$

The series converges absolutely and $|K(y)| \leq 8\|\Phi\|_2^2$. Reflection invariance makes the correlation even in its e_j coordinate. Equation (4) therefore gives

$$K(y) = \int \Psi(e^{-2r\lambda}) e^{ik\cdot y}\, d\varrho(\lambda, k).$$

Hence K is the Fourier transform of the positive measure obtained by projecting $\Psi(e^{-2r\lambda})d\varrho$ onto momentum space.

If $y_i \geq 2r$ for $i \neq j$, use the i-th reconstruction in (4) before summing. This yields

$$K(y) = \varepsilon \langle \psi^i_{R_i\Phi}, e^{-(y_i - 2r)\mathsf{H}_i} B_z(2r\mathsf{P}_j) V_i(y_{i'}e_{i'})\psi^i_\Phi\rangle.$$

Where the filter is nonzero, $|k_j| > \pi/(8r)$, and the cone gives $\lambda \geq |k_j|$. Since $K(-y) = \overline{K(y)}$, it follows that

$$|K(y)| \leq 8e^{-\pi(|y|_\infty - 2r)/(8r)}\|\Phi\|_2^2 \quad (|y|_\infty \geq 2r).$$

The area of the square $|y|_\infty \leq s$ is $4s^2$, so

$$\int |K(y)|\, dy \leq [128 + 64(16/\pi + 64/\pi^2)]r^2\|\Phi\|_2^2.$$

Fourier inversion now identifies the projected measure as $h(k)\,dk$, where h is continuous and $0 \leq h \leq C_2 r^2\|\Phi\|_2^2$. For completeness, its integral against a Schwartz test equals the integral against $(2\pi)^{-2}\int K(y)e^{-ik\cdot y}dy$, by Fubini; this proves the asserted equality of measures.

Decay on cube vectors. The density gives no mass to $k = 0$. By (5), ϱ therefore gives no mass to $\{\lambda > 0, k = 0\}$. The cone, the monotonicity of $x \mapsto e^{-tx}/\tanh(2rx)$, and $\coth x \leq 1 + x^{-1}$ give

$$\begin{aligned}\langle \psi^j_\Phi, e^{-t\mathsf{H}_j}(1 - P_{0,j})\psi^j_\Phi\rangle &\leq (1+\sqrt{2})\int \frac{e^{-t|k|_\infty}}{\tanh(2r|k|_\infty)} h(k)\, dk \\ &\leq (1+\sqrt{2})C_2\|\Phi\|_2^2(8r^2/t^2 + 4r/t).\end{aligned}$$

Here the last integral is evaluated using the density $8s$ of $|k|_\infty$ under two-dimensional Lebesgue measure. For general Φ, decompose $\Phi = \Phi_+ + \Phi_-$, $\Phi_\pm = (\Phi \pm R_j\Phi)/2$. The triangle inequality for the seminorm defined by $e^{-t\mathsf{H}_j}(1 - P_{0,j})$ and $\|\Phi_\pm\|_2 \leq \|\Phi\|_2$ give

$$\langle \psi^j_\Phi, e^{-t\mathsf{H}_j}(1 - P_{0,j})\psi^j_\Phi\rangle \leq C_3\|\Phi\|_2^2(r/t + r^2/t^2). \tag{6}$$

The correlation bound. Choose j with $|a_j| = |a|_\infty$. If $a_j > 0$, apply (4) and split $e^{-t\mathsf{H}_j} = P_{0,j} + e^{-t\mathsf{H}_j}(1 - P_{0,j})$. Zero energy has zero transverse momentum, and the invariant projection identity gives exactly $\langle \Phi, P\Phi'\rangle$ for the first term. Cauchy–Schwarz and (6) bound the other term as stated. Negative a_j follows by taking the conjugate correlation with $-a$ and exchanged observables. If $|a|_\infty \geq 4r$, then $t \geq |a|_\infty/2$ and $r/t \leq 1/2$, so $r/t + r^2/t^2 \leq 3r/|a|_\infty$. Finally, in an ergodic law $P\Phi' = \mu(\Phi')$, which proves the last assertion. □

Appendix D. The spectral representation of the two-point function

In this appendix $\mathbb{B}(b) = X(*^{-1}b)$, $S_2(b,b') = \mu(\mathbb{B}(b)\mathbb{B}(b'))$, and $\widehat{b}(p) = \int e^{-ip\cdot x} b(x)\,dx$. Write $\Pi(p) = 1 - |p|^{-2}pp^T$ for $p \neq 0$.

Proposition D.1. Let μ be a Gaussian-dominated law of closed real tempered two-forms, invariant under all Euclidean motions and reflection positive at the coordinate and diagonal planes. There is exactly one finite positive measure kl on $[0,\infty)$ such that

$$S_2(\bar{b},b') = (2\pi)^{-3}\int \widehat{b}(p)^*\Pi(p)\widehat{b}'(p)\left(\int_{[0,\infty)} \frac{|p|^2}{|p|^2+m^2}\,d\mathrm{kl}(m)\right)dp.$$

Its mass is at most one. On disjoint compact supports the covariance has the real analytic kernel

$$S_{ab}(x) = (-\delta_{ab}\Delta + \partial_a\partial_b)\int_{[0,\infty)} \frac{e^{-m|x|}}{4\pi|x|}\,d\mathrm{kl}(m) \qquad (x \neq 0).$$

The free Maxwell covariance has measure δ_0; the zero field has measure zero. ∎

Proof. Gaussian domination gives a bounded linear map $b \mapsto \mathbb{B}(b)$ from complex L^2 to $L^2(\mu)$ with norm at most one, and S_2 is symmetric and real on real tests [4, Proposition 5.1]. Use the first coordinate reconstruction [4, Propositions 7.1–7.2]; its joint spectral measure $\mathfrak{E}$ is carried by $K = \{(\omega,p_2,p_3) : \omega \geq |p_2|,\ \omega \geq |p_3|\}$.

The one-field spectral matrix. For $u \in L^1 \cap L^2$ supported in $x_1 \geq 0$ set

$$\widetilde{u}(\omega,p_\perp) = \int u(x)e^{-x_1\omega + ip_\perp\cdot x_\perp}\,dx.$$

For such u and any $\alpha \in L^2$ supported in $x_1 \geq 0$, the reconstruction law and a Bochner integral give

$$[\mathbb{B}_a(u*\alpha)]_1 = \widetilde{u}(\mathsf{H},\mathsf{P})[\mathbb{B}_a(\alpha)]_1. \tag{7}$$

Indeed $u*\alpha = \int_{x_1\geq 0} u(x)\tau_x\alpha\,dx$ in L^2; the class map is bounded and $[\mathbb{B}_a(\tau_x\alpha)]_1 = e^{-x_1\mathsf{H}}V(x_\perp)[\mathbb{B}_a(\alpha)]_1$. Fubini is justified by $\|u\|_1$.

Take $P_1(v) = \pi^{-1}(1+v^2)^{-1}$ and $\alpha_0(x) = 1_{(0,1)}(x_1)P_1(x_2)P_1(x_3)$. Its transform is $a_0(\omega,p_\perp) = (1-e^{-\omega})\omega^{-1}e^{-|p_\perp|_1}$, with the quotient equal to one at zero. It is continuous and positive. Put $\psi_a = [\mathbb{B}_a(\alpha_0)]_1$, $m_{ab} = \langle\psi_a, \mathfrak{E}(\cdot)\psi_b\rangle$, and $\varrho_{ab} = a_0^{-2}m_{ab}$. These are Radon measures, since a_0 has a positive lower bound on bounded sets. The matrix m_{ab} has a positive semidefinite density relative to $m_{11}+m_{22}+m_{33}$: this follows by applying positivity first to the countable set $(\mathbb{Q}+i\mathbb{Q})^3$ and then using continuity. The same is true of ϱ_{ab}. In particular their integrals satisfy the matrix Cauchy–Schwarz inequality.

Commutativity of convolution and (7) give $a_0(\mathsf{H},\mathsf{P})[\mathbb{B}_a(u)]_1 = \widetilde{u}(\mathsf{H},\mathsf{P})\psi_a$. On $A_n = \{a_0 > 1/n\}$ one may divide by a_0 in spectral calculus. These bounded sets increase to the whole spectral space. Taking scalar products, then using monotone convergence for diagonal entries and matrix Cauchy–Schwarz for the others, gives

$$\langle[\mathbb{B}_a(u)]_1, [\mathbb{B}_b(u')]_1\rangle = \int \overline{\widetilde{u}}\widetilde{u}'\,d\varrho_{ab}. \tag{8}$$

The measures are carried by K, because the m_{ab} are.

Size and the off-origin kernel. For $c > 0$ set $u_c(x) = c^{-3}1_{(c,2c)}(x_1)P_1(x_2/c)P_1(x_3/c)$. Then $\|u_c\|_2^2 = (4\pi^2c^3)^{-1}$ and $\widetilde{u}_c \geq e^{-4c\omega}$ on K. The contraction of the quotient map and (8) therefore give

$$I(t) := \max_a \int e^{-t\omega}\,d\varrho_{aa} \leq 128/(\pi^2t^3) < 13t^{-3}. \tag{9}$$

For $x_1 > 0$ define

$$S_{ab}(x) = (-\theta_1)_{aa}\int e^{-x_1\omega + ip_\perp\cdot x_\perp}\,d\varrho_{ab}, \qquad -\theta_1 = \mathrm{diag}(1,-1,-1).$$

Matrix Cauchy–Schwarz bounds each entry by $I(x_1)$. The same integral is holomorphic where $\operatorname{Re} z_1 > |\operatorname{Im} z_2| + |\operatorname{Im} z_3|$, by the cone and (9). For compactly supported α, α' strictly ordered in the first coordinate, reflection in (8) gives

$$S_2(\alpha e_a, \alpha' e_b) = \iint \alpha(x)\alpha'(x')S_{ab}(x' - x)\,dx\,dx'.$$

The positive separation supplies an integrable factor $e^{-\delta\omega}$, so Fubini is valid.

Euclidean invariance transports this density to every rotated half-space. On overlaps the densities agree, since their integrals against all pairs of smooth tests in small product neighborhoods agree. They therefore define one real analytic matrix on $\mathbb{R}^3 \setminus \{0\}$. For arbitrary disjoint compact supports, partition both into cubes of side d with three times d smaller than their sup-norm distance. Each pair of occupied cubes is strictly ordered in some signed coordinate direction. Summing the preceding formula proves the density formula for those supports. Reality, orthogonal covariance, and closedness yield

$$S(gx) = gS(x)g^T, \qquad S(x) \in \mathbb{R}^{3\times 3}, \qquad \sum_a \partial_a S_{ab} = 0, \qquad |S_{ab}(x)| < 39|x|^{-3}.$$

For the last bound rotate x to $|x|e_1$ and use $\sum_{k,l} |g_{ak}g_{bl}| \le 3$. For divergence, insert a gradient in the first covariance entry and integrate by parts on disjoint supports.

A positive radial potential. Commutation with the stabilizer $1 \oplus O(2)$ gives $S(re_1) = \operatorname{diag}(a_L(r), a_T(r), a_T(r))$ and hence

$$S_{ab}(x) = a_T(r)\delta_{ab} + (a_L(r) - a_T(r))x_a x_b/r^2.$$

The first diagonal spectral measure gives $a_L \ge 0$. Divergence gives $a_L' = -2(a_L - a_T)/r$. Define

$$P(r) = \int_r^\infty \tfrac{s}{2} a_L(s)\,ds.$$

Then $0 \le P(r) \le 20/r$, $P' = -ra_L/2$, $P'' = a_L/2 - a_T$, and direct differentiation gives $S_{ab} = (-\delta_{ab}\Delta + \partial_a\partial_b)P(|x|)$.

Set $c(t) = -\int_{\mathbb{R}^2}(S_{22}+S_{33})(t, x_\perp)\,dx_\perp$. The bound $78(t^2+|x_\perp|^2)^{-3/2}$ proves integrability. Inserting $e^{-\epsilon|x_\perp|^2/2}$ first, the spectral formula gives

$$c_\epsilon(t) = \int e^{-t\omega}\frac{2\pi}{\epsilon}e^{-|p_\perp|^2/(2\epsilon)}\,d(\varrho_{22} + \varrho_{33}).$$

This is a positive Laplace transform, and dominated convergence gives $c_\epsilon(t) \to c(t)$ for every $t > 0$. The extended continuity theorem for Laplace transforms [6, Section XIII.1, Theorem 2a] therefore gives a positive measure ϱ_c with $c(t) = \int e^{-mt}\,d\varrho_c(m)$. Polar integration in the transverse plane yields

$$c(t) = -2\pi\int_t^\infty \left[r(a_T + a_L)(r) - \frac{t^2}{r}(a_L - a_T)(r)\right]dr = -4\pi(tP(t))'.$$

Indeed $r(a_T + a_L) = -(rP')' - 2P'$ and $(a_L - a_T)/r = (P'/r)'$; $P(r)$ and $rP'(r)$ tend to zero at infinity.

Let $k(t) = tP(t)$. It is nonnegative, at most twenty, and decreasing. Thus $k(\infty)$ exists and $\int_t^\infty c(s)ds = 4\pi(k(t) - k(\infty)) < \infty$. This forces $\varrho_c(\{0\}) = 0$. Tonelli now gives

$$4\pi k(t) = 4\pi k(\infty) + \int_{(0,\infty)} m^{-1}e^{-mt}\,d\varrho_c(m).$$

Define $\mathrm{kl} = 4\pi k(\infty)\delta_0 + 1_{(0,\infty)}m^{-1}\varrho_c$. Monotone convergence at zero shows its mass is at most 80π and $P(r) = \int e^{-mr}(4\pi r)^{-1}d\mathrm{kl}(m)$.

No contact term. The bounded positive sesquilinear covariance has a Riesz operator commuting with translations. By [9, Chapter I, Theorem 3.18] its entries are bounded Fourier multipliers; write the matrix as M, with $0 \le M \le 1$. Closedness gives $M(p)p = 0$ almost everywhere, by testing a

gradient whose scalar Fourier transform is a nonvanishing Gaussian. The potential $P(|x|)$ is locally integrable and tempered, and its Fourier transform is $\int (|p|^2+m^2)^{-1} d\mathrm{kl}(m)$. Fubini follows from finiteness of the measure and local integrability of $|p|^{-2}$ in three dimensions. Its differentiated covariance has the bounded multiplier

$$M_{\mathrm{kl}}(p) = \Pi(p) \int \frac{|p|^2}{|p|^2+m^2}\, d\mathrm{kl}(m).$$

The two covariance distributions agree off the origin by the density formula. Their difference is supported at zero: apply their equality to two separated tests, then let one test approximate a point mass. A distribution supported at a point is a finite sum of derivatives of that point mass [7, Theorems 6.24–6.25]. Its Fourier transform is therefore a polynomial. Here that polynomial equals the bounded matrix $M - M_{\mathrm{kl}}$, so it is constant. Both matrices annihilate p almost everywhere; the constant is zero. This proves the stated covariance formula on all L^2 tests.

The mass and uniqueness. The continuous function $h(s) = \int s^2(s^2+m^2)^{-1} d\mathrm{kl}(m)$ is at most one because $M = M_{\mathrm{kl}} \le 1$ almost everywhere. Monotone convergence as $s \to \infty$ gives $\mathrm{kl}([0,\infty)) \le 1$. Two measures giving the same covariance have equal h, hence equal Stieltjes transforms after division by s^2. Using $(s^2+m^2)^{-1} = \int_0^\infty e^{-t(s^2+m^2)} dt$, these are Laplace transforms, at $s^2 - 1 \ge 0$, of the finite measures with continuous densities $e^{-t} \int e^{-tm^2} d\mathrm{kl}(m)$. Laplace uniqueness [6, Section XIII.1, Theorem 1a] first equates these densities and then the images of the two measures under $m \mapsto m^2$. That map is a homeomorphism, so the measures agree. The multipliers of the free Maxwell and zero covariances are Π and zero, which proves the final statements. □

D.1. **Gaussian laws and reflection positivity.**

Lemma D.2. Let C be a continuous positive semidefinite real symmetric bilinear form on the real Schwartz two-forms. There is a unique centered Gaussian law γ_C with covariance C, characterized by

$$\Phi_{\gamma_C}(g) = \exp(-C(g,g)/2).$$

It satisfies (G) if and only if $C(g,g) \le \|g\|^2$. It is carried by closed two-forms if and only if $C(*^{-1}\nabla u, *^{-1}\nabla u) = 0$ for every real scalar Schwartz function u. A Euclidean motion preserves the law if and only if it preserves C.

Suppose C is invariant under reflection r in a plane. Then γ_C is reflection positive there if and only if $C(r\cdot g, g) \ge 0$ for every real smooth compactly supported test g in its positive half-space.

If C is translation invariant and $C(g, \tau_a h) \to 0$ as $|a| \to \infty$ for all real tests g, h, then γ_C is mixing and hence ergodic. ■

Proof. For real tests $g_1, \dots, g_n$, the matrix $M_{pq} = C(g_p, g_q)$ is positive semidefinite. Its entrywise exponential is positive semidefinite by the Schur product theorem and the exponential series. Since

$$e^{-C(g_q-g_p, g_q-g_p)/2} = e^{-C(g_p,g_p)/2} e^{-C(g_q,g_q)/2} e^{M_{pq}},$$

the function $e^{-C(g,g)/2}$ is positive definite, continuous, and equal to one at zero. Minlos' theorem, in the form (K21), gives the unique law. Evaluating its characteristic functional at $\sum t_j g_j$ identifies the finite-dimensional distributions as centered Gaussian vectors with covariance matrix $(C(g_i,g_j))$. Consequently

$$\gamma_C(e^{X(g)}) = e^{C(g,g)/2}.$$

This proves the assertion about (G). A gradient test has zero variance precisely when the corresponding random variable vanishes almost surely. Taking a countable dense family of scalar tests and using distributional continuity gives the assertion about closedness. The symmetry assertion follows by comparing the characteristic functionals, which determine the law.

For tests in the positive half-space put $M_{pq} = C(r \cdot g_p, g_q)$. Reflection invariance makes M real symmetric. If the reflected covariance is nonnegative, it is positive semidefinite, and

$$\Phi_{\gamma_C}(g_q - r \cdot g_p) = e^{-C(g_p,g_p)/2} e^{-C(g_q,g_q)/2} e^{M_{pq}}$$

is a positive semidefinite matrix by the same Schur-product argument. These are exactly the character inequalities for reflection positivity. Conversely, reflection positivity applied to the L^2 random variable $X(g)$ gives $C(r \cdot g, g) \geq 0$; the passage from characters to $X(g)$ is obtained by the difference quotients $(e^{itX(g)} - 1)/(it)$, which converge in L^2 for a Gaussian variable.

Finally,

$$\gamma_C(e^{iX(g)} U(a) e^{iX(h)}) = \exp(-C(g,g)/2 - C(h,h)/2 - C(g, \tau_a h)).$$

The assumed covariance decay proves mixing for characters. Their linear span is dense in $L^2(\gamma_C)$, and translations are isometries there, so the same limit holds for all L^2 functions. □

For the covariances C_ϱ in Proposition 4.4, the multiplier has operator norm at most one. Thus the product of that multiplier with two Schwartz Fourier transforms is integrable. The Riemann–Lebesgue lemma gives the decay required in the last assertion. In particular, the mixing conclusion here uses the displayed covariance of those laws; it is not an assertion about arbitrary translation-invariant Gaussian laws.

D.2. **The one-field spectral subspace.**

Lemma D.3. Let σ_1, σ_2 be complex Borel measures on $[0, \infty) \times \mathbb{R}^d$. Suppose that, for some $t_0 \geq 0$,

$$\int e^{-t\omega} \, d|\sigma_i|(\omega, p) < \infty \qquad (t > t_0, \ i = 1, 2).$$

If their Fourier–Laplace transforms agree for every $t > t_0$ and $y \in \mathbb{R}^d$, then $\sigma_1 = \sigma_2$. ■

Proof. Fix $t_1 > t_0$, and set $\sigma = e^{-t_1\omega}(\sigma_1 - \sigma_2)$, a finite measure. For every integer $n \geq 0$, Fourier uniqueness in p implies that the p marginal of $e^{-n\omega}\sigma$ is zero. After $x = e^{-\omega}$, extend the resulting measure by zero to $\{0\} \times \mathbb{R}^d$. For each Borel set $B \subset \mathbb{R}^d$, its measure in the x variable annihilates all monomials, hence all of $C([0,1])$ by polynomial approximation. It is therefore zero. Rectangles determine finite measures, so $\sigma = 0$; multiplication by $e^{t_1\omega}$, first on bounded strips, gives the claim. □

Lemma D.4. Suppose that a centred law has the two-point function

$$C(g, g') = \int_{[0,\infty)} C_{\delta_s}(g, g') \, \rho(ds), \qquad \widehat{C}_{\delta_s}(p) = \frac{|p|^2 I - pp^T}{|p|^2 + s^2},$$

where two-forms are identified with vectors by $*$, ρ is a finite positive measure, and the Fourier inversion factor is $(2\pi)^{-3}$. Assume that its reflection-positive reconstruction has commuting operators $H \geq 0$ and $P = (P_2, P_3)$, with $V(y) = e^{iy \cdot P}$, and that the Euclidean translations give $e^{-tH} V(y)$ on the reconstructed classes. Put $z = \rho(\{0\})$ and

$$\mathcal{H}^{(1)} = \overline{\{[X(g)] : g \in C_c^\infty(\{x_1 > 0\}; \Lambda^2 \mathbb{C}^3)\}}.$$

If E is the joint spectral measure of (H, P) and $\partial V = \{(\omega, p) : \omega = |p|\}$, then, for $z > 0$, the map

$$E(\partial V)[X(g)] \longmapsto z^{1/2} a_g, \qquad a_g(k) = \int_0^\infty dt \int_{\mathbb{R}^2} d^2y \, e^{-|k|t - ik \cdot y} \, \mathrm{div}(*g)(t, y),$$

extends to a unitary map from $E(\partial V)\mathcal{H}^{(1)}$ onto $L^2(\mathbb{R}^2, d^2k/((2\pi)^2 \, 2|k|))$. Under this map, $F(H, P)$ becomes multiplication by $F(|k|, -k)$ for every bounded Borel function F. If $z = 0$, then $E(\partial V)\mathcal{H}^{(1)} = \{0\}$. For the boundary vector $\psi_h = [B_0(h)]$ of Proposition 4.2, the same map sends $E(\partial V)\psi_h$ to $z^{1/2}|k|\hat{h}(k)$. ■

Proof. Write $j = \operatorname{curl}(*g)$, $E_s(k) = (|k|^2 + s^2)^{1/2}$, and

$$v_s(k) = \int e^{-E_s(k)t - ik\cdot y} j(t,y)\, dt\, d^2y.$$

The one-dimensional resolvent integral gives

$$\langle [X(g)], e^{-tH} V(y)[X(g)]\rangle = \int \rho(ds) \int \frac{d^2k}{(2\pi)^2} \frac{|v_{s,2}|^2 + |v_{s,3}|^2 - |v_{s,1}|^2}{2E_s(k)} e^{-tE_s(k) - ik\cdot y}.$$

Integration by parts in $\operatorname{div} j = 0$ gives $E_s v_{s,1} + ik \cdot v_{s,\perp} = 0$, so the density is nonnegative. For $s = 0$, integration by parts in $j = \operatorname{curl}(*g)$ also gives $|v_{0,2}|^2 + |v_{0,3}|^2 - |v_{0,1}|^2 = |a_g|^2$. Lemma D.3 identifies the joint spectral measure with the image of this positive measure under $(s,k) \mapsto (E_s(k), -k)$. Only $s = 0$ contributes to ∂V, and the origin has zero mass. Polarization therefore gives the asserted isometry and its intertwining identity on the linear span of the one-field classes.

For $u_{\epsilon,t}$ in Proposition 4.2, integration by parts gives

$$a_{u_{\epsilon,t}}(k) = |k|\hat{h}(k) \int_0^\infty e^{-|k|r} \varrho_\epsilon(r - t)\, dr.$$

First let $\epsilon \downarrow 0$ at fixed $t > 0$, then let $t \downarrow 0$. Proposition 4.2 gives convergence of the classes; the displayed formula gives the image $z^{1/2}|k|\hat{h}(k)$. These functions have dense span in the stated L^2 space: if a function is orthogonal to all of them, its pairing with every $\hat{h}$, $h \in C_c^\infty(\mathbb{R}^2)$, vanishes; Fourier uniqueness for tempered distributions makes that function zero. The isometry is consequently onto. □

Appendix E. The finite lattice and parameter ranges

E.1. **Configurations and operators.**

Definition E.1. The lattice in the fixed-spacing statements has unit mesh. A real p-cochain is a real function on oriented p-cells of $\mathbb{Z}^3$, odd under orientation reversal. Write C^p for these cochains, C^p_{c} for those of finite support, and $C^p(\Gamma)$ for the periodic cochains on $\mathbb{Z}^3/\Gamma$. The inner products are the sums over positively oriented cells, and d^* is the adjoint of the coboundary d. In particular,

$$(dA)_{\mu\nu}(x) = A_\mu(x) + A_\nu(x + e_\mu) - A_\mu(x + e_\nu) - A_\nu(x).$$

Lattice isometries act by pullback with the orientation signs. Set $\Omega = \{F \in C^2 : dF = 0\}$, with the product topology. For a law ν on Ω, condition (G) means $\nu(e^{\langle \beta, F\rangle}) \le e^{\|\beta\|^2/2}$ for every real $\beta \in C^2_{\mathrm{c}}$.

Let O_{cub} be the group of signed permutation matrices, $W = \mathbb{Z}^3 \rtimes O_{\mathrm{cub}}$. The lattice mirrors are $x_\mu = c$, $c \in \frac{1}{2}\mathbb{Z}$, and $x_\mu \pm x_\nu = c$, $c \in \mathbb{Z}$. For $\ell(x) = n \cdot x$, the reflection at $\ell = c$ is $\theta_{\ell,c}x = x - 2(\ell(x) - c)n/|n|^2$. Thus $\mathrm{Adm}(x_\mu) = \frac{1}{2}\mathbb{Z}$. Fix $v_0 = (v_1, v_2, v_3)$ with $0 < v_3 < v_2 < v_1 < 1/2$. Let $\mathbb{X}$ be the set of fermion sites, which have the unique form $x = k + rv_0$, $k \in \mathbb{Z}^3$, $r \in O_{\mathrm{cub}}$; put $\pi(x) = k$. Let s_0 change the sign of coordinate three, s_1 interchange coordinates two and three, s_2 interchange coordinates one and two, and let r_{e_1} change the sign of coordinate one. The four neighbours of (k, r) are

$$(k, rs_i) \quad (0 \le i \le 2), \qquad (k + re_1, rr_{e_1}) \quad (i = 3).$$

The geometric site in each case is its first coordinate plus its second coordinate applied to v_0. Write $\widehat{d}_i(x)$ for the unit vector from x to $x^{(i)}$. The Pauli convention is $\gamma_\mu = \operatorname{diag}(\sigma_\mu, -\sigma_\mu)$, $\gamma_n = \sum_\mu n_\mu \gamma_\mu$, and $\gamma_5 = \left(\begin{smallmatrix} 0 & 1_2 \\ 1_2 & 0 \end{smallmatrix}\right)$. For real ρ, M, $w_i > 0$ and $q > 0$, put

$$U_A(x, x') = e^{iqA(\pi(x),\pi(x'))}, \qquad w(x, x^{(i)}) = \frac{w_i}{2}(\rho - \gamma_{\widehat{d}_i(x)}),$$

$$(D_A\psi)(x) = M\psi(x) - \sum_i w(x, x^{(i)}) U_A(x, x^{(i)}) \psi(x^{(i)}),$$

where $A(y, y) = 0$. The standard weights are $(w_0, w_1, w_2, w_3) = (6, 6\sqrt{2}, 6\sqrt{2}, 6)$.

A torus $\mathbb{T} = (\Gamma, \chi)$ has a rank-three period lattice with $\min_{g\neq 0} |g|_\infty \geq 4$ and a sign character χ. Spinors obey $\psi(x+g) = \chi(g)\psi(x)$. The antiperiodic cubic torus $C(n)$ has $\Gamma = n\mathbb{Z}^3$ and $\chi(nv) = (-1)^{v_1+v_2+v_3}$. Introduce Grassmann generators with these same sign conditions. Write $\mathrm{Gr}(\mathbb{X})$ for the Grassmann algebra on the covering generators and $\mathrm{Gr}_\mathbb{T}$ for its quotient by these boundary relations. Normalize the Berezin integral by $\int_\mathbb{T} \prod_{x\in\mathbb{X}/\Gamma,\alpha} \psi_\alpha(x)\bar\psi_\alpha(x) = 1$, and put $S_\mathbb{T}(A) = \sum_{x\in\mathbb{X}/\Gamma} \bar\psi(x)(D_A\psi)(x)$. For a real periodic zero-cochain λ, gauge transformations act by $A \mapsto A - d\lambda$, $\psi(x) \mapsto e^{iq\lambda(\pi(x))}\psi(x)$ and $\bar\psi(x) \mapsto \bar\psi(x)e^{-iq\lambda(\pi(x))}$; the integer two-cochain z is unchanged.

The photon configurations and their measure are

$$\mathrm{Conf}(\Gamma) = \{(A,z) : A \in C^1(\Gamma),\ z \in C^2(\Gamma;\mathbb{Z}),\ dz = 0\}/\!\sim, \quad (A,z) \sim (A + 2\pi k/q, z - dk),$$

$$\int \Phi \,\mathrm{d}(A,z) = \sum_{dz=0} \int_{[0,2\pi/q)^{3|\mathbb{Z}^3/\Gamma|}} \Phi(A,z)\,\mathrm{d}A, \qquad F = dA + 2\pi z/q.$$

One may use any measurable fundamental domain. The unnormalized functional, its normalization, and its curvature law are

$$Z_\mathbb{T}[G] = \int e^{-\|F\|^2/2} \int_\mathbb{T} e^{-\bar\psi D_A\psi} G \,\mathrm{d}(A,z), \qquad \mathbb{E}_\mathbb{T} = Z_\mathbb{T}/Z_\mathbb{T}[1], \qquad \mu_\mathbb{T} = \mathrm{Law}(\widetilde F).$$

Here $\widetilde F$ is the periodic extension of F to the covering lattice. For scalar G, the inner integral is $G \det D_A^\mathbb{T}$. The normalization and the integrability used below are proved in Proposition E.5. ■

The unit-mesh construction just defined is the chamber construction of [4, Section 2]. The definition here permits all real ρ; the precise algebraic, positivity and Gaussian ranges used below are distinguished in Lemma E.2 and Proposition E.6. In the fixed-spacing statements, β denotes a lattice two-cochain and ν a law on Ω. Write $\mathrm{sw} = \sum_i w_i$ and $\mathbf{m} = M - \rho\,\mathrm{sw}/2$; D_A includes this mass. For standard weights it is $D^{\mathrm{ch}}[qA] + \mathbf{m}$. At a continuum cutoff $q = e\eta^{1/2}$ and $\mathbf{m} = \boldsymbol{\mu}_N$. For $\rho \geq 0$ the norm of a type-three hop is $\mathsf{h}_3 = w_3(1+\rho)/2$, equal to $3(1+\rho)$ for the standard weights. The graph distance is denoted $d_\mathbb{T}$. Only type-three hops change π, by a unit coordinate vector; the graph is connected, and π is W-equivariant [4, Proposition 2.1].

E.2. **Algebra, normalization and reflection positivity.**

Lemma E.2. Use the chamber graph and the cochain conventions of [4, Section 2]. Let $\rho, M \in \mathbb{R}$, $w_i > 0$, $q > 0$, and $\mathbf{m} = M - \rho\sum_i w_i/2$. For real A, on the covering ℓ^2 space and on a torus with unitary boundary character, the massive chamber matrix satisfies

$$D_A = M - \frac{\rho}{2}H_A + \frac{1}{2}J_A, \qquad H_A^* = H_A, \qquad J_A^* = -J_A,$$

where

$$(H_A\psi)(x) = \sum_i w_i U_A(x,x^{(i)})\psi(x^{(i)}), \qquad (J_A\psi)(x) = \sum_i w_i \gamma_{\widehat{d_i(x)}} U_A(x,x^{(i)})\psi(x^{(i)}).$$

and $(\nabla_A\psi)(x,i) = \sqrt{w_i/2}\,(\psi(x) - U_A(x,x^{(i)})\psi(x^{(i)}))$. Consequently

$$\mathrm{Re}\, D_A = \mathbf{m} + \frac{\rho}{2}\nabla_A^*\nabla_A, \qquad \|D_A\| \leq |M| + \frac{1+|\rho|}{2}\sum_i w_i. \tag{10}$$

In the splitting $\mathbb{C}^4 = \mathbb{C}^2 \oplus \mathbb{C}^2$,

$$D_A = \mathrm{diag}(\check D_A, \check D_A^*), \qquad \det D_A^\mathbb{T} = |\det \check D_A^\mathbb{T}|^2. \tag{11}$$

These identities also hold for compressions to a set of sites. The gauge, lattice and charge-conjugation identities in [4, Section 2.4] hold for these real parameters. ■

Proof. Each neighbour map is an involution. Reversing a bond conjugates its phase and changes $\widehat{d}_i$ to $-\widehat{d}_i$. Reindexing the site sum therefore gives $H_A^* = H_A$ and $J_A^* = -J_A$. The identity $\nabla_A^*\nabla_A = \sum_i w_i - H_A$ proves the real-part formula. The norm of the type-i hopping matrix is $w_i(1+|\rho|)/2$, and its phase-weighted neighbour map is unitary; summing these four bounds proves the norm estimate. The two Pauli blocks of H_A coincide and those of J_A have opposite signs, which proves (11). Compression preserves these relations. Gauge covariance is the telescoping of the endpoint phases; lattice covariance follows from the Clifford reflection identity; charge conjugation follows from $\sigma_2\sigma_v^{\mathsf{T}}\sigma_2^{-1} = -\sigma_v$. None of these computations imposes a sign condition on ρ or $\mathbf{m}$. □

Lemma E.3. For finite-dimensional complex Hilbert spaces, maps j_a, j_b from one boundary space, and nonnegative Hermitian matrices Q^+, Q^-, put

$$\mathfrak{D}(a,b) = \det \begin{pmatrix} a & -j_a Q^+ j_b^* \\ j_b Q^- j_a^* & b^* \end{pmatrix}.$$

Then $\mathfrak{D}(a,a) \geq |\det a|^2$, $|\mathfrak{D}(a,b)|^2 \leq \mathfrak{D}(a,a)\mathfrak{D}(b,b)$, and $\mathfrak{D}(a,a)$ does not decrease when either Q^+ or Q^- increases in the Hermitian order. ■

Proof. For invertible a, b, set $\widehat{a} = (Q^-)^{1/2} j_a^* a^{-1} j_a (Q^+)^{1/2}$ and define $\widehat{b}$ similarly. Schur complementation and $\det(1-XY) = \det(1-YX)$ give

$$\mathfrak{D}(a,b) = \det a \, \overline{\det b} \, \det(1 + \widehat{a}\widehat{b}^*).$$

Cauchy–Binet expresses the last determinant as the inner product of the vectors of maximal minors of $[1\ \widehat{a}]$ and $[1\ \widehat{b}]$. Schwarz gives the second assertion. With $a = b$ it is a sum of squared minors containing the term 1, which gives the first assertion. For monotonicity, let $Z = (j_a^* a^{-1} j_a) Q^+ (j_a^* a^{-1} j_a)^* \geq 0$. Sylvester's identity rewrites the last factor as $\det(1 + Z^{1/2} Q^- Z^{1/2})$; increasing Q^- increases this determinant. The form $\det(1 + (Q^-)^{1/2} Z (Q^-)^{1/2})$ gives monotonicity in Q^+. Replace a, b by $a+\epsilon, b+\epsilon$ and let $\epsilon \downarrow 0$ through values for which both are invertible. Continuity proves all assertions without invertibility. □

Lemma E.4. For $\rho = \pm 1$, real M, positive w_i, and the free matrix on an antiperiodic coordinate torus with positive integer sides,

$$\det D_0^{C(n_1,n_2,n_3)} \geq (\det D_H)^{8n_1n_2n_3} > 0, \qquad \det D_H = \left(M^6 + \left(\frac{3w_1w_2}{4} \right)^3 \right)^4.$$

Here H consists of the six permutations of v_0 in $(0,1/2)^3$, and D_H is the matrix obtained by retaining only hops whose two endpoints belong to H. For the auxiliary tori with short periods, every term of the four-hop formula is retained with its boundary character, even if two quotient endpoints coincide. ■

Proof. We record the cutting argument, which also holds for $-1 \leq \rho \leq 1$. Compress the free operator to a set Y invariant under the coordinate mirrors in a direction μ, with period n and antiperiodic character in that direction; other directions may already have been restricted to strips. Let d_k be the determinant on $Y \cap \{0 < x_\mu < k\}$, with this period removed, and set $d_0 = 1$. Identifying reflected half-spaces by $\psi(x) \mapsto \gamma_\mu \psi(\theta x)$ puts a glued matrix in the form of Lemma E.3, with boundary matrices $w^\times(1 \pm \rho\gamma_\mu)/2 \geq 0$. Thus $d_k \geq 0$ and $d_k^2 \leq d_{k-1} d_{k+1}$. In the latter comparison, the two upper strips have a common boundary layer: every crossing bond has length less than one, so its upper endpoint lies in $c < x_\mu < c + 1/2$ for the mirror $c = (k-1)/2$. The same lemma gives $d_1 \geq |\det D_{Y \cap \{0 < x_\mu < 1/2\}}|^2$.

For the periodic matrix, reflect across $x_\mu = 0$. The two off-diagonal blocks are $-(R_0 + R_1)$ and $R_0' + R_1'$, where all four matrices are nonnegative. The lower seam contributes (R_0, R_0') with crossing matrices (Q^+, Q^-); the upper seam contributes (R_1, R_1') with (Q^-, Q^+). At the upper seam the normal reverses and the antiperiodic character contributes -1, giving these same block signs.

Deleting the lower-seam matrices does not increase the determinant, by monotonicity in Lemma E.3. The resulting matrix is the compression to one open period, so the original determinant is at least d_n. Log-convexity and $d_0 = 1$ give $d_n \geq d_1^n$. If $d_1 = 0$ this is immediate; otherwise all terms are positive and $d_{k+1}/d_k \geq d_k/d_{k-1} \geq d_1$. We have proved

$$\det D_Y \geq |\det D_{Y\cap\{0<x_\mu<1/2\}}|^{2n}.$$

Apply this successively in the three coordinate directions. The last compression is D_H, and the exponent is $8n_1n_2n_3$.

For completeness, number the vertices of H cyclically by

$$(v_1,v_2,v_3),\ (v_1,v_3,v_2),\ (v_2,v_3,v_1),\ (v_3,v_2,v_1),\ (v_3,v_1,v_2),\ (v_2,v_1,v_3).$$

The alternating edge weights are w_1, w_2. If $\gamma_{(k)}$ is the Clifford matrix of the directed edge $k \to k+1$, then $\gamma_{(k+2)} = \gamma_{(k+1)} - \gamma_{(k)}$ and $\gamma_{(k)}\gamma_{(k+1)} = U$, where $U^2 - U + 1 = 0$. Set $R = (1+U)/\sqrt{3}$. Then $R^2 = U$, $R^6 = -1$, and $\gamma_{(k)} = R^{-k}\gamma_{(0)}R^k$. The change of variables $\widehat{\psi}(k) = R^k\psi(k)$ gives antiperiodic sequences of length six. Their two-step momenta are $t = \pi/3, \pi, -\pi/3$. With

$$B_+ = (\rho - \gamma_{(0)})R^{-1}, \quad B_- = R(\rho + \gamma_{(0)}), \quad p = \frac{w_1w_2}{2},$$

the matrix at momentum t is $\left(\begin{smallmatrix} M & -X \\ -Y & M \end{smallmatrix}\right)$, where $X = (w_1B_+ + w_2e^{-it}B_-)/2$ and $Y = (w_2e^{it}B_+ + w_1B_-)/2$. At $\rho^2 = 1$, $B_+B_- = B_-B_+ = 0$. On the two eigenspaces of U with eigenvalues $e^{\pm i\pi/3}$, direct multiplication gives

$$\det(M^2 - XY) = (r_+r_- + \kappa^2)^2, \quad r_\pm = M^2 - p\cos t - p\cos(t \mp \pi/3), \quad \kappa = \sqrt{3}p\rho\sin t.$$

At $t = \pi$ this is $(M^2 + 3p/2)^4$. At either remaining momentum it is $(M^4 - 3pM^2/2 + 9p^2/4)^2$. Their product is $\left(M^6 + (3p/2)^3\right)^4$, which is strictly positive. □

Proposition E.5. For real ρ, M, positive weights and $q > 0$, on every torus $\mathbb{T} = (\Gamma, \chi)$ the determinant is a nonnegative trigonometric polynomial in the link phases and is not identically zero. Its zero set in the real link variables has Lebesgue measure zero, and $0 < Z_{\mathbb{T}}[1] < \infty$. Polynomial curvature insertions with bounded trigonometric coefficients are integrable. More generally, for every $c > 0$,

$$\int_{\mathrm{Conf}(\Gamma)} e^{-c\|F\|^2}\,\mathrm{d}(A,z) < \infty.$$

■

Proof. The algebraic identities in Lemma E.2 prove nonnegativity for every real ρ. For $\rho^2 \neq 1$, the extreme coefficient calculation of [4, Proposition 2.2, equations (2.19)–(2.21)] uses only the four-hop formula and gives a coefficient of modulus

$$\left[(w_0/2)^{32}(w_1/2)^{64}(w_2/2)^{64}(w_3/2)^{32}|1-\rho^2|^{96}\right]^{[\mathbb{Z}^3:\Gamma]}.$$

It is nonzero. This calculation is independent of the sign of ρ, and boundary characters have modulus one.

For $\rho = \pm 1$, put $v = [\mathbb{Z}^3 : \Gamma]$, so that $\Gamma' = v\mathbb{Z}^3 \subset \Gamma$. Equip Γ' with the character equal to -1 on its three coordinate generators. Translations by Γ commute with the free matrix on this cover. Their joint eigenspaces are precisely the spinor spaces of the characters $\widetilde{\chi}$ of Γ extending this character. Therefore

$$0 < \det D_0^{C(v,v,v)} = \prod_{\widetilde{\chi}} \det D_0^{(\Gamma,\widetilde{\chi})}.$$

The inequality is Lemma E.4; hence every factor is nonzero. Fix one such $\widetilde{\chi}$ and choose $\xi \in \mathbb{R}^3$ with $\widetilde{\chi}(g) = e^{i\xi\cdot g}\chi(g)$ on Γ. Multiplication by $e^{i\xi\cdot\pi(x)}$ conjugates $D_A^{(\Gamma,\chi)}$ to $D_{A-\xi/q}^{(\Gamma,\widetilde{\chi})}$. At the constant cochain $A = \xi/q$, its determinant is consequently nonzero.

A nonzero real-analytic trigonometric polynomial has a null zero set. On a fundamental link box the determinant and all of its derivatives are bounded. The flux sum is bounded by the product of the scalar Gaussian sums in [4, equation (2.5)], with parameter c. This proves finiteness of the Gaussian integral. Polynomial factors are absorbed by decreasing c, giving the asserted integrability. The determinant is positive on a nonempty open set of link values; the zero-flux term alone then proves $Z_{\mathbb{T}}[1] > 0$. □

Proposition E.6. Let $-1 \le \rho \le 1$, $M \in \mathbb{R}$, $w_i > 0$ and $q > 0$. Use the upper, mirror and cut cell classes of [4, Section 2.7]. Let $\mathcal{B}^+$ have upper Grassmann generators and bounded measurable coefficients depending only on upper and mirror curvatures and link phases. Define $\mathcal{G}^+$ with the same generator support and coefficient dependence, replacing boundedness by the following requirement for each coefficient Φ: for some $C < \infty$ and $\epsilon > 0$,

$$W_+|\Phi| \le C \prod_{\sigma \in P_+} e^{-\epsilon F_\sigma^2}, \qquad W_+ = \prod_{\sigma \in P_+} e^{-F_\sigma^2/2} \prod_{Q \in Q_+} 1_{\{(dz)(Q)=0\}}.$$

There is no growth factor on mirror plaquettes. On a torus adapted to the mirror family, the reflected functional is a positive semidefinite Hermitian form on $\mathcal{G}^+$. In particular it applies to polynomials in strictly upper curvature variables and upper Grassmann generators. On the antiperiodic coordinate torus $C(n)$, $n \ge 4$, every real two-cochain β satisfies

$$E_n e^{\langle \beta, F \rangle} \le e^{\|\beta\|^2/2}. \tag{12}$$

Consequently

$$E_n\langle \beta, F\rangle = 0, \quad E_n\langle \beta, F\rangle^2 \le \|\beta\|^2, \quad E_n\langle \beta, F\rangle^4 \le \tfrac{3}{2}\exp(2)\|\beta\|^4, \tag{13}$$

and $P_n(|\langle \beta, F\rangle| \ge r\|\beta\|) \le 2e^{-r^2/2}$ for $r \ge 0$ and $\beta \ne 0$. The scalar reflection is conjugation followed by pullback of the two-cochain, including its orientation signs. On Grassmann generators it reverses products and satisfies, for the upward unit normal n,

$$\Theta_\Pi \psi_\alpha(x) = -\sum_\beta \bar\psi_\beta(\theta_\Pi x)(\gamma_n)_{\beta\alpha}, \qquad \Theta_\Pi \bar\psi_\alpha(x) = -\sum_\beta (\gamma_n)_{\alpha\beta}\psi_\beta(\theta_\Pi x).$$

If $\theta x = Rx + a$, write $\theta\mathbb{T} = (R\Gamma, \chi \circ R^{-1})$. The generator formulas descend to a map $\mathrm{Gr}_{\mathbb{T}} \to \mathrm{Gr}_{\theta\mathbb{T}}$: indeed $\theta(x+g) = \theta x + Rg$, so they preserve the respective boundary relations. ■

Proof. We check the parameter dependence in the reflected representation and domination proof of [4, Proposition 2.4, Theorem 2.5 and Theorem 3.5]. The modes at a fermion crossing have coefficients $w_i(1+\rho)/2$ or $w_i(1-\rho)/2$. They are nonnegative throughout the interval in this statement. The finite crossing expansion uses products of these coefficients, with no division by them. At the second seam the antiperiodic character cancels the sign from the reversed normal. The photon Poisson representation and its Gauss-law indicator are independent of ρ. Thus the positive boundary measure and half-transform formula of that theorem remain valid with these parameters. A vanishing crossing coefficient simply gives a zero term. The Gaussian majorant used to extend the formula to polynomial insertions is supplied by Proposition E.5.

The all-flux slab trace used for Gaussian domination has the same crossing factors. Its Hilbert–Schmidt bounds require their nonnegativity and finiteness, not their strict positivity. The unitary electric shifts and the two Schatten–Hölder comparisons in that proof therefore give $0 < Z_{C(n)}(\beta) \le Z_{C(n)}(0)$ on this full interval. Strict positivity and normalization are justified by Proposition E.5. Completing the square gives (12). These statements at $\rho = \pm 1$ can also be obtained by taking a limit from the open interval: the determinants and their coefficient insertions vary continuously, the Gaussian flux majorant is uniform on a compact parameter interval, and the limiting partition function is positive. Differentiating the one-dimensional generating bound at zero gives the first two moment assertions. Its entire-function bound and Cauchy's formula with radius $2/\|\beta\|$ give the fourth-moment bound. Applying Markov's inequality to $\pm r\beta/\|\beta\|$ gives the tail estimate. Finally

the displayed reflection formulas and their two-form signs are algebraic and impose no further restriction on ρ. □

Proposition E.7. Fix $0 < \rho < 1$, positive weights, $q > 0$ and real M. The curvature laws $\mu_{C(n)}$, $n \geq 4$, are tight on Ω. Every weak limit along any prescribed sequence $n_i \to \infty$ satisfies (G), is invariant under W and $F \mapsto -F$, and is reflection positive on the full open-side L^2 space at every coordinate site and link plane. ■

Proof. The one-plaquette tail bound in Proposition E.6 is uniform in n. Enumerating the covering plaquettes and choosing bounds with summable tail probabilities gives compact product boxes carrying arbitrarily large probability, uniformly in n. Since Ω is a closed subspace of a countable product of real lines, it is Polish; Prokhorov's theorem gives tightness and weak limit points. Every fixed finitely supported cochain embeds without overlap for large n, with the same norm and curvature pairing. Truncating the positive exponential, taking the weak limit, and removing the truncation proves (G). The finite laws have the stated symmetries by Lemma E.2; their actions on Ω are continuous, so the symmetries pass to the limit.

A bounded continuous cylinder function of plaquettes in the open coordinate half-space lies in the corresponding open half-period slab for all large n. Finite-torus reflection positivity therefore passes to the weak limit. Such cylinder functions are dense in the L^2 space of the countably many upper-half coordinates. Reflection invariance and Schwarz give continuity of the reflected form in both L^2 arguments, proving the full open-side assertion. This is the coordinate part of [4, the proof of Theorem 4.7]; that part uses no divisibility condition on the side sequence. □

E.3. **The discrete transfer operator.**

Lemma E.8. Let ν be a probability law on the closed lattice two-cochains Ω, invariant under lattice translations and the coordinate reflections, and reflection positive at the coordinate site and link planes. Fix $\mu \in \{1, 2, 3\}$ and put $\theta_\mu = \theta_{x_\mu, -1/2}$. Let $\mathcal{O}_\mu^+$ be the bounded Borel functions of finitely many plaquette values, with all corners of these plaquettes in $\{x_\mu \geq 0\}$. The form

$$(G, H)_{\mathrm{OS}} = \nu\big(\overline{\theta_\mu G}\, H\big)$$

is positive semidefinite. Its quotient completion is a Hilbert space $\mathcal{H}_\mu$. The class map extends continuously to the $L^2(\nu)$-closure of $\mathcal{O}_\mu^+$, and $\|[G]\|_{\mathcal{H}_\mu} \leq \|G\|_{L^2(\nu)}$. There is a unique operator T_μ such that

$$T_\mu[G] = [\tau_{e_\mu} G], \qquad 0 \leq T_\mu \leq 1.$$

For G, H in that closure and every integer $j \geq 0$,

$$([G], T_\mu^j[H])_{\mathcal{H}_\mu} = \nu\big(\overline{\theta_\mu G}\, \tau_{j e_\mu} H\big). \tag{14}$$

■

Proof. Reflection positivity and polarization give a positive semidefinite Hermitian form. Reflection invariance and the ordinary Schwarz inequality give

$$0 \leq (G, G)_{\mathrm{OS}} \leq \|\theta_\mu G\|_2 \|G\|_2 = \|G\|_2^2.$$

This proves the assertion about the class map and its extension. Write $\tau = \tau_{e_\mu}$. Translation moves the supporting plaquettes into the same upper half-space. The relations $\theta_\mu \tau = \tau^{-1} \theta_\mu$ and translation invariance imply

$$(G, \tau H)_{\mathrm{OS}} = (\tau G, H)_{\mathrm{OS}}.$$

The other coordinate mirror, $x_\mu = 0$, gives positivity of this shift: since $\theta_\mu = \theta_{x_\mu, 0} \tau$ and τG is supported strictly above that plane,

$$(G, \tau G)_{\mathrm{OS}} = \nu\big(\overline{\theta_{x_\mu, 0}(\tau G)}\, \tau G\big) \geq 0.$$

For bounded G set $a_j = \|[\tau^j G]\|$. Symmetry and the Schwarz inequality for the reflected form give $a_j^2 = (G, \tau^{2j} G)_{\mathrm{OS}} \le a_0 a_{2j}$. Iteration yields

$$a_1 \le a_0^{1-2^{-r}} a_{2^r}^{2^{-r}}, \qquad a_{2^r} \le \|G\|_\infty.$$

Letting $r \to \infty$ gives $a_1 \le a_0$; if $a_0 = 0$, this follows already from $a_1^2 \le a_0 a_2$. Thus translation preserves null vectors and defines a contraction on the quotient. Its continuous extension is self-adjoint by symmetry and nonnegative by the preceding site-plane reflection identity. This proves $0 \le T_\mu \le 1$. Iteration proves (14) for bounded local functions. Approximation in $L^2(\nu)$ proves it on their closure, because translations and reflection are isometries there. □

For the slab observable Φ_0 in Proposition 5.3, truncation at height R gives bounded local functions converging to Φ_0 in $L^2(\nu)$, by Gaussian domination. With the reflection at $-1/2$, $\theta_1 \Phi_0 = \Phi_{-1}$. Translation invariance and (14) therefore give

$$\nu(\Phi_0 \Phi_t) = ([\Phi_0], T_1^{t-1}[\Phi_0])_{\mathcal{H}_1} = \int_{[0,1]} \lambda^{t-1} \, \mathrm{d}\mathfrak{e}(\lambda) \quad (t \ge 1),$$

where $\mathfrak{e}$ is the spectral measure of $[\Phi_0]$ for T_1. Its mass is $\nu(\Phi_0 \Phi_1) \le \nu(\Phi_0^2)$, and reflection invariance gives the corresponding formula for negative t.

E.4. **Lattice probes and continuum tests.**

Definition E.9. For a unit plaquette $\sigma = (x; \mu\nu)$ put

$$Q_\eta(\sigma) = \eta(x + (e_\mu + e_\nu)/2 + [-1/2, 1/2)^3), \qquad f_{[\eta]}(\sigma) = \eta^{-3/2} \int_{Q_\eta(\sigma)} f_{\mu\nu}(y) \, dy.$$

The step field $\sharp_\eta u$ has component $\eta^{-3/2} u(\sigma)$ on $Q_\eta(\sigma)$, so that

$$(\sharp_\eta u)(f) = \sum_\sigma f_{[\eta]}(\sigma) u(\sigma).$$

For a curvature u on $C(n)$, with periodic extension $\widetilde{u}$ and physical side $R = n\eta$, write $X_{\eta,R}(f) = (\sharp_\eta \widetilde{u})(f)$. The cubes of each orientation partition $\mathbb{R}^3$; hence $\|f_{[\eta]}\|_{\ell^2} \le \|f\|_2$ by Cauchy–Schwarz.

For the measure-theoretic meaning of the step map, set

$$\Omega' = \left\{ u \in \Omega : \sup_\sigma \frac{|u(\sigma)|}{1 + |x(\sigma)|_\infty} < \infty \right\}, \qquad \Omega^{(\eta)} = \{ u \in \Omega' : \sharp_\eta u \in H_{-3} \}.$$

Use the indicated step field on $\Omega^{(\eta)}$ and set it to zero elsewhere, obtaining a Borel map $\sharp_\eta : \Omega \to H_{-4}$. Choose a real smooth two-form b^σ, supported in the interior of $Q_\eta(\sigma)$, with only component $\mu\nu$ nonzero and with its integral equal to $\eta^{3/2}$. Define

$$J_\eta(F)(\sigma) = F(b^\sigma).$$

This map is continuous into the space of cochains with the product topology, and $J_\eta \sharp_\eta u = u$ on $\Omega^{(\eta)}$. Any law λ on Ω satisfying

$$\lambda(e^{\langle \beta, u \rangle}) \le e^{\|\beta\|_{\ell^2}^2/2} \quad \text{for every finitely supported real cochain } \beta$$

gives $\Omega^{(\eta)}$ full measure. Indeed, the coordinate Gaussian tail bound and Borel–Cantelli give Ω' full measure, and

$$\int \|\sharp_\eta u\|_{-3}^2 \, d\lambda(u) \le \sum_\kappa \lambda_\kappa^{-6} < \infty.$$

These maps identify fixed-spacing volume limits: if $P_{N,T_j} \to \mu^{(N)}$ weakly on H_{-4}, then

$$\mu_{C(n_j)} \longrightarrow \lambda = (J_\eta)_* \mu^{(N)}, \qquad \mu^{(N)} = (\sharp_\eta)_* \lambda.$$

The first limit follows from continuity of J_η; the second is the fixed-spacing identification proved in [4, Section 6.3]. In particular it does not require continuity of $\sharp_\eta$ on the product space. ■

Lemma E.10. For a real $\beta \in C_c^\infty(\mathbb{R}^3;\Lambda^2)$, put $f = dd^*\beta$. Let A_η be averaging over the cube of side η centred at the argument, and set

$$\delta_{\eta,\mu}g(x) = \frac{g(x+\eta e_\mu/2) - g(x-\eta e_\mu/2)}{\eta}, \qquad a'_{\eta,\nu} = -\sum_{\mu\neq\nu}\delta_{\eta,\mu}\beta_{\mu\nu},$$

$$f^{(\eta)}_{\mu\nu} = \delta_{\eta,\mu}a'_{\eta,\nu} - \delta_{\eta,\nu}a'_{\eta,\mu}, \qquad \widehat{\mathfrak{a}}_\eta(x,x+e_\nu) = \eta^{1/2}(\mathsf{A}_\eta a'_{\eta,\nu})(\eta x + \eta e_\nu/2).$$

Periodize this cochain when it is used on $C(n)$, with physical side $R = n\eta$. It is orthogonal to the constant cochains, depends linearly on β, and

$$d\widehat{\mathfrak{a}}_\eta = (f^{(\eta)})_{[\eta]}, \qquad \langle d\widehat{\mathfrak{a}}_\eta, F^{\mathrm{u}}\rangle = X_{\eta,R}(f^{(\eta)}), \qquad \|f^{(\eta)} - f\|_2 \le \eta^2\|\beta\|_{H^4}. \tag{15}$$

For fixed probes β,β' and $\eta_i \downarrow 0$, $R_i \to \infty$, the counting inner products satisfy

$$\langle d\widehat{\mathfrak{a}}_i, d\widehat{\mathfrak{a}}'_i\rangle \longrightarrow \langle dd^*\beta, dd^*\beta'\rangle.$$

Once $(-R_i/2, R_i/2)^3$ contains $\operatorname{supp}\beta + [-1,1]^3$, Gaussian domination gives

$$E_i\,|\langle d\widehat{\mathfrak{a}}_i, F^{\mathrm{u}}\rangle - X_{\eta_i,R_i}(f)|^2 \le \|f^{(\eta_i)} - f\|_2^2. \tag{16}$$

If $P_{N_i,T_i} \to \mu$ weakly on H_{-4}, with $N_i, T_i \to \infty$, all fixed finite families $X_{\eta_i,R_i}(g)$ converge jointly in law to $X(g)$ under μ; their entire generating functions converge locally uniformly and all joint moments converge. These statements hold at every charge and for every real mass assignment in the chamber model. ■

Proof. The averaged physical cochain is $a^o_\eta = d^*_\eta b_\eta$, where $b_{\eta,\mu\nu}(c) = (\mathsf{A}_\eta\beta_{\mu\nu})(c + \eta(e_\mu+e_\nu)/2)$. Summation by parts makes it orthogonal to every closed one-cochain, including constants, before and after periodization. Evaluation at the bond and plaquette midpoints gives (15). The Fourier symbols satisfy $|2\eta^{-1}\sin(\eta p_\mu/2)| \le |p_\mu|$ and $|2\eta^{-1}\sin(\eta p_\mu/2) - p_\mu| \le \eta^2|p_\mu|^3/24$. These give the displayed H^4 bound; this is [4, equations (19.1)–(19.5)]. The counting norm of $d\widehat{\mathfrak{a}}_\eta$ is the L^2 norm of the componentwise cell average of $f^{(\eta)}$. That average converges to f in L^2; equivalently use the quantitative bound in [4, equation (19.10)]. Polarization proves the assertion about two probes. The support of $f^{(\eta)}$ is contained in $\operatorname{supp}\beta + [-\eta,\eta]^3$. On the stated period cube, the test-map contraction and the variance consequence of (12) prove (16). The weak-limit assertion and the generating-function and moment convergence are the argument of [4, Section 6.2]: Gaussian domination gives uniform exponential bounds on each compact set of generating parameters. Uniform integrability passes the expectations to the weak limit; Cauchy's formula passes all their derivatives. The finite Gaussian bound holds for every real lattice mass, so no tuning or positivity assumption on that mass is used here. □

Lemma E.11. For a closed Gaussian-dominated continuum law and every real $g \in L^2$, one has $X(g) = X(\Pi_{\mathrm{c}}g)$ in every L^p, $p \ge 1$. ■

Proof. Closedness gives $X(d^*\gamma) = 0$ for smooth compactly supported three-forms γ. The closure of these coexact forms is the orthogonal complement of the closed L^2 two-forms, by the Fourier Hodge decomposition. The $L^2 \to L^p$ continuity from Gaussian domination therefore proves the claim. □

E.5. **The lattice spectral cone.** For $Q \subset \mathbb{R}^3$, let $\mathcal{O}(Q)$ be the bounded Borel functions of finitely many curvature plaquettes all of whose corners belong to Q. The closure of this algebra in $L^2(\lambda)$ is the local space used in Proposition 5.17. In this subsection λ is W-invariant and reflection positive at all nine lattice mirror families. These are assumptions on the law, not conclusions from coordinate reflection positivity alone.

Lemma E.12. Let η be a finite positive measure on the closed unit disc. Suppose that, for every polynomial p,

$$\int |p(z)|^2\,d\eta(z) \le C \sup_{[0,1]}|p| \sup_{|z|=1}|p|.$$

Then $\eta(\{z : |1+z|^2 < 4|z|\}) = 0$. ∎

Proof. Uniform Taylor approximation extends the inequality to functions holomorphic on a disc of radius greater than one. For $|\xi| < 1$ put $w(\xi) = (1+\xi)/(1-\xi)$. Its real part is positive and its principal logarithm is odd. Thus

$$f_s(\xi^2) = \frac{\cos(s \operatorname{Log} w(\xi))}{\cosh(\pi s/2)} \qquad (s > 0)$$

defines a holomorphic function on the unit disc. If $\alpha(z) = |\arg w(\sqrt{z})|$, independent of the square-root choice, then $0 \le \alpha(z) < \pi/2$ and

$$|f_s(z)| \le 1, \qquad |f_s(x)| \le \frac{1}{\cosh(\pi s/2)} \quad (0 \le x < 1), \qquad |f_s(z)|^2 \ge \frac{\sinh^2(s\alpha(z))}{\cosh^2(\pi s/2)}.$$

These follow from $|\cos(x+iy)|^2 = \cos^2 x + \sinh^2 y$. Apply the assumed inequality to $f_s(rz)$, $r < 1$, and let $r \uparrow 1$. On the open disc $\alpha(rz) \to \alpha(z)$; on the unit circle other than 1 it tends to $\pi/2$. Denote this extension by $\widetilde{\alpha}$. Fatou's lemma gives

$$\int_{z \ne 1} \sinh^2(s\widetilde{\alpha}(z))\, d\eta(z) \le C \cosh(\pi s/2).$$

For every $\epsilon > 0$, the set $\{\widetilde{\alpha} \ge \pi/4 + \epsilon\}$ consequently has measure at most $16Ce^{-2s\epsilon}$ for all sufficiently large s, and hence has measure zero. Finally, writing $z = \xi^2$ gives

$$|1+z|^2 - 4|z| = (1-|\xi|^2)^2 - 4(\operatorname{Im}\xi)^2, \qquad \tan\alpha(z) = \frac{2|\operatorname{Im}\xi|}{1-|\xi|^2}.$$

In the open disc the strict inequality in the statement is therefore equivalent to $\alpha > \pi/4$. Its boundary points lie on the unit circle other than 1, where $\widetilde{\alpha} = \pi/2$. Taking a countable union of the preceding null sets proves the claim. □

Proposition E.13. For each j, form the quotient at $\theta_j = \theta_{x_j,-1/2}$ as in Lemma E.8. Translation by e_i, $i \ne j$, induces a unitary $U_{j,i}$, and the three operators $T_j, U_{j,i}, U_{j,i'}$ commute. Their joint spectral measure is carried by

$$(1-t)^2 \ge t|1-u_i|^2, \qquad (1-t)^2 \ge t|1-u_{i'}|^2.$$

Let P_j project in $L^2(\lambda)$ onto the observables invariant under τ_{e_j}, and let P project onto those invariant under all lattice translations. Then $P_j = P$, and, with $\widehat{P}_j = \mathbf{1}_{\{1\}}(T_j)$,

$$\langle [G]_j, \widehat{P}_j [H]_j \rangle = \langle \theta_j G, PH \rangle_2.$$

∎

Proof. Transverse translations preserve the positive half-space and its reflected form. They therefore induce unitaries, with inverses given by the opposite translations; commutation follows on the dense local classes and then by continuity. The joint spectral measure exists by the spectral theorem for commuting bounded normal operators.

The proof of Lemma E.8 also applies to a lattice mirror $\ell = c$ and a lattice translation v normal to it, provided $\ell(v) > 0$ and $c + \ell(v)/2$ is an admissible mirror height. Indeed the first reflection makes the translation symmetric, reflection in the shifted mirror makes it nonnegative, and iterated Cauchy–Schwarz makes it a contraction. A lattice translation tangent to the mirror induces a commuting unitary, by the preceding argument.

Fix $i \ne j$ and abbreviate $T = T_j$, $U = U_{j,i}$. Let G be a bounded upper-half local observable. Choose $r \ge 1$ so that all its plaquette corners satisfy $0 \le x_j \le r$ and $|x_i| \le r$. Use the diagonal mirror $\ell = x_j + x_i = r$, with reflection θ', normal translation $v_+ = e_j + e_i$ and tangent translation

$v_- = e_j - e_i$. The mirror at $r+1$ is also admissible, so this quotient has a positive contraction T_+ and a unitary V_-. The observables

$$\Phi = \theta' \theta_j G, \qquad G_0 = \tau_{(2r+1)e_j} G$$

lie strictly above $\ell = r$: each has its plaquette corners in $\{\ell \geq r+1\}$. The extra unit in the translation keeps the support off the mirror. The two reflected forms give, for $k, l \geq 0$,

$$\langle [G]_j, T^{2r+1+k+l} U^{k-l} [G]_j \rangle = \langle [\Phi]', T_+^k V_-^l [G_0]' \rangle.$$

Let $\mathfrak{e}_G$ be the joint spectral measure in this matrix element and let η be the image of $t^{2r+1} d\mathfrak{e}_G(t,u)$ under $(t,u) \mapsto tu$. For $p(z) = \sum_k a_k z^k$ put $p^*(z) = \sum_k \overline{a_k} z^k$. Summing the preceding identity with coefficients $a_k \overline{a_l}$ yields

$$\int |p(z)|^2 \, d\eta(z) = \langle [\Phi]', p(T_+) p^*(V_-) [G_0]' \rangle \leq \|[\Phi]'\| \|[G_0]'\| \sup_{[0,1]} |p| \sup_{|z|=1} |p|.$$

Lemma E.12 applies. Since

$$|1 + tu|^2 - 4t = (1-t)^2 - t|1-u|^2,$$

the measure $t^{2r+1} d\mathfrak{e}_G$ gives no mass to the set where the right side is negative. The factor is positive for $t > 0$, and the right side equals 1 at $t = 0$. Thus $\mathfrak{e}_G$ itself gives that set no mass. Local classes are dense, so the corresponding spectral projection vanishes. Applying this to each transverse direction proves both cone inequalities.

We identify the invariant projection. The mean ergodic theorem and the reflected correlation identity give

$$\langle [G]_j, \widehat{P}_j [H]_j \rangle = \langle \theta_j G, P_j H \rangle_2.$$

Indeed $T_j^n \to \widehat{P}_j$ strongly, while the Cesàro means of τ_{ne_j} tend to P_j. Every vector in $\operatorname{ran} P_j$ can be approximated by bounded local observables translated into the positive half-space. Reflection positivity then shows that θ_j restricts to a positive involution on this subspace, and hence is the identity. The last display consequently equals $\langle P_j G, P_j H \rangle_2$. It follows that

$$\widehat{P}_j [H]_j \longmapsto P_j H$$

is an isometry on the dense set of such zero-energy classes. At $t = 1$ the cone forces every $u_i = 1$, so $U_{j,i} \widehat{P}_j = \widehat{P}_j$. Under this isometry it follows that $\tau_{e_i} P_j H = P_j H$ for upper-half local H. Such $P_j H$ are dense in $\operatorname{ran} P_j$, by the same translated approximations. Thus every vector invariant in direction j is invariant in all directions, and $P_j = P$. □

E.6. **Coordinate reflections in arbitrary joint limits.** Here the mass assignment need not be the tuned assignment of the main construction. Fix $0 < \rho < 1$, $v_0 = (v_1, v_2, v_3)$ with $0 < v_3 < v_2 < v_1 < 1/2$, the standard weights, an odd integer $L \geq 3$, $0 < \bar{m} \leq 1$, $e > 0$, and a real sequence $(m_{0,N})_{N \geq 0}$. Use the chamber law of Definition E.1 on $C(n)$, with

$$\eta = L^{-N}, \qquad h = L^{\lfloor \log_L \bar{m}^{-1} \rfloor}, \qquad R = 8hL^T, \qquad n = R/\eta,$$
$$q = e\eta^{1/2}, \qquad M = \bar{m}\eta + m_{0,N} + \frac{\rho}{2} \sum_i w_i.$$

Let $P_{N,T}$ be its physical step-field law, with the test map of [4, Section 6.1]. Thus the value of the physical field on a plaquette averaging cube is $\eta^{-3/2} F(\sigma)$.

Proposition E.14. Every weak limit μ on H_{-4} of P_{N_i,T_i}, where both N_i and T_i tend to infinity, satisfies (G) and (dF), is invariant under $\mathbb{R}^3 \rtimes O_{\mathrm{cub}}$, and is invariant and reflection positive at every coordinate plane. Almost every translation-ergodic component has (G) and (dF), is invariant under $\mathbb{R}^3 \rtimes \{\pm 1\}^3$, and is reflection positive at every coordinate plane. Iterated subsequential limits, in either order, are among these joint limits. For the parent law and almost every component, the

bounded two-point Fourier multiplier in the Hodge-dual variables is real and symmetric almost everywhere. ■

Proof. The compactness, moment convergence, closedness and cubic-invariance arguments of [4, Sections 6.1–6.3] apply to any real mass assignment independent of T. Their finite inputs are Gaussian domination, closedness of the lattice curvature and cubic invariance, not the tuned renormalization trajectory. In particular the step-field laws are uniformly tight on H_{-4}, their joint limits have (G) and (dF), and these limits are translation and cubically invariant.

We give the coordinate-reflection passage, which does not require taking volume before spacing. Let $f_1, \dots, f_s$ be real compact tests supported in $\{x_j > 0\}$. Choose $\delta, r > 0$ so that their supports lie in $\{x_j \geq \delta\} \cap [-r, r]^3$. For all sufficiently large i, $2\eta_i < \delta$ and $R_i > 2r + 4\eta_i$. The torus test cochains then lie strictly inside the positive half-period slab; no plaquette in their support meets either bounding mirror. The test map commutes with the tensor reflection. The finite reflected form of Proposition E.6 therefore gives, for arbitrary $z_1, \dots, z_s \in \mathbb{C}$,

$$\sum_{a,b} \overline{z_a} z_b \int e^{iF(f_b - \theta_j \cdot f_a)} \, dP_{N_i,T_i}(F) \geq 0.$$

Each integrand is bounded and continuous on H_{-4}, so this inequality passes to μ. Density of compact-test character sums and Cauchy–Schwarz extend it to the full positive-side L^2 space. Translation invariance moves the reflection to every real height.

For completeness, the component argument only uses the three coordinate reflections. Let $\mathcal{I}$ be the invariant event sigma algebra and write $\mu_F(\Phi) = E_\mu[\Phi \mid \mathcal{I}](F)$ for its regular conditional laws. The construction and ergodicity proof in [4, Section 7.5] require only translation invariance. If h is a bounded $\mathcal{I}$-measurable function, then h is fixed by translations in direction e_j. Translating compact-test approximations to h into either half-space shows that h has a representative in both half-spaces. On this invariant subspace, the geometric reflection is a positive self-adjoint involution, by reflection positivity; hence it is the identity. For bounded measurable A, the first identity below holds; for A in the positive-side algebra the second holds as well:

$$\mu(h\,\hat{\theta}_j A) = \mu(hA), \qquad \mu(h\,\overline{\hat{\theta}_j A} A) \geq 0 \quad (h \geq 0),$$

the second identity following by applying reflection positivity to $\sqrt{h}A$. Testing against all bounded invariant weights transfers reflection invariance and positivity to the conditional laws. A countable determining character algebra provides one full-measure set on which all three assertions hold; density then extends their forms to L^2. The three coordinate reflections generate $\{\pm 1\}^3$. Gaussian domination passes to these conditional laws by the translation-average argument in the same section, which uses only parent domination and translation invariance. A parent event of probability one has conditional probability one almost surely, so closedness passes as well.

For an iterated volume-first limit, choose spacing indices N_i and fixed-spacing limits $\mu^{(N_i)} \to \mu$. In a metric d for weak convergence on H_{-4} choose, along the defining volume subsequence, $T_i \geq i$ with $d(P_{N_i,T_i}, \mu^{(N_i)}) < 1/i$. The triangle inequality gives $P_{N_i,T_i} \to \mu$. Interchanging the two indices gives the same conclusion for the other iterated order.

Finally, (G) makes the two-point form a bounded positive form on L^2. Translation invariance makes its operator a Fourier multiplier [9, Chapter I, Theorem 3.18]. For a real random field its symbol satisfies $M(p)^* = M(p)$ and $M(-p) = \overline{M(p)}$. The inversion $x \mapsto -x$ belongs to $\{\pm 1\}^3$ and acts trivially on the Hodge-dual pseudovector components, so inversion invariance gives $M(-p) = M(p)$. Thus M is real and symmetric almost everywhere. □

E.7. **The free symbol.**

Definition E.15. Use the standard weights and $0 < \rho < 1$. The fibre consists of maps $u : O_{\mathrm{cub}} \to \mathbb{C}^4$. Define the constant injection and the normalized average by

$$(Jv)(r) = v, \qquad J^*u = \frac{1}{48}\sum_r u(r), \qquad P = JJ^*.$$

The star in this formula denotes that specified averaging map. For the fibre inner product $48^{-1}\sum_r$ it is the adjoint of J. Multiplying the fibre inner product by 48 changes neither endomorphism norms nor adjoints of endomorphisms. Thus we may use the counting norm on $\mathcal{V} = \mathbb{C}^{48} \otimes \mathbb{C}^4$ while retaining these two explicit maps. For $m > 0$, let $D^{(p)}$ be the symbol of the massive free chamber matrix, obtained from the four-hop formula in Definition E.1 on $e^{ip\cdot k}u(r)$. Put $\mathrm{Rsy}(p) = (D^{(p)})^{-1}$ and $\mathrm{Vx}_\nu(p) = \partial_{p_\nu} D^{(p)}$. Writing $s_3^{\mathrm{lin}} = \mathrm{diag}(-1, 1, 1)$, define

$$(\mathrm{Ehp}_{\pm e_\nu} u)(r) = 1_{\{re_1 = \pm e_\nu\}}\, 3(\rho \mp \gamma_\nu) u(r s_3^{\mathrm{lin}}).$$

Then

$$D^{(p)} = D_{\mathrm{loc}} - \sum_\nu (e^{ip_\nu}\mathrm{Ehp}_{e_\nu} + e^{-ip_\nu}\mathrm{Ehp}_{-e_\nu}), \qquad \mathrm{Vx}_\nu(p) = -i(e^{ip_\nu}\mathrm{Ehp}_{e_\nu} - e^{-ip_\nu}\mathrm{Ehp}_{-e_\nu}). \quad (17)$$

In particular $\mathsf{X}_\nu = \mathrm{Vx}_\nu(0) = -i(\mathrm{Ehp}_{e_\nu} - \mathrm{Ehp}_{-e_\nu})$. In the singular-value proof the mass in this symbol is the auxiliary mass $\mathbf{m}'$, not the lattice mass $\mathbf{m}$ of the interacting law. In the later mass-recursion proof J, J^* are the same injection and average applied at each cluster of a spinor field. The integer J in the first localization proof has a different meaning. ■

Lemma E.16. There is $C_E = C_E(\rho) < \infty$ such that, for every $m > 0$ and $p \in [-\pi, \pi]^3$,

$$\mathrm{Rsy}(p) = J(m + i\gamma \cdot p)^{-1} J^* + \mathrm{Er}(p, m), \qquad \|\mathrm{Er}(p, m)\| \le C_E. \quad (18)$$

For $0 < m \le 1$, with $c_E = (\sqrt{2} + 7C_E)^{-1}$,

$$\|\mathrm{Rsy}(p)\| \le \frac{1}{c_E(m + |p|_2)}, \qquad |D^{(p)}| \ge c_E\sqrt{m^2 + \mathsf{s}(p)^2}, \qquad |D^{(p)*}| \ge c_E\sqrt{m^2 + \mathsf{s}(p)^2},$$

where $\mathsf{s}(p)^2 = \sum_\mu 4\sin^2(p_\mu/2)$. For every multi-index α there is $C_{\alpha,\rho}$ such that

$$\|\partial^\alpha \mathrm{Rsy}(p)\| \le C_{\alpha,\rho}(\mathrm{dist}_2(p, 2\pi\mathbb{Z}^3) + m)^{-1-|\alpha|} \quad (0 < m \le 1,\ p \in \mathbb{R}^3). \quad (19)$$

The full symbol and its inverse are periodic in p. ■

Proof. The inverse decomposition and its bounded remainder for $0 < m \le 1$ are [4, equations (18.4)–(18.5)]. Their proof uses the full block inverse, including its mixed and heavy blocks, not merely the light Schur complement. For $m \ge 1$, accretivity gives $\|\mathrm{Rsy}(p)\| \le m^{-1}$, and the injected continuum inverse has norm at most m^{-1}. Increasing C_E to at least 2 proves (18) on this remaining range as well. This is the same mass-independent remainder estimate. Since $|p|_2 + m < 7$ when $0 < m \le 1$,

$$C_E + (m^2 + |p|_2^2)^{-1/2} \le \frac{7C_E + \sqrt{2}}{m + |p|_2}.$$

Taking inverse singular values and using $\mathsf{s}(p) \le |p|_2$ proves the two lower bounds. The differentiated inverse identity and the bounded derivatives of the four-hop symbol give (19), as proved in [4, equations (18.6)–(18.8)]. Periodicity follows from the displayed external-hop formula. □

Appendix F. Inputs for the tuned construction

F.1. **RG notation and terminal inequalities.**

Definition F.1. Let $\eta = L^{-N}$, $q = e\sqrt{\eta}$, and set

$$a_j = L^j\eta, \qquad e_j = e\sqrt{a_j}, \qquad \mu_j = \bar{m}a_j, \qquad w_j = e_j^{3/4-8\varepsilon}, \qquad w_j^{\mathrm{E}} = e_j^{1/4-7\varepsilon}.$$

The scale construction and stopping index $k_*(R)$ are those of [4, Proposition 16.1]. Put $K = N + \log_L h$, $v_j = -\log e_j$, and $r_j = \min\{L^i : i \in \mathbb{N}\cup\{0\},\ L^i \geq v_j^r\}$, where $r \geq 2$ is the fixed exponent in the scale construction. Its top and reference sides are $R_{\mathrm{top}} = hMr_K$ and $R^* = R_{\mathrm{top}}/L$. The admissible physical sides are $\mathrm{Tor}(e) = \{hL^T : T \in \mathbb{N}\cup\{0\},\ hL^T \leq R_{\mathrm{top}}\}$. For each such side, $\ell_j(R) = R/a_j$ and $k_*(R)$ is the first level with $\log_L(\ell_j/(Mr_j)) = 0$. The mesh threshold is $N_{\min}(R,e)$ of that proposition. The allowed unit sides form $\mathrm{Cl} = \{\ell : \ell/(LM)$ is an odd integer at least $13\}$. On the top torus $a_* \in \{h, h/L\}$; on the reference torus $a_* \in \{h/L, h/L^2\}$. The symbol $*$ always denotes the stop on the torus currently under consideration.

We use the zero-external-slot activity spaces of its Section 12.1. For an activity family E their norm is

$$\|E\|_j = \sup_X e^{\kappa d_M(X)} \sup_{A\in\widetilde{R}_j} \|E(X;A)\|_{h_j}, \qquad h_j = (e_j^{-1/4}, e_j^{-1/4+\varepsilon}),$$

where the coefficient norm $\|\cdot\|_{h_j}$ and its index measures are exactly those of equations (12.1)–(12.3) there. The notation $E \in K_j$ includes its locality, analyticity, gauge, coordinate and charge-conjugation symmetries, and reality. The maps $m(E)$, $L(RE)^\natural$ and $E_j^*(m,E)$ are defined in Propositions 12.1, 12.5 and 14.7 there. The recursion is (16.9)–(16.10), with $E_0 = 0$. The flow data (FL) mean these scale data together with the primitive families constructed in Sections 9–15 of [4], using their top-torus families on R_{top} and the projected families on a smaller torus. Their common estimates are Proposition 15.2, and their use on these scales is justified at the end of its Section 16.1. ∎

The scale bounds also give $v_j = -\log e_j \geq 8$ at all required levels. The fixed choices in [4, Proposition 24.1] give $\bar{\varepsilon} = 1/200$, $L \geq 13$ and $c_{\mathrm{fl}}L^{-3/8+6\bar{\varepsilon}} \leq 1/16$. Throughout a flow, its charge condition (16.13) gives

$$\sum_{j=0}^{k_*}(c_{\mathrm{fl}}+2)e_j^{\varepsilon} \leq \frac{1}{16}. \tag{20}$$

Thus $(c_{\mathrm{fl}}+2)e_j^{\varepsilon} \leq 1/16$ and $\sum_{j<k} c_{\mathrm{fl}}e_j^{2\varepsilon} \leq e_k^{\varepsilon}/16$ for $k \leq k_*$. The terminal conditions (22.36) of that construction include

$$\begin{aligned} 8C_0^2\ell_*^6(b_-^{-1}+\mu_*^{-1}) &\leq e_*^{-\varepsilon/2}, \\ 4(2C_0 + c_\kappa + 5c_{\mathrm{fl}} + 6)\ell_*^3 \leq e_*^{-\varepsilon}, \qquad & e_*^{1/4-9\varepsilon} \leq \frac{1}{8}. \end{aligned} \tag{21}$$

Here C_0 is the terminal field-map constant, b_- the positive lower bound for the block penalty, and $c_\kappa = \sup_\square \sum_{X\supset\square} e^{-\kappa d_M(X)}$. These three inequalities are the ones used below under the terminal smallness condition; the remaining inequalities of (22.36) are retained whenever the terminal remainder or partition-function estimates are invoked.

Definition F.2. On a fine unit torus with n sites per side, put $F_N = dC^1(n)$, with the counting inner product, and let γ_N be its standard Gaussian law. For $F \in F_N$, let $A(F)$ be its unique coclosed, mean-zero potential: $dA(F) = F$. For $\theta \in [-\pi,\pi]^3$, put $H_{\theta,\mu} = \theta_\mu/(qn)$, and define

$$w_N(B) = e^{-\varepsilon_{\mathrm{in}}^{\mathrm{v}}n^3}\frac{\det(D^{\mathrm{ch}}[qB] + \bar{m}\eta + m_{\mathrm{in}})}{\det(D^{\mathrm{ch}}[0] + \bar{m}\eta)}, \qquad \nu^\theta(dF) = w_N(A(F)+H_\theta)\,\gamma_N(dF).$$

The bracket argument is the bond phase. Write

$$\bar{\nu} = \int \nu^{\theta} \, \frac{d^3\theta}{(2\pi)^3}, \qquad \bar{\mu} = \frac{\bar{\nu}}{\bar{\nu}(1)}.$$

The adjoint spin blocks make each ν^{θ} positive at every real lattice mass. The zero-sector identity gives $0 < \bar{\nu}(1) < \infty$ and identifies $\bar{\mu}$ with the harmonic-average zero-flux curvature law; see [4, Section 22.2 and Proposition 21.2]. ■

F.2. **The tuned data and their component bounds.** Fix a real $\beta \in C_c^{\infty}(\mathbb{R}^3; \Lambda^2)$, put $f = dd^*\beta$, and assume $b(f) > 0$. Choose a support radius r_β and a point x_0 satisfying (22). Put

$$\beta_0 = \beta, \qquad \beta_{\pm} = \beta \pm \tau_{x_0}\beta, \qquad g_{\mathrm{ix}} = dd^*\beta_{\mathrm{ix}} \quad (\mathrm{ix} = 0, +, -).$$

Use the tuned sequence of [4, Theorems 16.3–16.4] and the reference-scale construction of its Sections 24–25 for these three fixed probes. The scale and charge conditions are those of that construction, applied to each of the three probes: the support-fit condition (19.6), the terminal comparison (25.28), and the charge conditions of Section 24. In particular, the same scale data give an exponent $\sigma > 0$. Choose C_U to be the maximum of the three constants in (25.35), including their free reference bounds. Set

$$C_{\bar{\kappa}} = 2(8C_U + 3C_U^2), \qquad C'_{\bar{\kappa}} = 8C_{\bar{\kappa}} + 3C_{\bar{\kappa}}^2.$$

The reference estimate (25.30) is bounded uniformly in the charge, so we may take $\mathfrak{e}_2 = 0$; all the conclusions below remain valid with any $\mathfrak{e}_2 \in [0, 1)$ for which the weaker bound $|c^{\mathrm{inf}}_{2,\mathrm{ix}}| \le C_U e^{-\mathfrak{e}_2}$ is used. Fix $e_2 > 0$ so that the preceding construction applies for $0 < e \le e_2$, and so that

$$e_2 \le 1, \qquad C'_{\bar{\kappa}} e_2^{\sigma} \le b(f)/10.$$

All constants and the three probes are fixed before varying the charge in this interval. The mesh thresholds may depend on the charge. These probe-dependent restrictions are used only for the tuned conclusions, not for the arbitrary-mass statements.

Proposition F.3. Let $0 < e \le e_2$, and let μ be a volume-first limit of the tuned laws. For each $m \in \{2, 4\}$ and $\mathrm{ix} \in \{0, +, -\}$ there are real $c^{\mathrm{inf}}_{m,\mathrm{ix}}$ and nonnegative $\mathsf{u}^{\mathrm{inf}}_{m,\mathrm{ix}}$ such that

$$\mathsf{u}^{\mathrm{inf}}_{4,\mathrm{ix}} \le C_U e^{\sigma}, \qquad (1 + |c^{\mathrm{inf}}_{2,\mathrm{ix}}|)\mathsf{u}^{\mathrm{inf}}_{2,\mathrm{ix}} \le C_U e^{\sigma},$$
$$|c^{\mathrm{inf}}_{2,\mathrm{ix}}| \le C_U e^{-\mathfrak{e}_2}, \qquad |c^{\mathrm{inf}}_{4,\mathrm{ix}} - 3(c^{\mathrm{inf}}_{2,\mathrm{ix}})^2 - b(g_{\mathrm{ix}})| \le C_U e^{\sigma}.$$

For the polynomials

$$\mathrm{Wk}_{m,\mathrm{ix}} = e^{-m} H_m(X(g_{\mathrm{ix}}); \|g_{\mathrm{ix}}\|^2) - c^{\mathrm{inf}}_{m,\mathrm{ix}},$$

where $H_2(x; v) = x^2 - v$ and $H_4(x; v) = x^4 - 6vx^2 + 3v^2$, one has

$$\left| \mu \left(\prod_{y \in Y} U(y) \mathrm{Wk}_{m,\mathrm{ix}} \right) \right| \le (\mathsf{u}^{\mathrm{inf}}_{m,\mathrm{ix}})^{|Y|} \qquad (Y \subset (R^*/L)\mathbb{Z}^3 \text{ finite}).$$

Fix any one family with these properties. In the ergodic decomposition $\mu = \int \mu_\xi \, \nu(d\xi)$ there is one measurable set Ξ_μ of full ν measure on which all of the following statements hold simultaneously:

(D1) μ_ξ is translation invariant and ergodic, has the sharp Gaussian bound (G), is supported on closed two-forms, and is invariant and reflection positive at every plane in each of the nine coordinate and diagonal families.

(D2) μ_ξ is invariant under every Euclidean map and is reflection positive on the full positive-side L^2 space at every plane. The joint spectral measures in its three coordinate reconstructions are supported on $\omega \ge |p|$. Its two-point function has the Källén–Lehmann representation, with a unique positive measure of mass at most one.

(D3) In each coordinate reconstruction, $\ker \mathsf{H}_j = \mathbb{C}\Omega_j$.

(D4) Put

$$c_{\mathrm{cl}} = \frac{96(1+\sqrt{2})}{4\pi^2}(128 + 64(16/\pi + 64/\pi^2)).$$

For $r > 0$, $\Phi, \Psi \in L^2(\Sigma_{(-r,r)^3}, \mu_\xi)$, and $|a|_\infty \geq 4r$,

$$\left|\mu_\xi(\overline{\Phi}\, U(a)\Psi) - \overline{\mu_\xi(\Phi)}\mu_\xi(\Psi)\right| \leq c_{\mathrm{cl}}\|\Phi\|_2\|\Psi\|_2 \frac{r}{|a|_\infty}.$$

Moreover the left side tends to zero as $|a| \to \infty$ for all $\Phi, \Psi \in L^2(\mu_\xi)$.

(D5) The tensor Schwinger functions obey the OS growth bound $|S_n(\Psi)| \leq 2 \cdot 66^n \sqrt{n!}\, q_{6n}(\Psi)$, Euclidean covariance, reflection positivity on positive-side tensor tests, permutation symmetry, and clustering.

(D6) For $\mathrm{ix} = 0, +, -$,

$$|\kappa_4^\xi(g_{\mathrm{ix}}) - e^4 b(g_{\mathrm{ix}})| \leq C'_{\bar\kappa} e^{4+\sigma},$$
$$|S_2^\xi(g_{\mathrm{ix}}, g_{\mathrm{ix}}) - \|g_{\mathrm{ix}}\|^2| \leq C'_{\bar\kappa} e^{2-\mathfrak{e}_2},$$
$$\mathrm{Ext}_{x_0}^\xi(f) \geq \tfrac{1}{2}e^4 b(f), \qquad \kappa_4^\xi(f) \geq \tfrac{1}{2}e^4 b(f) > 0.$$

The last cumulant is unchanged by any Euclidean map applied to f.

In addition, on Ξ_μ, $|\mu_\xi(\mathrm{Wk}_{m,\mathrm{ix}})| \leq \mathsf{u}_{m,\mathrm{ix}}^{\mathrm{inf}}$ for all six pairs. More precisely, Ξ_μ consists of all component laws for which the three assertions (translation invariance and ergodicity, (G), and closedness), the nine family-wise reflection assertions, and the six displayed Wick expectation bounds hold. It depends on the chosen family. In particular it is invariant under the image-law map $\lambda \mapsto \mathrm{Cc}_*\lambda$, where $\mathrm{Cc}(F) = -F$. ■

Proof. Apply [4, Proposition 25.4 and Section 26.2] to the three probes and take one common subsequence for their finitely many reference coefficients. Its weak limit is still μ. The reference estimate (25.30), the Wick bounds (25.31), and the product estimate (25.32) give the first two displays and the product bound. The proof in Section 26.2 justifies the replacement of the lattice tests by the smooth tests in every finite Wick product.

The invariant-product argument of [4, Lemma 26.1 and equation (26.2)] gives the six component inequalities. Theorem A of [5] gives the Gaussian, closedness, invariance, ergodicity, and nine reflection statements on one common full-measure set. The set of all laws with these eighteen properties is measurable: the countable tests and continuity argument in Section 5 of that companion express each assertion by countably many measurable conditions. It has full measure by the preceding argument. This is the set Ξ_μ in the statement, and proves (D1). Every one of its defining properties is preserved by Cc_*: this map commutes with translations and reflections, preserves closedness, and changes the sign of all tests in (G); the six Wick polynomials are even. Thus it also preserves Ξ_μ. The independent reflection and reconstruction results in [4, Sections 7–8], together with Lemma B.1, Section B.2, and Proposition D.1, give (D2); the invariant-projection identity in its Section 7.3 gives (D3). Proposition C.1 gives the displayed constant in (D4). Approximation by bounded local functions gives its last assertion. The tensor estimates and reflection argument in [4, Sections 5.3 and 27], and (D4), give (D5).

For clarity, the numerical implication needed in (D6) is recalled. Write $c_m = c_{m,\mathrm{ix}}^{\mathrm{inf}}$ and $d_m = \mu_\xi(\mathrm{Wk}_{m,\mathrm{ix}})$. The Gaussian bound implies centering, and hence

$$e^{-4}\kappa_4^\xi(g_{\mathrm{ix}}) = c_4 + d_4 - 3(c_2 + d_2)^2.$$

Since $|d_m| \leq \mathsf{u}_{m,\mathrm{ix}}^{\mathrm{inf}}$, the first display gives

$$|e^{-4}\kappa_4^\xi(g_{\mathrm{ix}}) - b(g_{\mathrm{ix}})| \leq (8C_U + 3C_U^2)e^\sigma = \tfrac{1}{2}C_{\bar\kappa}e^\sigma.$$

Similarly, $|S_2^{\xi}(g_{\mathrm{ix}}, g_{\mathrm{ix}}) - \|g_{\mathrm{ix}}\|^2| \le 2C_U e^{2-\mathfrak{e}_2}$. Both bounds imply the stated weaker bounds with $C'_{\bar\kappa}$. Lemma F.4 gives the two lower bounds in (D6), and Euclidean invariance gives its last assertion. □

F.3. **The three-probe implication.**

Lemma F.4. Fix the probe β, with support in a ball of radius r_β, put $a = d^*\beta$, $f = da$, and suppose $b(f) > 0$. Set $g_0 = f$, $g_\pm = f \pm \tau_{x_0} f$, and $d_0 = |x_0| - 2r_\beta > 0$. Suppose

$$\frac{6}{\bar m^2}\|a\|_\infty^4 \left(\frac{4\pi r_\beta^3}{3}\right)^2 \left(\bar m^2 + (\bar m + d_0^{-1})^2\right) \frac{e^{-2\bar m d_0}}{4\pi^2 d_0^2} \le \frac{b(f)}{12}. \tag{22}$$

Let λ be a translation-invariant law with (G). If

$$|\kappa_4^\lambda(g_{\mathrm{ix}}) - e^4 b(g_{\mathrm{ix}})| \le C'_{\bar\kappa} e^{4+\sigma} \quad (\mathrm{ix} = 0, +, -), \qquad C'_{\bar\kappa} e^{\sigma} \le b(f)/10,$$

then

$$\mathrm{Ext}_{x_0}^\lambda(f) \ge \tfrac{1}{2} e^4 b(f), \qquad \kappa_4^\lambda(f) \ge \tfrac{9}{10} e^4 b(f) > 0.$$

■

Proof. Write $m = \bar m$, $T_a = S_m i\gamma\cdot a$, and $\mathsf{b}(a) = -6\,\mathrm{Tr}\, T_a^4 = b(f)$. These products are trace class; the Fourier symbol of S_m is $(m - i\gamma\cdot p)/(m^2 + |p|^2)$ and its entries lie in L^4, as do the entries of the compactly supported multiplier $i\gamma\cdot a$. The Schatten estimate used in [4, Section 17, equation (17.4)] therefore gives $T_a \in \mathfrak{S}_4$ and permits cyclic permutation under each fourfold trace. For another real test a' define

$$\mathsf{b}(a, a, a', a') = -4\,\mathrm{Tr}(T_a^2 T_{a'}^2) - 2\,\mathrm{Tr}(T_a T_{a'} T_a T_{a'}).$$

Expanding the two fourth powers and grouping the four cyclic words of type a, a, a', a' and the two of type a, a', a, a' gives

$$\mathsf{b}(a + a') + \mathsf{b}(a - a') = 2\mathsf{b}(a) + 2\mathsf{b}(a') + 12\mathsf{b}(a, a, a', a').$$

All these quantities are real. Indeed, the matrix exchanging the two Pauli blocks anticommutes with each γ_j; conjugating an adjoint product by this matrix reverses its factors under the trace. The three words above are cyclically equivalent to their reversals. Translations conjugate T_a unitarily, so $\mathsf{b}(\tau_x a) = \mathsf{b}(a)$.

We record the decay estimate that gives the role of the separation condition. If the supports of a, a' have distance $d > 0$, put $N = 1_{\mathrm{supp}\, a} S_m 1_{\mathrm{supp}\, a'}$ and let N' have the two supports reversed. The explicit Dirac kernel is

$$S_m(u) = \left[m + \frac{\gamma\cdot u}{|u|}(m + |u|^{-1})\right] \frac{e^{-m|u|}}{4\pi|u|}.$$

Consequently both Hilbert–Schmidt norms satisfy

$$\|N\|_{\mathrm{HS}}^2,\ \|N'\|_{\mathrm{HS}}^2 \le |\,\mathrm{supp}\, a|\,|\,\mathrm{supp}\, a'| \left(m^2 + (m + d^{-1})^2\right) \frac{e^{-2md}}{4\pi^2 d^2}.$$

This follows by integrating $\mathrm{tr}\, S_m(u)^2 = 4(m^2 + (m + |u|^{-1})^2) e^{-2m|u|}/(4\pi|u|)^2$, which decreases with $|u|$. Since $\|S_m\| = m^{-1}$ and $\|\gamma\cdot a\| = \|a\|_\infty$, insertion of the two support projections yields

$$\|T_a T_{a'}\|_{\mathrm{HS}} \le m^{-1}\|a\|_\infty \|a'\|_\infty \|N\|_{\mathrm{HS}},$$

and the analogous bound with a, a' exchanged. Cauchy–Schwarz for the two traces, whose coefficients add to six, gives

$$|\mathsf{b}(a, a, a', a')| \le \frac{6}{m^2}\|a\|_\infty^2 \|a'\|_\infty^2 |\,\mathrm{supp}\, a|\,|\,\mathrm{supp}\, a'| \left(m^2 + (m + d^{-1})^2\right) \frac{e^{-2md}}{4\pi^2 d^2}.$$

With $a' = \tau_{x_0} a$, condition (22) bounds this by $b(f)/12$.

Fourth-order polarization and translation invariance give

$$\mathrm{Ext}_{x_0}^\lambda(f) = 2\kappa_4^\lambda(f) - \tfrac{1}{4}(\kappa_4^\lambda(g_+) + \kappa_4^\lambda(g_-)).$$

The same polarization of b, followed by the three assumed error bounds, therefore gives

$$\mathrm{Ext}^{\lambda}_{x_0}(f) \geq e^4\,(b(f) - 3\mathsf{b}(a,a,\tau_{x_0}a,\tau_{x_0}a)) - \tfrac{5}{2}C'_{\bar\kappa}e^{4+\sigma} \geq \left(\tfrac{3}{4} - \tfrac{1}{4}\right) e^4 b(f).$$

The assertion for $\kappa^{\lambda}_4(f)$ follows directly from the error bound at $g_0 = f$ and $C'_{\bar\kappa}e^{\sigma} \leq b(f)/10$. □

Definition F.5. At level j the chamber spinor space has mesh $\eta_j = L^{-j}$ and inner product $\eta_j^3 48^{-1}\sum_{c,g}\bar u(c,g)v(c,g)$. The unit spinor space has the counting inner product. The massless operator is

$$D_A u(x) = -\eta_j^{-1}\sum_{i=0}^{3}\frac{w_i}{2}(\rho - \gamma_{\widehat{d_i(x)}})(U_A(x,x^{(i)})u(x^{(i)}) - u(x)).$$

Its connection phases on external hops are $U_A(x,x^{(3)}) = e^{ie_j\eta_j A(c,c_3)}$ and the internal phases are one. Thus this D_A contains no mass; it is the operator D_A of Definition E.1 after subtracting its scalar lattice mass and applying the indicated mesh dilation. Write

$$\nabla_A^*\nabla_A = \sum_{i=0}^{3} w_i(1 - R_i(A)), \qquad (R_i(A)u)(x) = U_A(x,x^{(i)})u(x^{(i)}).$$

This convention makes $\mathrm{Re}(u, D_A u) = \frac{\rho}{2\eta_j}(u, \nabla_A^*\nabla_A u)$.

The cluster injection and average are $(Jv)(c,g) = v(c)$ and $J^*u(c) = 48^{-1}\sum_g u(c,g)$. Let $\tau A(c,c')$ be the mean of the oriented bond sums over the six coordinate-order paths, and let $\tau_j A(z,c)$ be the sum of these means along the nested block centres from c to z. Put

$$\varphi_A(z,c) = e^{ie_j\eta_j\tau_j A(z,c)}, \qquad (Q_j(A)v)(z) = \eta_j^3\sum_{c\in\Delta_z}\varphi_A(z,c)v(c),$$

$$(Q_j^T(-A)v')(c) = \varphi_A(z,c)^{-1}v'(z), \qquad P_j(A) = Q_j^T(-A)Q_j(A).$$

Here $\Delta_z = \{c : |c-z|_\infty < 1/2\}$. The next average $Q(A)$ is given by the same formula on the unit lattice, with blocks of side L and normalization L^{-3}. For real A one has $Q_j^T(-A) = Q_j(A)^*$. All averages and injections commute with constant spin matrices, and $Q_j(A)Q_j^T(-A) = 1$, $J^*J = 1$.

Set

$$b = 1, \quad b_L = b/L, \quad b_- = b(1-L^{-1}), \quad b_j = \frac{b(1-L^{-1})}{1-L^{-j}} \quad (j \geq 1).$$

For $j \geq 1$, on the invertibility domains of [4, Sections 9 and 11], define

$$S_j(A) = (D_A + \mu_j + b_j JP_j(A)J^*)^{-1}, \qquad H_j(A) = b_j S_j(A)JQ_j^T(-A),$$

$$D_j(A) = b_j - b_j^2 Q_j(A)J^*S_j(A)JQ_j^T(-A), \qquad \bar H_j(A) = \mathsf{C}^{-1}H_j(-A)\mathsf{C}, \quad \mathsf{C} = i\gamma_2.$$

The matrix C is real. These are the exact coarse operators in [4, Section 11.2, equations (11.8)–(11.9)]. At level zero the field maps $H_0(A), \bar H_0(A)$ are identities and the corresponding density uses $D_A + \mu_0$.

For distinct sites the covariant Hölder quotient is

$$(\delta_{\alpha,A}v)(x,x') = \frac{e^{ie_j\eta_j\tau A(c,c')}v(x') - v(x)}{d_X(x,x')^{\alpha}}, \qquad d_X(x,x') = \max(|c-c'|_\infty, \eta_j).$$

The exponent is fixed with $1-\varepsilon < \alpha < 1$, where $0 < \varepsilon \leq 1/200$ is the exponent of the RG estimates. Insertion into an activity means $\psi = H_j(A)\Psi$, $\bar\psi = \bar H_j(A)\bar\Psi$, $\chi = \delta_{\alpha,A}\psi$ and $\bar\chi = \delta_{\alpha,-A}\bar\psi$. We write $H_j^{\#}(A)\Psi$ for this full set of inserted fields. The photon minimizer, written H_jA' with no connection argument, is a different map: it is the minimizer of [4, Proposition 10.1]. ■

The following identities explain the operator references in the sum rule. For real A, the hop involutions are self-adjoint, so [4, equations (9.2)–(9.4)] gives $D_A^* = \gamma_5 D_A \gamma_5$ and the displayed real-part identity. With $\Sigma_3 = \mathrm{diag}(1_2, -1_2)$, D_A commutes with Σ_3, and γ_5 exchanges its two spin blocks. Real gauge changes conjugate D_A by their diagonal unitary phases. A covariantly constant spinor is annihilated by D_A. For real A, A', the four-hop formula gives

$$\|D_A - D_{A'}\| \le 6 e_j \|A - A'\|_\infty .$$

The nested paths have total length at most $3/2$ after multiplication by η_j, as in [4, Section 9.5]; hence

$$\|(Q_j(A) - Q_j(A'))J^*\| \le \tfrac{3}{2} e_j \|A - A'\|_\infty .$$

Definition F.6. At zero external slot, put

$$\begin{aligned}\mathscr{F}_j[\varepsilon^{\mathrm{v}}, m, E](A;\Psi) = {} & \det D_j(0)^{-1} \exp\Big\{ -\langle \bar\Psi, D_j(H_j A)\Psi\rangle - m\langle \bar\psi, \psi\rangle \\ & + \sum_X E(X; H_j A, H_j^{\#}(H_j A)\Psi) - \varepsilon^{\mathrm{v}} \ell_j^3 \Big\}.\end{aligned}$$

The mass pairing is the chamber pairing and the first pairing is the unit-spinor pairing. At $j = 0$ this means

$$\mathscr{F}_0(A;\Psi) = \det(D_0 + \bar m \eta)^{-1} e^{-\langle \bar\Psi, (D_A + \bar m\eta + m_{\mathrm{in}})\Psi\rangle^{\mathrm{ch}} - \varepsilon^{\mathrm{v}}_{\mathrm{in}} n^3}.$$

For a unit spinor v, its dilation is $v_L(y) = L^{-1} v(y/L)$. Write $\mathcal{A}_j(B) = Q(B)$ for $j \ge 1$ and $\mathcal{A}_0(B) = Q(B)J^*$. The exact block transformation is the linear map

$$\begin{aligned}(\mathbf{T}_j F)(A';\Psi') = c_j \int_{\mathrm{Box}_j} \mu_I(d\zeta_j) \int \mathrm{D}\Psi \, F(A_j;\Psi) \\ \times \exp\Big\{ -b_L \Big\langle e^{-ie_j\omega_j}\bar\Psi'_L - \mathcal{A}_j(-\mathscr{B}_j)\bar\Psi, e^{ie_j\omega_j}\Psi'_L - \mathcal{A}_j(\mathscr{B}_j)\Psi \Big\rangle_L \Big\}.\end{aligned}$$

It is equation (13.19) of [4], at zero external slot, and its integrated marginal is (13.20). Its photon field is $A_j = A_j^{\min}(A') + \mathrm{Sq}_j \zeta_j$, $\mathscr{B}_j = H_j A_j^{\min}(A')$, $\mathrm{Sq}_j = C C_j^{1/2}$, and its real gauge function is ω_j from (10.29). Here C is the coordinate completion of (10.21)–(10.22), and $C_j^{1/2}$ is the positive square root of the covariance, as in (10.26). The coefficient c_j is the reciprocal outgoing free Berezin normalization; thus it depends on the scales and b, and not on the incoming mass or activity. The standard Gaussian fluctuation law is μ_I and

$$\mathrm{Box}_j = \{\zeta_j : |\zeta_j(\Upsilon)| \le p_{0,j} \text{ for all } \Upsilon\}, \qquad p_{0,j} = (-\log e_j)^{p_0}.$$

All these definitions agree with (13.15)–(13.20) there. ∎

F.4. **Comparison of two flows.**

Lemma F.7. Use the activity spaces, unit balls and RG maps of [4, Sections 12.1, 14.7 and 16.2], at zero external slot. The mass functional is complex linear, and

$$|m(E)| \le c_{\mathrm{fl}} e_j^{1/2} \|E\|_j, \qquad \|L(RE)^\natural\|_{j+1} \le c_{\mathrm{fl}} L^{-1/4+2\varepsilon} \|E\|_j .$$

For two unit-ball data at the same scale,

$$\|E_j^*(m,E) - E_j^*(m',E')\|_{j+1} \le e_j^{-1/2+2\varepsilon} |m - m'| + e_j^{\varepsilon} \|E - E'\|_j .$$

The second term is omitted at $j = 0$. Consequently, for two flows whose preceding data remain in those unit balls, the differences $x_j = m_j - m'_j$ and $Y_j = \|E_j - E'_j\|_j$ satisfy

$$\begin{aligned} x_{j+1} &= L(x_j + m(E_j - E'_j)), \\ Y_{j+1} &\le (c_{\mathrm{fl}} L^{-1/4+2\varepsilon} + e_j^{\varepsilon}) Y_j + e_j^{-1/2+2\varepsilon} |x_j|. \end{aligned}$$

∎

Proof. The mass estimate follows from [4, Proposition 12.1, equation (12.12)]; the chosen c_{fl} is larger than its coefficient. The second estimate is [4, Proposition 12.5, equation (12.62)]. For the correction, substitute $w_m = e_j^{3/4-8\varepsilon}$ and $w_E = e_j^{1/4-7\varepsilon}$ into [4, Proposition 14.7, equation (14.41)]. These are estimates for the actual zero-slot maps, as used in the fixed-point proof in its Section 16.3; no extension of an arbitrary activity to nonzero slots is needed. Subtract the two recursions (16.9)–(16.10). Their determinant activity is the same, so it cancels. □

F.5. **Observables of all terminal coordinates.**

Lemma F.8. Use the photon coordinates (A_*, ζ) and the hypotheses of [4, Proposition 22.1 and Theorem 22.2]. Thus $A_0^{\downarrow}(A + c_\theta, \zeta) + a$ is the reconstructed fine connection, m_* is the terminal Gaussian measure, and w_N is the normalized fine determinant density. For every bounded Borel function $\Phi(A, \zeta)$,

$$\nu^{\theta,a}\big(\mathbf{1}_{\mathfrak{G}}\Phi(A_*,\zeta)\big) = \int \mathsf{m}_*(dA)\,\chi_*(A)\int \prod_{j<k}\big(\mathbf{1}_{\mathrm{Box}_j}(\zeta_j)\,\mu_I(d\zeta_j)\big)\Phi(A,\zeta)\,w_N\big(A_0^{\downarrow}(A+c_\theta,\zeta)+a\big).$$

If a is real, the identity also holds for nonnegative Borel functions, with values in $[0,\infty]$. ■

Proof. Proposition 22.1 of [4] gives a linear bijection from (A_*, ζ) to the fine axial slice, and identifies its Gaussian measure with $\mathsf{m}_*(dA)\prod_j \mu_I(d\zeta_j)$. The event $\mathfrak{G}$ is exactly the terminal cutoff together with the fluctuation boxes. Apply this change of variables to the determinant density defining $\nu^{\theta,a}$. The finite-dimensional change-of-variables formula proves the identity for every bounded Borel Φ, rather than only for a function of A_*. If a is real, all factors except Φ are nonnegative, and monotone convergence gives the last assertion. □

Appendix G. Measurable component laws

Definition G.1. Let $\mathcal{P}(\mathcal{S}')$ carry the sigma algebra generated by $\lambda \mapsto \lambda(B)$, $B \in \Sigma$. For a translation-invariant law μ, write

$$\mu = \int \mu_\xi\,\nu(d\xi)$$

for the ergodic decomposition of [5, Theorem A and Sections 5–7]. Here the index space is $\mathcal{P}(\mathcal{S}')$, $\mu_\xi = \xi$, and ν is the distribution of the conditional law given the translation-invariant sigma algebra. In particular, the symbol μ_ξ denotes a probability kernel, not a choice made separately for each observable. ■

Lemma G.2. For every nonnegative Borel Φ, the map $\xi \mapsto \mu_\xi(\Phi)$ is measurable and $\mu(\Phi) = \int \mu_\xi(\Phi)\,\nu(d\xi)$. For integrable complex Φ, the same assertion follows by decomposing its real and imaginary parts into positive and negative parts. ■

Proof. The claims follow first for indicators from the kernel property, then for simple functions and finally by monotone convergence. □

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